

Nullexit: Unified Project Document

Omar Alaaeldein

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1 Project Directory Tree

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|-- .gitignore
|-- .host_ips
|-- LICENSE
|-- README.md
|-- Recover-Gateway.applescript
|-- Toggle-Gateway.applescript
|-- agent.md
|-- benchmarks
|   |-- benchmark.sh
|   `-- test_load.py
|-- black_list.txt
|-- cloud-init.yaml
|-- devref.md
|-- diagrams.md
|-- docker
|   `-- tor
|       |-- Dockerfile
|       `-- entrypoint.sh
|-- docker-compose.yml
|-- launchd
|   `-- com.nullexit.wake-recovery.plist
|-- post-rules.txt
|-- recover.sh
|-- scripts
|   |-- Dockerfile.routing-fix
|   |-- Dockerfile.rule-compiler
|   |-- Toggle-Gateway.bat
|   |-- common.sh
|   |-- crypto.sh
|   |-- diagnose-host-leak.sh
|   |-- dns-proxy.py
|   |-- fix-docker-bridge-collision.sh
|   |-- fix-ssh-delay.sh
|   |-- generate_tex.py
|   |-- logger
|   |   |-- go.mod
|   |   `-- main.go
|   |-- pf.conf
|   |-- recover-linux.sh
|   |-- reinstall-host-tailscale.sh
|   |-- routing-fix.sh
|   |-- rule-compiler
|   |   |-- go.mod
|   |   `-- main.go
|   |-- setup-common.sh
|   |-- setup-linux.sh
|   |-- toggle-linux.sh
|   |-- unlock-files.sh
|   `-- watcher.sh
|-- setup.sh
```

```
|-- toggle.sh  
|-- tor-bridges.txt  
`-- white_list.txt
```

2 README.md

```
1 # nullexit: Tailscale + Cloudflare WARP Docker Gateway
2
3 > **Last updated:** July 11, 2026 [Architecture & Flow Diagrams ](./diagrams.md) [Full Dev Reference ](./
  devref.md)
4
5 **nullexit** is a chained network gateway that routes all Tailscale exit-node traffic through a Cloudflare WARP
  VPN tunnel double-encrypting every packet, hiding your ISP metadata, and providing network-wide DNS ad-
  blocking (AdGuard Home) and kernel-level IP threat blocking (`ipset`/`iptables`) for every device on your
  mesh.
6
7 ---
8
9 ## 1. Prerequisites
10 - Docker and Docker Compose installed.
11 - A Tailscale account.
12 - **macOS:** Install the standalone Tailscale CLI via Homebrew (`brew install tailscale`) and register `
  tailscaled` as a system service (`brew services start tailscale`). No `.app` GUI install required.
13 - **OS & Python:** Developed and tested against **macOS 26.5.2**. The project has zero external Python
  dependencies (no `requirements.txt`) and explicitly targets the default Apple-provided **Python 3.9.6** (`/
  usr/bin/python3`). While the scripts could function on newer versions, `nullexit` explicitly avoids
  experimenting with or modifying the built-in system Python environment to prevent unexpected OS-level side
  effects.
14
15 ## 2. Installation & Setup
16
17 Run the interactive setup script for your OS. It downloads `wgcf`, generates fresh Cloudflare WARP WireGuard
  keys, prompts for a Tailscale Auth Key, and writes your `.env`.
18
19 ```bash
20 ./setup.sh # macOS
21 ./scripts/setup-linux.sh # Linux
22 ```
23
24 ### Reference `.env`
25 ```env
26 WIREGUARD_PRIVATE_KEY=<Generated>
27 WIREGUARD_PUBLIC_KEY=<Generated>
28 WIREGUARD_ADDRESSES=<Generated>
29 TS_AUTHKEY=tskey-auth-...
30 GATEWAY_RULE_PROFILE=medium # light | medium | heavy
31 # Safe MSS for double-tunneled traffic (Tailscale + WARP). Change to 1180 if you experience slow speeds and
  have a healthy path.
32 GATEWAY_MSS=1120
33 # Set to false to run as a 'Headless Server' (Docker acts as an exit node, but the host Mac/Linux's own
  internet is NOT hijacked).
34 GATEWAY_HIJACK_HOST=true
35 GATEWAY_USE_EXIT_NODE=true
36 WARP_FAIL_THRESHOLD=6 # consecutive warp=off polls before auto-shutdown (default 6 = 30s)
37 KILL_SWITCH=false # enforce strict PF lock that breaks SSH if VPN fails
38 # On campus/enterprise networks (WPA2-Enterprise / 802.1x), AP Client Isolation blocks direct
39 # LAN traffic between devices. Tailscale attempts a local P2P upgrade anyway, which causes
40 # macOS gvproxy to SNAT-mangle the reply's source port poisoning the phone's WireGuard
41 # endpoint and blackholing 100% of its traffic. Setting this to false drops those local
42 # hole-punch packets so the phone safely falls back to Tailscale DERP relays.
43 # watcher.sh auto-detects 802.1x and AP isolation on every Wi-Fi change and overrides
44 # this setting automatically you only need to set this manually if auto-detection fails.
45 TAILSCALE_ALLOW_LAN_P2P=false # auto-managed; true only on trusted hotspots/home networks
46 ADGUARD_PASSWORD=nullexit
47 BLOCKED_COUNTRIES="kp il" # dynamically block country IP ranges via ipdeny.com
48 ```
49
50 ## 3. Deploy the Gateway
51
52 **macOS (Recommended):**
53 ```bash
54 ./toggle.sh
55 ```
56 Running it again stops the gateway and restores your network. Handles the full lifecycle: Colima VM, containers
  , DNS hijacking, Tailscale exit-node routing, rule compilation, and sleep prevention.
57
58 **Linux / Manual:**
59 ```bash
60 docker compose up -d
61 ```
62
63 ## 4. Post-Deployment
64
65 ### Approve the Exit Node
```

```

66 1. Go to [Tailscale Admin Machines](https://login.tailscale.com/admin/machines).
67 2. Find the `tailscale` device `...` **Edit route settings** enable as Exit Node.
68
69 ### Verify Traffic Flow
70 On any Tailscale client device, select this gateway as the exit node, then visit [api.ipify.org](https://api.
71 ipify.org) or run:
72 ```bash
73 curl -s https://www.cloudflare.com/cdn-cgi/trace | grep -E '^(warp|ip|colo)='
74 # expect: warp=on
75 ```
76 ### AdGuard DNS Setup
77
78 To ensure client devices correctly route DNS through the AdGuard container (and do not bypass it via `
79 tailscaled`'s internal resolver), you **must** configure your Tailscale Admin Console:
80 1. Go to the **DNS** tab in the [Tailscale Admin Console](https://login.tailscale.com/admin/dns).
81 2. Enable **MagicDNS**.
82 3. Under **Nameservers**, add a **Custom** nameserver and enter the gateway's Tailscale IPv4 address (run `
83 tailscale ip -4` on the gateway machine to find it, e.g., `100.x.x.x`).
84 4. Click the **three dots (...)** next to the nameserver you just added and enable **"Use with exit node"**.
85 5. Delete any other Global Nameservers (like Google or Cloudflare).
86 6. Toggle ON **Override local DNS**.
87
88 Traffic will now be correctly forced into the AdGuard sinkhole, and mobile devices will be blocked from
89 bypassing it via encrypted DNS (DoH/DoT).
90
91 > [!WARNING]
92 > **Disable "Private DNS" on mobile devices.** Android's Private DNS (DoT/DoH) bypasses port 53 interception
93 entirely. Go to `Settings Connections More connection settings Private DNS Off`.
94
95 > [!WARNING]
96 > **Bandwidth doubling:** Streaming 1GB of video on a remote device consumes 1GB download + 1GB upload on the
97 host's network. Be mindful on metered connections.
98
99 ### AdGuard Dashboard
100 Navigate to `http://<gateway-tailscale-ip>:3000` credentials: **admin** / **nullexit**.
101
102 ### Access Any Mesh Device From Anywhere (SFTP/SSH)
103
104 Once the gateway exit node is online, **any device on your Tailscale mesh can reach any other mesh device,
105 anywhere in the world** as long as both devices are on the mesh and the target device has an SSH server
106 running. No port forwarding, no public IPs, no firewall holes. You can SSH into any device using its
107 MagicDNS hostname (e.g. `ssh username@my-macbook`) or its assigned Tailscale IP (`100.x.x.x`).
108
109 > **Security Tip (Zero Local Attack Surface):** It is highly recommended to disable macOS's native "Remote
110 Login" and "Screen Sharing" in System Settings to prevent exposing those ports (22, 5900) to physical Wi-Fi
111 networks (which scanners can fingerprint). The gateway enforces `tailscale up --ssh=true` and the `pf`
112 Kill-Switch automatically blocks local Wi-Fi access to these ports. Since both MagicDNS and Tailscale IPs
113 route through the exact same encrypted tunnel, they are equally securebut **MagicDNS is recommended simply
114 for ease of use.**
115
116 ### Extreme OPSEC: The Secret Tor-over-WARP Proxy
117 `nullexit` includes a completely isolated, headless Tor infrastructure proxy designed for terminals and hacking
118 tools (e.g., `curl`, `sqlmap`, `nmap`).
119 To protect against local malware fingerprinting:
120 1. **Procedural Ports:** The proxy ports are randomized into the ephemeral range (10000-65000) on every single
121 boot using a strict kernel socket collision-avoidance check (`netstat`).
122 2. **Honey-Port Tripwire:** The gateway spawns a fake trap port. If any malware attempts a sequential port-scan
123 on `localhost` to find the proxy, it trips the Honey-Port, instantly triggering a macOS desktop
124 notification and a critical log alert.
125
126 > [!NOTE]
127 > **Threat Model limitation:** Port randomization and the Honey-Port only defeat blind, generic port scanners.
128 They do not protect against targeted local malware that runs `lsof -i` or reads the `/tmp/nullexit-ports.
129 env` file directly. To actually prevent malicious local processes from reaching the proxy, you must run a
130 host-level outbound firewall (like LuLu 4.3.0+ or Little Snitch) with **localhost/loopback filtering
131 explicitly enabled** to gate access by process identity.
132
133 To use the Tor proxy, run `cat /tmp/nullexit-ports.env` to find your **$TOR SOCKS_PORT** for the current session,
134 and point your hardened browser (LibreWolf/Tor Browser) or CLI tool to **127.0.0.1:$TOR SOCKS_PORT**.
135
136 #### Hardening: Using obfs4 Tor Bridges
137 If you want to disguise your Tor traffic from the WARP exit node provider, you can optionally enable Tor
138 bridges:
139 1. Obtain private `obfs4` bridge lines from [bridges.torproject.org](https://bridges.torproject.org) or the
140 official Telegram bot.
141 2. Create a file named `tor-bridges.txt` in the root of the `nullexit` folder and paste your bridge lines
142 inside it (one per line).
143 3. Set **TOR_USE_BRIDGES=true** in your **`.env` file and restart the gateway. The proxy will refuse to start if
144 the bridges file is empty or missing.

```

```

120 ##### Example: Browse your Android phone's files from your Mac
121
122
123 1. **On your Android phone**, install [Termux](https://termux.dev/) and [Termux:Boot](https://f-droid.org/
124     packages/com.termux.boot/) from F-Droid.
125 2. In Termux, install and start the SSH server:
126     ```bash
127     pkg install openssh
128     sshd
129     ```
130     Termux defaults to **port 8022** (ports below 1024 require root on Android).
131 3. Find your phone's Tailscale IP:
132     ```bash
133     tailscale ip -4    # e.g. 100.87.42.87
134     ```
135 4. **Stop Android from killing Termux** do ALL of these (Samsung One UI is especially aggressive):
136     - **Settings Apps Termux Battery Unrestricted**
137     - **Device Care Battery Background usage limits Sleeping apps / Deep sleeping apps** remove Termux if
138       listed
139     - In Termux, run `termux-wake-lock` to hold a persistent wakelock
140 5. **On your Mac/PC**, open [Cyberduck](https://cyberduck.io/) (or any SFTP client: WinSCP, FileZilla, FE File
141     Explorer on iOS) and connect to:
142     - **Server:** `100.87.42.87` (your phone's Tailscale IP)
143     - **Port:** `8022`
144     - **Protocol:** SFTP (SSH File Transfer Protocol)
145     - **Username:** (your Termux username, usually `u0_a123` run `whoami` in Termux to check)
146     - **Password:** (your Termux password set with `passwd` in Termux if you haven't already)
147
148 That's it you can now browse, download, and upload files to your phone from any device on the mesh, no matter
149 where either device is physically located. This works identically for any mesh device: Mac Mac, Windows
150 Linux, phone NAS, etc.
151
152 ---
153
154 ## 5. Multi-Layer Firewall & Kill-Switch
155
156 ### Layer 1 DNS Sinkhole (AdGuard Home)
157 Blocks ads and tracking domains before they are downloaded. In automated Lighthouse tests on ad-heavy sites:
158 - **Time to Interactive:** ~22% faster (51.0s 41.8s)
159 - **Speed Index:** ~20% faster (15.3s 12.8s)
160
161 Customize blocking via `black_list.txt` and `white_list.txt`. Rule profiles (`light` / `medium` / `heavy`) are
162 set in `.env`.
163
164 ### Layer 2 Kernel IP Firewall (`ipset`/`iptables`)
165 On every startup, the `rule-compiler` fetches threat-intelligence feeds (Spamhaus DROP, Feodo Tracker, Emerging
166 Threats, CINS) and compiles **~16,700 unique malicious IPs/CIDRs** into the kernel `FORWARD` chain
167 blocking both outbound C2 connections and inbound attack traffic. Zero configuration required.
168
169 ### Layer 3 Geo-IP Blocking
170 Dynamically blocks all traffic to and from specific countries using live IP ranges from `ipdeny.com`.
171 To add countries to your blocklist, open your `.env` file and set the `BLOCKED_COUNTRIES` variable with 2-
172 letter ISO country codes (e.g. `BLOCKED_COUNTRIES="kp il cn ru"`), then restart the gateway.
173
174 ### Layer 4 macOS PF Kill-Switch
175 A native macOS Packet Filter (`pf`) kill-switch that enforces a strict default-deny policy on your physical Wi-
176 Fi interface (`en0`). When `KILL_SWITCH=true` is set in your `.env`, it drops all outgoing traffic except
177 the Cloudflare WARP endpoints and Tailscale DERP relays.
178
179 - **Failsafe Design:** If the VPN tunnel crashes or Docker dies, your Mac completely loses internet access
180   rather than leaking your real IP.
181 - **Remote-Access Safe:** Carefully designed to whitelist Tailscale's control plane (`192.200.0.0/24`), `utun*`
182   tunnel traffic, and local LAN subnets. This guarantees that your remote SSH sessions and local AirDrop
183   will survive even when the kill switch engages.
184 - **Setup Requirement:** The toggle scripts need permission to manipulate the network and firewall in the
185   background without prompting you for a password. You must configure your passwordless sudo config by
186   running exactly:
187     ```bash
188     echo "$USER ALL=(root) NOPASSWD: /sbin/pfctl, /usr/sbin/networksetup, /usr/bin/dscacheutil, /usr/bin/killall,
189     /usr/bin/pkill, /bin/kill, /sbin/route, /sbin/ifconfig, /usr/bin/true, /opt/homebrew/bin/brew, /usr/local/
190     bin/brew, /usr/bin/python3 -I $PWD/scripts/dns-proxy.py, /opt/homebrew/bin/python3 -I $PWD/scripts/dns-
191     proxy.py, /usr/local/bin/python3 -I $PWD/scripts/dns-proxy.py" | sudo tee /etc/sudoers.d/nullexit
192     ```
193
194 ---
195
196 ## 6. Toggle Scripts

```

```

180
181 ### macOS `toggle.sh`
182 Fully automated lifecycle script. Also supports a native `.app` launcher:
183 ```bash
184 osacompile -o "Toggle Gateway.app" Toggle-Gateway.applescript
185 ```
186
187 Optional `.env` settings:
188 - `GATEWAY_BYPASS_PING=true` proceed even if pre-flight connectivity checks fail.
189 - `GATEWAY_USE_EXIT_NODE=false` DNS-only mode, skip exit-node routing.
190 - `GATEWAY_HIJACK_HOST=false` skip DNS hijacking on the host (provides VPN/adblocking only to external
    Tailscale peers).
191 - `GATEWAY_MSS=1360` override TCP Maximum Segment Size for the tunnel (default is calculated automatically).
192 - `HOST_LEAK_PROBE=false` disable background host-egress leak probe (default: true).
193 - `HOST_MTU=1200` override the host Tailscale interface MTU. Defaults to 1200 to fit inside the 1280-byte WARP
    tunnel.
194 - `KILL_SWITCH=true` enforce strict network lock that breaks SSH if the VPN fails.
195 - `STOP_COLIMA_ON_EXIT=true` fully shut down the Colima VM on toggle-off (saves battery on dedicated hosts).
196 - `TAILSCALE_ALLOW_LAN_P2P=true` allow direct Tailscale LAN P2P connections between devices on the same
    network.
197 - **Default: `false`** On campus or enterprise Wi-Fi (WPA2-Enterprise / 802.1x), AP Client Isolation blocks
    local device-to-device traffic. When Tailscale still attempts a local P2P upgrade, macOS's `gvproxy` layer
    SNAT-mangles the reply's source port. WireGuard's endpoint roaming treats this as a legitimate address
    change and updates the phone's outbound path to the newly-mangled port which has no listener. This
    blackholes 100% of the phone's traffic while the DERP relay path is abandoned. Setting this to `false`
    preemptively drops all RFC1918-destined Tailscale UDP from the gateway so the phone receives a clean
    failure and safely falls back to DERP.
198 - **Auto-managed:** `watcher.sh` automatically detects WPA2-Enterprise via `system_profiler SPAirPortDataType`
    (the `airport` binary was removed in macOS 14.4) and probes for AP isolation by pinging the default
    gateway (`route -n get 1.1.1.1`) on every network change. It writes the detected value to `lan_p2p_detected`
    in the repo root, which `routing-fix.sh` re-reads every 30 seconds **no restart required**
    Only set this manually if auto-detection fails. See `devref.md 36` for the full SNAT endpoint poisoning
    analysis and `devref.md 40` for the `airport` removal background. *(Note: Direct P2P between the gateway
    container and the host Mac itself is always permitted via the Docker bridge to bypass DERP relays,
    regardless of this setting).*
199 - Set to `true` on trusted home networks or Windows/mobile hotspots (no AP isolation) for a faster, lower-
    latency direct path.
200 - `WARP_FAIL_THRESHOLD=3` number of consecutive failed checks before forcing a gateway teardown (default: 6
    checks = 30s).
201 - `WARP_ENDPOINT_1` / `WARP_ENDPOINT_2` override the Cloudflare WARP WireGuard endpoints.
202
203 ### Linux `scripts/toggle-linux.sh`
204 ```bash
205 ./scripts/toggle-linux.sh # toggle ON/OFF
206 ./scripts/recover-linux.sh # recovery tool
207 ```
208
209 ### Sleep Prevention (macOS)
210 When the gateway is ON, `caffeinate -i` runs in the background to prevent idle system sleep, keeping Docker and
    all services alive even on battery. The display can still sleep normally.
211
212 ### Auto-Recovery (macOS post-wake / post-roam)
213 Install the launchd LaunchAgent so the gateway self-heals after sleep/wake and Wi-Fi roams:
214 ```bash
215 cp ./launchd/com.nullexit.wake-recovery.plist ~/Library/LaunchAgents/
216 sed -i '' "s|_NULLEXIT_HOME_|$(pwd)|" ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
217 launchctl load -w ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
218 ```
219 See `devref.md 11.16` for the full deep dive.
220
221 ---
222
223 ## 7. Networking Notes
224
225 - **Port conflicts:** None. Tailscale uses dynamic UDP ports; WARP uses outbound UDP:2408. The gateway binds
    `0.0.0.0:41642/udp` on the host for Tailscale's WireGuard listener this is intentional and required to
    receive inbound hole-punch packets from peers on the internet. All other internal ports (AdGuard :5335,
    SOCKS5 :1080) are bound to `127.0.0.1` only and are not reachable from the LAN.
226 - **Tailscale P2P vs DERP:** On the same Wi-Fi, Tailscale establishes a fast P2P connection. On cellular (CGNAT
    ), it always falls back to a DERP relay (~80-200ms). See `devref.md 5.1`.
227 - **AirDrop / AirPlay:** Unaffected the gateway only touches standard Wi-Fi/Ethernet interfaces, not `awdl0`.
228 - **VPN coexistence:** Do **not** run a local VPN client (WARP, Mullvad, NordVPN) simultaneously with the exit
    node. Both fight for the default route and will blackhole your connection.
229 - **Banking & financial sites:** May intermittently block logins through WARP. Banks use anti-fraud databases
    that flag datacenter IP ranges (like Cloudflare's `104.28.x.x`) as VPN/proxy traffic. Success fluctuates
    depending on which WARP IP you land on. If a site blocks you, either disable the exit node temporarily for
    that session or use the bank's mobile app (which often uses different fraud detection).
230 - **MSS clamping (Bufferbloat):** Double-tunnelling adds ~120-160 bytes of overhead per packet. This is now
    automatically handled natively via the macOS `pf` firewall, which universally applies TCP MSS Clamping to
    all outbound packets to strictly prevent MTU fragmentation and bufferbloat. No manual `.env` configuration
    is required. Additionally, the host's Tailscale interface MTU is explicitly lowered to 1200 (configurable

```

```

via `HOST_MTU` in `.env`) to guarantee packets fit inside the container's WARP tunnel without fragmenting.
231 ---
232
233
234 ## 8. Privacy & Security Hardening
235
236 The stack shifts trust across four layers: WARP hides traffic from your ISP, AdGuard blocks tracking at DNS,
    Tailscale encrypts device-to-device, and `ipset` blocks known malicious IPs at the kernel.
237
238 For higher security, the gateway supports these drop-in upgrades:
239
240 | Layer | Default | Hardened |
241 |---|---|---|
242 | VPN Exit | Cloudflare WARP (US) | Mullvad (Swedish, no-logs, anonymous payment) |
243 | DNS Resolver | AdGuard via WARP | AdGuard via Mullvad DoH |
244 | Coordination | Tailscale (US) | Headscale (self-hosted) |
245 | Auth | Persistent OAuth keys | Ephemeral auth keys |
246 | Content | WireGuard asymmetric | WireGuard + Pre-Shared Keys (PSK) |
247
248 #### Python Runtime Airgap (Isolated Mode)
249 **Never** install third-party `pip` packages (like `requests` or `pyyaml`) into the Python environment that `
    nullexit` uses. All python scripts in this project (e.g., the `dns-proxy.py` daemon) are strictly
    engineered to run on the zero-dependency Python Standard Library. To strictly isolate the environment from
    supply-chain attacks, `nullexit` executes these scripts using **Python's Isolated Mode** (`python3 -I`).
    This deliberately blinds the execution environment to any malicious user-installed `pip` packages on the
    host, permanently air-gapping the gateway from PyPI.
250
251 See `devref.md 8` for the full threat model, Snowden programme analysis, and trust assumptions.
252
253 ---
254
255 ## 9. Debugging
256
257 - **`output.log`** All stderr from `toggle.sh`, `recover.sh`, `setup.sh`, and the rule-compiler is written
    here. The WARP Watcher (`WARP_FAIL_THRESHOLD`) and the Host Leak Probe (`HOST_LEAK_PROBE`) also write all
    their events here `LEAK`, `ROTATE`, `HOST-PROBE failed`, and WARP shutdown notices. Check this first.
258 - **`bash scripts/diagnose-host-leak.sh`** One-shot host-routing diagnostic. Runs 8 checks and classifies your
    state into one of four known scenarios (System Extension conflict, SOCKS5 fallback, IPv6 leak, route-table
    freeze) or OK, and prints the exact fix command. Check 5 uses `route -n get 1.1.1.1` to ask the kernel
    which interface real traffic actually resolves through (more accurate than `netstat -rn | grep default` on
    macOS). Pass `--fix` to apply the remediation automatically. Pass `--watch` (or `--watch 30`) to run a full
    baseline diagnostic then continuously monitor warp/IPv6/default-route for leaks, alerting on any state
    change.
259 - **`bash scripts/host-leak-probe.sh`** Sub-second host-egress prober (enabled automatically via `
    HOST_LEAK_PROBE=true` in `.env`). Polls `cdn-cgi/trace` every 300ms directly from the host NIC not via `
    docker exec` so it catches flash-leaks invisible to the in-container WARP Watcher. All state changes, curl
    errors, and probe timeouts land in `output.log`. Grep with `grep LEAK output.log`.
260 - **`bash scripts/fix-docker-bridge-collision.sh`** One-shot fix for Docker bridge IP conflicts. If your local
    Wi-Fi router assigns you an IP in the `172.17.0.0/16` range, it will violently collide with Docker's
    default internal bridge, permanently killing your internet. This script detects the collision and auto-
    patches your Colima VM to use a safe subnet (e.g. `10.200.0.1/24`).
261 - **`devref.md`** Complete architecture reference, routing deep-dives, and full resolved-issues log. Read this
    before modifying any routing or firewall logic.
262
263 > [!CAUTION]
264 > **Two infinite routing loops are possible.** (1) If a hotspot device providing internet to the gateway host
    is also* using the gateway as its exit node, packets bounce forever between the two. (2) When the host's
    default route is redirected to `utun*`, WARP endpoint packets can loop back into the tunnel. Both are
    documented in `devref.md 11.10` and `11.18` and are automatically handled by `toggle.sh`.
265
266 ---
267
268 ## 10. Unified Project Document (Quine)
269
270 The project includes a `generate_tex.py` script that compiles the entire source code, configurations, and
    documentation into a single, unified LaTeX document (`nullexit_unified.tex` / `.pdf`).
271
272 Funnily enough, this document acts as a "project-level quine." It encapsulates the entirety of its own source
    code, its documentation, and the exact Python code required to generate itself. It serves as a self-
    contained, long-term archiving strategy providing everything needed to reconstruct and understand the `
    nullexit` system from a single file.
273
274 ---
275
276 ## 11. Acknowledgements
277 - **[SyameimaruKoa](https://github.com/SyameimaruKoa):** Dual-stack TCP MSS clamping, `SIGHUP` state-tracking
    in the routing sidecar, and `TS_AUTH_ONCE` integration.
278
279 ## 12. License
280 GNU Affero General Public License v3. See [LICENSE](./LICENSE).
281

```

```

282 ## 13. Changelog
283
284 - **July 10, 2026** **Bug fixes:**
285 1. Resolved a race condition where Wi-Fi roaming caused the host IP to leak as the raw ISP IP (`132.x`) after
    every network switch. The WARP Watcher was firing nuclear shutdown while `recover.sh --post-wake` was
    still intentionally recreating the warp container. Fixed by adding a `/tmp/nullexit-warp-inhibit.marker`
    that pauses the watcher's failure counter during post-wake recovery. See `devref.md 25`.
286 2. Resolved a startup race condition where the pre-flight connectivity check `[1/3] Gateway reachable via
    Tailscale` would fail because the host's `tailscaled` had not yet propagated the netmap immediately after `
    tailscale up --reset`. Upgraded the check to a 5-attempt retry loop with a 1-second delay. See `devref.md
    26`.
287 3. Resolved a concurrency collision where the network changes from a nuclear `recover.sh` teardown triggered
    `watcher.sh` to run `recover.sh --post-wake` in parallel, corrupting container states. Implemented a shared
    lifecycle lock (`/tmp/nullexit-toggle.lock`) across both scripts to safely skip post-wake logic during
    teardowns. See `devref.md 27`.
288 4. Resolved an infinite auto-recovery loop after a tunnel failure shutdown. When the WARP Watcher triggered
    nuclear `recover.sh` to teardown the gateway, the active-state marker file `/tmp/nullexit-gateway-active.
    marker` was leaked on disk. The resulting network configuration changes from the teardown then triggered `
    watcher.sh`, which saw the marker and immediately restarted the gateway, looping indefinitely. Fixed by
    explicitly deleting the marker file at the start of nuclear `recover.sh`. See `devref.md 28`.
289 5. Resolved a pre-flight check false negative during cold boots where the `[1/3] Gateway reachable via
    Tailscale` check would fail. Because gluetun blocks UDP STUN traffic to the public internet, Tailscale is
    forced to use a slower DERP-assisted handshake which takes ~30-35 seconds on cold boots. Increased the
    retry loop limit to 45 attempts (45s) to give the handshake sufficient time to complete. See `devref.md
    29`.
290 6. Resolved a startup race condition during restarts where `docker compose build` would fail with DNS
    resolution errors because physical Wi-Fi was power-cycled during teardown and Colima booted before the host
    obtained a DHCP lease. Implemented a unified `wait_for_dhcp_settle` helper (configured with a 30s timeout
    to tolerate macOS DHCP lease latency) in `scripts/common.sh` and added it to the start of `toggle.sh`. See
    `devref.md 30`.
291 7. Resolved a bug where enabling the PF Kill-Switch during SOCKS5 fallback mode blocked all host traffic (
    including `tailscaled` daemon connections to the control plane) because SOCKS5 does not route non-TCP/
    system traffic. Removed kill-switch activation from the SOCKS5 fallback path in `toggle.sh`. See `devref.md
    31`.
292 8. Resolved a bug where host-side `tailscale up` calls passed the `--reset` flag, which aggressively cleared
    macOS Tailscale credentials and logged the user out. Removed the `--reset` flag from all host-side
    invocations to preserve authenticated sessions during exit-node state toggling. See `devref.md 32`.
293 9. Resolved a bug where container connectivity check failures in `toggle.sh` and `recover.sh` were discarded
    to `/dev/null`. Redirected stderr of Docker exec checks to `output.log` to prevent diagnostic blind spots
    during tunnel outages. See `devref.md 33`.
294 - **July 6, 2026** Edge-case security and stability hardening: DNS Watcher interface independence, robust lock
    -file PID verification, fail-closed crypto integrity check, strict `iptables` pinning for tailscale0tun0,
    Docker port binding to `127.0.0.1` to prevent LAN exposure, routing verification via `netstat -nr`. See `
    devref.md 12`.
295 - **July 4, 2026** Colima bridged networking, SOCKS5 proxy migrated natively to Tailscale, headless prompt
    crashes fixed, proxy bypass domains added to fix LAN P2P. Python dependencies completely eliminated: `
    logger` and `rule-compiler` ported to blazing-fast Go multi-stage Docker builds. Added `crypto.sh` to
    enforce cryptographic signature validation on all core bash scripts. Massively refactored `toggle.sh` and `
    recover.sh` to eliminate ~200 lines of duplicated code by migrating network and process logic to `scripts/
    common.sh`. Added a concurrency mutex to `toggle.sh`, resolved unbound `TS_AUTHKEY` check crashes on setup,
    and integrated Go validation to the CI pipeline.
296 - **July 3, 2026** **Critical Security Updates:** Addressed the overnight silent IP leak and improved geo-
    blocking. Introduced `scripts/host-leak-probe.sh` (a sub-second host-egress leak detector) and a
    statistical threshold auto-shutdown watcher. Added `unlock-files.sh` to safely reset file permissions.
    Resolved numerous startup regressions and system bugs.
297 - **July 2, 2026** Two infinite routing loops resolved (`devref.md 11.18`): host-VM tunnel loop (WARP bypass
    routes now via gateway IP + manual `utun` redirection) and Docker Compose subnet takeover (`10.200.1.0/24`
    lock). `--accept-routes=true` now explicit on all `tailscale up --reset` calls.
298 - **July 2, 2026** Auto-recovery daemon documented in 6 above (`scripts/watcher.sh` + launchd plist). See `
    devref.md 11.16`.
299 - **July 2, 2026** Linux scripts (`scripts/toggle-linux.sh`, `recover-linux.sh`, `setup-linux.sh`) documented
    in 6.
300 - **July 1, 2026** `recover.sh --post-wake` ships; wired to watcher daemon. Triggered on every sleep/wake or
    network-state change.
301 - **June 28, 2026** 8 internal scripts moved to `scripts/`. User-facing files (`toggle.sh`, `recover.sh`, `
    setup.sh`, `docker-compose.yml`) stay at repo root.
302 - **June 28, 2026** All hardcoded install paths removed. `SCRIPT_DIR`-relative resolution and `
    __NULLEXIT_HOME__` placeholder used throughout.
303
304 ### Cryptographic Integrity Verification
305
306 nullexit uses a built-in cryptographic integrity checker designed to provide tamper-*evidence* against
    accidental edits, logic bugs, or non-privileged malware. During setup, a 256-bit `NULLEXIT_SEED` is
    injected into your `.env` file. The core entrypoint scripts (`toggle.sh`, `recover.sh`, etc.) are hashed
    using HMAC-SHA256, and their signatures are stored in `.signatures`.
307
308 *(Note: This is not a strict boundary against an attacker who already has write access to the repo, as they
    could simply read the seed and re-sign the scripts. It is designed as a strict footgun-catcher).*

```

```

311 ```bash
312 ./scripts/crypto.sh --sign
313 ```

```

3 agent.md

```

1 # nullexit Detailed Agent Analysis
2
3 > Complete per-file breakdown of every function, constant, duplication, and bug found.
4 > **Note (July 2026):** `sync-rules.py` and `logger.py` have been fully ported to Go (`scripts/rule-compiler/
   main.go` and `scripts/logger/main.go`) to dramatically improve performance and reduce container footprint
   via multi-stage Alpine builds. The analysis below reflects their previous Python states.
5
6 ## Important Agent Instruction
7 - **NEVER use `cat << EOF` or terminal redirections to write or modify files.** You must always use your native
   file-editing tools (`replace_file_content` or `multi_replace_file_content`). Using `cat EOF` leads to
   duplicated text, broken markdown headers, and structural corruption (which previously destroyed the section
   numbering in `devref.md`).
8 - **Always run `bash scripts/crypto.sh --sign` after making any modifications to the core bash scripts.** This
   is required to update the cryptographic HMAC-SHA256 signatures; otherwise, the startup integrity checks
   will fail and block execution.
9 - **Always run `git diff` and print the changes to the user *before* requesting or executing any git commit
   command.** The user must be shown the diff so they can review and understand the changes. Based on this
   diff, formulate a highly precise, detailed, and descriptive commit message (explaining exactly what was
   changed and why) rather than a generic summary, and present it to the user alongside the diff.
10 - **Always use the `--restart` flag when asked to restart the toggle (e.g., run `./toggle.sh --restart`).**
11 - **NEVER execute any `sudo` commands (especially `sudo route`, `sudo dscacheutil`, or any network-modifying
   commands) without explicitly explaining to the user what the command does and why it is needed FIRST. Do
   not assume implicit permission. Wait for the user's approval before running it.**
12 - **When fixing an error and using logs to debug, always show the user the reasoning and the specific logs that
   support this theory.**
13 - **When the user tells you to "push", this means read git diffs and prepare commit messages that correspond to
   these changes (do not ignore any changes). If the changes can be atomized into multiple commits for easier
   understanding, do so.**
14 - **When the user says "latex this", it means you should run `python3 scripts/generate_tex.py` to regenerate
   the unified LaTeX document.**
15
16 ## How to Verify Gateway is Working
17 Whenever modifications are made to the gateway scripts, routing, or containers, execute these verification
   checks in order:
18 1. **Container Health:** Run `docker compose ps` to ensure all containers (`warp`, `adguardhome`, `routing-fix
   `, `tailscale`) are `running` and `healthy`.
19 2. **DNS Hijacking Status:** Run `networksetup -getdnsservers "Wi-Fi"` (or appropriate network service) and
   ensure it is set exclusively to the gateway's Tailscale static IP (typically `100.100.21.8`).
20 3. **DNS Query Interception:** Run `dig @100.100.21.8 google.com` (using the gateway static IP from `
   gateway_ip`) to ensure AdGuard Home is active, intercepting, and resolving queries.
21 4. **Double-Tunnel Egress:** Run `curl -s https://www.cloudflare.com/cdn-cgi/trace | grep warp` to verify the
   egress is routed through Cloudflare WARP (`warp=on`).
22 5. **Host Leak Monitoring:** Run `tail -n 30 output.log` and verify the background watchers and host leak
   probe are reporting healthy status without `LEAK` warnings.
23
24 ---
25
26 ## Part 1: macOS Main Scripts
27
28 ### toggle.sh (1281 lines, 56KB)
29
30 **Functions (27 total):**
31
32 | # | Function | Lines | Purpose |
33 |---|---|---|---|
34 | 1 | `run_with_timeout()` | L3064 | Run a command with a timeout watchdog; kills after N seconds |
35 | 2 | `run_gui_cmd()` | L6880 | Run a GUI command as the logged-in console user |
36 | 3 | `write_gateway_active_marker()` | L8890 | Write timestamp to `/tmp/nullexit-gateway-active.marker` |
37 | 4 | `clear_gateway_active_marker()` | L9295 | Remove gateway active marker + TUNNEL_FAILED_CLOSED.marker |
38 | 5 | `start_sleep_prevention()` | L108154 | Start `caffeinate` in background with shutdown trap to revert DNS
   |
39 | 6 | `stop_sleep_prevention()` | L157167 | Stop caffeinate process via PID file |
40 | 7 | `start_dns_watcher()` | L173191 | Background loop to re-hijack DNS every 30s for Wi-Fi roaming |
41 | 8 | `stop_dns_watcher()` | L193203 | Stop DNS watcher via PID file |
42 | 9 | `start_warp_watcher()` | L220279 | Background WARP liveness monitor; triggers recover.sh after N
   consecutive failures |
43 | 10 | `stop_warp_watcher()` | L281291 | Stop WARP watcher via PID file |
44 | 11 | `start_host_leak_probe()` | L297310 | Start host-egress leak probe (300ms polling) |
45 | 12 | `stop_host_leak_probe()` | L312322 | Stop host leak probe via PID file |
46 | 13 | `cleanup_handler()` | L325382 | Error/signal handler: resets DNS, stops all watchers, tears down
   containers |

```

```

47 | 14 | `restart_tailscaled_daemon()` | L449463 | Restart tailscaled via brew services (user then system
    | fallback) |
48 | 15 | `disconnect_tailscale_host()` | L465477 | Disconnect host Tailscale with retry on failure |
49 | 16 | `get_active_service()` | L480508 | Detect active macOS network service name (e.g., "Wi-Fi") |
50 | 17 | `add_warp_bypass_routes()` | L524546 | Add static routes for WARP endpoints (162.159.192.1,
    | 162.159.193.1) |
51 | 18 | `remove_warp_bypass_routes()` | L548552 | Remove WARP bypass routes |
52 | 19 | `setup_exit_node_routing()` | L558577 | Override default route to point through Tailscale utun interface
    |
53 | 20 | `reset_dns()` | L588595 | Reset DNS to 1.1.1.1 on ACTIVE_SERVICE + ENO_SERVICE |
54 | 21 | `cleanup_network_state()` | L600645 | Nuclear cleanup: disable proxies, flush DNS cache, flush routes,
    | power-cycle Wi-Fi, kill sharing services |
55 | 22 | `enable_socks_proxy()` | L663682 | Enable system-wide SOCKS5 proxy on macOS |
56 | 23 | `disable_socks_proxy()` | L684689 | Disable SOCKS5 proxy |
57 | 24 | `stop_local_dns_proxy()` | L702710 | Kill local Python DNS proxy |
58 | 25 | `start_local_dns_proxy()` | L713768 | Start Python DNS proxy (UDP:53 TCP:5354) |
59 | 26 | `force_dns_to_gateway()` | L780820 | Force DNS to a specific IP with 3-retry verification loop |
60 | 27 | `is_gateway_active()` | L833854 | Check if gateway is running (containers, DNS, SOCKS proxy) |
61
62 **Constants:**
63
64 | Constant | Value | Line |
65 |-----|-----|-----|
66 | `LOG_FILE` | `$PWD/output.log` | L8 |
67 | `PID_FILE` | `/tmp/nullexit-cafeinate.pid` | L105 |
68 | `DNS_WATCHER_PID_FILE` | `/tmp/nullexit-dns-watcher.pid` | L169 |
69 | `WARP_WATCHER_PID_FILE` | `/tmp/nullexit-warp-watcher.pid` | L170 |
70 | `HOST_LEAK_PROBE_PID_FILE` | `/tmp/nullexit-host-leak-probe.pid` | L171 |
71 | `SOCKS_PROXY_PORT` | `1080` | L661 |
72 | `PATH` export | `/opt/homebrew/bin:/opt/homebrew/sbin:/usr/local/bin:$PATH` | L396 |
73 | Gateway active marker | `/tmp/nullexit-gateway-active.marker` | L89 |
74 | TUNNEL_FAILED_CLOSED marker | `${SCRIPT_DIR}/TUNNEL_FAILED_CLOSED.marker` | L94 |
75 | WARP endpoints | `162.159.192.1`, `162.159.193.1` | L535-544 |
76 | DNS fallback | `1.1.1.1` | L592 |
77 | DNS search domain | `ts.net` | L794 |
78 | `LOCK_FILE` | `/tmp/nullexit-toggle.lock` | L62 |
79
80 ---
81
82 ### recover.sh (566 lines, 23KB)
83
84 **Functions (7 total):**
85
86 | # | Function | Lines | Purpose |
87 |---|-----|-----|-----|
88 | 1 | `step()` | L40 | Print bold step header |
89 | 2 | `ok()` | L41 | Print green success message |
90 | 3 | `warn()` | L42 | Print yellow warning message |
91 | 4 | `fail()` | L43 | Print red failure message |
92 | 5 | `die()` | L44 | Print red error and exit |
93 | 6 | `run_with_timeout()` | L5674 | Run command with timeout (bash-native, no GNU coreutils) |
94 | 7 | `restart_tailscaled_daemon()` | L118129 | Restart tailscaled daemon via brew services |
95
96 **Constants:**
97
98 | Constant | Value | Line |
99 |-----|-----|-----|
100 | Color codes | `RED`, `GREEN`, `YELLOW`, `BOLD`, `NC` | L37-38 |
101 | `SCRIPT_DIR` | `${dirname "${BASH_SOURCE[0]}"}` | L50 |
102 | `PATH` export | `/opt/homebrew/bin:/usr/local/bin:$PATH` | L77 |
103 | `PID_FILE` | `/tmp/nullexit-cafeinate.pid` | L387 |
104 | `DNS_WATCHER_PID_FILE` | `/tmp/nullexit-dns-watcher.pid` | L407 |
105 | `HOST_LEAK_PROBE_PID_FILE` | `/tmp/nullexit-host-leak-probe.pid` | L423 |
106 | WARP endpoints | `162.159.192.1`, `162.159.193.1` | L329-336 |
107 | TUNNEL_FAILED_CLOSED marker | `${SCRIPT_DIR}/TUNNEL_FAILED_CLOSED.marker` | L51 |
108
109 **Duplicated logic from toggle.sh:**
110 *(Note: As of July 2026, **ALL** of the below duplicated logic has been successfully extracted to `scripts/
    | common.sh`!)*
111 - `run_with_timeout()` simpler subshell version vs toggle's PID-tracking version
112 - `restart_tailscaled_daemon()` near-identical (uses warn()/ok() vs echo)
113 - Active network service detection inline at L217-233 (toggle has `get_active_service()` function)
114 - ENO service resolution inline at L230-233 (same as toggle L515-516)
115 - WARP bypass routes inline at L329-337 (toggle has `add_warp_bypass_routes()`)
116 - Exit node routing setup inline at L340-345 (toggle has `setup_exit_node_routing()`)
117 - DNS cache flushing L296-302 (same as toggle L611-616)
118 - Wi-Fi power-cycling L442-461 (similar to toggle L626-637)
119 - Proxy disabling L282-288 (same as toggle L604-608)
120 - Sharing services reset L586 (same as toggle L642)
121 - PID file stop pattern L387-399, L407-419, L423-435 (same as toggle L157-167, L193-203, L312-322)
122 - KILL_SWITCH check L194 (same as toggle L141, L469)

```

```

123 - .gateway_ip reading L138-142, L209-213 (same pattern as toggle L1080)
124
125 ---
126
127 ### setup.sh (410 lines, 18KB)
128
129 **Functions (4 total):**
130
131 | # | Function | Lines | Purpose |
132 |---|-----|-----|-----|
133 | 1 | `step()` | L10 | Print bold blue step header |
134 | 2 | `ok()` | L11 | Print green success message |
135 | 3 | `warn()` | L12 | Print yellow warning message |
136 | 4 | `die()` | L13 | Print red error and exit |
137
138 **Constants:**
139
140 | Constant | Value | Line |
141 |-----|-----|-----|
142 | Color codes | `RED`, `GREEN`, `YELLOW`, `BLUE`, `BOLD`, `NC` | L7-8 |
143 | `RECOMMENDED_TS_VERSION` | `1.98.5` | L71 |
144 | `ADGUARD_PASSWORD` | `nullexit` | L215 |
145 | Default .env values | `GATEWAY_RULE_PROFILE=medium`, `GATEWAY_MSS=1120`, etc. | L240-248 |
146
147 ---
148
149 ## Part 2: Linux Scripts
150
151 ### scripts/toggle-linux.sh (931 lines, 39KB)
152
153 **Functions (17 total):**
154
155 | # | Function | Line | Purpose |
156 |---|-----|-----|-----|
157 | 1 | `run_with_timeout()` | L19 | Runs command with timeout watchdog, kills on expiry |
158 | 2 | `run_gui_cmd()` | L57 | macOS code uses `stat -f '%Su' /dev/console`, DEAD CODE on Linux |
159 | 3 | `start_sleep_prevention()` | L74 | Uses `systemd-inhibit` (Linux) to block sleep |
160 | 4 | `stop_sleep_prevention()` | L116 | Kills sleep prevention process via PID file |
161 | 5 | `start_dns_watcher()` | L130 | Background daemon polling DNS every 30s, re-hijacks if changed |
162 | 6 | `stop_dns_watcher()` | L149 | Kills DNS watcher process via PID file |
163 | 7 | `cleanup_handler()` | L162 | Trap handler for ERR/INT/TERM/HUP restores DNS, kills bg processes |
164 | 8 | `disconnect_tailscale_host()` | L273 | Disconnects host Tailscale from mesh |
165 | 9 | `get_active_service()` | L283 | Gets active network interface via `ip route` (Linux) |
166 | 10 | `reset_dns()` | L304 | Restores DNS via `resolvectl revert` (Linux) |
167 | 11 | `cleanup_network_state()` | L314 | Flushes DNS cache, routes, power-cycles network interface |
168 | 12 | `enable_socks_proxy()` | L361 | Stub prints message that Linux global SOCKS is DE-specific |
169 | 13 | `disable_socks_proxy()` | L367 | Stub prints skip message |
170 | 14 | `stop_local_dns_proxy()` | L383 | Kills Python DNS proxy processes |
171 | 15 | `start_local_dns_proxy()` | L394 | Starts Python DNS proxy (UDP:53 TCP:5354) + hijacks DNS |
172 | 16 | `force_dns_to_gateway()` | L461 | 3-attempt loop to set + verify DNS via `resolvectl` |
173 | 17 | `is_gateway_active()` | L495 | Checks if containers running or DNS was hijacked |
174
175 **Shared constants with macOS toggle.sh:**
176
177 | Constant | Value | Both |
178 |-----|-----|-----|
179 | `SOCKS_PROXY_PORT` | `1080` | toggle-linux L359, toggle.sh L661 |
180 | `PID_FILE` | `/tmp/nullexit-caffeinate.pid` | toggle-linux L71, toggle.sh L105 |
181 | `DNS_WATCHER_PID_FILE` | `/tmp/nullexit-dns-watcher.pid` | toggle-linux L128, toggle.sh L169 |
182 | ARP threshold | `15` devices | toggle-linux L518, toggle.sh L863 |
183 | Colima memory | `0.6` (600MB) | toggle-linux L578, toggle.sh L928 |
184 | Colima swap | `400MB` | toggle-linux L587, toggle.sh L937 |
185 | Tailscale wait loop | `60` iterations `1s` | toggle-linux L613, toggle.sh L1010 |
186 | NoState stuck threshold | `40` consecutive | toggle-linux L628, toggle.sh L1025 |
187
188 ---
189
190 ### scripts/recover-linux.sh (257 lines, 10KB)
191
192 **Functions (6 total):**
193
194 | # | Function | Line | Purpose |
195 |---|-----|-----|-----|
196 | 1 | `step()` | L24 | Print formatted step header |
197 | 2 | `ok()` | L25 | Print green success message |
198 | 3 | `warn()` | L26 | Print yellow warning message |
199 | 4 | `fail()` | L27 | Print red failure message |
200 | 5 | `die()` | L28 | Print red error + exit 1 |
201 | 6 | `run_with_timeout()` | L33 | Simplified timeout (immediate SIGKILL, no PID tracking) |
202
203 ---

```

```

204
205 ### scripts/setup-linux.sh (416 lines, 18KB)
206
207 **Functions (4 total):**
208
209 | # | Function | Line | Purpose |
210 |---|-----|-----|-----|
211 | 1 | `step()` | L10 | Print formatted step header |
212 | 2 | `ok()` | L11 | Print green success message |
213 | 3 | `warn()` | L12 | Print yellow warning message |
214 | 4 | `die()` | L13 | Print red error + exit 1 |
215
216 **~95% identical to macOS setup.sh.** Only differences:
217 1. Desktop shortcuts (L366-391 Linux `.desktop` files vs macOS `.app` via `osacompile`)
218 2. Final instructions text
219 3. .env template: macOS adds `GATEWAY_USE_EXIT_NODE`, `WARP_FAIL_THRESHOLD`, `HOST_LEAK_PROBE`, `KILL_SWITCH`
    Linux omits these
220
221 ---
222
223 ## Part 3: Utility Scripts
224
225 ### scripts/diagnose-host-leak.sh (722 lines, 38KB)
226
227 **Purpose:** Comprehensive host-routing diagnostic. 8 checks classify host IP leaks. Supports `--fix` and `--
    watch` modes.
228
229 **Duplicated logic:**
230 - `resolve_active_service()` identical pattern as toggle.sh, recover.sh, fix-docker-bridge-collision.sh
231 - `run_with_timeout()` reimplemented timeout
232 - Color codes RED/GREEN/YELLOW/BOLD/NC with tty detection
233 - WARP check via `cloudflare.com/cdn-cgi/trace`
234 - `export PATH="/opt/homebrew/bin:/usr/local/bin:$PATH"`
235 - .gateway_ip reading pattern
236
237 ### scripts/fix-docker-bridge-collision.sh (402 lines, 18KB)
238
239 **Purpose:** Fixes Docker bridge subnet collisions with `10.200.1.0/24`.
240
241 **Duplicated logic:**
242 - `resolve_active_iface()` same interface detection as diagnose-host-leak.sh
243 - Color codes, step()/ok()/warn()/fail()/die() formatting helpers
244
245 ### scripts/fix-ssh-delay.sh (63 lines, 2.7KB)
246
247 **Purpose:** One-shot fix for ~20s SSH delay. Standalone, minimal duplication (uses hardcoded / instead of
    color functions).
248
249 ### scripts/host-leak-probe.sh (110 lines, 5.5KB)
250
251 **Purpose:** Sub-second host-egress leak prober at 300ms intervals.
252
253 **Duplicated logic:**
254 - WARP check via cdn-cgi/trace (same curl + awk as toggle.sh WARP Watcher)
255 - Color codes (hardcoded; status updates and warnings are conditioned on TTY detection to prevent log pollution
    )
256 - LOG_FILE = `PROJECT_ROOT/output.log`
257
258 ### scripts/reinstall-host-tailscale.sh (740 lines, 31KB)
259
260 **Purpose:** Complete uninstall + reinstall of host Tailscale.
261
262 **Duplicated logic:**
263 - `with_timeout()` yet another bash timeout implementation (polling-based, different from diagnose and toggle
    versions)
264 - Color codes + logging functions
265 - System Extension detection logic (similar to diagnose-host-leak.sh)
266
267 ### scripts/routing-fix.sh (266 lines, 10.5KB)
268
269 **Purpose:** Runs INSIDE warp container. Maintains routing table 200, FORWARD rules, country blocking, IP
    blocklists, tunnel health checks.
270
271 **Duplicated logic:**
272 - (Fixed) WARP_ENDPOINT defaults are now centralized in common.sh
273 - WARP health check via cdn-cgi/trace (another instance)
274 - iptables dual-backend pattern (for-loop over `iptables iptables-legacy`)
275
276 ### scripts/watcher.sh (269 lines, 13KB)
277
278 **Purpose:** Long-running launchd daemon for post-wake/post-roam recovery.

```

```

279
280 **Key function `detect_lan_p2p_mode()`:**
281 Added July 10, 2026. Called on startup and on every network state change (Listener 2 NET: events). Determines
    whether the current Wi-Fi network safely allows direct Tailscale LAN P2P connections:
282 1. **WPA2-Enterprise check:** `system_profiler SPAirPortDataType` reads the `Security:` field for the current
    association. Enterprise = 802.1x = always AP-isolated sets `allow=false`.
283 2. **AP isolation probe:** `route -n get 1.1.1.1` finds default gateway IP, then `ping -c 3 -t 1 -q "$gw"` if
    0 responses on a non-empty network, AP isolation is active sets `allow=false`.
284 3. **Output:** Writes `true` or `false` to `.lan_p2p_detected` in the repo root.
285 4. **Consumer:** `routing-fix.sh` reads this file every 30 seconds and enforces/relaxes the RFC1918 DROP rule
    accordingly no restart required.
286
287 **Duplicated logic:**
288 - PATH export (similar to toggle.sh/recover.sh)
289 - Marker file pattern (`/tmp/nullexit-gateway-active.marker`)
290
291 ### scripts/dns-proxy.py (92 lines, 2.9KB) Clean, standalone
292
293 ### scripts/logger.py (144 lines, 5.9KB) Clean, standalone
294
295 ---
296
297 ## Part 4: Cross-Cutting Issues
298
299 ### Functions Identical Across macOS Linux
300
301 | Function | macOS toggle | macOS recover | macOS setup | Linux toggle | Linux recover | Linux setup |
302 |-----|-----|-----|-----|-----|-----|-----|
303 | `run_with_timeout()` | | | identical | identical | | |
304 | `stop_local_dns_proxy()` | | | identical | | | |
305 | `start_local_dns_proxy()` | | | ~95% similar | | | |
306 | `is_gateway_active()` | | | ~80% similar | | | |
307 | `step/ok/warn/die` | | | identical | identical | | |
308
309 ### Platform-Adapted Functions (same logic, different OS commands)
310
311 | Function | Linux Command | macOS Command |
312 |-----|-----|-----|
313 | `get_active_service()` | `ip route get 1.1.1.1 \| awk '{print $5}'` | `route get default` + `networksetup -
    listnetworkserviceorder` |
314 | `reset_dns()` | `resolvectl revert` | `networksetup -setdnsservers` |
315 | `cleanup_network_state()` | `ip link set dev down/up` | `networksetup -setairportpower off/on` + proxy
    disable |
316 | `force_dns_to_gateway()` | `resolvectl dns` + `resolvectl domain -ts.net` | `networksetup -setdnsservers` +
    `--setsearchdomains ts.net` |
317 | `start_sleep_prevention()` | `systemd-inhibit --what=sleep` | `caffeinate -i` |
318 | `enable/disable_socks_proxy()` | Stub (no-op on Linux) | `networksetup -setsocksfirewallproxy` |
319 | DNS cache flush | `resolvectl flush-caches` | `dscacheutil -flushcache` + `killall -HUP mDNSResponder` |
320
321 ### Features Missing From Linux Scripts
322
323 These exist in macOS toggle.sh but NOT in toggle-linux.sh:
324
325 1. `write_gateway_active_marker()` / `clear_gateway_active_marker()`
326 2. `restart_tailscaled_daemon()`
327 3. `start_warp_watcher()` / `stop_warp_watcher()` WARP liveness monitor
328 5. `add_warp_bypass_routes()` / `remove_warp_bypass_routes()` / `setup_exit_node_routing()`
329 6. `LOG_FILE` structured logging (timestamps to output.log)
330 7. MSS clamping injection (step 4c)
331 8. Rule count reporting
332 9. `--post-wake` mode in recover
333 10. Container teardown on failure in cleanup_handler
334
335 ---
336
337
338
339 ## Part 6: Constant Duplication Map
340
341 | Value | Files |
342 |-----|-----|
343 | `162.159.192.1` (WARP endpoint) | docker-compose.yml, routing-fix.sh (now dynamically loaded via .env/common.
    sh!) |
344 | `5335` (AdGuard DNS port) | docker-compose.yml, AdGuardHome.yaml, post-rules.txt |
345 | `5354` (mapped DNS port) | docker-compose.yml, dns-proxy.py |
346 | `41642` (Tailscale WireGuard) | docker-compose.yml, routing-fix.sh, post-rules.txt |
347 | `1080` (SOCKS5 port) | docker-compose.yml, toggle.sh |
348 | `1120` (MSS clamp) | post-rules.txt, .env |
349 | `admin:nullexit` (AdGuard creds) | AdGuardHome.yaml (bcrypt) |
350 | `10.200.1.0/24` (Docker subnet) | (Fixed) no longer duplicated |
351 | `America/New_York` (timezone) | docker-compose.yml (4 services) |

```

```

352 | `/tmp/nullexit-caffeinate.pid` | (Fixed) centralized in common.sh |
353 | `/tmp/nullexit-dns-watcher.pid` | (Fixed) centralized in common.sh |
354 | `/tmp/nullexit-host-leak-probe.pid` | toggle.sh, recover.sh |
355 | `/opt/homebrew/bin:...` (PATH) | (Fixed) centralized in common.sh |
356
357 ---
358
359 ## Part 7: Refactoring Outcome (Implemented July 2026)
360
361 **Decision: "Scripts-only" architecture**
362 Instead of introducing a new `lib/` directory, all shared code was moved into the existing `scripts/` directory
  to keep the project hierarchy flat and simple.
363
364 ### 1. `scripts/common.sh` (288 lines, 10KB)
365 Extracts the fundamental primitives and networking logic used across orchestrator scripts:
366 - **Formatters:** `step()`, `ok()`, `warn()`, `fail()`, `die()`
367 - **Timeout Wrapper:** `run_with_timeout()` (Canonical version with PID tracking, graceful SIGTERM, and SIGKILL
  fallback)
368 - **Daemon Lifecycle:** `restart_tailscaled_daemon()`, `stop_pidfile_daemon()` (collapsed caffeinate, DNS
  watcher, WARP watcher, and host leak prober teardown logic)
369 - **Network Interface/Routing:** `get_active_service()`, `get_en0_service()`, `bounce_wifi_interfaces()`, `
  add_warp_bypass_routes()`, `remove_warp_bypass_routes()`
370 - **Proxy/DNS Cleanup:** `disable_all_proxies()`, `flush_dns_cache()`, `reset_sharing_services()`
371 - **Configuration Helpers:** `read_adguard_ip()`, `is_kill_switch_enabled()`, `get_warp_endpoint_1()`, `
  get_warp_endpoint_2()` (which centralizes the hardcoded 162.159.192.1 / .193.1 into .env logic)
372
373 ### 2. `scripts/setup-common.sh` (~350 lines)
374 Extracts the heavy, duplicated installation logic shared between `setup.sh` (macOS) and `scripts/setup-linux.sh`
  :
375 - Docker and Colima prerequisite checks
376 - `wgcf` binary download and WARP key generation
377 - `.env` configuration file generation
378 - Interactive Tailscale Auth Key and AdGuard Rule Profile prompts
379
380 ### Sourcing Pattern
381 - macOS root scripts (`toggle.sh`, `recover.sh`, `setup.sh`) source via:
382 `source "$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/scripts/common.sh"`
383 - Linux sibling scripts (`scripts/toggle-linux.sh`, etc.) source via:
384 `source "$(dirname "${BASH_SOURCE[0]}")/common.sh"`
385
386 ## Agent Verbs / Protocols
387
388 ### `/sweep` (or `sweep the gateway`)
389 When the user invokes this verb, the agent MUST perform a full end-to-end connectivity and health audit of the
  gateway:
390 1. **WARP Validation:** Run `curl -s https://www.cloudflare.com/cdn-cgi/trace | grep warp=` (must equal `on`).
391 2. **Tor Validation:** Identify the dynamically generated `SOCKS_PROXY_PORT` from `docker compose ps`, then run
  `curl -s --socks5-hostname 127.0.0.1:<PORT> https://check.torproject.org/api/ip` to confirm the proxy
  works.
392 3. **Tailscale Validation:** Run `ping -c 1 100.100.21.8` to ensure the gateway is reachable, and check `
  tailscale status` to ensure other mesh devices are visible/pingable.
393 4. **Log Audit:** Run `tail -n 100 output.log | grep -iE "(error|fail|warn)"` to catch any underlying component
  failures.
394
395 ### `/latex` (or `generate latex`)
396 When the user invokes this verb, the agent MUST run `python3 -I scripts/generate_tex.py` to regenerate the
  documentation source code (`nullexit.unified.tex`). The agent MUST NOT attempt to compile the `.tex` file
  into a PDF (the user handles PDF compilation locally).

```

4 devref.md

```

1 # nullexit Development Reference & Resolved Issues
2
3 > **Last updated:** July 10, 2026
4 > **Purpose:** Provide any LLM or developer with complete project understanding, debugging history, and
  resolved issues so they can make informed changes without re-reading every file.
5 > **Diagrams:** See [diagrams.md](./diagrams.md) for system architecture, toggle.sh flowcharts, monitoring
  layer, traffic sequence, recover.sh decision tree, and the full failureself-healing map.
6
7 ---
8
9 ## 1. What Is nullexit?
10
11 nullexit is a **Tunnel-in-Tunnel VPN gateway** that chains **Tailscale** (mesh VPN) through **Cloudflare WARP**
  (exit VPN). It runs on a macOS host inside Docker containers managed by **Colima** (lightweight Linux VM).
  The goal: any device on the user's Tailscale mesh can use this gateway as an exit node, and all traffic

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```

    exits through Cloudflare WARP achieving double encryption, ISP invisibility, and network-wide ad-blocking
    via **AdGuard Home**.
12
13 ### Traffic Flow
14 ---
15 Phone/Laptop  Tailscale mesh (WireGuard)  Mac host  Docker container
16 Tailscale decrypts  Gluetun re-encrypts  WARP tun0  Cloudflare  Internet
17 ---
18
19 ### SOCKS5 Failover Path (if exit node is unavailable)
20 ---
21 DNS:  Browser  host 127.0.0.1:53  Python proxy  TCP:5354  AdGuard  WARP  DNS
22 TCP:  Apps  macOS SOCKS5 proxy (127.0.0.1:1080)  container  tun0  WARP  Internet
23 ---
24
25 ---
26
27 ## 2. Architecture
28
29 ### Containers (all share `warp`'s network namespace via `network_mode: service:warp`)
30 | Container | Image | Role |
31 |-----|-----|-----|
32 | `warp` | `qmcgaw/gluetun:v3.41.1` | Gluetun WireGuard client  Cloudflare WARP. Owns the network namespace.
    Strict firewall. |
33 | `tailscale` | `tailscale/tailscale:v1.98.4` | Advertises as exit node on the mesh. Also provides the built-in
    SOCKS5 proxy fallback via `TS_SOCKS5_SERVER=:1080`. |
34 | `routing-fix` | `alpine:3.20` | Sidecar that maintains routing tables + iptables rules every 30 seconds. |
35 | `rule-compiler` | `golang:1.22-alpine` / `alpine:3.20` | One-shot startup container that compiles 500k+ DNS
    block rules in <2s and immediately exits. |
36 | `adguardhome` | `adguard/adguardhome:v0.107.77` | DNS sinkhole for ads/trackers. Listens on port 5335.
    Upstream DNS: `127.0.0.1:53` (through WARP). |
37
38 ### Port Mappings (on host via Colima)
39 | Host Port | Container Port | Protocol | Purpose |
40 |-----|-----|-----|-----|
41 | 5354 | 5335 | TCP+UDP | AdGuard DNS |
42 | (Tailscale IP):3000 | 3000 | TCP | AdGuard web UI (Internal to Mesh) |
43 | 41641 | 41641 | UDP | Tailscale WireGuard direct |
44 | 1080 | 1080 | TCP | SOCKS5 proxy |
45
46 ### Host Components
47 - **Colima VM** Linux VM running Docker (600MB RAM + 400MB swap)
48 - **tailscaled** Standalone Tailscale daemon via `brew install tailscale` + `brew services start tailscale` (
    NOT the GUI `.app`)
49 - **macOS network settings** DNS, SOCKS proxy, routing all manipulated by `toggle.sh`
50
51 ---
52
53 ## 3. Key Files
54
55 | File | Lines | Purpose |
56 |-----|-----|-----|
57 | `toggle.sh` | ~1265 | **Main script.** Detects state, toggles gateway ON/OFF. Handles DNS hijacking,
    Tailscale exit node, SOCKS proxy, sleep prevention, timeouts, cleanup. |
58 | `setup.sh` | 406 | **One-time setup.** Installs deps (Docker, Tailscale, wgcf), generates WARP keys, writes
    `.env`, configures AdGuard via API, starts containers. |
59 | `docker-compose.yml` | 194 | Service definitions for all 5 containers. |
60 | `scripts/routing-fix.sh` | 201 | Maintains routing tables (table 200 for SOCKS5, table 52 for CGNAT) and
    FORWARD iptables rules in both nftables and legacy backends. Loads and enforces the IP blocklist via `ipset`
    . Runs in a 30-second loop. |
61 | `post-rules.txt` | 18 | Gluetun iptables rules loaded at container start. FORWARD accept for tailscale0, NAT
    MASQUERADE on tun0, DNS redirect to port 5335, IPv6 drop, TCP MSS clamping. |
62
63 | `scripts/dns-proxy.py` | 91 | Local DNS proxy. UDP:53 TCP:5354 with proper 2-byte length prefix. Fallback
    when exit node is unavailable. |
64 | `scripts/rule-compiler/` | ~800 | Unified threat compiler (ported to Go from Python). **DNS pipeline:**
    fetches remote blocklists, deduplicates against AdGuard native filters, optimizes subdomains (~50%
    reduction), compiles to AdGuard syntax. **IP pipeline:** fetches threat-intel feeds (Spamhaus, Feodo, ET,
    CINS), strips private/Tailscale ranges, outputs atomic `ipset` restore file. Built dynamically in Docker
    multi-stage build. |
65 | **`adguard/work/`** | | **Bind-mounted container data folder. Off-limits from outside Docker READ-ONLY-
    EXCEPT-VIA-`sync-rules`. See README 6.** |
66 | `recover.sh` | ~557 | Nuclear recovery script (gain: use `--post-wake` for non-destructive refresh after
    sleep/wake or Wi-Fi roam keeps the gateway live while still re-hijacking DNS + refreshing Tailscale exit-
    node + force-recreating warp if unhealthy). Resets DNS on ALL services, disables proxies, flushes routes,
    stops containers, kills sleep prevention, power-cycles Wi-Fi. Cryptographically verified. |
67 | `scripts/diagnose-host-leak.sh` | 580+ | **One-shot host-routing diagnostic.** Runs 7 checks (Tailscale,
    SOCKS5, CDN-cgi/trace, IPv6 leak, default route, output.log hints, gateway IP), classifies into ONE of
    three known leak scenarios (A=SOCKS5 fallback, B=IPv6 leak, C=route freeze) or OK, writes a timestamped
    report to `host-leak-diagnostic-<UTC>.txt`, and prints a ready-to-run fix on stdout. Pass `--fix` to apply
    the matched remediation and re-verify egress. Pass `--watch` (or `--watch 30`) to run a full baseline then

```

```

continuously monitor warp/IPv6/default-route every N seconds, alerting on any state change. See 10.30. |
68 | `scripts/host-leak-probe.sh` | ~110 | **Continuous sub-second host-egress prober.** Launched automatically by
    `toggle.sh` when `HOST_LEAK_PROBE=true` in `.env` (default). Polls `cdn-cgi/trace` every 300ms directly
    from the host NIC via `curl` not via `docker compose exec` so it detects flash-leaks invisible to the in-
    container WARP Watcher. Logs only on state change (`LEAK`, `ROTATE`, `HOST-PROBE failed/timeout`) to `
    output.log`. curl errors (`2>>output.log`) also land there. Stopped gracefully (SIGTERM) by `toggle.sh`
    STOP path, `cleanup_handler`, and `recover.sh`. See 10.30.1. |
69 | `scripts/fix-docker-bridge-collision.sh` | ~290 | **One-shot fix for 9.13.** Detects Docker's `172.17.0.0/16`
    bridge colliding with the host Wi-Fi subnet (the symptom that produces `default 172.17.0.1 UGScg en0` in `
    netstat -rn`), picks a non-colliding `docker.bip` from a small candidate set, idempotently edits `~/colima
    /default/colima.yaml` with a timestamped backup, restarts Colima, rebinds Wi-Fi DHCP, verifies the new
    default route is no longer Docker's bridge, and re-runs `toggle.sh` + `scripts/diagnose-host-leak.sh`.
    Supports `--dry` and `--skip-toggle`. |
70 | `scripts/watcher.sh` | 194 | **Long-running post-wake + post-roam daemon.** Launched by launchd LaunchAgent;
    subscribes to `com.apple.powermanagement` events via `log stream` and to network-state changes via `scutil
    n.watch`. Both fire `recover.sh --post-wake`. Single-instance lock + SCRIPT_DIR-relative resolve of `
    RECOVER`. See 10.29. |
71 | `scripts/common.sh` | ~40 | **Shared script primitives.** Centralizes formatted echo statements (`step`, `ok
    `, `warn`, `die`) and the safe background `run_with_timeout` execution wrapper. Sourced by all orchestrator
    scripts across macOS and Linux. |
72 | `scripts/setup-common.sh` | ~350 | **Shared installation logic.** Centralizes logic for verifying Docker,
    installing Tailscale/wgcf, generating `.env`, and prompting for auth keys. Sourced by `setup.sh` and `setup
    -linux.sh`. |
73 | `scripts/toggle-linux.sh` | ~900 | **Linux sibling of `toggle.sh`.** Sources `common.sh`. Native Linux
    equivalents for every macOS-specific primitive `nmcli radio wifi off/on` instead of `networksetup -
    setairportpower`, `ip link set ... down/up` fallback, systemd-resolved wiring. Paired with `.desktop`
    shortcuts. |
74 | `scripts/recover-linux.sh` | ~250 | **Linux sibling of `recover.sh`.** Sources `common.sh`. Same nuclear
    semantics (flush, restart, verify) but using `nmcli` / `ip link` instead of macOS `networksetup`. |
75 | `scripts/setup-linux.sh` | ~80 | **Linux sibling of `setup.sh`.** Sources `setup-common.sh`. Distro detection
    (apt/dnf/pacman), installs Linux-specific dependencies, creates `.desktop` shortcuts. |
76 | `scripts/Toggle-Gateway.bat` | ~300 | **Windows Toggle helper.** Moved from root to `scripts/`. Helps invoke
    the framework on Windows if applicable. |
77 | `scripts/reinstall-host-tailscale.sh` | ~740 | **Nuke-and-reinstall tool for host Tailscale.** Completely
    purges Tailscale, its settings, macOS Background Items (`.btm` database), and stale System Extensions.
    Wipes ALL Login Items via `sfltool reset-login-items` (requires sudo). Useful for crisis recovery but
    destructive to other login items. |
78 | `scripts/fix-ssh-delay.sh` | ~100 | **Fixes SSH connection delays.** One-shot script to address SSH delays
    caused by DNS or routing issues, useful in crisis situations. |
79 | `scripts/logger/` | ~100 | **Internal logger piped from `scripts/routing-fix.sh`** inside the `warp`
    container. Ported to Go from Python. Built via Docker multi-stage build. |
80 | `black_list.txt` | 71 | Custom domains to block (ads, trackers, telemetry). Supports `$important` modifier. |
81 | `white_list.txt` | 222 | Domains to force-allow (YouTube, Apple services, etc.). Always wins over blocks. |
82 | `.env` | ~14 | WARP WireGuard keys, Tailscale auth key, rule profile. **Contains secrets & NULLEXIT_SEED.** |
83 | `.gateway_ip` | 1 | Static gateway Tailscale IP (fallback for dynamic resolution). |
84 | `scripts/unlock-files.sh` | ~20 | **One-shot stale-permission fix.** Uses atomic rename (`cp` + `mv`) to
    replace locked inodes (mode `000`/`0444` from old `chmod` decisions) with fresh writable ones no `chmod`
    called. |
85 | `scripts/crypto.sh` | ~50 | **Cryptographic integrity enforcement.** Uses `NULLEXIT_SEED` from `.env` to sign
    core bash scripts (HMAC-SHA256) and strictly verify them on start to prevent manipulation. Pass `--sign`
    or `--verify`. |
86 | `scripts/pf.conf` | 35 | **macOS native firewall ruleset.** Enforces the default-deny kill-switch, allows
    local LAN / Tailscale traffic, and applies TCP MSS Clamping (`max-mss 1160`) to prevent WireGuard MTU
    fragmentation. |
87
88 ---
89
90 ## 4. toggle.sh Flow
91
92 ### State Detection
93 `is_gateway_active()` returns true if EITHER containers are running OR host DNS is not `1.1.1.1`. This dual
    check prevents the script from starting when it should stop (e.g., containers crashed but DNS is still
    hijacked).
94
95 ### START Path (gateway OFF ON)
96 1. **Reset DNS to 1.1.1.1** Prevents deadlocks during startup
97 2. **Disconnect host Tailscale** `tailscale down` (prevents exit-node routing during container boot)
98 3. **Compile DNS rules** `docker compose run --rm rule-compiler` (Go binary inside multi-stage Docker build)
99 4. **Boot Colima VM** `colima start --memory 0.6 --vm-type vz --network-address --network-mode bridged` if not
    running
100 - **Why Bridged Networking?** The `--network-mode bridged` and `--network-address` flags are critical. By
    default, Colima creates an isolated network. Bridged mode forces the VM to receive its own IP address
    directly from your local router's DHCP server, placing it on the same physical subnet as your host machine.
    This bypasses macOS network isolation and is absolutely required for peer-to-peer (P2P) connections, mDNS,
    and local LAN service discovery to function properly across the container boundary.
101 - **Enterprise Wi-Fi Fallback & Dynamic Roaming:** `toggle.sh` dynamically queries `system_profiler
    SPAirPortDataType` on boot. If it detects a **WPA2-Enterprise** connection (802.1X), it forces Colima to
    fall back to `--network-mode shared`. This prevents aggressive network intrusion systems on corporate/
    university Wi-Fi switches from terminating the host's Wi-Fi connection due to "MAC spoofing" (detecting
    multiple MAC addresses originating from a single authenticated port). P2P connections are impossible on
    these networks anyway due to AP Client Isolation.

```

```

102 - **Dynamic Roam Downgrading:** `toggle.sh` writes its selected network mode to `/tmp/nullexit_colima_mode
    .txt`. When roaming between networks (e.g., Home to University), `recover.sh` actively checks this state
    against the new network's security requirement (`.lan_p2p_detected`). If a mismatch is detected (e.g.,
    bridged Colima on an Enterprise network), `recover.sh` automatically forces a full `toggle.sh --restart` in
    the background to safely transition the Virtual Machine's network mode and protect the host from de-
    authentication.
103 5. **Configure VM swap** 400MB swap file inside the VM to prevent OOM
104 6. **Clean corrupted AdGuard config** Remove empty `AdGuardHome.yaml`
105 7. **Start containers** `docker compose up -d`
106 8. **Wait for gateway Tailscale** Poll `tailscale status` for "offers exit node" (up to 60s, abort at 40
    consecutive NoState)
107 9. **Resolve gateway IP** From `.gateway_ip` or `docker compose exec tailscale tailscale ip -4`
108 10. **Connect host to mesh** Verify `tailscaled` is running (auto-start if needed), then `tailscale up --reset
    --ssh=true --accept-dns=false --accept-routes=true --exit-node=` ( `--accept-routes=true` must be explicit
    `--reset` reverts it to `false` otherwise, silently preventing the default route from switching to `utun
    *`)
109 11. **Pre-flight checks** [1/3] `tailscale ping` gateway, [2/3] `dig +tcp` AdGuard DNS, [3/3] WARP container
    internet
110 12. **If all pass** Set exit node + hijack DNS to gateway IP (single server, no fallback)
111 13. **If any fail** Enable SOCKS5 proxy + local DNS proxy as fallback
112 14. **Final DNS re-force** `force_dns_to_gateway` called again after exit-node transition (tailscaled clobbers
    DNS briefly)
113 15. **Enable sleep prevention** `caffeinate -i` in background to prevent idle sleep
114
115 ### STOP Path (gateway ON OFF)
116 1. Disconnect host Tailscale (`tailscale down`)
117 2. `docker compose down -t 5`
118 3. Reset DNS to 1.1.1.1
119 4. Stop local DNS proxy
120 5. **Stop sleep prevention** Kill caffeinate process
121 6. Full network state cleanup (proxies off, DNS cache flush, route flush, Wi-Fi power cycle)
122
123 ### Safety Mechanisms
124 - `run_with_timeout()` Pure-bash watchdog for every system command (15s120s)
125 - `cleanup_handler()` Trap on ERR/INT/TERM/HUP restores DNS to 1.1.1.1, kills caffeinate, and tears down
    Tailscale
126 - `force_dns_to_gateway()` Sets DNS and VERIFIES via `networksetup -getdnsservers` (3 attempts with backoff)
127 - `SUCCESS_RUN` flag Cleanup only runs if script didn't complete successfully
128 - `start_sleep_prevention()` / `stop_sleep_prevention()` Manages `caffeinate -i` via PID file at `/tmp/
    nullexit-caffeinate.pid`
129
130 ---
131
132 ## 5. Routing & Firewall Architecture (Inside Container)
133
134 ### IP Rules (`ip rule show`)
135 | Priority | Match | Table | Purpose |
136 |-----|-----|-----|-----|
137 | 99 | `to 100.64.0.0/10` | 52 | CGNAT range Tailscale routes (injected by routing-fix) |
138 | 100 | `from 172.18.0.2` | 200 | Container's own traffic (SOCKS5 proxy) tun0 |
139 | 101 | all | 51820 | Gluetun's WireGuard fwmark tun0 |
140
141 ### Table 200 (SOCKS5 proxy traffic)
142 - `162.159.192.1` via $DOCKER_GW dev eth0` WARP endpoint exception (prevents tunnel loop)
143 - `default dev tun0` Everything else through WARP
144
145 ### iptables (TWO separate stacks!)
146 - **nftables backend** (`iptables`) Used by Gluetun's `post-rules.txt`. FORWARD policy DROP.
147 - **legacy backend** (`iptables-legacy`) Used by Tailscale's `ts-forward`. FORWARD policy ACCEPT.
148 - Both stacks apply to every packet. FORWARD RELATED,ESTABLISHED must exist in BOTH.
149
150 ### post-rules.txt (loaded by Gluetun)
151 ```
152 FORWARD: ACCEPT for tailscale0 in/out
153 INPUT: ACCEPT for tailscale0
154 NAT POSTROUTING: MASQUERADE on tun0
155 MANGLE: TCP MSS clamp to 1180
156 NAT PREROUTING: Redirect DNS (53) 5335 on tailscale0 and eth0
157 ip6tables: DROP FORWARD on tailscale0
158 ```
159
160 ---
161
162 ### 5.1 Tailscale Local Network Discovery (DERP Bypass)
163 By default, Gluetun's strict NAT swallows all of Tailscale's UDP hole-punching packets, forcing Tailscale to
    fall back to incredibly slow DERP relay servers (often adding 500ms+ latency).
164
165 To fix this, we use `FIREWALL_OUTBOUND_SUBNETS=192.168.0.0/16,172.16.0.0/12,10.0.0.0/8` in `docker-compose.yml`
    (on the `warp` container). This creates `ip rule` bypasses (Priority 99, Table 199) that route all packets
    destined for private IP ranges *outside* the WARP tunnel directly to the host's `eth0`.

```

```

166 - **172.16.0.0/12** is especially critical because many modern Wi-Fi routers (and Docker itself) assign IPs in
    this block (e.g., `172.17.x.x`).
167 - This bypass allows Tailscale's UDP packets to reach devices on the same Wi-Fi directly, establishing a fast
    peer-to-peer connection while the actual web traffic inside the tunnel remains fully routed through WARP.
168
169 ---
170
171 ### 5.2 Firewalling & Per-Device Access Control
172 Because every mesh device's traffic passes through the `warp` container's network namespace, this namespace
    serves as the ultimate choke point for network-wide firewall rules.
173
174 **Where to Put Rules**
175 The right place to add firewall rules is `scripts/routing-fix.sh`. It runs a 30-second loop inside the
    namespace and already handles re-injecting rules that Gluetun resets on reconnect. **Do not use `post-rules
    .txt` for dynamic rules**, as Gluetun flushes and resets it upon VPN reconnects.
176
177 **The Dual iptables Backend Constraint (CRITICAL)**
178 The container runs two simultaneous iptables backends: `iptables` (for the nftables backend used by Gluetun)
    and `iptables-legacy` (for Tailscale). Every rule **must** be added to both stacks, or it will silently
    fail for certain traffic.
179
180 **Avoiding Duplication**
181 Because `scripts/routing-fix.sh` runs in a loop, you must always check for a rule's existence with `-C` before
    appending with `-A`. Otherwise, your rules will duplicate every 30 seconds.
182
183 **Per-Device Identification & Filtering**
184 Each mesh device has a stable `100.x.x.x` Tailscale IP, which is visible as the `src` IP in the `FORWARD` chain
    . This is how you identify devices for filtering.
185 * **Domain-level:** AdGuard Home (port 3000, credentials `admin/nullexit`) provides a REST API that supports
    per-client blocking rules based on their Tailscale IP.
186 * **IP-level:** Use `iptables` in `scripts/routing-fix.sh` (e.g., `iptables -I FORWARD -s 100.x.x.x -d <
    blocked_ip> -j DROP`).
187 * **Country-level:** Add `ipset` to the `routing-fix` apk install step in `docker-compose.yml`, and load CIDR
    ranges from `ipdeny.com` into an ipset, then block that set in the `FORWARD` chain.
188
189 ## 6. macOS Quirks to Remember
190
191 1. **Colima SSH tunnel is TCP-only** UDP ports (like 5354/udp) are NOT accessible from host. Use `dig +tcp` or
    the Python DNS proxy (UDPTCP converter).
192 2. **docker compose exec -T` injects `\r`** Always `tr -d '\r'` when capturing output for comparisons or `
    networksetup` commands.
193 3. **brew services` can get stuck** Fix with `sudo brew services restart tailscale`. Common state: `brew
    services list` shows `started` but `tailscale status` shows "Tailscale is stopped."
194 4. **No `timeout` command** macOS lacks GNU `timeout`. Use the `run_with_timeout()` bash function.
195 5. **sudo -v` prompts even with NOPASSWD** Use `sudo -n` (non-interactive) everywhere.
196 6. **DNS is scoped to en0** macOS scutil DNS resolver is scoped to en0 (Wi-Fi). DNS changes on other
    interfaces (like USB Ethernet) are ignored unless en0's service is also updated. The script always sets DNS
    on BOTH `$ACTIVE_SERVICE` and `$ENO_SERVICE`.
197 7. **tailscaled clobbers DNS during exit-node transition** `force_dns_to_gateway` is called a second time at
    the end of the ENABLE branch to counteract this.
198 8. **tailscale up --reset** resets all unspecified flags to defaults Every `--reset` call MUST also pass `--
    accept-dns=false` explicitly or tailscaled re-enables DNS management.
199 9. **docker compose ps` CWD pitfall** `docker compose` commands require a `docker-compose.yml` in the current
    working directory. Use `docker ps` for raw container listing from any CWD; `docker compose ps` requires
    the project directory.
200
201 ---
202
203 ## 7. Environment Variables
204
205 ### `.env` File
206 | Variable | Purpose |
207 |-----|-----|
208 | `WIREGUARD_PRIVATE_KEY` | WARP WireGuard private key (from `wgcf generate`) |
209 | `WIREGUARD_PUBLIC_KEY` | WARP WireGuard public key |
210 | `WIREGUARD_ADDRESSES` | WARP WireGuard address (e.g., `172.16.0.2/32`) |
211 | `TS_AUTHKEY` | Tailscale auth key for the container |
212 | `NULLEXIT_SEED` | 256-bit seed for HMAC-SHA256 script integrity signing (generated by `setup.sh`) |
213 | `GATEWAY_RULE_PROFILE` | Rule compilation tier: `light`, `medium`, `heavy` |
214 | `GATEWAY_BYPASS_PING` | (Optional) `true` to proceed even if pre-flight checks fail |
215 | `GATEWAY_USE_EXIT_NODE` | (Optional) `false` to skip exit node, DNS-only mode |
216 | `GATEWAY_HIJACK_HOST` | (Optional) `false` to skip DNS hijacking on the host (VPN/adblocking for Tailscale
    peers only) |
217 | `GATEWAY_MSS` | (Optional) TCP MSS clamp value (default 1120); 1180 for speed on healthy paths |
218 | `WARP_FAIL_THRESHOLD` | (Optional) Consecutive `warp=off` polls before auto-shutdown (default 6 = 30s; 3 = 15
    s) |
219 | `WARP_ENDPOINT_1` | (Optional) Override Cloudflare WARP WireGuard endpoint IP 1 (default `162.159.192.1`) |
220 | `WARP_ENDPOINT_2` | (Optional) Override Cloudflare WARP WireGuard endpoint IP 2 (default `162.159.193.1`) |
221 | `KILL_SWITCH` | (Optional) `true` to enforce strict PF lock drops all host traffic if VPN fails. Breaks SSH
    if VPN dies. |
222 | `HOST_LEAK_PROBE` | (Optional) `false` to disable the 300ms host-egress leak prober (default: `true`) |

```

```

223 | `STOP_COLIMA_ON_EXIT` | (Optional) `true` to fully shut down the Colima VM on toggle-off (saves battery on
    | dedicated hosts) |
224 | `ADGUARD_PASSWORD` | AdGuard Home web UI password (default: `nullexit`; username always `admin`) |
225 | `BLOCKED_COUNTRIES` | Space-separated 2-letter ISO codes to block via ipdeny.com CIDR ranges (e.g. `kp il cn
    | ru`) |
226
227 ---
228
229 ## 8. Diagnostic Commands
230
231 ### Container Health
232 ```bash
233 docker ps --format '{{.Names}} {{.Status}}'
234 docker compose exec -T tailscale tailscale status
235 docker compose exec -T warp wget -q0- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace
236 ```
237
238 ### SOCKS5 Proxy
239 ```bash
240 curl --socks5-hostname 127.0.0.1:1080 --max-time 5 -s https://www.cloudflare.com/cdn-cgi/trace
241 ```
242
243 ### AdGuard DNS
244 ```bash
245 dig +tcp @127.0.0.1 -p 5354 google.com +short +timeout=5
246 ```
247
248 ### Firewall & Routing
249 ```bash
250 # nftables FORWARD chain
251 docker compose exec -T warp iptables -L FORWARD -v --line-numbers
252 # Legacy FORWARD chain
253 docker compose exec -T warp iptables-legacy -L FORWARD -v --line-numbers
254 # IP rules + routing tables
255 docker compose exec -T warp ip rule show
256 docker compose exec -T warp ip route show table 199
257 docker compose exec -T warp ip route show table 200
258 ```
259
260 ### Host macOS
261 ```bash
262 tailscale status
263 brew services list | grep tailscale
264 networksetup -getdnsservers Wi-Fi
265 networksetup -getsocksfirewallproxy Wi-Fi
266 sysctl net.inet.ip.forwarding
267 ```
268
269 ---
270
271 ## 9. Development Issue Tracker
272
273 This section documents issues encountered during development, their status, and resolutions.
274
275 ### 9.1 Exit Node Return Path (`rx 0`) RESOLVED
276 **Symptom:** Gateway showed `tx 936 rx 0` transmitted but never received return traffic.
277
278 **Root Cause:** `docker-compose.yml` included `FIREWALL_OUTBOUND_SUBNETS=100.64.0.0/10`. Gluetun created a
    strict routing rule (`ip rule add to 100.64.0.0/10 lookup 199`, priority 99) that forced all Tailscale
    CGNAT traffic to bypass the VPN via the Docker bridge. Return packets were sent to the macOS host instead
    of back through `tailscale0`.
279
280 **Fix:** Removed `FIREWALL_OUTBOUND_SUBNETS` entirely. `scripts/routing-fix.sh` now injects `lookup 52`, and
    return packets correctly flow back into `tailscale0`. The SOCKS5 proxy was repurposed as a bulletproof
    failover.
281
282 ### 9.2 Gluetun Resets nftables FORWARD Rules
283 **Symptom:** FORWARD RELATED,ESTABLISHED rule disappears after Gluetun health check.
284
285 **Fix:** `scripts/routing-fix.sh` re-adds to both iptables backends every 30 seconds. Legacy backend (policy
    ACCEPT) acts as safety net.
286
287 ### 9.3 docker compose exec `\r` Injection
288 **Fix:** All `docker compose exec` output piped through `tr -d '\r'`.
289
290 ### 9.4 Colima VM Memory / Swap Thrashing Latency
291 **Symptom:** After running nullexit flawlessly for over 24 hours straight to test stability, the 0.5GB VM RAM
    configuration began experiencing severe network latency on the Tailscale mesh later in the day.
292
293 **Root Cause:** Services (like AdGuard, Tailscale, Docker logs) slowly accumulate cache and state over extended
    periods. Under the tight 512MB physical RAM limit, this slow creep forced the Linux kernel to aggressively

```

```

swap memory to the 512MB SSD swap file. The constant disk I/O thrashing blocked process execution, causing
DNS resolutions and packet forwarding to delay significantly (making the internet feel "very slow").
294
295 **Proof (Empirical Observation):** While in this OOM-thrashing state, direct DNS queries through the Tailscale
mesh (`dig @100.100.21.8 google.com`) would intermittently hang or take several seconds to resolve. After
restarting with the 600MB configuration, the exact same query immediately dropped to a lightning-fast **41
ms** response time. (Note: Querying the mapped host port via `127.0.0.1:5354` will always time out due to
Gluetun's strict leak-prevention firewall blocking non-VPN ingress traffic).
296
297 **Fix:** Increased VM base physical RAM to 600MB (`--memory 0.6`) and reduced the swap file to 400MB. This
provides enough native RAM headroom for long-running state caches without forcing heavy I/O swapping.
Subdomain deduplication still reduces blocklists by ~60%. Memory profiles (`light`/`medium`/`heavy`) let
users trade off coverage for memory.
298
299 ### 9.5 tailscaled Daemon Broken State
300 **Symptom:** `brew services list` shows `started` but `tailscale status` shows "stopped".
301
302 **Fix:** `sudo brew services restart tailscale`. Toggle script now detects and auto-restarts.
303
304 ### 9.6 Stderr Invisibly Swallowed
305 **Symptom:** Failures were silent due to stderr redirected to `output.log`.
306
307 **Fix:** Tailscale wait loop now shows output. Critical commands show errors inline.
308
309 ### 9.7 Missing iptables in routing-fix Container
310 **Root Cause:** `alpine:3.20` does not have iptables installed. Commands failed silently with `2>/dev/null ||
true`.
311
312 **Fix:** Updated `docker-compose.yml` to `apk add --no-cache iptables iproute2` before `scripts/routing-fix.sh`
`.
313
314 ### 9.8 DOCKER_GW Variable Parsing Bug
315 **Root Cause:** `ip route show default` printed two lines; `awk '{print $3}'` picked up both, causing route
commands to fail.
316
317 **Fix:** Added `head -1` to the parsing chain.
318
319 ### 9.9 Native SSH & SFTP Security (Zero Local Attack Surface)
320 **Context:** macOS "Remote Login" (`ssh`) and "File Sharing" (`smbd`) open ports on the local network (e.g.
`172.x.x.x`), exposing the host to brute-force attacks on public Wi-Fi.
321
322 **Implementation:** The script strictly enforces `tailscale up --ssh=true` whenever the mesh state is reset.
This empowers the user to completely disable native macOS Remote Login and File Sharing. This provides an
incredible zero-trust security advantage: even if an attacker steals your Mac username and password, they
**cannot** access your files remotely. Disabling the native services completely closes the listening ports
on the local Wi-Fi interface (`172.x.x.x`), rendering the Mac inaccessible to local network logins.
Meanwhile, Tailscale intercepts port 22 traffic exclusively over the encrypted mesh and authenticates
cryptographically based on your Tailscale SSO identity. Stolen Mac passwords are mathematically useless
without first bypassing your Tailscale Two-Factor Authentication and registering a device onto the mesh.
323
324 **Cross-Platform File Exchange:** To easily browse and exchange files securely, simply use an SFTP client and
connect to your Mac's MagicDNS name on port 22. Recommended clients: **WinSCP** or **FileZilla** (Windows),
**Cyberduck** (macOS), and **FE File Explorer** or **Solid Explorer** (iOS/Android).
325
326 ### 9.10 Changing macOS Local Hostname
327 **Context:** Users may wish to change their Mac's computer name/hostname in macOS System Settings.
328 **Effect:** Changing the Mac's hostname will **not** break Nullexit (which relies on internal routing/localhost
). However, Tailscale will automatically detect this change and update the machine name on the Tailnet.
329 - The underlying Tailscale `100.x.x.x` IPv4 address will remain exactly the same.
330 - The **MagicDNS name** will change to match the new hostname. Users must remember to update their SFTP/SSH
clients to use the new MagicDNS address.
331
332 ### 9.11 Logging Architecture
333 For debugging, logs are strictly segmented based on the component's lifecycle:
334 - **`output.log` (Host-side):** Contains all standard error (`stderr`) and verbose output from the host scripts
(`toggle.sh`, `recover.sh`, `setup.sh`). Since the `rule-compiler` container is ephemeral and deleted
after running, its logs (and any Python errors from `scripts/sync-rules.py`) are extracted and appended to
this file before deletion.
335 - **Log Rotation Policy:** On startup, `toggle.sh` checks if `output.log` exceeds **50MB** (`52,428,800`
bytes). If it does, it performs a metadata-only rename (`mv output.log output.log.old`), preserving history
while resetting the active log to 0 bytes. This rename-based rotation avoids the heavy disk I/O and CPU
overhead of rewriting a massive file line-by-line.
336 - **Egress Probe Logging:** The background host-egress leak prober checks if stdout is a TTY (`[ -t 1 ]`) and
will only print live status updates (using carriage return `\r`) or duplicate console warnings in
interactive mode. It automatically detects macOS/BSD vs. Linux environments to dynamically construct
timestamps with date and time (omitting milliseconds on macOS to prevent high CPU utilization from spawning
sub-second processes).
337 - **`docker logs <container>` (Guest-side):** The persistent containers (`warp`, `tailscale`, `routing-fix`, `
adguardhome`) use the Docker `json-file` logging driver with a strict `max-size` (1m-10m) to prevent VM
disk exhaustion. Use standard `docker logs` to view them.
338

```

```

339 ### 9.12 Chrome Remote Desktop Connection Failures
340 **Context:** When attempting to access a remote machine via Chrome Remote Desktop (CRD), the connection fails
    or hangs if Tailscale is active **on the remote device**. The connection works fine even if Tailscale (and
    the exit node) is active **on the local client/viewer device**, but only if Tailscale is disabled on the
    remote machine.
341
342 **Why:** Having Tailscale active on the remote machine creates virtual network interfaces and DNS/routing
    modifications that conflict with Chrome Remote Desktop's WebRTC bindings and signaling traffic.
343
344 **Workaround:**
345 - **Use Tailscale Native RDP/RustDesk (Recommended):** Connect directly to the remote device's Tailscale IP
    (`100.X.Y.Z`) via an RDP or RustDesk client. This is faster and avoids third-party relay conflicts.
346 - **Disable Tailscale on the Remote Device:** If you must use CRD, disable Tailscale on the remote machine
    before connecting.
347
348 ### 9.13 Docker Default Bridge vs Host Wi-Fi Subnet Collision (The 172.17.0.0/16 Subnet Overlap)
349 **Context:** When attempting to ping a local Windows client (`172.17.22.187`) or the default gateway
    (`172.17.0.1`) directly over Wi-Fi, the Mac host returns `No route to host` (displaying a `REJECT` / `!`
    flag in the routing table). Tailscale on the client also drops offline shortly after starting when using
    the exit node, failing to establish direct P2P connections and falling back to slow DERP relays.
350
351 **Root Cause & Subnet Overlap Mechanism:**
352 1. **Docker Default Subnet:** By default, the Docker daemon initializes its default bridge network (`docker0`)
    on the `172.17.0.0/16` IP range.
353 2. **Subnet Conflict:** The host's physical Wi-Fi network happens to be assigned the exact same
    `172.17.0.0/16` range by the local network router.
354 3. **Routing Clash:** Colima launches a Linux VM on the Mac host to run the Docker engine. Because Docker
    inside that VM creates `docker0` and claims the `172.17.0.0/16` subnet, any packet sent to `172.17.x.x`
    from within the containers (such as Tailscale local discovery) is captured by the VM's internal virtual
    bridge interface (which is down/empty) and is blackholed. On the Mac host, macOS dynamically marks routes
    as `REJECT` (!) when local ARP resolution fails, causing local pings to immediately abort with `No route
    to host`.
355 4. **Tailscale Relay Saturation:** Because local peer-to-peer discovery is blackholed, Tailscale falls back to
    public DERP relay servers (e.g. `relay "tor"` in Toronto). When exit-node routing is enabled, all client
    web traffic is routed through the relay. Public relays impose strict rate limits and throttle high-volume
    traffic, resulting in packet drops. This saturation drops Tailscale's control packets, causing the
    connection to time out and showing the client as offline.
356
357 **Fix:**
358 The canonical script for this is `scripts/fix-docker-bridge-collision.sh`. **One command applies the entire fix
    :**
359
360 ```bash
361 bash scripts/fix-docker-bridge-collision.sh          # apply
362 bash scripts/fix-docker-bridge-collision.sh --dry   # preview changes without applying
363 ```
364
365 It auto-detects the host's actual Wi-Fi subnet (no more guessing whether the campus DHCP handed out `172.x`,
    `10.x`, or `192.168.x`), picks a non-conflicting `docker.bip` from a small candidate set (defaulting to
    `172.26.0.1/24` if no overlap is matched), backs up `~/colima/default/colima.yaml` with an ISO-8601
    timestamp, edits the file in place (idempotent replaces an existing `docker.bip` if present, appends
    under an existing `docker:` block, or creates a fresh block), restarts Colima, rebinds Wi-Fi DHCP, verifies
    the new default route is no longer Docker's bridge, and re-runs both `./toggle.sh` and `scripts/diagnose-
    host-leak.sh` to print the post-fix verdict.
366
367 For reference, the underlying manual fix is: edit the Colima configuration file (`~/colima/default/colima.yaml
    `) on your Mac to define:
368 ```yaml
369 docker:
370   bip: 172.26.0.1/24   # any /24 that does NOT overlap your Wi-Fi subnet
371 ```
372 Then, restart Colima to apply the new subnet inside the VM:
373 ```bash
374 colima restart
375 ```
376 This forces Docker to use `172.26.x.x` for its default bridge, freeing the physical `172.17.x.x` range and
    letting pings and direct Tailscale peer-to-peer connections flow normally.
377
378 ---
379
380 ### 9.14 Cloudflare WARP 24-Second Startup Delay (UDP Session Rate-Limiting)
381
382 **Context:** When running `toggle.sh` to turn the gateway ON, `docker compose up -d` often hangs for exactly
    24-25 seconds waiting for the `warp` container to become `healthy`. Users might mistakenly blame the Go
    `rule-compiler` processing 400k+ rules for this delay, as Docker prints `rule-compiler Exited 24.3s` at the
    end of the wait.
383
384 **Root Cause:** The `rule-compiler` actually finishes its work in <2 seconds. The 24-second blockage is
    entirely due to Cloudflare WARP's edge server rate-limiting. WireGuard is a stateless protocol with no
    "disconnect" packet. When the toggle is turned OFF, the old UDP session state remains active on Cloudflare's
    backend for 20-30 seconds. When the toggle is instantly turned back ON, Docker starts a new WireGuard
    connection from a new ephemeral UDP source port but uses the **same static cryptographic key** (from `.env

```

```

    `). Cloudflare detects this as a potential replay attack or key-sharing abuse, and intentionally drops all
    packets (rate-limits) for the new connection until the old session's ghost state fully expires (-20-25
    seconds).
384 **Result:** Gluetun's healthcheck fails repeatedly every 2 seconds until Cloudflare lifts the penalty box, at
    which point traffic flows and Docker unblocks the rest of the startup sequence. This is a known, expected
    behavior of reusing static keys rapidly on Cloudflare's network and cannot be bypassed.
385
386 ### 9.15 Background Sudo Failures
387 **Context:** The `--post-wake` script is intentionally designed to skip the `sudo -v` interactive prompt
    because it is launched by `watcher.sh` running via `launchd` in the background (which has no TTY and cannot
    accept password input). Thus, it relies entirely on `sudo -n` (non-interactive sudo) for all its internal
    routing changes.
388 **Problem:** If the user has not configured the `/etc/sudoers.d/nullexit` file properly, any automated
    background wake event will silently fail: `sudo -n` commands drop out, the WARP bypass routes fail to apply
    , and the `warp` container loses internet connectivity.
389
390 ### 9.16 Tailscale Hijacking the Default Route (PHYSICAL_GW Bug)
391 **Problem:** When `--post-wake` runs, it clears the routing table using `sudo route -n flush` to ensure a clean
    slate after the Wi-Fi card roams or wakes up. Because it skips bouncing the Wi-Fi interface (to make the
    reconnect faster), macOS's DHCP client does not automatically rebuild the physical default route. To
    counteract this, the script must manually restore the physical default route. Originally, the script tried
    to capture the physical gateway via `route get default`. However, because the gateway is already running
    and Tailscale is active, the default route is often already pointing to `utunX`. When this happens, `route
    get default` does not output a `gateway:` field (it only outputs the `interface:`), causing the captured `
    PHYSICAL_GW` variable to be empty.
392 When `PHYSICAL_GW` is empty, the physical default route is never restored, causing the `en0` interface to be
    isolated from the physical network. The WARP bypass routes then fall back to targeting `interface en0` (
    instead of routing via the gateway), which breaks WireGuard's remote connections completely.
393 **Fix:** The script now bypasses the OS routing table entirely and directly queries the hardware DHCP lease (
    `ipconfig getpacket en0`) to reliably extract the physical router IP, ensuring the physical route is
    correctly restored even when Tailscale is active.
394
395 ### 9.17 Tailscale MagicDNS Bypassing the AdGuard PREROUTING Intercept
396 **Context:** The `routing-fix.sh` script applies an iptables NAT rule (`PREROUTING -i tailscale0 -p udp --dport
    53 -j REDIRECT`) to intercept all incoming DNS queries from connected Tailscale peers and force them into
    the AdGuard container on port 5335.
397 **Problem:** When remote clients (like Android or iOS) have "Use Tailscale DNS settings" turned on, they query
    Tailscale's MagicDNS IP (`100.100.100.100`). This packet enters the exit node, but the `tailscaled` daemon
    running *inside* the gateway container intercepts the `100.100.100.100` packet internally. `tailscaled`
    then generates a brand new, local DNS request to the upstream DNS server to resolve the query. Because this
    new request is generated *locally* by the daemon (not arriving externally via the `tailscale0` interface),
    it bypasses the `PREROUTING` chain completely. The request glides straight out the WARP tunnel to
    Cloudflare/Google, entirely bypassing the AdGuard sinkhole.
398 **Fix:** The only way to fix this blindspot is to force `tailscaled` to use AdGuard as its upstream. In the
    Tailscale Admin Console, the gateway's Tailscale IPv4 address (e.g., `100.x.x.x`) must be added as the sole
    Custom Global Nameserver, and "Override local DNS" must be enabled. Additionally, to block Android devices
    from bypassing this via Private DNS (DNS-over-TLS), outbound Port 853 traffic is explicitly REJECTED in
    both `routing-fix.sh` and `post-rules.txt`.
399
400 ### 9.18 Network Bufferbloat & MTU Optimizations (Double-VPN UDP Crash)
401 **Symptom:** When running aggressive UDP bandwidth tests (such as macOS `networkQuality`, which uses QUIC/HTTP3
    ), the `nullexit` tunnel completely crashes or exhibits severe lag (6,000ms+ bufferbloat).
402
403 **Root Cause:** The bug is a cascading MTU mismatch. Tailscale generates a UDP packet. Cloudflare WARP (also
    WireGuard) receives the packet, wraps it in another layer of encryption, pushing the packet size beyond the
    standard 1500 byte limit. Unlike TCP, UDP packets cannot be resized dynamically via MSS negotiation,
    resulting in massive IP packet fragmentation. The Apple Virtualization framework (`vz`), which Colima uses
    for bridged networking, silently drops heavily fragmented UDP packets under high load. This blackholes the
    entire UDP upload stream, instantly dropping the connection.
404
405 **Fix (Partial):** To prevent MTU fragmentation for standard web traffic, **TCP MSS Clamping** is strictly
    enforced via the macOS `pf` firewall. By adding `scrub out all max-mss 1160` to `scripts/pf.conf`, macOS
    unilaterally shrinks all outgoing TCP headers to fit comfortably inside the double-encrypted tunnel,
    bypassing Path MTU Discovery blackholes. `TS_DEBUG_MTU=1200` was also added to `docker-compose.yml` to
    minimize internal fragmentation.
406 **Unresolved:** The only way to solve the UDP crash bug natively is to artificially cap the maximum tunnel
    bandwidth using SQM (`fq_codel`). Because a static bandwidth cap would artificially throttle high-speed
    networks, `nullexit` leaves the bandwidth uncapped, optimizing perfectly for TCP while accepting UDP
    fragmentation under extreme edge-case load tests.
407 *Why not Dynamic SQM (Aurate)?* Implementing an EMA/PID-controlled aurate daemon (like `sqm-aurate`)
    requires a constant 100ms polling loop to measure the kernel packet queue, which severely degrades laptop
    battery life by preventing deep C-state CPU idling. Furthermore, macOS's native firewall (`dummynet`) only
    supports `fq_codel` and does not support the newer Linux `CAKE` algorithm, which is the only algorithm that
    offers kernel-native, event-driven `aurate-ingress` (zero polling penalty). Therefore, dynamic SQM is
    computationally hostile to battery-powered macOS hosts.
408
409 ## 10. Resolved Issues (from README)
410
411 These bugs and edge cases were discovered and resolved during development.
412
413 ### 10.1 DNS Proxy: TCP Wire Format Mismatch (socat / dnsmasq)

```

```

414
415 **Problem:** When the Tailscale data plane is unavailable (DERP relay only), the script falls back to a local
DNS proxy that forwards queries from host UDP:53 to Docker's TCP:5354 (AdGuard). The `socat` and `dnsmasq`
approaches both failed.
416
417 - **socat** forwards raw bytes it does not prepend the 2-byte length prefix that DNS-over-TCP requires.
AdGuard parses the stream incorrectly and returns garbage.
418 - **dnsmasq** tries UDP first to reach the upstream (`127.0.0.1#5354`), but Colima's SSH tunnel only forwards
TCP. With a single upstream server, dnsmasq doesn't fall back to TCP even with `--timeout=1`.
419
420 **Solution:** Replaced both with a **Python DNS proxy** (`scripts/dns-proxy.py`, ~25 lines) that properly
handles the DNS-over-TCP wire format: reads a UDP query from the host, prepends the 2-byte length prefix,
sends it over TCP to AdGuard, strips the prefix from the response, and sends it back over UDP.
421
422 *Architectural Note: Why is the DNS proxy in Python while the SOCKS5 proxy was moved to a compiled Go Docker
container?
423 The SOCKS5 proxy handles heavy data streaming (e.g., video, downloads). Python introduces massive CPU/battery
overhead for raw socket streaming, which is why we rely on the compiled Go implementation built directly
into the `tailscaled` daemon via `TS_SOCKS5_SERVER=:1080`.
424 Conversely, the DNS proxy only handles intermittent UDP queries (a few bytes per minute) generated solely by
the local host machine. The Python overhead for this is effectively 0.001 Watts. We deliberately kept `dns-
proxy.py` in Python because it runs natively on the macOS/Linux host rewriting it in Go would force users to
install a Go compiler (`brew install go`) on their host machine for zero noticeable battery gain. (Note:
If this architecture is ever adapted to serve as a public DNS resolver for thousands of users or for
massive automated web-scraping clusters, `dns-proxy.py` must be rewritten in Go to survive the concurrent
packet flood).
425
426 ### 10.2 Exit Node Routing Conflict & The SOCKS5 Failover
427
428 **Problem:** For days, Tailscale's exit node refused to return traffic back to the client (`rx 0`), leading to
the assumption that Tailscale's userspace forwarding was bypassing `iptables` and ignoring the WARP `tun0`
interface. In a desperate attempt to fix this, a **SOCKS5 proxy** (`scripts/socks5-proxy.py`) was
introduced as a replacement.
429
430 **The Real Root Cause:** Tailscale's userspace forwarding was not the problem. The issue was an environment
variable in Gluetun (`FIREWALL_OUTBOUND_SUBNETS=100.64.0.0/10`) that instructed the VPN container to hijack
all Tailscale CGNAT traffic and forcibly route it out the unencrypted Docker bridge (`eth0`). This
blackholed all return packets back to the client.
431
432 **Solution:** Removed the offending `FIREWALL_OUTBOUND_SUBNETS` variable, immediately restoring the Tailscale
exit node. Instead of removing the SOCKS5 proxy, we repurposed it into a **bulletproof failover** for when
restrictive Wi-Fi networks block Tailscale UDP ports.
433
434 Architecture after the fix:
435 ```
436 PRIMARY (Exit Node):
437 DNS/TCP/UDP: Apps Tailscale Exit Node gateway container tun0 WARP internet
438
439 FAILOVER (SOCKS5 - if Tailscale is blocked by local Wi-Fi):
440 DNS: Browser 127.0.0.1:53 Python proxy TCP:5354 AdGuard WARP
441 TCP: Apps SOCKS5:1080 gateway container tun0 WARP internet
442 Mesh: SSH/ping 100.x.x.x utun5 WireGuard peers (independent of proxy)
443 ```
444
445 ### 10.3 Pre-Flight Check 1: Wrong `tailscale ping` Flag
446
447 **Problem:** Check 1 of the pre-flight connectivity checks used `--c 1` (double dash) but `tailscale ping`
accepts `-c 1` (single dash). The invalid flag caused the command to exit with code 1, marking the check as
FAIL even though the gateway was reachable.
448
449 **Fix:** Changed `--c 1` to `-c 1`.
450
451 ### 10.4 Pre-Flight Check 1: DERP Relay Exit Code
452
453 **Problem:** Even with the correct `-c 1` flag, `tailscale ping` exits with code 1 when the connection goes
through a DERP relay (`direct connection not established`) even though a pong was successfully received
(28ms via DERP(tor)). The check treated relayed connections as failures.
454
455 **Fix:** Changed the check from relying on exit code to piping stdout through `grep -q "pong"`. This accepts
relayed connections as valid (they work fine for the exit node use case), while still failing on true
timeouts or unreachable peers.
456
457 ### 10.5 Container Default Route Bypasses WARP
458
459 **Problem:** Inside the WARP container's network namespace, the main routing table's default route was via `
eth0` (Docker bridge), not `tun0` (WARP tunnel). While gluetun uses policy routing (rule 101 table 51820
`tun0`) for most traffic, traffic originating from the container's own IP (`172.18.0.2`) hits rule 100 (
from 172.18.0.2 lookup 200), whose default was also via `eth0`. This caused the SOCKS5 proxy's outbound
connections to bypass WARP.
460
461 **Fix:** Updated the `routing-fix` container to:

```

```

462 1. Change the main table's default route to `default dev tun0`
463 2. Change table 200's default route to `default dev tun0`
464 3. Add a specific route for the WARP WireGuard endpoint (`162.159.192.1`) through `eth0` via the Docker gateway
    in table 200, preventing a tunnel loop
465 4. Re-assert all routes every 30 seconds (gluetun may reset them on health checks)
466 5. Dynamically detect the Docker subnet from `ip route show dev eth0` instead of hardcoding `172.18.0.0/16`
467
468 ### 10.6 IPv6 Exit-Node Leak
469
470 **Problem:** Tailscale supports IPv6, but WARP/glutun is IPv4-only. IPv6 packets forwarded through the exit
    node would leak through the Docker bridge unencrypted.
471
472 **Fix:** Added `ip6tables -A FORWARD -i tailscale0 -j DROP` to `post-rules.txt`, blocking all IPv6 forwarded
    traffic from the Tailscale interface.
473
474 ### 10.7 Hardcoded Docker Subnet in Routing Fix
475
476 **Problem:** The `routing-fix` container hardcoded `172.18.0.0/16` and `172.18.0.1` for the Docker bridge
    subnet and gateway. If Docker's bridge network changed (after `docker network prune`, Colima restart, or on
    a different machine), these routes would be wrong.
477
478 **Fix:** Detection is now dynamic using `ip route show`:
479 - `DOCKER_NET=$(ip route show dev eth0 | grep -v default | head -1 | awk '{print $1}')
480 - `DOCKER_GW=$(ip route show default | awk '{print $3}')
481
482 This extracts the actual subnet and gateway directly from the routing table, working with any subnet size
    (`/16`, `/24`, `/20`, etc.).
483
484 ### 10.8 SOCKS5 Proxy Threading Race Condition
485
486 **Problem:** The `forward` function in `scripts/socks5-proxy.py` called `select.select([src, dst], [], [])`
    which raised `ValueError: file descriptor cannot be a negative integer (-1)` when one thread closed a
    socket while the other thread was selecting on it.
487
488 **Fix:** Added `ValueError` exception handling around `select.select()` and `recv()` calls. When a file
    descriptor becomes negative, the thread breaks out of the forwarding loop cleanly instead of crashing.
489
490 ### 10.9 Docker Healthcheck: Multi-line Python in CMD Array
491
492 **Problem:** The socks-proxy healthcheck used a `CMD` array with a multi-line Python string. YAML collapsed the
    newlines, causing the `#` comment to comment out the rest of the code and the `assert` statement to be
    mangled into `as`.
493
494 **Fix:** Changed from `["CMD", "python3", "-c", "..."]` to `["CMD-SHELL", "python3 -c '...']` with a single-
    line Python expression. Uses a real SOCKS5 handshake (sends `[5, 1, 0]`, expects `[5, 0]`) instead of a
    simple TCP port check.
495
496 ### 10.10 Sudo Credential Caching Removed
497
498 **Problem:** The script used `sudo -v` at startup to cache sudo credentials, plus a background `SUDO_KEEPER_PID`
    loop refreshing them every 60 seconds. This required interactive password entry even with NOPASSWD
    sudoers configured, because `sudo -v` itself prompts.
499
500 **Fix:** Removed the entire `sudo -v` + `SUDO_KEEPER_PID` section. All privileged commands now use `sudo -n` (
    non-interactive), which works silently because the user has NOPASSWD rules in `/etc/sudoers.d/nullexit` for
    the specific commands needed (`networksetup`, `python3`, `dscacheutil`, `killall`, `pkill`, `route`, `
    ifconfig`).
501
502 ### 10.11 macOS Application Firewall Silently Blocking AirDrop & Continuity
503
504 **Problem:** Users of complex network extensions (Tailscale, LuLu, Docker, etc.) often run into an issue where
    AirDrop, AirPlay, and Universal Clipboard silently fail, even with the gateway inactive and Wi-Fi/Bluetooth
    correctly configured. The internal Apple Wireless Direct Link (`awdl0`) interface appears healthy, but
    connections never establish.
505
506 **Root Cause:** The built-in macOS Application Firewall has a specific list of explicitly blocked applications.
    Apple's own core networking services (e.g., `/usr/libexec/sharingd`, `/usr/libexec/rapportd`, `/usr/
    libexec/avconferenced`) can be silently added to the "Block incoming connections" list either through
    accidental "Deny" clicks on permission prompts, execution of aggressive privacy-hardening scripts, or a
    known macOS bug that retains strict blocks even after toggling the "Block all incoming connections" setting
    off. Since these are background daemons, macOS provides no visible error.
507
508 **Fix:** The firewall rules must be cleared via the terminal (or System Settings > Network > Firewall > Options
    ):
509 ```bash
510 sudo /usr/libexec/ApplicationFirewall/socketfilterfw --unblockapp /usr/libexec/sharingd
511 sudo /usr/libexec/ApplicationFirewall/socketfilterfw --unblockapp /usr/libexec/rapportd
512 ```
513 Once unblocked, `sharingd` immediately resumes broadcasting mDNS/Bonjour discovery packets, and AirDrop
    restores instantly without requiring a reboot.
514

```

515 **### 10.12 Hard Reboots Leave Stale Docker Socket in Colima VM**
516
517 ****Problem:**** If the macOS host machine is subjected to a hard reboot, sudden power loss, or a kernel panic, the Colima VM running in the background is abruptly killed. Upon next boot, running `toggle.sh` resulted in the contradictory output:
518 ``Colima is already running.`` followed by ``Cannot connect to the Docker daemon at unix:///-.colima/default/docker.sock.``
519
520 ****Root Cause:**** The hard crash prevents the inner Docker daemon from executing its graceful shutdown routines. It leaves behind a corrupted ``docker.sock`` file inside the VM. On boot, the outer VM spins up (hence "already running"), but the inner Docker daemon sees the stale socket, assumes another instance is running, and crashes.
521
522 ****Fix:**** Added an Auto-Healing routine directly into `toggle.sh` (Step 4b). The script now explicitly tests the Docker daemon with ``docker ps``. If the daemon is unresponsive despite Colima reporting as active, it assumes a stale socket and automatically runs ``colima restart`` in the background to cleanly rebuild the VM and socket before proceeding.
523
524 **### 10.13 Graceful Shutdowns Leave DNS Hijacked**
525
526 ****Problem:**** The gateway overrides the macOS host's DNS settings (via ``networksetup``) to route queries to the AdGuard container. Because macOS persists ``networksetup`` changes to disk, shutting down the Mac while the gateway is running causes the Mac to boot up later with hijacked DNS. Since the Colima VM and Docker containers don't auto-start on boot, the user would have no internet access until they manually run the `toggle script`.
527
528 ****Root Cause:**** The `toggle.sh` script exits after the gateway starts. There was no background daemon listening for macOS system shutdown signals to revert the network settings.
529
530 ****Fix:**** Wrapped the background ``caffeinate`` process (used to prevent system sleep) in a bash ``trap`` command. The trap listens for ``SIGTERM``, ``SIGINT``, and ``SIGHUP``. When the user initiates a graceful shutdown, macOS sends ``SIGTERM`` to the background trap, which instantly executes ``recover.sh`` to flush the host DNS back to ``1.1.1.1`` right before the machine powers off.
531 ***Note:*** We explicitly use ``recover.sh`` instead of `toggle.sh` because during a shutdown, the OS limits process cleanup time and is already independently sending termination signals to Docker and Colima. Trying to run ``docker compose down`` inside a shutdown trap creates a race condition that can hang the script until macOS forcefully ``SIGKILL``s it. ``recover.sh`` abandons container management and surgically flushes the macOS network settings in milliseconds, guaranteeing completion before the OS pulls the plug.
532
533 **### 10.14 Wi-Fi Edge Packet Loss via QUIC (UDP) Fragmentation**
534
535 ****Problem:**** Applications that heavily utilize HTTP/3 (QUIC) over UDP such as Facebook, Instagram, and Google services experience massive stuttering and stalled image loading when the client device is far away from the router (weak Wi-Fi signal). The issue did not occur when close to the router.
536
537 ****Root Cause:**** The ``nullexit`` gateway uses a double-encrypted tunnel (Tailscale + WARP). To prevent packets from fragmenting when entering these tunnels, a firewall rule in ``post-rules.txt`` uses ``--set-mss 1120`` to safely clamp TCP packets. However, because QUIC runs entirely over UDP, it bypasses the TCP MSS handshake completely. QUIC blasts maximum-MTU (1280 byte) UDP packets. The inner ``tailscale0`` interface (MTU 1200) forces these large UDP packets to fragment into two pieces. When the client is on the edge of Wi-Fi range (where 5-10% packet loss is common), losing *either* UDP fragment destroys the *entire* image frame. This exponential amplification of packet loss stalled QUIC streams.
538
539 ****Fix & Trade-offs:**** Appended an ``iptables`` rule to ``post-rules.txt`` that explicitly blocks ``UDP --dport 443`` (QUIC). When modern web apps detect QUIC is blocked, they instantly fall back to HTTP/2 (TCP 443). Because TCP is routed through the MSS clamp, the packets are resized *before* they leave, entirely eliminating IP fragmentation and restoring high-speed loading over weak Wi-Fi.
540 ***Consequences:*** This adds a minor latency penalty (TCP 3-way handshake) compared to QUIC's zero-RTT connection. It may also force WebRTC video calls (Google Meet/Discord) to fall back to TCP TURN servers, slightly inflating real-time video latency. However, in a constrained double-tunnel mesh, the absolute stability gained from eliminating fragmentation vastly outweighs these trade-offs.
541
542 **### 10.15 Threat Model: Local Proxy Discovery**
543
544 The Tor SOCKS proxy and Honey-Port Tripwire only bind to ``127.0.0.1`` (loopback). The real security boundary here is "can an unprivileged local process reach the loopback interface," not "does the malware know the port number."
545
546 The port-randomization and the Honey-Port trap only catch blind or generic port-scanners that do not already know where to look. They **do not protect** against a process that directly reads the ``/tmp/nullexit-ports.env`` file, greps the `toggle.sh` memory space, or runs ``lsof -i`` to inspect open socket bindings.
547
548 Because modifying file ownership on ``/tmp/nullexit-ports.env`` via ``chown`/`chmod`` to restrict read-access introduces an unavoidable Time-Of-Check to Time-Of-Use (TOCTOU) race condition (the file is created user-owned before privilege escalation, meaning file descriptors can be snatched), it is deliberately left as a standard user-owned file.
549
550 The actual, proper mitigation for preventing a malicious local process from reaching the proxy is not hiding the port; it is running a host-level outbound firewall with strict localhost/loopback filtering enabled (e.g., LuLu 4.3.0+ or Little Snitch). These firewalls gate access by cryptographic process identity (binary signature) at the kernel/network-extension level, rather than by port secrecy.

```

551
552 ##### Verifying Localhost Filtering (The Rogue Binary Test)
553
554 To empirically validate that the host firewall properly intercepts local proxy hijacking, you can compile and
    execute a completely unknown, un-whitelisted "rogue" binary to attempt a connection to the proxy port:
555
556 ```c
557 // lulu-test.c
558 #include <stdio.h>
559 #include <stdlib.h>
560 #include <arpa/inet.h>
561
562 int main(int argc, char *argv[]) {
563     int port = atoi(argv[1]);
564     int sock = socket(AF_INET, SOCK_STREAM, 0);
565     struct sockaddr_in server = { .sin_family = AF_INET, .sin_port = htons(port) };
566     server.sin_addr.s_addr = inet_addr("127.0.0.1");
567
568     printf("Rogue binary attempting connection to 127.0.0.1:%d...\n", port);
569     if (connect(sock, (struct sockaddr *)&server, sizeof(server)) < 0) {
570         printf("Connection BLOCKED by firewall.\n");
571         return 1;
572     }
573     printf("Connection SUCCESSFUL (Firewall bypassed!).\n");
574     return 0;
575 }
576 ```
577 Compile with `gcc -o lulu-test lulu-test.c`. When executed against the `$TOR SOCKS_PORT`, a properly configured
    host firewall (with "Allow Loopback" disabled in LuLu Preferences) will immediately pause the socket
    execution and generate a desktop alert for the unrecognized binary signature. This physically stops
    targeted local malware from exfiltrating data through the Tor tunnel, proving the threat model mitigation.
578
579 ---
580
581 ## 11. Deep Dive: Exit Node Return Path Analysis (June 26, 2026)
582
583 This section preserves the detailed packet-level analysis that led to identifying and fixing the `rx 0` bug.
584
585 ### The exit node was probably never going to work in this container topology
586
587 Here's the packet path traced by reading `ip rule show` and the routing tables live:
588
589 **Outbound (Phone-1 internet):**
590 1. Phone-1 sends packet to gateway via Tailscale mesh
591 2. Gateway's tailscale0 (kernel TUN, `TS_USERSPACE=false`) decrypts packet enters FORWARD chain
592 3. Packet has src=`100.87.42.87` (Phone-1), dst=some internet IP
593 4. FORWARD chain (nftables): Rule 1 matches (`-i tailscale0 -j ACCEPT`)
594 5. Routing decision for the forwarded packet. Which ip rule matches?
595 - Rule 98: `to 172.18.0.0/16` no (dst is internet)
596 - Rule 99: `to 100.64.0.0/10` no (dst is internet)
597 - Rule 100: `from 172.18.0.2` **NO** (src is `100.87.42.87`, not the container IP!)
598 - Rule 101: `not fwmark 0xca6c lookup 51820` **YES** (no fwmark on forwarded packets)
599 6. Table 51820: `default dev tun0` packet goes out through WARP
600 7. NAT POSTROUTING: `MASQUERADE on tun0` src becomes WARP IP
601 8. Packet exits through WARP to internet (in theory)
602
603 **Return (internet Phone-1):**
604 1. Return packet arrives on tun0, conntrack de-MASQUERADES dst back to `100.87.42.87`
605 2. Routing decision for dst=`100.87.42.87`:
606 - Rule 99: `to 100.64.0.0/10 lookup 199` **YES**
607 3. Table 199: `100.64.0.0/10 via 172.18.0.1 dev eth0` **sends it to the Docker gateway!**
608 4. **This is wrong.** The packet should go to tailscale0 (to be re-encrypted and sent to Phone-1), but table
    199 sends it out eth0 to the Docker bridge host.
609
610 **Table 199 is the smoking gun.** It has one route: `100.64.0.0/10 via 172.18.0.1 dev eth0`. This sends ALL
    Tailscale CGNAT return traffic out the Docker bridge to the host not back through tailscale0. The return
    packet leaves the container's network namespace entirely, hits the host's routing stack, and gets dropped
    because the host has no idea what to do with a raw packet for `100.87.42.87`.
611
612 ### Why table 199 exists and why it's wrong for exit node traffic
613
614 Table 199 + the CGNAT rule (pref 99) was designed so that the **container itself** can reach other Tailscale
    peers (e.g., for mesh management, DERP relay). For that use case, routing CGNAT traffic via the Docker
    bridge to the host (which has its own tailscaled) makes sense.
615
616 But for **forwarded exit node traffic**, the return packet needs to go back through tailscale0 inside the
    container not out to the host. The CGNAT rule intercepts the return packet before it can be routed back to
    tailscale0 through the main table or Tailscale's own routing (table 52, which is at pref 5270).
617
618 ### The SOCKS5 path works because it dodges all of this
619

```

```

620 SOCKS5 proxy creates connections **from the container's own IP** (`172.18.0.2`). So:
621 - ip rule 100 (`from 172.18.0.2 lookup 200`) matches
622 - Table 200 has `default dev tun0`
623 - Return traffic comes back to `172.18.0.2` on tun0, conntrack de-NATs, proxy sends response to client
624
625 No FORWARD chain involved. No CGNAT rule involved. No table 199. It just works.
626
627 ### Summary of findings
628
629 | Finding | Status | Evidence |
630 |-----|-----|-----|
631 | Table 199 CGNAT route hijacks exit node return traffic | **Fixed** | Changed `lookup 199` to `lookup 52` in `
  scripts/routing-fix.sh`. Return packets now route via tailscale0. |
632 | FORWARD RELATED,ESTABLISHED missing | **Fixed** | Installed `iptables` via apk in `docker-compose.yml`. Rule
  now injects successfully. |
633 | DOCKER_GW variable parsing error | **Fixed** | Added `head -1` in `scripts/routing-fix.sh`. |
634 | `FIREWALL_OUTBOUND_SUBNETS` was the root cause | **Fixed** | Removed from `docker-compose.yml`. Gluetun no
  longer hijacks CGNAT routing. |
635 | Host tailscaled error state from aggressive `tailscale down` | **Mitigated** | Toggle script now detects and
  auto-restarts. |
636
637 ### 11.1 Double-Tunneling MSS Clamping (Stalling Bug)
638
639 **Problem:** Traffic double-wrapped through Tailscale (WireGuard) and WARP (WireGuard) suffered 120 bytes of
  MTU overhead (60 + 60). The default MSS clamp of 1180 was still slightly too large, causing large packets
  to fragment or drop, which resulted in mysterious web page stalls on strict-MSS endpoints.
640
641 **Fix:** Lowered the TCP MSS clamp in `post-rules.txt` to `1120` to guarantee double-tunneled payloads fit
  safely within standard internet MTUs.
642
643 ### 11.2 Linux Native Wi-Fi Bouncing
644
645 **Problem:** The generated `scripts/recover-linux.sh` incorrectly retained the macOS `networksetup` commands,
  which would crash and fail to bounce Wi-Fi on Linux hosts.
646
647 **Fix:** Swapped the macOS logic for native Linux commands. The script now attempts `nmcli radio wifi off/on`
  first, and gracefully falls back to `ip link set wlan0 down/up` if NetworkManager is unavailable.
648
649 ### 11.3 TS_AUTH_ONCE Cloud Footgun & Ephemeral Keys
650
651 **Decision:** The compose file uses `TS_AUTH_ONCE=true` to prevent authentication loops. However, on headless
  cloud VPS deployments, if a standard auth key expires (usually 90 days), the container will silently drop
  off the mesh. We documented this trade-off explicitly in `docker-compose.yml` and recommended the use of **
  Tailscale Ephemeral Keys** for cloud deployments, which automatically clean themselves up and bypass this
  issue.
652
653 ### 11.4 Hardcoded AdGuard Credentials
654
655 **Decision:** AdGuard is deployed with the hardcoded credentials `admin / nullexit`. This is perfectly safe
  behind a NAT on a local macOS laptop. However, if deployed on a cloud host with exposed ports, this becomes
  a severe security risk. We added a stark warning comment in `docker-compose.yml` to ensure cloud users
  change this before deployment.
656
657 ### 11.5 Routing Fix: 30-second polling vs Netlink Socket
658
659 **Decision:** Instead of building a complex Netlink socket listener (`ip monitor`) to react instantly to
  iptables and routing flushes from Gluetun, we retained the 5-second `sleep` loop in `scripts/routing-fix.sh`
  .
660 - **Reasoning:** Building a netlink listener in pure bash on Alpine is incredibly brittle and complex (
  especially for iptables events). The 30-second polling loop is bulletproof. The only trade-off is a
  potential 1-4 second stutter if Gluetun reconnects, which was deemed an acceptable edge case for absolute
  reliability.
661
662 ### 11.6 Cache Poisoning and URL Rot in scripts/sync-rules.py
663
664 **Problem:** If a remote blacklist URL went permanently offline (404 Not Found), the Python `urllib` request
  would return the 404 HTML string. The script would aggressively regex-parse this HTML, find no domains, and
  overwrite the healthy local cache with a 0-domain file. This effectively disabled ad-blocking until the
  URL was fixed.
665
666 **Fix:** Implemented a defensive programming sanity check in `scripts/sync-rules.py`. Before overwriting the
  cache, the script now verifies that the compiled domain count is greater than 10 (and 1 for IP feeds). If
  it falls below this threshold (indicating a catastrophic 404 failure across multiple URLs), it aborts the
  cache overwrite, raises a `ValueError`, and keeps the previous day's healthy cache intact.
667
668 ### 11.7 Cellular Networks & PMTUD Blackholes (The 1280 Byte Limit)
669
670 **Experiment:** Conducted a live packet sweep from the PC-1 host to Phone-1 connected via a cellular network +
  Tailscale DERP relay using `ping -D -s <size>`.
671
672 **Result:**

```

```

673 - Packets with a payload of `1252` bytes (+ 28 bytes IP/ICMP headers = **1280 bytes total**) succeeded
    perfectly with ~60ms latency.
674 - Packets with a payload of `1253` bytes (**1281 bytes total**) and above resulted in a silent timeout.
675
676 **Analysis:** The cellular network (or the Tailscale DERP relay) enforces a strict MTU of 1280 bytes.
    Critically, it acts as a "PMTUD Blackhole" it silently drops packets larger than 1280 bytes instead of
    returning an ICMP "Fragmentation Needed" warning. If the exit node tries to send a standard 1500-byte
    internet packet to the phone over this link, it vanishes, causing web pages to stall permanently.
677
678 **Conclusion:** This empirically validates the absolute necessity of the TCP MSS clamping rule in `post-rules.
    txt`. By artificially clamping the MSS to `1120` (or `1180`), we ensure the TCP payload + headers + double-
    WireGuard overhead never exceeds the strict 1280-byte ceiling of the cellular mesh link.
679
680 ### 11.8 Gluetun nf_tables Parser Crash (Bash Interpolation Failure)
681
682 **Problem:** Attempting to make the TCP MSS dynamically configurable by using a standard bash variable inside `
    post-rules.txt` (`iptables ... --set-mss ${GATEWAY_MSS:-1120}`) caused the Gluetun container to
    catastrophically crash during startup. Gluetun's internal parser executes `post-rules.txt` line by line
    without a shell, meaning the variable was never expanded. It literally attempted to inject the string `${
    GATEWAY_MSS:-1120}` into iptables, resulting in a fatal `bad value for option "--set-mss"` error.
683
684 **Fix:** Removed the bash variable from `post-rules.txt` and reverted it to a pure hardcoded integer. To retain
    dynamic `.env` configuration, we moved the interpolation logic to `toggle.sh`. Right before Docker starts,
    the host script parses `GATEWAY_MSS` from the `.env` file and uses a cross-platform `sed` command to
    dynamically overwrite the integer directly inside `post-rules.txt`. This perfectly mimics manual
    configuration and completely bypasses Gluetun's strict parser.
685
686 ### 11.9 Seamless Wi-Fi Roaming & The macOS DNS Wipeout Loophole
687
688 **Problem:** The Tailscale, WireGuard, and Colima NAT stacks are natively designed for connectionless roaming
    and will seamlessly survive when the macOS host switches Wi-Fi networks (e.g., roaming from home Wi-Fi to a
    cellular hotspot). However, the internet completely breaks upon network transition. This occurs because
    macOS intentionally wipes out all custom DNS settings (`networksetup -setdnsservers`) and reverts to the
    new network's default DHCP nameserver whenever the active Wi-Fi BSSID changes. Since the Mac is locked to
    the exit node, its DNS requests are swallowed by WARP and dumped onto the public internet, which cannot
    route private DHCP IPs. This results in a total DNS failure.
689
690 **Solution:** Implementing a silent background "DNS Watcher" daemon (`nullexit-dns-watcher`) in `toggle.sh`.
    When the gateway starts, it spawns a background polling loop that checks the active Wi-Fi DNS every 30
    seconds. If macOS resets the DNS during a network change, the watcher instantly detects the drift and
    forcefully re-injects `100.100.21.8` via `networksetup`. When the gateway stops, the watcher process is
    cleanly killed. This allows true, seamless roaming across physical networks without ever dropping the
    gateway state.
691
692 ### 11.10 The Infinite Recursive Routing Loop (Hotspot Paradox)
693
694 **Problem:** A critical network topology crash occurs if the host Mac connects to a Mobile Hotspot (e.g., a
    Windows PC or Phone) that is *also* connected to the Tailscale mesh and has "Use Exit Node" enabled.
    Because the Mac is hosting the Exit Node, it routes its internet traffic to the Hotspot. The Hotspot
    intercepts the Mac's traffic on its way to the cellular tower and routes it *back* to the Exit Node (the
    Mac) via Tailscale. The Mac receives it, tries to send it out again, and the Hotspot intercepts it again.
    This creates an infinite, recursive encryption loop that instantly destroys network bandwidth and drops all
    connections.
695
696 **Fix:** This is a physical routing paradox that cannot be resolved in software. The upstream physical router
    providing internet to the Mac must be allowed to route its traffic directly to the physical WAN
    interface. The user must manually disable "Use Exit Node" on the upstream device providing the hotspot.
697
698 **Advanced Workaround:** If the upstream hotspot is a Windows machine and the user explicitly wants it to
    remain connected to the Exit Node, a permanent static route can be injected into the Windows routing kernel
    to forcefully bypass the Tailscale WFP interception for WARP packets. By binding the route to the physical
    adapter's Interface Index (IF), it becomes a permanent, roaming-aware bypass.
699
700 On Windows (Admin Command Prompt):
701 ```cmd
702 route print (find Interface List, note the IF number for the active Wi-Fi adapter, e.g. 15)
703 route -p add 162.159.192.1 mask 255.255.255.255 0.0.0.0 IF 15
704 route -p add 162.159.193.1 mask 255.255.255.255 0.0.0.0 IF 15
705 ```
706
707 On Linux (Root Terminal):
708 ```bash
709 ip route add 162.159.192.1 dev wlan0
710 ip route add 162.159.193.1 dev wlan0
711 ```
712
713 On macOS (Root Terminal):
714 ```bash
715 route add -host 162.159.192.1 -interface en0
716 route add -host 162.159.193.1 -interface en0
717 ```

```

```

718
719 *(Note: If the upstream router is an iPhone or Android device, you cannot inject static routes without
    jailbreak/root. You must simply disable "Use Exit Node" in the mobile Tailscale app).*
720
721 This punches a tiny hole straight through the paradox, letting the Mac's WARP packets escape to the physical
    internet while keeping the rest of the upstream router's traffic securely trapped in the Tailscale Exit
    Node.
722
723 ### 11.11 AdGuard Filter Redundancy & Memory Optimization
724
725 **Problem:** AdGuard Home does not perform cross-list deduplication in memory. When it loads its own native
    subscription filters (e.g., `AdGuard DNS filter`) alongside our massive `compiled_rules.txt`, overlapping
    rules are loaded twice, wasting significant RAM in the Colima VM. The UI would show ~500k rules,
    representing the sum of all lists rather than unique domains.
726
727 **Fix:** Updated `scripts/sync-rules.py` to intelligently cross-reference AdGuard's configuration and
    deduplicate our list against it.
728 1. The script now reads `adguard/conf/AdGuardHome.yaml` to dynamically fetch the exact URLs of any enabled
    native AdGuard filters.
729 2. The parsing engine was upgraded to normalize basic AdGuard syntax (`||domain`) back into raw base domains
    during the build process.
730 3. The script subtracts the native AdGuard domains from our custom blocklist *before* compiling, immediately
    purging ~83,000 completely redundant rules.
731 4. The normalized base domains then pass through our subdomain optimizer, which squashes an additional ~50% of
    the remaining rules.
732
733 This reduces the final compiled output from ~325k to ~281k rules on the `heavy` profile, saving ~45k rules from
    being loaded into memory twice and reducing the overall RAM footprint of the `adguardhome` container.
734
735 ### 11.12 Kernel-Level IP Blocklist (Threat Intelligence Firewall)
736
737 **Problem:** DNS sinkholing cannot stop malware or botnets that bypass DNS entirely by hardcoding direct IP
    addresses (a common technique for C2 communication). These connections pass straight through AdGuard unseen
    .
738
739 **Fix:** Added a second compilation pipeline to `scripts/sync-rules.py` that runs on every startup alongside
    the DNS pipeline.
740 1. The compiler concurrently fetches four curated threat-intelligence feeds: **Feodo Tracker** (abuse.ch
    botnet C2 IPs), **Spamhaus DROP** (IPs allocated to criminal organizations), **Emerging Threats** (
    Proofpoint active C2 and scanners), and **CINS** (active brute-force sources).
741 2. All entries are normalized via Python's `ipaddress` module. Any IP or CIDR that overlaps with RFC1918
    private ranges, loopback, link-local, or the Tailscale CGNAT range (`100.64.0.0/10`) is stripped to prevent
    accidentally locking users out of their own LAN or mesh.
742 3. The cleaned list (16,721 unique IPs/CIDRs) is written as an `ipset restore` file using an **atomic swap
    pattern**: `create_new populate swap with live destroy_new`. This guarantees the live `block_malicious`
    ipset is never empty during a reload.
743 4. `scripts/routing-fix.sh` watches the file's mtime every 30 seconds. On change it runs `ipset restore` and
    idempotently re-injects `FORWARD DROP` rules for both `src` and `dst` into both iptables backends (nftables
    + legacy).
744 5. `docker-compose.yml` mounts `adguard/work/userfilters/` into `routing-fix` as `/userfilters:ro` and adds `
    rule-compiler: service_completed_successfully` to its `depends_on`, eliminating the race condition where
    routing-fix could start before the file existed.
745
746 **Memory cost:** The entire 16,721-entry ipset costs only ~1.6 MiB of kernel memory negligible in the 600MB VM
    budget.
747
748 ### 11.13 Standalone Tailscale Daemon Freeze after macOS Sleep/Wake
749
750 **Problem:** When the macOS host goes to sleep and wakes up, the network interfaces and routing tables are
    rebuilt. The standalone `tailscaled` daemon (installed via Homebrew) occasionally fails to handle this
    transition and becomes completely frozen/unresponsive. In this state, any command like `tailscale down` or
    `tailscale status` hangs indefinitely, and all DNS requests sent to the local Tailscale resolver
    (`100.100.21.8`) time out, causing a total internet blackout on the host.
751
752 **Fix:** Enhanced the Tailscale verification and teardown logic in `toggle.sh` and `recover.sh`:
753 1. **Unresponsive Daemon Detection:** Instead of assuming the daemon is healthy just because the Unix socket `
    var/run/tailscaled.socket` exists, the scripts now actively run a status check with a 5-second timeout (`
    run_with_timeout 5 tailscale status`). If the check times out (exit code 143), the daemon is classified as
    unresponsive.
754 2. **Auto-Recovery Loop:** If the daemon is unresponsive, the script now automatically attempts to restart the
    service using `brew services restart tailscale` (falling back to `sudo -n brew services restart tailscale`)
    .
755 3. **Graceful Retry:** Once restarted, the script pauses for 3 seconds for the daemon to initialize before
    proceeding with connection or teardown commands.
756
757 ### 11.14 macOS Sharing Services (AirDrop/AirPlay) Freeze on Network Transitions
758
759 **Problem:** When the gateway is turned on or off, the scripts perform network configuration cleanups which
    include flushing the routing tables (`route -n flush`), restarting the `mDNSResponder` daemon (`killall -
    HUP mDNSResponder`), and power-cycling the Wi-Fi interface (`setairportpower` or `ifconfig en0 down/up`).
    While AirDrop BLE discovery continues to function, the actual file transfers stall at "Waiting..."

```

```

indefinitely.
760
761 **Root Cause:** The rapid tear-down and rebuild of the local routing table and Wi-Fi interface causes Apple's
    core sharing and connection daemons (`sharingd` and `rapportd`) to lose their socket bindings to the mDNS/
    Bonjour discovery interface. Instead of reconnecting gracefully, they enter a wedged state and try to route
    peer-to-peer (AWDL) IPv6/TCP transfer traffic into the default gateway (the Tailscale interface `utun`)
    instead of the direct wireless link, causing the TLS verification handshake to time out.
762
763 **Fix:** Integrated an automatic sharing services reset into both `toggle.sh` and `recover.sh`:
764 1. The script now automatically runs `sudo -n killall sharingd rapportd` at the end of both the **START** path
    and the **STOP** path (as well as inside `recover.sh`).
765 2. Killing these daemons forces macOS to immediately relaunch them, binding them fresh to the newly initialized
    network interfaces and routing tables, allowing AirDrop to work seamlessly while the gateway is active.
766
767 ### 11.15 Unlocking Mode-Locked Output Files Without `chmod` (`unlock-files.sh`)
768
769 **The one-command fix:** `bash scripts/unlock-files.sh` replaces every known locked inode in the project with
    a fresh writable one. No `chmod`. Covers `.env` (mode `000`), `adguard/work/userfilters/compiled_rules.txt`
    , `ip_blocklist.ipset`, `cache/*.txt`, `cache/ip/*.txt`, and `adguard/work/data/filters/*.txt` (mode
    `0444`). Run it once after setup or after pulling an old repo; `toggle.sh` and `sync-rules.py` will then
    work without permission errors.
770
771 **Context:** The nullexit project has a strict project-wide policy of `no chmod from scripts` (see README 6).
    This rule exists because every prior `chmod 0444` (post-write tamper-proof lock) and `chmod 000` (post-
    toggle `.env` lock) call from `scripts/sync-rules.py` / `toggle.sh` could leave a file permanently
    unreadable on disk if the script exited early, was killed mid-flight, or simply set a restrictive mode
    without ever being re-flipped. We've stripped every `chmod` from the codebase. But files **written by older
    versions of those scripts are still on disk in those restrictive modes (`0444` for `compiled_rules.txt`,
    `ip_blocklist.ipset`, `data/filters/<id>.txt`; `000` for `.env` after end-of-toggle lock). Re-running `
    toggle.sh` or `scripts/sync-rules.py` against those leftovers hits `PermissionError: [Errno 13] Permission
    denied` immediately.
772
773 **Why the stale permissions also block `mv`/`rm` of the parent folder:** The restrictive modes (`000` on `.env`
    , `0444` on output files) don't just block writes they prevent the kernel from unlinking the file's
    dentry during a `mv` or `rm` of the **containing directory** (`adguard/work/`). macOS enforces this at the
    VFS layer: you own the directory, but you can't delete a directory that contains entries you can't write to
    . This made the entire `adguard/work/` tree unmovable/undeletable until the locked inodes inside it were
    replaced. `unlock-files.sh` resolves this permanently by swapping each locked inode for a fresh `0644` one
    via atomic rename (`cp` + `mv`), which operates on the directory entry rather than the file contents no `
    chmod` called, no permission bypass needed.
774
775 **Problem:** You (the owner) cannot `cat`, `cp`, `rm`, `>` redirect, or any normal POSIX write against a `mode
    =000` file even though you own it the basic mode bits block ALL access for owner, group, and other. The
    script does not call `chmod` and you want a permanent fix that never relies on a chmod from any future run
    either.
776
777 **Solution:** Replace the locked inode with a fresh readable one. `mv` is atomic; mode bits live in the inode,
    not the directory entry. Two flavors cover every case we hit on a real machine:
778
779 **Flavor A locked file is mode `000` (owner can't even READ it):**
780 NOPASSWD sudoers (`/etc/sudoers.d/nullexit`) already includes `/usr/bin/python3`. So we read via `sudo python3`
    (which has the DAC override root has) and pipe the bytes around as base64 in user space, where the
    redirect happens with the user's normal umask = `0644`:
781 ```bash
782 sudo -n /usr/bin/python3 -c "import base64,sys; sys.stdout.write(base64.b64encode(open('<FILE>','rb').read()).
    decode())" > /tmp/snap.b64
783 base64 -d < /tmp/snap.b64 > <FILE>.new
784 mv -f <FILE>.new <FILE>
785 ls -la <FILE>          # mode is now 0644
786 ```
787 The old `mode=000` inode is unlinked by `mv`; the new `mode=0644` inode (created by base64-decode via the
    calling user's umask) takes the path. No chmod ever called. The parent directory just needs to be writable
    by you (it always is, since you own it).
788
789 **Flavor B locked file is mode `0444` (you can READ but cannot WRITE; `scripts/sync-rules.py` needs to re-
    write):**
790 You own the file and can read it, you just cannot truncate/overwrite it. The simplest path is to rename it
    aside and let the writer create a fresh inode from scratch (which gets the umask-default `0644`):
791 ```bash
792 mv <FILE> /tmp/old.<FILE>.$$
793 python3 scripts/sync-rules.py      # now writes <FILE> cleanly at mode 0644
794 ```
795 Or apply Flavor A if you'd prefer to preserve the contents while resetting the mode.
796
797 **Why this is the canonical recovery, not a hack:**
798 - Mode bits live in the inode, not the path. `mv -f` replaces the path's inode reference; the old locked inode
    drops out of the directory entirely (dentry eviction) and loses its last link, so the kernel reclaims it.
    The new inode (whatever it is: a freshly-written one, or a base64-decoded copy) gets the path.
799 - The only prerequisite to apply Flavor B's `mv` is: you own the parent directory (so you can unlink entries
    from it) AND you have a NOPASSWD-capable binary that can read the byte stream if mode prohibits user read.

```

```

800 - This pattern generalizes. Any time a script ends up producing a "locked file that the next run can't update"
      situation, the same trick applies without a single chmod anywhere.
801
802 **Empirical verification (user machine, July 1, 2026):**
803 - `.env` was mode=000` (owner $USER, i.e. you): Flavor A unblocked it in 3 commands, no chmod.
804 - Before: ----- 1 $USER staff 899 Jun 28 07:07 .env`
805 - After: ~-rw-r--r--@ 1 $USER staff 899 Jul 1 20:39 .env`
806 - `docker compose config` immediately picked up `TS_AUTHKEY` + `WIREGUARD_PRIVATE_KEY` substitution.
807 - `compiled_rules.txt`, `ip_blocklist.ipset`, `data/filters/1782645604.txt` were mode=0444`: Flavor B (`mv`-
      aside) unblocked all three.
808 - scripts/sync-rules.py then ran clean: 279587 block rules + 171 allow rules + 16710 IP entries compiled; new
      files came out at `0644`.
809 - `toggle.sh` then went end-to-end in 48s (warp / routing-fix / adguardhome / tailscale all healthy, gateway
      mesh-joined, DNS hijack verified single-server).
810
811 **When this trick is appropriate vs. inappropriate:**
812 - You own the parent directory and the file (typical desktop scenario).
813 - The lock is a leftover from a removed `chmod` call that you want off disk forever.
814 - NOPASSWD sudo includes at least one interpreter (`python3` on this system) that can be used to read mode-0
      files. If it does not, add it first: `<USER> ALL=(ALL) NOPASSWD: /usr/bin/python3`.
815 - The file is owned by another user (root, another account) and the lock is intentional respect it; do not
      bypass another user's security boundary.
816 - The parent directory is owned/writable only by another user escalate properly via sudo, or stop and ask the
      user.
817 - You do not actually own the file and there is a legitimate reason for its lock state restore the file via a
      trusted source rather than circumventing the perms.
818
819 **Reusable cookbook:**
820
821 ```bash
822 # Quick diagnostic: figure out which flavor you need.
823 WHOAMI=$(whoami)
824 ls -la <FILE>
825 stat -f 'mode=%Sp owner=%Su group=%Sg' <FILE>
826 # mode=----- or mode=----r----- : you cannot even read it Flavor A.
827 # mode=r--r--r-- but writer fails : you are blocked from write but can read Flavor B.
828 # mode=rw-r--r-- : not actually locked at all; unrelated write failure.
829
830 # Flavor A (mode=000):
831 sudo -n /usr/bin/python3 -c "import base64,sys; sys.stdout.write(base64.b64encode(open('<FILE>','rb').read()).
      decode())" > /tmp/snap.b64
832 base64 -d < /tmp/snap.b64 > <FILE>.new && mv -f <FILE>.new <FILE> && rm -f /tmp/snap.b64
833
834 # Flavor B (mode=0444, file is readable):
835 mv <FILE> /tmp/old.<FILE>.$(date +%s)
836 # (let the original writer recreate <FILE>; new file lands at mode 0644)
837
838 # Verify after either flavor:
839 ls -la <FILE> # mode should be 0644
840 <your normal validation command>
841 ```
842
843
844 ### 11.16 Gateway Breakage on Lid Close / Wi-Fi Roam Post-Wake + Post-Roam Auto-Recovery
845
846 **Context / Symptom.** Closing the lid (sleep/wake) OR losing + reconnecting Wi-Fi (e.g. in an elevator, caf,
      or roaming between APs) leaves the gateway in a partly-dead state. Symptoms range from a ~30s window of
      dead DNS to a permanent outage that only full `recover.sh` or `toggle.sh` fixes. The root cause is that
      nullexit has **no daemon watching macOS sleep/wake or network-state-change events** the existing DNS
      Watcher inside `toggle.sh` only runs in-process and is suspended along with the rest of the user shell on
      lid close, so the gateway cannot self-heal after either event.
847
848 **Diagnosis.** Two distinct failure modes share the same root gap.
849
850 * *(A) Lid close sleep wake.** `caffeinate -i` (used by `toggle.sh`) blocks `idle` sleep but **does not
      block forced sleep from lid close**. Closing the lid hard-suspends Colima VM + every container + every
      Tailscale-quit-shells-except-tailscaled. On wake the VM clock needs ~5-30s to resync via NTP, during which
      TLS cert validation fails: Cloudflare WARP `162.159.192.1:2408` rejects the handshake gluetun's
      healthcheck (`interval: 2s, retries: 15`) eventually flips unhealthy Docker `unless-stopped` restarts the
      `warp` container (~30s gap). `mDNSResponder`'s in-memory cache still holds pre-sleep entries (stale A
      records for tailnet hosts) so the first dozen DNS queries after wake time out. `tailscaled` on the host
      often keeps its stale DERP relay preference and routes packets through a dead relay for another 30-60s.
851 * *(B) Wi-Fi roam / loss in elevators, captive portals.** WARP is a UDP tunnel to Cloudflare's anycast edge.
      Carrier NATs and captive-portal networks silently invalidate the UDP binding on roam and on hotspot-bound
      re-association. macOS `networksetup` **should* preserve DNS settings on the same Wi-Fi service across a roam
      , but most captive-portal networks DHCP-replace DNS to the captive-portal resolver **during the captive-
      portal dance itself**, before the user has authenticated so even DNS hijack survives a clean roam and
      quietly breaks on captive-portal handoff.
852
853 **Resolution.** A single **launchd LaunchAgent** (`com.nullexit.wake-recovery`) runs a long-running shell
      daemon (`scripts/watcher.sh`) that listens for both event sources and, on each fire, executes `bash recover`

```

```

.sh --post-wake` a *lighter-touch* recovery mode that's the opposite of the existing `--nuclear` semantics
: it keeps the gateway live while still refreshing every stale subsystem.
854
855 ##### Apple Power Management Primer (for the unfamiliar)
856
857 macOS exposes several relevant surfaces; the right primitive depends on what you actually need to detect. None
of these are obvious and none of them have a standard splash screen in the docs.
858
859 * **`caffeinate <flags>`** creates I/O Kit power assertions via `IOPMAssertionCreateWithName`. It does NOT
block forced sleep (lid close, low-battery shutdown, sleep timer firing on a per-set Energy Settings policy
). Flags:
860 * `-i` PreventIdleSleep (the only one `toggle.sh` uses; protects against the OS going idle while the user is
active but a clamshell close still hard-puts it to sleep)
861 * `-s` PreventSystemSleep (only honored on AC power; the user may be on battery)
862 * `-u` DeclareUserActive (resets the idle timer every 30s by default; equivalent to keeping the cursor
moving)
863 * `-d` PreventDisplaySleep
864 * `-m` PreventDiskIdleSleep
865 * There is NO `-l` (lid-block) flag in any modern macOS. Lid close is a kernel-firmware-level event that
ignores user-space assertions.
866 * To prove any of this on live hardware: `pmset -g assertions` lists every active assertion with type /
process / reason / timeout.
867 * **`pmset`** is the read-side of the same system:
868 * `pmset -g log | grep -E 'Sleep|Wake|DarkWake'` shows the recent timeline.
869 * `pmset -g history` is the per-day summary.
870 * `pmset -g assertions` is the live assertion list (matches `-i -s -d -u` to processes).
871 * **`launchd`** does NOT have a native "on-wake" hook.** Not in any version of macOS as of Sonoma. The canonical
recipe is `StartCalendarInterval` with a sub-minute cadence and let launchd queue missed intervals for
replay-on-wake, or use a third-party daemon (Sleepwatcher). For our use case a long-running shell daemon
listening to the unified log is more responsive than a polling agent.
872 * **`com.apple.system.powermanagement.*` Darwin notifications** (`willSleep`, `didWake`, `didDim`, `didUndim`)
are the C-level signal. From a shell, they are best reached through the unified log with `log stream --
predicate` see the watcher implementation.
873 * **`scutil n.watch`** is the canonical CLI for live-following network-state changes. Pairs with `n.add State:/
Network/Global/IPv4` to get a single tap that fires on **any** IPv4 link/route/DNS/address change across **
all** interfaces. This is what `scripts/watcher.sh`'s network-change listener uses.
874 * **`com.apple.networkChange` Darwin notification** is the C-level equivalent but has no shell-friendly
listener; use `scutil` instead.
875 * **`SCNetworkReachability`** is the Objective-C API used by SOCKS5/HTTP clients to detect route changes per
target. Never a fit for shell scripts.
876
877 ##### The Fix (component-by-component)
878
879 1. **`recover.sh --post-wake`** (new flag, **non-destructive** by design). Adding `--post-wake` to the existing
nuclear recovery script inverts the semantics. The default mode still tears down Tailscale, resets DNS to
empty, stops caffeinate, runs `docker compose down`, power-cycles Wi-Fi. The new mode does:
880 * Skip Tailscale disconnect (we WANT it to keep the mesh connection)
881 * **Re-hijack** DNS to the gateway Tailscale IP read from `gateway_ip` (instead of resetting to empty)
applies to both `ACTIVE_SERVICE` and `ENO_SERVICE` because the system resolver is scoped to en0
882 * **Refresh** the exit-node preference with `tailscale up --reset --ssh=true --accept-dns=false --exit-node
=\"$TS_IP\" --exit-node-allow-lan-access=true` (the exact flags `toggle.sh` START uses). This re-asserts
DERP mapping without dropping the mesh
883 * Inspect `docker inspect --format '{{.State.Health.Status}}' nullexit-warp-1` and **only** `docker compose
up -d --force-recreate warp` if gluetun is unhealthy this single targeted recreate nudges the UDP NAT
binding back to Cloudflare without disturbing Tailscale or AdGuard
884 * Run the 10.27 sharingd-reset step (existing fix from `toggle.sh` START path) so AirDrop doesn't freeze on
the new IP lease
885 * Verify the gateway with `dig +tcp @127.0.0.1 -p 5354 google.com` (AdGuard via TCP through Colima's SSH
tunnel) and a 4-second `tailscale ping` to the gateway Tailscale IP, instead of the default mode's direct-
to-1.1.1.1 check
886 * Crucially: **skip `sudo -v`** because the LaunchAgent has no TTY to prompt on; all privileged calls use `
sudo -n` and lean on `/etc/sudoers.d/nullexit` NOPASSWD entries (`networksetup`, `dnsmasq`, `dscacheutil`,
`killall`, `pkill`).
887 2. **`toggle.sh`** writes and clears `/tmp/nullexit-gateway-active.marker` (an ISO-8601 UTC timestamp) at the
bottom of the START path and the top of the STOP path / `cleanup_handler`. Two new helpers: `
write_gateway_active_marker` and `clear_gateway_active_marker`. The watcher uses this file as its **only
signal** that the gateway is live: if `toggle.sh` was stopped (or never started), the watcher no-ops. This
prevents waking-up-only-on-counterfactual events from churning DNS.
888 3. **`scripts/watcher.sh`** (long-running daemon, sibling-style source file). Two backgrounded pipelines:
889 * **Wake listener**: `log stream --predicate 'composedMessage CONTAINS[c] \"didWake\" OR composedMessage
CONTAINS[c] \"Wake from Sleep\" OR composedMessage CONTAINS[c] \"Waking from\" OR composedMessage CONTAINS[c] \"
DarkWake\"' --style compact --info` piped through `while read; do run_recover \"WAKE: ...\"; done`. `[c]`
makes the predicate case-insensitive (the unified log writes phrases in mixed case across versions).
890 * **Network listener**: `(echo \"n.add State:/Network/Global/IPv4\"; echo \"n.watch\"; while true; do sleep
86400; done) | scutil 2>/dev/null` piped through a `case` match on `n.state` / `SCEventUpdate`.
891 * **`run_recover`** checks the marker, debounces via `/tmp/nullexit-watcher.last-recovery` (default 10s,
configurable via `NULLEXIT_DEBOUNCE_SECONDS`), and shells out to `bash recover.sh --post-wake`. The 10s
debounce prevents a single wake event (which typically emits 2-3 log lines) from triggering multiple
recoveries.
892 * `trap ... TERM INT` `kill 0` cleanly tears down the process group on launchctl `bootout` so the `sleep
86400` inside the scutil pipe also dies.

```

```

893 * `wait` blocks indefinitely while both pipelines feed it.
894 4. **`launchd/com.nullexit.wake-recovery.plist`** (LaunchAgent in the user domain runs as the console user at
    every login without needing sudo).
895 * `Label: com.nullexit.wake-recovery`
896 * `ProgramArguments: ["/bin/bash", "<absolute-path-to-your-nullexit-install>/scripts/watcher.sh"]`
897 * `RunAtLoad: true` starts immediately on `launchctl load` (no need to log out and back in)
898 * `KeepAlive: { SuccessfulExit: false, Crashed: false }` **don't** restart on clean SIGTERM exit (so `
    launchctl bootout` doesn't get stuck in a relaunch loop). Also **don't** restart on hard crash: we observed
    that macOS Sonoma's sleep/wake suspend-resume path accumulates orphan watcher.sh PIDs because SIGCONT
    does not reliably clean up the prior instance's listener grandchildren, and a `KeepAlive.Crashed=true`-
    driven relaunch actively races with the still-live prior process. The script enforces single-instance on
    its own via a `flock`-style PID-file lock, so a relaunch-on-crash would be skipped by the lock and produce
    0 listener coverage until the live instance happened to die. Trade-off: a real crash leaves the user
    without auto-recovery until they run `launchctl load -w` manually acceptable because (a) gateway still
    works without auto-recovery, (b) a relaunch wouldn't fix whatever caused the crash, and (c) the gateway-
    recovery itself is observably broken (no DNS, no exit-node) if it does go down.
899
900 ### 11.17 Host-Side Traffic Leaking Past the Exit Node (with Remote Clients Routing Correctly)
901
902 **Symptom.** `toggle.sh` reports success. Remote tailnet clients (e.g. an S24 phone) correctly egress through
    Cloudflare WARP and see a Cloudflare IP on `whatismyip.com`. But on the **HOST Mac running the gateway*, `
    whatismyip.com` reports the underlying physical ISP's ASN (e.g. `campus_isp` for a university campus Wi-Fi)
    . The gateway container itself is healthy; the leak is specifically on the host.
903
904 **Root causes.** Three distinct mechanisms can each produce this exact symptom on the host alone, while leaving
    remote clients unaffected. They are mutually rankable so the diagnostic picks the active one
    deterministically.
905
906 * **(A) SOCKS5 fallback lane active.** `toggle.sh`'s three Phase-B pre-flight checks (`tailscale ping` AdGuard
    via `dig +tcp` WARP internet, `toggle.sh` -lines 750-820) sometimes flake during the first minute. When ANY
    of the three fails, the script sets `SKIP_EXIT_NODE=true` (toggle.sh:849) and instead activates the SOCKS5
    fallback path (toggle.sh:894). The SOCKS5 fallback hijacks DNS to `127.0.0.1` (so AdGuard filtering still
    works) but **modern browsers on macOS do NOT honor the system SOCKS5 proxy** they ignore `networksetup -
    setsocksfirewallproxy`. Result: DNS gets ad-filtered but actual HTTP/S traffic exits direct through the
    campus ISP. This branch is invisible from a remote tailnet client because the client's tunnel is unaffected
    .
907
908 * **(B) IPv6 leak over campus Wi-Fi.** The Tailscale exit node and WARP tunnel are both IPv4-only (gluetun + `
    post-rules.txt` drop all IPv6 forwarded traffic; devref 10.6). But many university and enterprise APs are
    dual-stack the host happily accepts AAAA DNS responses and sends IPv6 traffic straight out `en0`. macOS
    happily encrypts that traffic through Tailscale ONLY if Tailscale has IPv6 advertised; with an IPv4-only
    exit node, IPv6 traffic skips Tailscale entirely. This branch is invisible from a phone because iOS/Android
    Tailscale apps actively sinkhole IPv6 in their VPN profile, but the macOS host's `tailscaled` does not.
909
910 * **(C) Route-table freeze and `--accept-routes` reset after `tailscaled` wake/toggle.** Running `tailscale up
    --reset` clears the exit-node preference and reverts `--accept-routes` back to its default (`false`).
    Without explicitly passing `--accept-routes=true`, `tailscaled` will not attempt to install the default
    route through the `utun*` tunnel interface even if the exit node preference is reconciled. Additionally,
    macOS occasionally freezes and fails to apply route-table changes (devref 10.26). In both cases, `netstat -
    rn` shows the default route still pointing to the physical Wi-Fi gateway (e.g. `172.17.0.1`) instead of the
    Tailscale `utun*` interface, causing host traffic to exit direct while remote clients route correctly.
911
912 **Why remote clients don't see this.** Each remote phone/laptop uses its OWN Tailscale tunnel to reach the
    container's `100.x.x.x:443` over the mesh they're end-to-end encrypted regardless of how the host Mac
    itself is configured. Only the HOST's own egress sockets (HTTP requests from the host's browser, curl, etc
    .) are affected.
913
914 **Resolution.** A single script: `scripts/diagnose-host-leak.sh`. It runs the 7 checks, classifies into one of
    A / B / C / OK, prints a verdict + a ready-to-run fix command on stdout, AND writes the full report to `
    host-leak-diagnostic-<UTC>.txt` so the user can paste the file back instead of scrolling-and-copying from
    the terminal. With `--fix`, the matched remediation is applied and the two checks that could change (warp +
    ipv6) are re-verified. With `--watch` (or `--watch 30`), it runs the full baseline diagnostic then enters
    a continuous loop checking warp/IPv6/default-route every N seconds, alerting only on state changes logs to
    `host-leak-watch.log`.
915
916 ```bash
917 # Diagnose only:
918 bash scripts/diagnose-host-leak.sh
919
920 # Diagnose + apply matched fix + re-verify:
921 bash scripts/diagnose-host-leak.sh --fix
922 ```
923
924 **What it prints.** A 7-section report (`tailscale status`, SOCKS5 state, `cdn-cgi/trace` warp=, IPv6 leak
    probe, host default route, last toggle.sh output.log hints, resolved gateway Tailscale IP), followed by a
    classification block (`Scenario: A | B | C | OK`) with the exact command(s) the user needs to paste into
    their terminal they don't have to figure out which `tailscale up` flags match their environment.
925
926 **What it does NOT do.** It does not touch `toggle.sh` or `toggle.sh`'s pre-flight logic, does not modify `.env
    ` (except `--fix` mode appends `GATEWAY_BYPASS_PING=true` when fixing Scenario A), and does not restart
    Docker. It is a read-only diagnostic with an opt-in remediation flag. Safe to run on a healthy gateway the

```

```

    worst case is a 30-line report that says "OK".
927
928 **Why cdn-cgi/trace over whatismyip.com for the verdict.** `whatismyip.com`'s ISP database routinely mislabels
    campus IPv6 ranges as residential ISPs (that's why the local ISP's ASN appears even when you're routing
    through Cloudflare). `cdn-cgi/trace` is hosted by Cloudflare ITSELF and reports `warp=on|off` plus the
    exact edge colo serving the request. It is the only public endpoint that can truthfully confirm whether the
    WARP tunnel is in use.
929
930 **Common false alarms.**
931
932 * **`warp=on` but whatismyip.com still shows the local ISP.** `whatismyip.com`'s ISP-detection database is
    unreliable on academic networks. Trust `cdn-cgi/trace`'s `warp=on` over `whatismyip.com`'s ISP string. Use
    `https://api.ipify.org` (returns just the IP, no ISP DB lookup) as a third confirmatory check.
933 * **`tailscale status` shows the host online but exit node preference is missing.** `tailscale up --reset --
    exit-node=` (empty value) clears any stale preference; `tailscale up --reset --exit-node=<100.x.x.x>` re-
    establishes it. The diagnostic prints the exact command with the resolved TS_IP filled in.
934 * **Both IPv6 leak AND route-freeze flags set simultaneously.** Scenario B outranks Scenario C fixing IPv6
    makes IPv4 the only viable egress, after which the route-freeze usually self-resolves within ~30s (or one
    more `toggle.sh` cycle).
935
936 #### 10.30.1 Host Leak Probe Continuous Sub-Second Egress Monitor
937
938 **What it is.** `scripts/host-leak-probe.sh` is a lightweight background daemon (~1.4 MB RSS, ~01% CPU) that
    polls `https://www.cloudflare.com/cdn-cgi/trace` every 300ms directly from the host via `curl` not via `
    docker compose exec`. This is a fundamentally different vantage point from the WARP Watcher (10.32): the
    WARP Watcher asks the WARP *container* whether it sees `warp=on`; the Host Leak Probe asks what a real
    browser request leaving the host's physical NIC looks like. The two blind spots it covers that the WARP
    Watcher cannot:
939
940 1. **Host-side flash-leaks** a Cloudflare anycast edge re-route, a browser reusing a stale pooled connection
    across a tunnel state change, or a split-second routing gap during a Gluetun healthcheck restart (see 10.18
    acknowledged 14s window).
941 2. **Host egress while container is healthy** the container can report `warp=on` while the host's own traffic
    exits direct (the three-scenario leak class documented in 10.30).
942
943 **Lifecycle.** Started by `toggle.sh`'s START path (`start_host_leak_probe`) immediately after `
    start_warp_watcher`. Controlled by:
944
945 | Signal path | What happens |
946 |---|---|
947 | `toggle.sh` STOP | `stop_host_leak_probe()` SIGTERM + PID file cleanup |
948 | `toggle.sh` `cleanup_handler` (error/INT/TERM/HUP) | Same |
949 | `recover.sh` (default, non `--post-wake`) | 6d block kill by PID file |
950 | `recover.sh --post-wake` | Left running the gateway is still live |
951
952 PID file: `/tmp/nullexit-host-leak-probe.pid`. Toggle with `HOST_LEAK_PROBE=false` in `.env` to disable at next
    gateway start.
953
954 **Output format.** All events are written to `output.log` (repo root) with UTC timestamps. Only state *changes*
    are logged the probe is silent during normal healthy operation.
955
956 ```
957 [09:10:26] LEAK warp=off ip=45.23.1.1 prev_warp=on prev_ip=104.28.246.50
958 [09:10:26] ROTATE warp=on ip=104.28.248.11 prev_ip=104.28.246.50
959 [09:10:26] HOST-PROBE failed/timeout (count=1)
960 ```
961
962 curl transport errors (`TLS handshake failed`, `Connection refused`, etc.) are also appended to `output.log`
    via `2>>output.log` nothing is silenced to `/dev/null`.
963
964 **Grep cheatsheet.**
965
966 ```bash
967 grep LEAK output.log          # real warp=off events from the host
968 grep ROTATE output.log        # Cloudflare anycast edge rotations (harmless)
969 grep 'HOST-PROBE' output.log   # curl failures / timeouts
970 ```
971
972 **Resource budget.** ~1.4 MB RSS, ~0% CPU at rest, ~3 HTTPS requests/second to Cloudflare. Safe to leave
    running for hours. Back off the polling interval (`bash scripts/host-leak-probe.sh 1.0`) if you observe
    probe-failure pile-ups indicating Cloudflare rate-limiting.
973
974 **Detection vs prevention.** This script detects leaks; it does not prevent them. A sub-300ms flash-leak can
    still slip between two polls undetected. If `grep LEAK output.log` shows confirmed leak events, the next
    step is a kill-switch (prevention) an `iptables` rule on the host's `en0` that drops non-Tailscale, non-
    local egress whenever the gateway is active. That is a separate engineering task not yet implemented.
975 * `ProcessType: Background` hint to App Nap not to suspend us.
976 * `StandardOutPath/StandardErrorPath: /tmp/nullexit-watcher.{out,err}.log` separates launchd-level
    diagnostics from the script's own `/tmp/nullexit-watcher.log` (which `exec >>>` redirects).
977
978 #### Install (one-time)

```

```

979
980 The plist lives in the repo at **`launchd/com.nullexit.wake-recovery.plist`** for version control. The
    installed (live) copy must land in ~/Library/LaunchAgents/ so launchd picks it up on the user session.
981
982 ```bash
983 # Copy the plist to the user-launchd location (run from repo root)
984 cp ./launchd/com.nullexit.wake-recovery.plist \
985     ~/Library/LaunchAgents/
986
987 # Substitute __NULLEXIT_HOME__ with your local install absolute path.
988 # The plist uses WorkingDirectory + a relative program arg, so this
989 # single substitution is the only place it references your machine.
990 sed -i '' "s|__NULLEXIT_HOME__|$(pwd)|" \
991     ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
992
993 # Validate the XML
994 plutil -lint ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
995
996 # Load + persist this user session + every future login
997 launchctl load -w ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
998
999 # Confirm it actually started
1000 launchctl list | grep nullexit
1001 ```
1002
1003 To **stop** the watcher (without uninstalling):
1004
1005 ```bash
1006 launchctl bootout gui/$UID/com.nullexit.wake-recovery
1007 # or:
1008 launchctl unload -w ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
1009 ```
1010
1011 To **uninstall** completely:
1012
1013 ```bash
1014 launchctl bootout gui/$UID/com.nullexit.wake-recovery 2>/dev/null || true
1015 rm ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
1016 ```
1017
1018 ##### Why this fix was preferred over alternatives
1019
1020 * **Polling with `StartCalendarInterval`** would have given ~30-60s detection latency per wake. The unified-log
    stream is sub-second.
1021 * **Sleepwatcher** would have covered wake events but not network changes. We'd still need a separate `scutil`
    listener, and introducing a third-party dependency for what amounts to ~150 lines of shell is unjustified.
1022 * **Forcing `caffeinate -l`** doesn't exist and the closest equivalent (`caffeinate -s`) only works on AC power
    . The whole point of the gateway is to stay usable on battery while traveling lid close must NOT silently
    kill the gateway.
1023 * **A kernel extension** is impossible (no entitlement) and out of scope.
1024
1025 ##### Reproduction recipe
1026
1027 Test both events in isolation to confirm the fix actually closes the gap.
1028
1029 * ** (A) Lid-only repro**: with the gateway up, run in a terminal: `pmset displaysleepnow && sleep 30 && pmset
    wake` (without physically closing the lid). Watch the wake wake-fires the watcher post-wake routine runs
    login should still work in <5s after wake. Compare to baseline pre-fix: pre-fix the gateway would be dead
    for 30-90s.
1030 * ** (B) Roam-only repro**: with the gateway up: `networksetup -setairportpower off && sleep 30 && networksetup
    -setairportpower on` (or simply walk out of Wi-Fi range and back). The captive-portal WILL repave DHCP DNS
    for a few seconds; the watcher fires on the second `State:/Network/Global/IPv4` change (after the Wi-Fi re-
    associates) and the post-wake routine re-hijacks DNS within 2-3s.
1031
1032 ##### Operative files
1033
1034 * `recover.sh` added arg parsing; wrapped destructive sections behind `POST_WAKE`; added re-hijack DNS /
    refresh exit-node / force-recreate-if-unhealthy path.
1035 * `toggle.sh` added `write_gateway_active_marker` / `clear_gateway_active_marker` called at the END of START
    and TOP of STOP / cleanup_handler.
1036 * `scripts/watcher.sh` new long-running daemon.
1037 * `launchd/com.nullexit.wake-recovery.plist` launchd LaunchAgent descriptor.
1038 * `~/Library/LaunchAgents/com.nullexit.wake-recovery.plist` the installed copy (one-time copy).
1039
1040 ##### Honest assessment (read before testing)
1041
1042 This fix has two asymmetric halves:
1043
1044 * **Network / roam / captive-portal path RELIABLE.** `scutil n.watch` on `State:/Network/Global/IPv4{4,6}`
    catches every Wi-Fi roam, captive-portal DHCP-rebind, hotspot handoff, and VPN change with zero missed
    events. To validate, drop Wi-Fi for 30 s and re-add it; `~/tmp/nullexit-watcher.log` will record a `NET:`

```

trigger and `recover.sh --post-wake` will run within ~10 s of the network coming back.

1045 * **Lid close / wake path** BEST-EFFORT on macOS Sonoma+.* Apple *suspends* LaunchAgents process-state during forced sleep and *resumes* them on wake; it does NOT re-launch them every wake. As a result: (a) `log stream` is a live subscription events emitted during the agent's suppressed window are DROPPED, not buffered, so the wake event that prompted the resume is invisible to `log stream` on resume; (b) the "fire on startup" guard runs once on `launchctl load -w` and after a `KeepAlive` crash-recovery, NOT on every successive wake; (c) `subsystem == "com.apple.powermanagement"` will catch FUTURE emissions (display dim/restore, manual sleep/wake) but not the lid-close/open wake directly.

1046

1047 In practice the lid-wake gap is short, because `toggle.sh`'s existing DNS Watcher already polls every 5 s and re-hijacks DNS (so DNS recovers within 5 s of any roam or wake), `tailscaled` self-heals via DERP fallback within 3060 s, and `gluetun` reconnects via its healthcheck retry within 30 s. The user-visible pain on lid open is ~30 s of wonky DNS, not a permanent outage.

1048

1049 **Test the ROAM path FIRST.** If ROAM works in your testing, the lid-wake gap is acceptable. If lid-wake handling is critical, the two clean follow-ups are:

1050 1. **Sleepwatcher** (one canonical install): `brew install sleepwatcher`; drop `recover.sh --post-wake` into `~/sleep` and `~/wake` so Sleepwatcher invokes it directly on every wake without the launchd propagation problem.

1051 2. **Move the watcher to a LaunchDaemon** (root) and use `IOPMSchedulePowerEvent` via a tiny C wrapper not portable to shell.

1052

1053 A truly reliable shell-only solution to "wake events from inside a LaunchAgent" does not exist on macOS Sonoma+.

1054

1055 **Empirical lessons from deployment**

1056

1057 Two findings came up while deploying this fix end-to-end; recording so the next operator doesn't lose an evening to each.

1058

1059 1. **Bash function-call-before-definition gotcha.** While introducing the gateway-active marker helpers, the defensive `clear\_gateway\_active\_marker` call was inserted at the very top of `toggle.sh` (right after `set -e`), but the function definition lived near `PID\_FILE="/tmp/nullexit-caffeinate.pid"` ~90 lines down. Bash parses top-to-bottom and resolves names at first invocation, so the call site exited with code 127 (`clear\_gateway\_active\_marker: command not found`). The gateway stayed unreachable from the START path even though `bash -n` passed (the parser doesn't catch order-of-resolution errors). Rule: define functions BEFORE any block that calls them. Verify with `grep -n 'name()'s\*{'` followed by `grep -n 'name\s\*\${'` the latter must appear on a line number AFTER the former.

1060

1061 2. **networksetup -setairportpower off+on** is not a faithful roam reproduction.* Synthetic Wi-Fi radio-cycling (`sudo networksetup -setairportpower Wi-Fi off` for 30 s, then back on) is expected to produce a `NET:` trigger in `scutil n.watch` but produced ZERO during testing because `setairportpower` powers the radio at the driver level while leaving the network SERVICE in the "up" state from scutil's perspective, so no `n.state State:/Network/Global/IPv4` event is generated. A real roam (macOS briefly loses the AP and re-associates into a different one) DOES fire the event because the link actually changes.

1062 Better reproduction recipes in priority order:

1063 * Walk between two real SSIDs via `networksetup -switchtolocation` (or actual physical station movement) guaranteed to fire the scutil event.

1064 * Force a DHCP rebind on the same SSID with `sudo ipconfig set en0 DHCP` triggers `State:/Network/Global/IPv4` to update.

1065 * Trust the existing 30-second DNS Watcher inside `toggle.sh` for the DNS path: it re-hijacks DNS automatically. The post-wake watcher adds clear value mostly for tailscale exit-node re-assertion and warp gluetun force-recreate, both of which self-heal within ~30 s anyway.

1066

1067 **11.18 Infinite Egress Routing Loops & Subnet Takeover Paradoxes**

1068

1069 **Problem:** Two distinct network loops can break host egress connectivity entirely when the exit node is active:

1070

1071 1. **The Recursive Host-VM Tunnel Loop:**

1072 - **Mechanism:** The host Mac's default route points to the Tailscale exit node (`utun\*`). The exit node container sends its encrypted tunnel packets (destined for the WARP endpoint `162.159.192.1`) back to the host via the VM NAT. Without a static route exception on the host Mac, the host Mac routes those packets right back into the exit node (`utun\*`), creating an infinite loop that blackouts all host network egress.

1073 - **ARP-Scoping Pitfall:** Simply binding the bypass route to the interface (`route add -host 162.159.192.1 -interface en0`) fails when the default route is deleted or redirected to `utun\*`. Since `162.159.192.1` is not local to the physical link, macOS fails to resolve it via ARP, dropping all tunnel packets.

1074 - **Fix:** Dynamically resolve the active physical gateway IP (e.g. `172.17.0.1`) on startup, and add the static host routes for the WARP endpoints via the gateway IP directly (`route add -host 162.159.192.1 <gateway\_ip>`). Additionally, because standalone `tailscaled` on macOS does not automatically modify the system default route via the CLI, we manually redirect the default gateway to `utun\*` using `setup\_exit\_node\_routing`.

1075

1076 2. **The Docker Compose Subnet Takeover Collision:**

1077 - **Mechanism:** To avoid collisions with the host's campus network, Colima's default bridge (`docker.bip`) is moved (e.g. to `10.200.0.1/24`). However, because `172.17.0.0/16` is now free inside the VM, Docker Compose's IPAM dynamically takes it over for the project's custom network (`nullexit\_default`). Because the containers receive IPs on `172.17.0.0/16`, the host Mac cannot send local peer discovery packets (e.g., Tailscale `magicsock` packets on `172.17.0.2:41641`) directly. The host routing table sends them out `en0` to the physical Wi-Fi, returning `host is down` or `no route to host`.

```

1078 - **Fix:** Explicitly configure a custom default network subnet `10.200.1.0/24` in `docker-compose.yml` to
1079 prevent Docker Compose from ever reclaiming the conflicting `172.17/172.18` ranges.
1080 > See also 10.23 (The Hotspot Paradox) for the related infinite loop that occurs when the physical upstream
1081 router is also a Tailscale exit-node client.
1082
1083 ### 11.19 In-Flight WARP Tunnel Liveness Monitor & Statistical Auto-Shutdown
1084 **Why.** nullexit's core promise is that your IP always appears as Cloudflare. If the WARP tunnel silently dies
1085 due to a UDP NAT timeout, a Cloudflare edge rotation, a Colima VM clock skew causing TLS cert rejection,
1086 or any of a dozen other failure modes the host silently falls back to egressing through the physical ISP.
1087 The user sees the gateway as "up" (containers running, Tailscale connected, DNS hijacked) but their real IP
1088 is exposed. This is the exact class of failure that Ingo Blechschmidt documented in his chilling 2024
1089 Linux kernel post-mortem: for over two years, LUKS disk encryption on suspend was silently broken because a
1090 refactored kernel syscall returned success while doing nothing ([source](https://mathstodon.xyz/@iblech
1091 /116769502749142438)). The lesson: **a security mechanism that fails silently is worse than no mechanism at
1092 all** it creates a false sense of safety that prevents the user from taking corrective action.
1093
1094 **Design principle.** The WARP Watcher (`start_warp_watcher()` in `toggle.sh`) is a background daemon that
1095 polls `cdn-cgi/trace` every 30 seconds and applies a **statistically calibrated threshold** before
1096 triggering any safety response. This threshold distinguishes transient network blips (which self-heal
1097 within seconds) from genuine outages (which require intervention). A single `warp=off` reading is
1098 meaningless Docker might be slow, the network might be congested, a container healthcheck might be
1099 restarting. But 6 consecutive off readings (30 continuous seconds of downtime) has vanishingly low
1100 probability of being a false positive: even at an unrealistically high 10% false-positive rate per poll, P
1101 (6 consecutive) = 0.1 0.0001%. In practice, the per-poll false-positive rate is far lower than 10%, making
1102 the real probability astronomically small.
1103
1104 **What it does.**
1105 1. **Always logs.** Every state transition (onoff, offon) is timestamped to `output.log`. The user can `grep
1106 WARP output.log` to audit the tunnel's entire lifetime. This is the "never fail silently" mandate.
1107 2. **Tracks consecutive failures.** A counter increments on each `off` reading and resets to 0 on any `on`
1108 reading. This ensures the watcher never fires on intermittent flapping.
1109 3. **Auto-shuts down at threshold.** When the counter reaches `WARP_FAIL_THRESHOLD` (default 6, configurable in
1110 `.env`), the watcher declares a statistically significant outage and runs `recover.sh` the nuclear
1111 recovery script that disconnects Tailscale, stops Docker containers, resets DNS to `1.1.1.1`, flushes
1112 routes, and power-cycles Wi-Fi. This instantly reverts the host to normal (un-encrypted) internet,
1113 guaranteeing no traffic leaks through a dead tunnel while the user thinks they're protected. A trigger file
1114 at `/tmp/nullexit-warp-shutdown-triggered` prevents double-firing.
1115 4. **Exits cleanly.** After shutdown, the watcher exits (the gateway is dead, there's nothing left to monitor).
1116 `stop_warp_watcher()` is also called by `toggle.sh`'s STOP path and `cleanup_handler` to terminate the
1117 daemon cleanly during normal teardown.
1118
1119 **Configuration.** Set `WARP_FAIL_THRESHOLD=3` in `.env` for a more aggressive 15s timeout, or `
1120 WARP_FAIL_THRESHOLD=12` for a conservative 60s window. The variable is parsed from `.env` using the same `
1121 grep` pattern as `GATEWAY_MSS` and `GATEWAY_BYPASS_PING`.
1122
1123 **Log sample.** After a real outage (or a forced test via `docker compose pause warp`):
1124
1125 ```
1126 [2026-07-03T21:15:17Z] WARP DOWN cdn-cgi/trace reports warp=off
1127 [2026-07-03T21:15:47Z] WARP SHUTDOWN 6 consecutive failures (threshold=6). Running recover.sh to kill the
1128 gateway and restore normal internet. Your IP is NO LONGER Cloudflare.
1129 [2026-07-03T21:15:52Z] WARP SHUTDOWN recover.sh completed, watcher exiting. Your IP is now your real ISP IP.
1130 ```
1131
1132 Or, if WARP self-recovers before the threshold:
1133
1134 ```
1135 [2026-07-03T21:15:17Z] WARP DOWN cdn-cgi/trace reports warp=off
1136 [2026-07-03T21:15:32Z] WARP RECOVERED warp=on after 3 consecutive off readings (threshold was 6)
1137 ```
1138
1139 **Relationship to other monitors.** The WARP Watcher is the **innermost safety layer** it runs as a child of `
1140 toggle.sh` and only while the gateway is active. It complements (does not replace) the `scripts/watcher.sh`
1141 daemon (10.29, which handles sleep/wake and Wi-Fi roam) and `scripts/diagnose-host-leak.sh` (10.30, which
1142 is a manual diagnostic tool). The WARP Watcher is the only component that actively verifies end-to-end
1143 tunnel liveness and takes autonomous action when it fails.
1144
1145 ---
1146
1147 ## 12. Recent Updates
1148
1149 Reflects the current branch's state. Each entry one line + commit hash; read the commit message for detail.
1150
1151 - **July 10, 2026 `fix(race): suppress WARP Watcher during post-wake force-recreate` Fixed a race condition
1152 where switching Wi-Fi caused the host IP to leak as the raw ISP IP (`132.x`) on every network roam. `

```

recover.sh --post-wake` force-recreating the warp container (-35s) caused the background WARP Watcher to accumulate 6 consecutive `warp=off` readings and fire nuclear `recover.sh`, killing a gateway that was healing itself. Fixed by adding a `/tmp/nullexit-warp-inhibit.marker`: `recover.sh` writes it at startup in `--post-wake` mode (with `trap ... EXIT` for guaranteed cleanup); the WARP Watcher loop in `toggle.sh` checks it first and resets `consec_off=0` while it exists. See 25 for full post-mortem.

- 1124
- 1125 - ****July 6, 2026 `fix: resolve edge-case vulnerabilities and system state leaks`**** Addressed a suite of subtle but critical edge cases identified in security review:
 - 1126 1. ****DNS Watcher Interface Independence:**** Fixed `start\_dns\_watcher` in `toggle.sh` to dynamically detect the active interface via `get\_active\_service()` rather than hardcoding a `grep` for "Wi-Fi", ensuring roaming on Ethernet/Thunderbolt stays protected.
 - 1127 2. ****Robust Lock Verification:**** Upgraded the `toggle.sh` lock-file logic to use `ps -p` verification, preventing permanent lockouts if a crashed script's PID is recycled.
 - 1128 3. ****Fail-Closed Crypto:**** Changed the `crypto.sh` integrity check in `toggle.sh`/`recover.sh` from an executable bit check (`[ -x ]`) to a pure file-existence check (`[ -f ]`), guaranteeing execution via `bash scripts/crypto.sh` even if git strips the `+x` bit.
 - 1129 4. ****Strict `iptables` Pinning:**** Upgraded `post-rules.txt` FORWARD-chain rules to explicitly pin traffic between `tailscale0` and `tun0` in both directions, mathematically preventing Tailscale traffic from escaping through `eth0` during WARP tunnel drops.
 - 1130 5. ****Docker Local Network Isolation:**** Modified `docker-compose.yml` to explicitly bind the exposed WARP container ports (`1080`, `5354`) to `127.0.0.1`. This prevents other devices on the local bridged Wi-Fi/LAN from accessing the SOCKS5 proxy or DNS resolver if Colima provisions a LAN IP.
 - 1131 6. ****Routing Verification & Sudoers:**** Added a `netstat -nr` check (handling macOS `0/1` syntax) to verify the default route (`0.0.0.0/1`) actually overrides to Tailscale, and updated the `README.md` and `setup.sh` sudoers templates to ensure `/usr/bin/python3` is permitted so the fallback DNS proxy doesn't silently fail.
- 1132
- 1133 - ****July 4, 2026 `cc789c5` (fix: resolve concurrency, setup variables, CI workflow, and untrack review file)**** Added concurrency protection on `toggle.sh` using `/tmp/nullexit-toggle.lock` to prevent parallel runs. Initialized `TS\_AUTHKEY` defensively in `setup-common.sh` to prevent unbound variable crashes under `set -u`. Added Go compilation (`go vet` and `go build`) steps to GitHub Actions CI workflow. Ported AdGuard Home container to cap logs. Fixed macOS BSD `grep` compatibility in `setup-common.sh` using `grep -oE`, resolved space stripping in `common.sh`, fixed quoting/PID file/routing text in `recover-linux.sh`, and removed redundant bypass routes block in `recover.sh`.
 - 1134
 - 1135 - ****July 2, 2026 `8c5fae1` + `181e1e1` (feat(routing): resolve recursive host-VM tunnel loop & Docker subnet collision)**** Two infinite routing loops fixed. (1) Host-VM tunnel loop: WARP endpoint bypass routes now injected via physical gateway IP (not interface scoping) to avoid ARP failures under default-route redirection; `setup\_exit\_node\_routing` added to manually force the host default route to `utun\*` since standalone `tailscaled` CLI on macOS does not update the routing table automatically. (2) Docker Compose subnet takeover: `docker-compose.yml` now locks the default project network to `10.200.1.0/24` to prevent IPAM reclaiming `172.17/172.18` after `docker.bip` is moved. `--accept-routes=true` added explicitly to all `tailscale up --reset` calls. Documented in 10.31.
- 1136
- 1137 - ****July 4, 2026 `fix: post-wake routing loop & destructive recovery` (networking: routing)**** Resolved a critical fatal bug where waking the Mac from sleep or roaming Wi-Fi permanently broke the internet and required a full system reboot to fix.
 - 1138 1. ****The Routing Loop Bug:**** `common.sh`'s `add\_warp\_bypass\_routes` was previously refactored to dynamically look up the default route without accepting explicit interface arguments. During `post\_wake`, Tailscale is already active, so `route get default` resolved to the Tailscale `utun` interface. This caused Cloudflare WARP traffic to be routed back into Tailscale, creating an infinite recursive routing loop that instantly killed the internet on every wake. Fixed by making `add\_warp\_bypass\_routes` accept explicit target arguments and updating `recover.sh` to pass the correct physical interface/gateway.
 - 1139 2. ****The Destructive Recovery Bug:**** When users ran `recover.sh` to fix the broken internet, the script blindly ran `sudo route -n flush`. On macOS, this doesn't just clear VPN routes it destroys essential system routes including loopback (`127.0.0.1`) and multicast (`224.0.0.0`). Bouncing Wi-Fi does not automatically restore these internal routes, meaning macOS network services (`configd`, `mDNSResponder`) remained permanently wedged until a reboot. Fixed by replacing `route -n flush` with targeted deletions of specific gateway routes (`0.0.0.0/1`, `128.0.0.0/1`) and WARP bypass routes.
- 1140
- 1141 - ****July 2, 2026 `feat: secure execution` (security: script lockdown & auth)**** Fixed a major privilege escalation vulnerability and added native macOS authentication prompts to `Toggle Gateway.app` and `Recover Gateway.app`. Previous setups granted `/usr/bin/python3` blanket `NOPASSWD` sudo access, allowing malware to bypass macOS security. We restricted `NOPASSWD` exclusively to `/usr/bin/python3 /Users/<USER>/Developer/nullexit/scripts/dns-proxy.py`. Additionally, `scripts/dns-proxy.py` was locked down with `chown root:wheel` and `chmod 755` so attackers cannot overwrite the script and exploit the restricted sudo rule. `toggle.sh` was also updated to capture `warp` container logs on failure before tearing down the environment.
- 1142 - ****July 3, 2026 `fix: resolve numerous system regressions`**** Addressed several breakages introduced in recent commits:
 - 1143 1. Fixed a truncated line syntax error in `/etc/sudoers.d/nullexit` that broke `NOPASSWD` fallback commands.
 - 1144 2. Fixed a `[Errno 13] Permission denied` error in `sync-rules.py` cache writes by using a temporary file and atomic `os.replace()` to bypass inherited mode locks (0444/000). **** (Later reverted see below.) ****
 - 1145 3. Fixed a quoting bug in `recover.sh` where `ts\_args=""` passed literal escaped quotes.
 - 1146 4. Repaired ANSI color outputs in `diagnose-host-leak.sh` and `fix-docker-bridge-collision.sh` using `%b` instead of `%s` and `-e` flags.
 - 1147 5. Fixed `toggle-linux.sh` calling a macOS-only `networksetup` command instead of `resolvectl dns`.
 - 1148 6. Corrected the AdGuard REST API refresh POST in `sync-rules.py` to correctly target port `3000`.
 - 1149 7. Restored the missing `START\_GATEWAY=true` variable in `toggle.sh` and purged a stale port `80` mapping from `docker-compose.yml`.
 - 1150 8. Verified `output.log` is untracked and safely in `.gitignore`.

```

1151 - **July 3, 2026 `unlock-files.sh` + revert atomic writes in `sync-rules.py`** The `.tmp` + `os.replace()`
    atomic-write pattern introduced in the commit above was a workaround for stale restrictive file permissions
    (mode `0444`/`000`) left on disk by old `chmod` calls that had already been removed from the codebase.
    Rather than keep the workaround, we created `scripts/unlock-files.sh` a one-shot script that permanently
    fixes the stale permissions by replacing each locked inode with a fresh writable one via atomic rename (`cp`
    + `mv`), operating on the directory entry instead of the file contents. No `chmod` used. All 5 `.tmp` + `
    os.replace()` sites in `sync-rules.py` were reverted back to direct writes. `scripts/unlock-files.sh`
    covers: `.env` (mode `000`), `adguard/work/userfilters/compiled_rules.txt`, `ip_blocklist.ipset`, `cache/*.
    txt`, `cache/ip/*.txt`, and `adguard/work/data/filters/*.txt` (mode `0444`). Run once after setup or
    pulling an old repo; `toggle.sh` and `sync-rules.py` then work without permission errors. Documented in
    10.28 + 3.
1152 - **July 1, 2026 `c5b92c0` (refactor: personal-leak cleanup)** eliminated every residual personal-username
    leak across source + config files. The launchd `com.nulllexit.wake-recovery.plist` now uses `
    WorkingDirectory` + a relative program arg with the `__NULLEXIT_HOME__` sentinel; the install recipe in
    10.29 gains a single `sed -i '' "s|__NULLEXIT_HOME__|$(pwd)|" step right after the `cp`. 8 `devref.md`
    prose sites genericized to `~/colima/...`, `~/Library/LaunchAgents/...`, `$USER` for sample output, `

```

```

1192 ### OPSEC: Tor-over-WARP Architecture
1193
1194 When extreme anonymity is required, integrating the Tor network into `nullexit` provides a powerful Tor-over-
VPN topology:
1195 * The University/ISP sees only an encrypted Cloudflare WireGuard tunnel, rendering them completely blind to
the fact that you are using Tor (effectively bypassing Enterprise firewall Tor blocks).
1196 * Cloudflare sees an encrypted TCP stream heading to a Tor Guard Node. They cannot see your DNS lookups,
the destination website, or the content of your traffic.
1197 * The Tor Exit Node sees your traffic, but not your real IP (only the Cloudflare IP).
1198
1199 ##### The "Transparent Proxy" Trap (What NOT to do)
1200 It is tempting to build a system-wide "Transparent Tor Proxy" that forces all host traffic (e.g., standard
Chrome/Safari browsers, background iCloud syncs) through Tor automatically. This is a massive OPSEC trap
and destroys anonymity.
1201 Modern operating systems and browsers will instantly leak your identity through background syncing (e.g., Apple
Mail, Google Drive) and browser fingerprinting (canvas fingerprints, WebRTC, screen resolution). The Tor
network protects your IP, but if your payload contains identifying cookies or unique fingerprints, the
Tor Exit Node (which could be run by malicious actors) will instantly tie your session to your real
identity.
1202
1203 ##### The nullexit Implementation: Isolated Procedural Proxy
1204 To safely integrate Tor without triggering the transparent proxy trap, `nullexit` implements an Isolated Tor
SOCKS5 Proxy with Procedurally Generated Ports:
1205 1. Isolated Container: A lightweight `tor` Docker container runs securely inside the `warp` network
namespace. It does not hijack system traffic.
1206 2. Opt-In Usage: You must consciously configure a highly-hardened browser (like the official Tor Browser or
a dedicated Firefox profile) to use the proxy. This prevents accidental background sync leaks.
1207 3. Procedurally Generated Ports: Rather than hardcoding the standard proxy ports (like `127.0.0.1:9050` or
`1080`), `toggle.sh` randomizes all internal proxy ports on every boot into the ephemeral range
(10000-65000) and silently writes them to `/tmp/nullexit-ports.env`. This completely defeats static
fingerprinting by malware.
1208 4. Kernel-Level Collision Avoidance: To prevent the randomizer from picking a port already in use by a root
daemon or Apple service (which would crash the gateway), the script directly queries the macOS kernel
socket table (`netstat -an -p tcp`) to verify the port is absolutely free before assigning it.
1209 5. The Honey-Port Tripwire: While procedural ports defeat static fingerprinting, advanced malware might
attempt a full sequential port scan of `localhost` to find the hidden proxy. To counter this, `toggle.sh`
spawns a "Honey-Port" (a fake random port with a netcat listener attached). If any local application
attempts to connect to the Honey-Port, it instantly logs a critical security alert to `output.log` and
fires a macOS desktop notification, serving as an early-warning Intrusion Detection System (IDS) for local
malware.
1210
1211 ##### Hardening: Tor Bridges and obfs4 Pluggable Transports
1212 For users operating under severe Deep Packet Inspection (DPI), `nullexit` supports configuring Tor to connect
exclusively through private bridges using the `obfs4` pluggable transport.
1213
1214 When `TOR_USE_BRIDGES=true` is set:
1215 - What it does: It hides the entry IP from being matched against the public Tor consensus list, and
disguises the traffic's protocol fingerprint from the WARP exit provider (Cloudflare).
1216 - What it does NOT do: It does not add any extra layers of anonymity beyond what Tor already provides. It
is purely a censorship-circumvention and obfuscation mechanism.
1217 - Limitations: `obfs4` is highly effective against passive DPI, but it is not guaranteed to defeat active-
probing-resistant detection by a highly sophisticated state-level adversary.
1218
1219 > Architecture Rule: The `nullexit` gateway deliberately does NOT bundle, hardcode, or automatically fetch
bridge lines. Bridge lines are highly sensitive and expire. Users must obtain their own bridge lines from
`https://bridges.torproject.org` and manually populate the `tor-bridges.txt` file. If bridges are enabled
but the file is missing or empty, the Tor container will intentionally crash and fail to start rather than
silently falling back to public guard relays.
1220
1221 The Bootstrapping Paradox (Out-of-Band Key Exchange)
1222 Automating the bridge acquisition process (e.g., scripting the gateway to log into the Telegram `@GetBridgesBot
` via MTPROTO API) introduces catastrophic OPSEC flaws:
1223 1. Identity Leaks: Baking a Telegram session into the container explicitly links the "anonymous" gateway to
a real-world physical phone number.
1224 2. The Chicken & Egg Deadlock: In heavily censored regimes, Telegram itself is usually blocked. If the
container requires Telegram to fetch bridges to bypass the firewall, it will fail because it cannot bypass
the firewall to reach Telegram.
1225 3. Bridge Exhaustion & CAPTCHAs: Automated scraping behaves identically to state-sponsored scrapers,
triggering Tor's anti-bot CAPTCHAs which leads to silent proxy failures and exhausts the limited public
bridge pool.
1226
1227 Instead, users should rely on a less secure layer to manually reach Telegram, obtain the bridges, and populate
`tor-bridges.txt`. This can be done via the standard `nullexit` WARP tunnel, a mobile phone on cellular
data, Mullvad VPN, or a secure remote VPS. (Note: We plan on fully integrating Mullvad and VPS
architectures directly into `nullexit` in the future, but have not had the time to build it yet). This
intentionally "air-gaps" the cryptographic key exchange from the proxy infrastructure, leaving zero API
tokens and zero network traces for an adversary to exploit.
1228
1229 ### OPSEC: Python Runtime Airgap (Isolated Mode)
1230 `nullexit` explicitly relies on the host's native system Python 3 runtime for critical components like `scripts
/dns-proxy.py`. Because this script runs with elevated privileges (`sudo -n`) to bind to the privileged UDP

```

```

/53 socket, modifying or polluting this Python environment is strictly forbidden.
1231
1232 **The Zero-Dependency Rule & Isolated Mode (`-I`)**
1233 You must **NEVER** install third-party `pip` packages (e.g., `requests`, `pyyaml`) into the system Python
environment that `nullexit` uses.
1234 - All Python scripts in this repository must be written using only the Python Standard Library (`socket`, `
urllib`, `json`, `ssl`).
1235 - Third-party packages introduce a massive supply-chain attack vector. A compromised `pip` package installed
globally could allow local malware to hook into the Python runtime and exploit the `sudo` privilege of the
DNS proxy to gain full root access.
1236 - To enforce strict immunity against user-installed malicious packages, `toggle.sh` explicitly invokes the
proxy using **Python's Isolated Mode** (`python3 -I`). This flag forces the interpreter to completely
ignore `PYTHONPATH`, `PYTHONHOME`, and the user's `site-packages` directory, executing exclusively from the
read-only, Apple-signed system framework.
1237
1238 ### Recommended Upgrades
1239
1240 | Layer | Default | Upgraded |
1241 |---|---|---|
1242 | VPN Exit | Cloudflare WARP (US) | Mullvad (Swedish, proven no-logs, anonymous payment) |
1243 | DNS Resolver | AdGuard via WARP | AdGuard via Mullvad DoH (Cloudflare-blind) |
1244 | Coordination | Tailscale (US, OAuth) | Headscale (self-hosted, no third-party) |
1245 | Auth | OAuth / persistent keys | Ephemeral auth keys (no identity persistence) |
1246 | Content | WireGuard asymmetric | WireGuard + PSKs (coordination server blind) |
1247
1248 #### Mullvad (Replacing WARP)
1249 Swedish-based. Police raided their offices in 2023 and left empty-handed no logs exist to seize. Accounts are
random numbers. Payment by cash or Monero. Replace `VPN_SERVICE_PROVIDER=custom` with `VPN_SERVICE_PROVIDER
=mullvad` in `docker-compose.yml`.
1250
1251 #### Mullvad DoH Upstreams
1252 Point AdGuard's upstream resolvers to `https://adblock.dns.mullvad.net/dns-query`. Cloudflare WARP only sees
encrypted HTTPS to Mullvad's IPs completely blind to your DNS queries.
1253
1254 #### Headscale
1255 Self-hosted Tailscale coordination server. Eliminates Tailscale the company, keeping all node topology under
your control.
1256
1257 #### Ephemeral Auth Keys
1258 Generate in the Tailscale admin panel. Used once to authenticate; node disappears from the admin panel after
disconnect. Breaks persistent identity linkage.
1259
1260 #### Pre-Shared Keys (PSKs)
1261 Add a WireGuard PSK between peer nodes. Adds a symmetric encryption layer that Tailscale's coordination server
has no knowledge of, blinding it from reading transit content.
1262
1263 ### Structural Limitations
1264
1265 - **Traffic Analysis / Metadata:** Passive fiber tapping (UPSTREAM) can observe timestamps, data volumes, and
IP ranges without reading content. Defeating traffic analysis requires mixing networks (Tor) or continuous
dummy packet padding both heavily degrade performance.
1266 - **Targeted State-Level Adversaries:** Consumer-grade VPNs are not a complete shield against a nation-state
actively targeting you. This stack is designed to defeat bulk passive surveillance and corporate dragnet
collection.
1267
1268 ---
1269
1270 ## 14. Per-Device Access Control
1271
1272 Because every mesh device routes internet-bound traffic through the `warp` container's `FORWARD` chain, they
all appear with their stable `100.x.x.x` Tailscale IP as the `src` address. This gives two powerful
enforcement surfaces:
1273
1274 ### IP & Port Level (`scripts/routing-fix.sh`)
1275 Write raw `iptables` rules targeting specific devices. The 30-second idempotent loop auto-survives Gluetun
reconnects:
1276
1277 ```bash
1278 # Block a specific device from a target subnet
1279 iptables -I FORWARD -s 100.x.x.x -d 203.0.113.0/24 -j DROP
1280
1281 # Restrict a device to only HTTP/HTTPS
1282 iptables -I FORWARD -s 100.x.x.x -p tcp ! --dport 3000 -j DROP
1283 iptables -I FORWARD -s 100.x.x.x -p tcp ! --dport 443 -j DROP
1284
1285 # Block a device from internet egress entirely (mesh only)
1286 iptables -I FORWARD -s 100.x.x.x -j DROP
1287
1288 # Time-based (e.g., cut off 10 PM 7 AM)
1289 iptables -I FORWARD -s 100.x.x.x -m time --timestart 22:00 --timestop 07:00 -j DROP
1290 ```

```

```

1291
1292 ### DNS Level (AdGuard REST API)
1293 AdGuard Home supports per-client rules keyed by Tailscale IP:
1294
1295 ```bash
1296 curl -X POST http://localhost:80/control/clients/add \
1297   -H "Content-Type: application/json" \
1298   -d '{
1299     "name": "kids-ipad",
1300     "ids": ["100.x.x.x"],
1301     "blocked_services": ["youtube", "tiktok", "instagram"],
1302     "safesearch_enabled": true
1303   }'
1304 ```
1305
1306 ---
1307
1308 ## 15. Packaging nullexit as a macOS Application (.dmg)
1309
1310 nullexit can be packaged into a native `.app` bundle, but this introduces nested virtualization requirements.
1311
1312 nullexit uses Colima (Apple Virtualization Framework `vz`) to run a Linux VM hosting Docker. Installing the
1313   `.app` inside a virtualized macOS environment (Parallels, cloud Mac) means running a Linux VM inside a
1314   macOS VM requiring hardware-level nested virtualization.
1315
1316 ### Hardware Support
1317 - Supported: Apple Silicon M3/M4 (macOS Sequoia 15+), Intel Macs with VMX passthrough.
1318 - Unsupported: M1, M2, A18 Pro. Colima crashes silently; the terminal (`setup.sh`) surfaces crash logs.
1319
1320 ### Build Steps
1321 ```bash
1322 # 1. Verify nested virtualization support
1323 sysctl -a | grep hv_nested_virt_supported # expect: 1
1324
1325 # 2. Compile the AppleScript launcher
1326 osacompile -o "Nullexit.app" "Toggle Gateway.applescript"
1327
1328 # 3. Embed scripts into app resources
1329 mkdir -p "Nullexit.app/Contents/Resources/scripts"
1330 cp toggle.sh setup.sh recover.sh docker-compose.yml "Nullexit.app/Contents/Resources/"
1331
1332 # 4. Package into DMG
1333 hdiutil create -volname "Nullexit Installer" -srcfolder "./Nullexit.app" -ov -format UDZO Nullexit.dmg
1334
1335 # 5. Bypass Gatekeeper (unsigned app)
1336 xattr -cr /Applications/Nullexit.app
1337 ```
1338
1339 Without a paid Apple Developer certificate ($99/yr), users see an "App is damaged" Gatekeeper warning and need
1340 to run step 5.
1341
1342 ---
1343
1344 ## 16. Moonlight/Sunshine + Tailscale Race Condition
1345
1346 When streaming from a remote Windows host running Sunshine over a Tailscale mesh (e.g., while the client is on
1347 a cellular hotspot), a race condition can occur on boot:
1348
1349 1. Both Tailscale and Sunshine are set to start automatically on boot.
1350 2. Tailscale takes a few seconds to fully initialize its `100.x.x.x` virtual adapter and establish a connection
1351 3. Sunshine starts too fast. It launches before Tailscale is fully ready. Since Sunshine only binds to
1352   network interfaces that are active at the exact moment of startup, it misses the Tailscale adapter entirely
1353 4. Moonlight clients (even when configured with the correct Tailscale IP) will fail to connect because Sunshine
1354   isn't listening on that interface.
1355
1356 The "Magic Toggle" Symptom:
1357 If the user manually enables or disables Wi-Fi on the host PC, it triggers a global Windows network change
1358 event. Sunshine listens for these events, re-enumerates all network adapters, and says, "Oh, there's a
1359 Tailscale adapter here!" and finally binds to it. Suddenly, Moonlight can connect.
1360
1361 The Permanent Fix:
1362 On the Windows host, open `services.msc`, locate the Sunshine Service, and change its Startup type from Automatic to Automatic (Delayed Start). This forces Windows to wait a minute or two after booting
1363 before launching Sunshine, ensuring Tailscale's virtual adapter is fully initialized first.
1364
1365 ---
1366
1367 ## 17. Future Work
1368
1369

```

```

1360 - **Direct P2P on VM Hosts:** Non-Linux hosts run Docker inside a VM, making true UDP hole-punching impossible.
    The only working solution is a native Linux host (Raspberry Pi, Intel NUC) where Docker runs without VM
    hypervisor translation.
1361 - **Native Linux Deployment:** Benchmark on a Raspberry Pi native container footprint is ~75MB without macOS
    hypervisor overhead.
1362 - **Mesh-Wide Filesystem (SFTP over Tailscale):** Any mesh device passively exposes its filesystem over SFTP.
    Android via Termux + OpenSSH + wakelock; iOS is client-only due to background process limits.
1363 - **Post-Quantum Cryptography (PQC):** Tailscale uses Curve25519, theoretically vulnerable to "harvest now,
    decrypt later" quantum attacks. A future iteration could replace the Tailscale container with raw WireGuard
    + [Rosenpass](https://rosenpass.eu/) to negotiate post-quantum PSKs.
1364
1365 ## 18. Geo-IP Blocking
1366
1367 To block traffic to and from specific countries, the system dynamically downloads country-specific IP ranges
    from [ipdeny.com](http://www.ipdeny.com/ipblocks/data/countries/) and drops them via iptables `FORWARD`
    rules.
1368
1369 To add a new country to the blocklist:
1370 1. Find the 2-letter ISO country code (e.g., `cn` for China, `ru` for Russia, `il` for Israel, `kp` for North
    Korea).
1371 2. Open your `.env` file.
1372 3. Update the `BLOCKED_COUNTRIES` variable: e.g., `BLOCKED_COUNTRIES="kp il cn ru"`.
1373 4. Run `toggle.sh` to restart the gateway and apply the new environment variables to the routing-fix container.
1374
1375 > [!NOTE]
1376 > **Limitations of Geo-IP Blocking:** IP-based blocking is highly effective at blocking servers, direct apps,
    and raw infrastructure physically located and registered in a specific country (e.g., `www.gov.il`).
    However, it will **not** block high-profile websites (e.g., `jpost.com`) that route their traffic through
    global Content Delivery Networks (CDNs) like Google Cloud, Cloudflare, or Fastly. Because the CDN's IP
    addresses are registered in the US or globally, the network traffic physically never routes to the blocked
    country, allowing it to bypass the Geo-IP firewall.
1377
1378 ## 19. Incident Post-Mortems
1379
1380 ### Incident: The Overnight Silent IP Leak (July 2026)
1381
1382 **The Problem:** The host leaked traffic over its raw, unencrypted `eth0` IP continuously for roughly 8 hours,
    completely bypassing the VPN. The `toggle.sh` state and the container liveness checks all reported the
    gateway as healthy and active.
1383
1384 **The Investigation:**
1385 1. **Sleep/Wake Checks:** We initially suspected macOS sleep cycles broke the routing table. A check of `pmset
    -g log` confirmed the laptop literally never slept (0 sleep events recorded since boot), ruling out all
    sleep/wake code paths.
1386 2. **Keepalive Expiry:** We discovered `docker-compose.yml` was missing `
    WIREGUARD_PERSISTENT_KEEPAIVE_INTERVAL`. Without a keepalive, standard router NAT tables (which typically
    garbage collect UDP after 30 to 120 seconds of silence) quietly expired the WireGuard port mapping
    overnight.
1387 3. **Container State Masking:** Because WireGuard is stateless, the `warp` container didn't immediately crash.
    It stayed "Up", which meant Docker's healthchecks and our `toggle.sh` liveness checks never noticed the
    tunnel inside it was totally dead.
1388
1389 **The Solution:**
1390 1. Added `WIREGUARD_PERSISTENT_KEEPAIVE_INTERVAL=25s` (the WireGuard standard to beat standard 30s NAT
    timeouts) to keep the UDP tunnel permanently pinned open through the router.
1391 2. Injected a **Fail-Closed Kill-Switch** into `routing-fix.sh`: A 30-second loop now actively verifies tunnel
    health by running `curl` over `tun0` to Cloudflare's trace endpoint. If it fails 3 consecutive times, it
    immediately rewrites the default route to `blackhole`, severing all traffic instead of letting it leak
    through `eth0`.
1392 3. Added a listener in `watcher.sh` that detects this fail-closed event and instantly pushes a native macOS UI
    notification to the desktop, alerting the user to manually intervene.
1393
1394 > [!NOTE]
1395 > **Why dual failure systems?**
1396 > The system now relies on two independent failure handlers that cover each other's blind spots:
1397 > 1. **Automated Self-Healing (WARP Watcher + `recover.sh`):** Triggered when the `warp` container logs `warp=
    off`. This happens when the server actively kicks the client. Because the container knows it is
    disconnected, it triggers `recover.sh` to safely reboot Docker and reconnect.
1398 > 2. **Fail-Closed Kill-Switch (`routing-fix.sh` + `watcher.sh`):** Triggered when a silent network failure
    occurs (e.g., NAT timeouts). The container incorrectly believes it is connected (`warp=on`), so auto-
    recovery never fires. Instead of auto-recovering (which risks falling back to raw Wi-Fi mid-process while
    the user isn't looking), this kill-switch actively blackholes the internet and forces a manual intervention
    via a desktop notification.
1399
1400 ## 20. Design Decisions: Exit Node IP Rotation
1401
1402 A common question for privacy architectures is whether the system should aggressively rotate its public exit IP
    (e.g., every 5 minutes, like a Tor circuit or a proxy rotator).
1403
1404 `nullexit` uses Cloudflare WARP via Gluetun. WARP provides **anonymity in a crowd** rather than aggressive
    rotation. Thousands of users share the same Cloudflare Edge IP simultaneously, making your traffic

```

```

indistinguishable from the herd. The IP may occasionally rotate on its own (which `host-leak-probe.sh` logs
as a `ROTATE` event), but it remains generally stable.
1405
1406 **Why aggressive IP rotation was intentionally excluded:**
1407 1. **Broken TCP Connections:** Every rotation instantly drops all long-lived TCP sessions (SSH, large downloads
, WebSockets, gaming, Zoom).
1408 2. **CAPTCHA & 2FA Hell:** Modern web security (e.g., Cloudflare, Akamai) aggressively flags IP-hopping as bot
behavior. Aggressive rotation guarantees the user will be bombarded with "Verify you are human" CAPTCHAs
and "New Login Detected" 2FA emails constantly.
1409 3. **WireGuard Architecture:** WireGuard is stateless and does not natively support IP hopping on the fly. To
rotate an IP, the VPN client must actively drop the tunnel, request a new endpoint, and re-handshake. This
introduces latency gaps where the kill-switch would repeatedly trigger and blackhole the user's internet.
1410
1411 **Verdict:** For a daily-driver secure gateway, shared-IP crowding (WARP) is superior to aggressive rotation.
As an added benefit, Cloudflare WARP IPs are actually highly static (tied to the persistent WireGuard
public key generated for your device identity). Because your IP stays static and is part of Cloudflare's
own trusted network, it builds a high "trust score," virtually eliminating CAPTCHA loops while still hiding
you within the massive crowd of other users in that data center. All mesh devices routing through your
gateway will reliably share this exact same trusted IP.
1412
1413 *(Note: The only exception where aggressive IP rotation is genuinely required is for specialized offensive
security tasks like high-volume web scraping or dark web research where avoiding IP bans or active tracking
overrides the need for TCP stability.)*
1414
1415 ## 21. Cloudflare WARP Privacy & Logging
1416
1417 Because `nullexit` uses Cloudflare WARP (via Gluetun) as its upstream exit node, the system inherits Cloudflare
's consumer privacy policy.
1418
1419 **What Cloudflare DOES NOT log (Strict No-Logs Policy):**
1420 * **Browsing History:** They do not log the domains or URLs you visit.
1421 * **Traffic Destinations:** They do not log the IP addresses of the servers you connect to.
1422 * **Content:** They do not inspect or log the payload/data of your traffic.
1423 * **Data Brokering:** They explicitly guarantee they will never sell or rent your data to advertisers.
1424
1425 **What Cloudflare DOES log (Minimal Telemetry):**
1426 * **Data Transfer Volume:** Total bandwidth used (required to enforce 10GB/mo quotas if you are not using a
paid WARP+ key).
1427 * **Performance Metrics:** Anonymized connection speeds and latency to optimize their network routing.
1428 * **Aggregate Volume:** Total traffic volumes to popular destinations in aggregate (e.g., "datacenter X sent 50
TB to Netflix"), stripped of all user IDs.
1429 * **Device ID (Public Key):** They store the persistent WireGuard public key generated by your container to
authorize the connection.
1430
1431 **Privacy Verdict:**
1432 Cloudflare WARP provides excellent **historical privacy**. Because they do not log your destinations, they
cannot comply with historical subpoenas asking what websites you visited yesterday. However, it is **not
absolute anonymity**. They still know your real ISP IP address while you are actively connected. For daily-
driver privacy against ISPs, hackers, and corporate trackers, it is top-tier. For nation-state evasion, use
Tor.
1433
1434 ---
1435
1436 ## 22. Battery & Power Measurement Quirks
1437
1438 When trying to optimize this framework (or any daemon) for battery life, developers often look for a way to
programmatically measure the exact Wattage consumed by a specific process (e.g., "How many Watts is `toggle
.sh` using?").
1439
1440 **This is a hardware impossibility on macOS and Linux.**
1441
1442 ### 22.1 Why Activity Monitor is Lying to You
1443 Your machine's logic board only has physical power sensors for the *entire* CPU package, the GPU, and the RAM.
It knows the CPU is pulling 5W total, but it has no physical hardware capability to know whether Docker is
using 3W and Chrome is using 2W.
1444
1445 The "Energy Impact" score you see in macOS Activity Monitor is a **synthetic, heuristic score** calculated by `
powerd`. It penalizes apps for waking up the CPU (idle wakes), doing disk I/O, or keeping the screen awake.
It is a relative score, not actual Wattage.
1446
1447 ### 22.2 The Proper A/B Testing Methodology
1448 If you want to truly know how many Watt-hours or mAh your background daemons (`routing-fix.sh`, `host-leak-
probe.sh`, etc.) are costing you, you must perform a hardware-level A/B drain test:
1449
1450 1. Unplug the laptop, run the gateway, and note your exact battery mAh using:
1451     ```bash
1452     ioreg -l | grep "AppleRawCurrentCapacity"
1453     ```
1454 2. Leave the laptop idle for exactly 1 hour.
1455 3. Run the command again to see exactly how many mAh were drained.
1456 4. Turn the gateway off and repeat the test for another hour.

```

```

1457
1458 The delta in mAh drained between the two tests is the true, exact cost of running the software. When optimizing
    polling loops (e.g., changing `sleep 5` to `sleep 30`), this is the only reliable way to measure the
    actual battery life returned to the user.
1459
1460 ---
1461
1462 ## 23. Architectural Limitations: Host Lockdown Mode
1463
1464 A common privacy feature request is a "Host Lockdown Mode" (Mode 3): A mode where the Docker exit node remains
    perfectly functional for remote devices, but the host machine itself is completely blocked from accessing
    the internet (to prevent even encrypted host traffic from leaking metadata).
1465
1466 **Why this is impossible on macOS:**
1467 On Linux (e.g., a Raspberry Pi), Docker runs natively. The host's traffic traverses the `OUTPUT` iptables chain
    , while Docker traffic traverses the `FORWARD` chain. We could easily drop all `OUTPUT` traffic to kill the
    host's internet, and Docker would continue routing traffic perfectly.
1468
1469 However, Docker on macOS runs inside a Linux Virtual Machine (via `qemu` or Apple `Virtualization.framework`).
    This VM runs as a standard macOS user-space application. If you configure the macOS firewall (`pf`) to drop
    all host outbound traffic, it will indiscriminately drop the VM application's traffic as well, instantly
    killing the Cloudflare WARP tunnel and the exit node.
1470
1471 Writing complex macOS `pf` rules to "allow the VM app, allow Cloudflare WARP IPs, allow Tailscale DERP IPs, but
    block everything else" is highly fragile and heavily discouraged. Since the current `toggle.sh` default
    mode already fully encrypts the host's traffic via the WARP tunnel, the host is already protected from ISP
    leaks.
1472
1473 ### 23.1 July 4, 2026: Multi-stage Docker Builds for Go Ports & Teardown Hang Fix
1474 **Symptom:** The Python implementations of `logger.py` and `sync-rules.py` were slow and required a heavy
    python runtime inside Alpine containers. Also, `routing-fix` container teardown was hanging for 30s during
    `toggle.sh` shutdown.
1475 **Root Cause:**
1476 - Python is slower than Go and installing it dynamically via `apk add` added overhead and complexity to the
    containers.
1477 - Docker containers ignore `SIGTERM` if the init process doesn't handle it, causing a full 30s timeout on
    shutdown.
1478 **Resolution:**
1479 - Ported `logger` and `sync-rules` to Go using Goroutines for massive concurrent speedups.
1480 - Implemented **Multi-stage Docker builds** (using `golang:1.22-alpine` as builder) to compile the static
    binaries inside Docker. The final containers are raw Alpine with zero dependencies.
1481 - Added `--build` to `docker compose up -d` in `toggle.sh` to ensure updates are consistently applied on boot.
1482 - Added `stop_grace_period: 1s` to `routing-fix` in `docker-compose.yml` which eliminates the 30-second
    teardown hang instantly.
1483 - Implemented a cryptographic integrity checker (`scripts/crypto.sh`) using HMAC-SHA256 and `NULLEXIT_SEED` in
    `.env` to prevent tampering of bash scripts.
1484
1485 ### 23.2 July 4, 2026: Consolidation of Core Bash Scripts (Refactoring)
1486 **Symptom:** Over 200 lines of bash logic were directly duplicated across `toggle.sh` and `recover.sh` (e.g. `
    restart_tailscaled_daemon`, `stop_sleep_prevention`, WARP bypass routes). This caused drift when one script
    was updated without the other.
1487 **Root Cause:**
1488 - Historical separation of responsibilities allowed `recover.sh` to grow complex alongside `toggle.sh`.
1489 **Resolution:**
1490 - Massively refactored both scripts by extracting all duplicated networking logic, PID lifecycle management,
    and environmental configuration down to `scripts/common.sh`.
1491 - Specifically, the `stop_sleep_prevention`, `stop_dns_watcher`, `stop_host_leak_probe`, and `stop_warp_watcher`
    functions were collapsed into a single `stop_pidfile_daemon()` abstraction.
1492 - Proxy disable loops were abstracted to `disable_all_proxies()`.
1493 - The `162.159.192.1/.193.1` WARP edge IPs are now dynamically resolved from `.env` instead of hardcoded.
1494
1495 ---
1496
1497 ### 23.3 July 4, 2026: The "Network Unreachable" Host Leak & Post-Wake Container Destructors
1498 **Symptom:**
1499 1. Running `docker compose config` with dummy keys corrupted the user's `.env` file by appending invalid
    WireGuard keys. This caused the `warp` container to become unhealthy on boot, which triggered a full
    teardown of the gateway by `toggle.sh`.
1500 2. After fixing `.env`, `toggle.sh` successfully booted the gateway, but the host's internet started bypassing
    the tunnel entirely (a massive host leak). The Cloudflare trace returned `warp=off` and exposed the host's
    physical ISP IP.
1501 3. Running `recover.sh --post-wake` would violently force-recreate all gateway containers and drop the host's
    internet connection.
1502
1503 **Root Cause:**
1504 - **Bug 1 (The Host Leak):** In `toggle.sh`, the logic to find the Tailscale `utun` interface used `ifconfig |
    grep -B4 "inet 100."`. Due to the 4 lines of before-context, this regex was accidentally capturing the
    interface *above* the actual Tailscale interface (e.g. grabbing `utun4` instead of `utun5`). Because `utun4`
    had no IP address, the subsequent `sudo route add -net 0.0.0.0/1 -interface utun4` silently failed with "
    Network is unreachable", leaving the host's default route exposed.

```

```

1505 - **Bug 2 (The Post-Wake Destructors):** The health check in `recover.sh --post-wake` relied on `docker
    compose exec -T warp curl -sf https://www.cloudflare.com/cdn-cgi/trace`. However, the newer `qmcgaw/glutun
    ` Alpine-based Docker image no longer ships with `curl` pre-installed. The health check failed with `
    executable file not found in $PATH`, tricking `recover.sh` into thinking the tunnel was completely dead and
    needed to be forcefully recreated via `docker compose up -d --force-recreate`.
1506 - **Bug 3 (Restart Flags):** Previous refactors incorrectly permitted `toggle.sh restart` instead of strictly
    enforcing `toggle.sh --restart`.
1507
1508 **Resolution:**
1509 - Cleaned the `.env` file of duplicate dummy entries.
1510 - Replaced the brittle `grep -B4` interface parsing in `toggle.sh` with a strict AWK parser: `awk '/^[a-z0
    -9]+:/ {iface=$1} /inet 100\./ {print iface; exit}' | tr -d ':'`. This mathematically isolates the exact
    interface possessing the Tailscale `100.x.x.x` IP address, preventing the `0.0.0.0/1` route from silently
    failing.
1511 - Changed the health check in `recover.sh` to use `wget -q0- --timeout=3` which is natively installed in the
    Gluetun image. This stopped the unnecessary destructive recreations of the containers during post-wake
    events.
1512 - Reverted the `restart` alias in `toggle.sh` to strictly require `--restart`.
1513
1514 ### 23.4 July 4-5, 2026: Tailscale Data-Plane Loop (The DERP Relay Deadlock)
1515 **Symptom:**
1516 After bypassing the Tailscale control plane (`192.200.0.0/16`) to fix the "offline" mesh status, the Mac
    appeared online. However, external peer devices (like the user's phone on cellular) could not successfully
    ping or SSH into the Mac. `tailscale netcheck` reported `Nearest DERP: unknown (no response to latency
    probes)`.
1517
1518 **Root Cause:**
1519 While bypassing the control plane kept the daemon connected to the coordination servers, Tailscale's actual
    peer-to-peer data plane relies on STUN (for NAT traversal) and DERP relays. These relays span ~80
    dynamically allocated IP addresses across multiple cloud providers. Because we were forcing `0.0.0.0/1`
    through the `utun*` exit node, the Mac's own outbound connection attempts to DERP relays were being routed
    back into the tunnel. To prevent an infinite loop, Tailscale's daemon dropped its own packets when it saw
    them returning on the TUN interface. This effectively left the daemon deaf and blind to peer connections.
1520
1521 **Resolution:**
1522 - Modified `add_warp_bypass_routes` in `scripts/common.sh` to execute a live API fetch (`curl -s https://login.
    tailscale.com/derpmap/default`) during startup.
1523 - The script parses out all active DERP relay IPv4 addresses (~80 IPs) and adds a physical host bypass route
    for every single one of them.
1524 - These IPs are written to a temporary file (`/tmp/nullexit-derp-ips.txt`) so they can be cleanly un-routed by
    `remove_warp_bypass_routes` when the gateway shuts down. Peer connectivity (ping, SSH, SFTP) was fully
    restored.
1525
1526 ### 23.5 July 5, 2026: Hardcoded WARP Endpoints in docker-compose
1527 **Symptom:**
1528 The bash scripts allowed overriding the default Cloudflare WARP IP endpoints via `.env` variables (`
    WARP_ENDPOINT_1` and `WARP_ENDPOINT_2`) to establish bypass routes. However, `docker-compose.yml`
    statically hardcoded `162.159.192.1` for the `warp` container's `WIREGUARD_ENDPOINT_IP`.
1529
1530 **Root Cause & Risk:**
1531 If a user set a custom endpoint in `.env`, the script would successfully bypass the custom IP on the host level
    . However, Gluetun would still attempt to connect to the hardcoded `162.159.192.1`. Since `162.159.192.1`
    was no longer explicitly bypassed, its packets would be sucked into the `utun*` exit node, causing an
    infinite WireGuard routing loop that instantly breaks the gateway.
1532
1533 **Resolution:**
1534 Modified `docker-compose.yml` to dynamically read the environment variable via `${WARP_ENDPOINT_1
    :-162.159.192.1}`, ensuring both the host routing scripts and the container use the exact same endpoint.
1535
1536 ### 23.6 July 6, 2026: Packet Filter (pfctl) Defaulting Local LAN Rules to TCP-Only
1537 **Symptom:**
1538 Devices on the exact same local Wi-Fi network as the gateway were experiencing ~500ms latency to the gateway
    instead of the expected ~80ms.
1539
1540 **Root Cause:**
1541 When writing the `pf.conf` rules to whitelist local LAN traffic (`pass out quick on $ext_if to {
    192.168.0.0/16, ... }`), the rule omitted an explicit protocol. The macOS `pfctl` compiler implicitly
    applies state tracking to all `pass` rules. However, because state tracking (`keep state`) natively
    evaluates TCP sessions, `pfctl` automatically appended the `flags S/SA` modifier to the compiled rule in
    the kernel. This silently restricted the whitelist to **TCP only**, causing all local UDP traffic to be
    dropped by the default-deny kill-switch. Since Tailscale relies on UDP port 41641 for direct P2P
    connections, the local devices were unable to hole-punch and were forced to fall back to routing traffic
    through a remote Tailscale DERP relay, massively inflating latency.
1542
1543 **Resolution:**
1544 Split the single implicit rule into explicit `proto tcp`, `proto udp`, and `proto icmp` rules in `scripts/pf.
    conf` to guarantee the macOS compiler allows all three core transport protocols on the local subnet without
    mutating the rule into a TCP-only lock.
1545 ### 23.7 July 8, 2026: Network Watcher Event Mismatch, Sudo-in-Background Crash, and Loop-Prevention
1546 **Symptom:**

```

```

1547 1. Switching Wi-Fi or waking the Mac from sleep did not trigger a post-wake recovery (`recover.sh --post-wake`)
    automatically.
1548 2. The network connection would eventually get completely sinkholed/blocked. If the user tried to restart or
    wait, it did not self-heal.
1549 3. During the recovery attempt, checking the public IP on the host would return the unencrypted campus/ISP IP (
    starting with `132.`) instead of the encrypted tunnel IP.
1550
1551 **Root Cause:**
1552 - **Bug 1 (Network Watcher Mismatch):** In `scripts/watcher.sh`, the network change listener watched `State:/
    Network/Global/IPv4` changes via `scutil n.watch` but matched them against `*n.state*|*SCEventUpdate*`. On
    macOS, the actual output is `changedKey [0] = State:/Network/Global/IPv4`. Because the pattern did not
    match, all network switches were silently ignored.
1553 - **Bug 2 (Sudo Caching Background Crash):** When the background WARP Watcher eventually detected the broken
    tunnel (after 6 failures/30s), it triggered the default nuclear `recover.sh` (with `POST_WAKE=false`).
    Because there was no active TTY in the background context, the `sudo -v` caching check aborted instantly
    with a terminal required error. Due to `set -e`, `recover.sh` died immediately before resetting DNS,
    disabling the packet-filter killswitch, or stopping containers, leaving the host network completely
    sinkholed.
1554 - **Bug 3 (Post-Wake Infinite Loop):** Once the watcher patterns were fixed, `recover.sh --post-wake` triggered
    successfully on network changes. However, because the script deleted the default routing entries at the
    start, spent ~15 seconds rebuilding 88 DERP relay bypass routes, and re-added the default routes at the end
    , this execution time exceeded the watcher's 10-second debounce threshold. Furthermore, the watcher
    recorded the `start` timestamp in the debounce file. Consequently, the route changes made by the script
    itself triggered the watcher again immediately after completion, trapping the system in an infinite loop of
    recovery runs. During this loop, the VPN routes were constantly deleted, resulting in the host leaking
    traffic through the real ISP interface (an IP starting with `132.`) for 15s out of every cycle.
1555
1556 **Resolution:**
1557 - **Fixed Watcher Matching:** Updated `scripts/watcher.sh` to match `*changedKey*` and `*State:/Network/Global/
    IPv*` to correctly catch network changes.
1558 - **Bypassed Background Sudo:** Added `--auto`/`--non-interactive` flags to `recover.sh` and added a TTY check
    `[ -t 0 ]` to skip interactive `sudo -v` authentication when running headlessly in the background. Updated
    the WARP Watcher call in `toggle.sh` to pass `--auto`.
1559 - **Prevented Infinite Loops:** Modified `scripts/watcher.sh` to write the **completion** timestamp of `recover
    .sh` to `DEBOUNCE_FILE` (instead of only the start timestamp). This ensures all buffered network config
    events generated during route reconstruction are evaluated against the end-of-run timestamp and
    successfully debounced, breaking the infinite loop.
1560 - **Centralized & Timestamped Logs:** Redirected `watcher.sh` stdout/stderr from `/tmp/nullexit-watcher.log` to
    the repository's main `output.log` and updated `scripts/common.sh`'s `step`, `ok`, `warn`, `fail`, and `
    die` helpers to prefix logs with a local `[YYYY-MM-DD HH:MM:SS]` timestamp.
1561
1562 ### 23.8 July 11, 2026: Tor Container Hardening & obfs4proxy Restoration
1563 **Symptom:**
1564 1. The Tor container entered a fatal crash loop upon boot, throwing `exec: "obfs4proxy": executable file not
    found in $PATH` when `TOR_USE_BRIDGES=true`.
1565 2. When the user pasted Tor bridge lines containing comments (`#`) or blank lines into `tor-bridges.txt`, the
    container failed to validate the file properly and crashed.
1566
1567 **Root Cause:**
1568 - **Bug 1 (Missing Pluggable Transports):** The Tor Dockerfile was built on `debian:bullseye-slim`, which had
    silently dropped or failed to resolve the `obfs4proxy` package in its latest apt repositories.
1569 - **Bug 2 (Fragile Validation):** The `entrypoint.sh` bridge validation check merely checked `[ -s tor-bridges.
    txt ]` (file size > 0). It did not verify if the file actually contained valid, uncommented bridge lines.
    Passing empty lines or comments tricked the validation script into appending invalid `Bridge` directives to
    the `torrc`, causing the Tor daemon to instantly exit.
1570
1571 **Resolution:**
1572 - **Upgraded Base Image:** Shifted the Tor Dockerfile base image to `debian:bookworm-slim`, restoring access to
    the `obfs4proxy` pluggable transport package.
1573 - **Robust Bridge Validation:** Replaced the fragile `[ -s ]` check with a strict `grep -vE '\s*(#|\$)'\`
    command that strips comments and empty lines. If no valid lines remain, the container safely falls back to
    standard public guard relays instead of crashing, ensuring Tor remains available even with a misconfigured
    bridge file.
1574 - **Image Pinning:** Explicitly pinned `image: nullexit-tor:v1.0.0` in `docker-compose.yml` to prevent
    arbitrary local builds from drifting to `:latest`.
1575
1576 ---
1577
1578 ## 24. Censorship-Resistant Transport & Egress Compartmentalization
1579
1580 ### 24.1 Why this is needed
1581 WARP can be blocked from deployment countries with strict internet controls. The most likely cause isn't that "
    encrypted traffic is blocked" in general it's that WireGuard has a recognizable handshake signature, and `
    WIREGUARD_ENDPOINT_IP=162.159.192.1` on `WIREGUARD_ENDPOINT_PORT=2408` is a fixed, easily-enumerable target
    (Cloudflare's known WARP range). That combination is exactly what IP/ASN-based and DPI-based blocking is
    good at catching.
1582
1583 ### 24.2 Important correction: Gluetun's `SHADOWSOCKS=on` is the wrong direction
1584 Gluetun's built-in Shadowsocks option runs a Shadowsocks **server** inside the already-connected VPN tunnel, so
    other LAN devices can reach out through Gluetun's **existing** connection via the SS protocol. It is not a
    way to make Gluetun itself connect **outbound** through a Shadowsocks server Gluetun only speaks OpenVPN

```

or WireGuard as its actual transport. There is no `VPN\_TYPE=shadowsocks`. Confirmed against Gluetun's own maintainer/community discussion: people have asked for a "Shadowsocks-client" mode specifically to chain through Gluetun, and the standing answer is "run a separate Shadowsocks client container."

1585

1586 **### 24.3 Diagnose before building anything**

1587 The right fix depends entirely on what's actually being blocked:

1588 - Point Gluetun at a ****different provider/IP/ASN**** (any other Gluetun-supported WireGuard/OpenVPN provider, or a self-hosted WireGuard server) and test. If that gets through, the block is Cloudflare/WARP-specific, not a general WireGuard block ****no new component needed****, just swap `VPN\_SERVICE\_PROVIDER` / the endpoint.

1589 - If WireGuard fails regardless of endpoint, the block is protocol-level (DPI fingerprinting the handshake) proceed to obfuscation.

1590

1591 **### 24.4 Path A: Swap WARP for a different Gluetun-native transport**

1592 Simplest fix, only applies if 24.3 shows the block is Cloudflare-specific. Change `VPN\_SERVICE\_PROVIDER` / `VPN\_TYPE` / endpoint in `docker-compose.yml` and in the now-centralized `WARP\_ENDPOINT\_\*` values in `.env`. Nothing else in the stack needs to change.

1593

1594 **### 24.5 Path B: Add an obfuscation hop**

1595 Needed if WireGuard itself is being fingerprinted/blocked. Two ways to slot it in:

1596

1597 ****B1 Wrap (recommended):**** Run a Shadowsocks (or obfs4/v2ray) client as a new sidecar container. Point Gluetun's `WIREGUARD\_ENDPOINT\_IP` at `127.0.0.1:<local-port>` instead of Cloudflare directly; the sidecar relays that traffic to a Shadowsocks server you run yourself abroad. Keeps Gluetun's kill-switch, healthchecks, and the entire rest of the stack (Tailscale, AdGuard, ipset, `routing-fix.sh`) untouched, since none of them know the transport changed underneath `warp`.

1598

1599 ****B2 Replace:**** Swap the `warp` service entirely for a Shadowsocks-client + `tun2socks` container that owns the network namespace instead of Gluetun. More invasive loses Gluetun's built-in kill-switch/healthcheck, which would need to be rebuilt by hand (see Section 24.9 for the pluggable architecture to support this natively).

1600

1601 ****Recommendation: B1.**** Smaller surface area, preserves the self-healing architecture already built and documented.

1602

1603 **### 24.6 Fingerprinting caveat**

1604 Plain Shadowsocks can still be caught by active-probing DPI in more aggressive censorship environments. If plain SS gets blocked too, the next step up is an obfuscation plugin (`v2ray-plugin`, `Cloak`) or a newer protocol designed against active probing (VLESS+Reality, Hysteria2). Test plain SS first don't over-build before confirming it's needed.

1605

1606 **### 24.7 AmneziaWG (The High-Speed Alternative)**

1607 If you want to maintain the bare-metal speeds of WireGuard while bypassing DPI (e.g. in Egypt or Russia), the best alternative to Shadowsocks/V2Ray is AmneziaWG. AmneziaWG is a fork of WireGuard that pads the handshake packets with randomized "junk" data, entirely destroying the 148-byte DPI signature that firewalls look for.

1608 - ****Pros:**** Because it runs entirely as UDP without TCP-over-TCP overhead, it completely avoids "TCP meltdown" latency spikes associated with Shadowsocks/V2Ray wrappers.

1609 - ****Cons:**** You cannot use Cloudflare WARP. Furthermore, it requires completely ripping Gluetun out of the `nullexit` stack and building a custom Docker container, meaning you lose Gluetun's built-in kill-switches and health-check logic.

1610

1611 **### 24.8 Infrastructure Costs & Privacy**

1612 To use either Shadowsocks (Path B) or AmneziaWG, you cannot use the free Cloudflare WARP infrastructure, because Cloudflare servers only speak standard WireGuard. The software for both custom protocols is 100% free and open-source, but you must host the remote server infrastructure yourself.

1613 - ****Free Tier (Oracle Cloud):**** You can host your remote server on Oracle Cloud's "Always Free" tier for \$0. However, this routes your highly sensitive, censorship-evading traffic directly through Oracle's massive corporate data broker. If privacy is the core objective, handing all your metadata to Oracle defeats the purpose.

1614 - ****Paid Tier (Hetzner / Mullvad):**** The recommended path is to rent a standard ~\$4/month Virtual Private Server (VPS) in a free country (e.g., Hetzner in Germany, subject to strict EU GDPR privacy laws) and self-host the server. Alternatively, Mullvad VPN (5/month) provides native v2ray/Shadowsocks bridges built into their network, allowing you to bypass censorship with a strict zero-logs policy. It is highly recommended to pay with untrackable payment methods to maintain complete anonymity, which these providers typically support without requiring local residency.

1615

1616 **### 24.9 The Future: Egress Compartmentalization (Pluggable Architecture)**

1617 To natively support invasive replacement paths like ****Path B2**** or ****AmneziaWG**** (Section 24.7) without breaking the core `nullexit` routing and monitoring logic, the architecture must be refactored into a modular, "pluggable" design.

1618

1619 1. ****The Egress Plugin System:**** Currently, WARP-specific logic (like `cdn-cgi/trace` health checks and hardcoded interface names) is scattered across `toggle.sh`, `scripts/routing-fix.sh`, and `scripts/host-leak-probe.sh`. This should be abstracted into a `scripts/plugins/` directory, where each egress method (`warp.sh`, `v2ray.sh`) implements standardized functions: `get\_egress\_interface()`, `check\_health()`, and `get\_bypass\_ips()`.

1620 2. ****Dynamic Docker Compose:**** Based on an `EGRESS\_TYPE` variable in `.env`, dynamic compose file generation would spin up only the required egress containers (e.g., skipping the `warp` container when `EGRESS\_TYPE=v2ray` is set).

1621 3. ****Agnostic Routing & Watchers:**** `routing-fix.sh` would dynamically read the `EGRESS\_INTERFACE` from the active plugin to apply `iptables` rules, and the background watchers in `toggle.sh` would become an

```

agnostic `Egress Watcher` that invokes `check_health()` periodically.
1622
1623 This compartmentalization transforms `nullexit` into a universal, censorship-resistant gateway where the entire
    egress layer can be swapped by changing a single `.env` variable and running `./toggle.sh --restart`.
1624
1625 ---
1626
1627 ## 25. Incident Post-Mortem: WARP Watcher Race vs Post-Wake (July 10, 2026)
1628
1629 ### Symptom
1630 After switching Wi-Fi networks, the host IP appeared as the raw ISP IP (`132.x.x.x`) instead of a Cloudflare/
    WARP IP. The gateway appeared to complete post-wake recovery successfully in the logs, but nuclear shutdown
    fired immediately after and killed everything.
1631
1632 ### Root Cause
1633 A fatal race condition between two concurrent processes:
1634
1635 1. Wi-Fi roam `watcher.sh` fires `recover.sh --post-wake`
1636 2. Post-wake finds the `warp` container missing/dead calls `docker compose up -d --force-recreate` (~35
    seconds)
1637 3. In parallel, the WARP Watcher (running as a child of `toggle.sh`) polls the warp container every 5s
    during failures
1638 4. While the container is being recreated it returns `warp=off` for those 35 seconds 7 consecutive failures
    WARP SHUTDOWN fires, calling nuclear `recover.sh`
1639 5. Nuclear `recover.sh` tears down Tailscale, resets DNS to 1.1.1.1, stops all containers gateway dead, IP
    exposed
1640 6. The post-wake `docker compose up` finishes at ~35s mark, container healthy but the gateway is already gone
1641
1642 The evidence in `output.log`:
1643 ```
1644 [2026-07-10T06:07:27Z] NET: ... bash recover.sh --post-wake
1645 [2026-07-10T06:07:47Z] WARP DOWN cdn-cgi/trace reports warp=off watcher starts counting
1646 ...
1647 add net 0.0.0.0: gateway utun5 post-wake successfully re-routes at 02:08:22
1648 ...
1649 [2026-07-10T06:09:04Z] WARP SHUTDOWN 6 consecutive failures watcher fires nuclear
1650 ```
1651
1652 ### Fix (July 10, 2026)
1653 Added a WARP Watcher inhibit marker (`/tmp/nullexit-warp-inhibit.marker`):
1654
1655 - `recover.sh --post-wake` writes the marker at startup before any container operations
1656 - The WARP Watcher loop checks for the marker at the top of every iteration; if present it resets `consec_off
    =0` and sleeps 5s (skips the poll entirely)
1657 - `recover.sh` removes the marker at the very end (on both success and failure via `trap cleanup_inhibit EXIT`)
1658 - This guarantees the watcher never counts failures during the intentional container-down window of a post-wake
    force-recreate
1659
1660 The marker file path is hardcoded to `/tmp/nullexit-warp-inhibit.marker` in both `toggle.sh` and `recover.sh`.
1661
1662 ## 26. Incident Post-Mortem: Startup Pre-Flight Check Tailscale Race (July 10, 2026)
1663
1664 ### Symptom
1665 Immediately after startup via `toggle.sh`, the pre-flight connectivity check `[1/3] Gateway reachable via
    Tailscale` returned `FAIL`, forcing `toggle.sh` to skip the full exit-node activation and fall back to
    SOCKS5/local DNS proxy mode.
1666
1667 ### Root Cause
1668 A timing race condition during startup. In `toggle.sh`, the host joins the mesh using `tailscale up --reset` (
    Phase A) right before performing the pre-flight check. Because `tailscale up` resets and reconfigures the
    host's `tailscaled` daemon, the local daemon must fetch the latest netmap asynchronously from the
    coordination servers.
1669 This network map propagation and route establishment takes a few seconds. Running `tailscale ping` immediately
    after `tailscale up` returns (with no delay) causes the check to fail because the local daemon has not yet
    discovered the gateway node `100.100.21.8`.
1670
1671 ### Fix (July 10, 2026)
1672 Upgraded pre-flight Check 1 in both `toggle.sh` and `scripts/toggle-linux.sh` to use a 5-attempt retry loop
    with a 1-second delay between checks. This allows up to 5 seconds for local tailnet routing paths to settle
    , ensuring the gateway is correctly reached and a pong is received before deciding whether to enable exit-
    node routing.
1673
1674 ## 27. Incident Post-Mortem: Concurrency Collision between Nuclear Teardown and Post-Wake (July 10, 2026)
1675
1676 ### Symptom
1677 When the background WARP Watcher triggers a nuclear shutdown (due to a real or simulated tunnel failure), the
    gateway is completely torn down. However, the network configuration changes made during the teardown
    trigger a `State:/Network/Global/IPv4` network change event. The LaunchAgent-based network watcher (
    watcher.sh) intercepts this event and concurrently spawns `recover.sh --post-wake` to heal/restore the
    gateway. This causes `docker compose down` (from nuclear teardown) and `docker compose up` (from post-wake)
    to run in parallel, resulting in container errors like `dependency failed to start: container warp exited

```

```

(0)` and the gateway becoming wedged.
1678
1679 ### Root Cause
1680 A lack of coordination/locking between the lifecycle control scripts. While `toggle.sh` used `/tmp/nullexit-
toggle.lock` to prevent concurrent `toggle.sh` runs, `recover.sh` (both nuclear and `--post-wake` modes)
did not utilize or check any lock files.
1681
1682 ### Fix (July 10, 2026)
1683 Implemented a shared lifecycle lock file (`/tmp/nullexit-toggle.lock`) across both scripts:
1684 - **`recover.sh` (all modes)**: Checks `/tmp/nullexit-toggle.lock` at startup. If the lock is held by another
running lifecycle script (`toggle.sh` or `recover.sh`):
1685 - In `--post-wake` mode, it logs a skip message to `output.log` and exits cleanly (`exit 0`). This safely
prevents the auto-recovery agent from recreating containers during a teardown.
1686 - In nuclear recovery mode, it exits with an error.
1687 - **`recover.sh` Lock Cleanup**:
```

Cleans up the lock file upon completion using `trap cleanup\_recover EXIT` (which checks if the lock file belongs to the current PID).

```

1688 - **`toggle.sh` update**:
```

The lock check in `toggle.sh` was updated to look for both `toggle.sh` and `recover.sh` in the running processes to enforce proper exclusion.

```

1689
1690 ## 28. Incident Post-Mortem: Infinite Auto-Recovery Loop after WARP Watcher Nuclear Shutdown (July 10, 2026)
1691
1692 ### Symptom
1693 When the background WARP Watcher triggered a nuclear shutdown (due to a tunnel outage), it successfully
executed `recover.sh` (nuclear). However, immediately after the teardown completed, the system started
spawning `recover.sh --post-wake` and rebuilding the containers again. This created an infinite loop of
tearing down and auto-restarting the gateway, keeping the host's internet route permanently wedged.
1694
1695 ### Root Cause
1696 An active-state marker file leak. The LaunchAgent-based network watcher (`watcher.sh`) only fires `recover.sh
--post-wake` when `/tmp/nullexit-gateway-active.marker` is present on disk. While `toggle.sh` (STOP path)
correctly deleted this marker, the nuclear `recover.sh` script did not. When the WARP Watcher ran `recover.
sh` (nuclear), the containers were stopped but the active-state marker was left on disk. The network
changes from the teardown then triggered `watcher.sh`, which saw the marker, believed the gateway was still
supposed to be active, and triggered `--post-wake` to start it back up.
1697
1698 ### Fix (July 10, 2026)
1699 Modified the start of `recover.sh` to explicitly delete `/tmp/nullexit-gateway-active.marker` if `POST_WAKE` is
`false` (nuclear mode). This ensures that any subsequent network configuration changes made during the
teardown are correctly ignored by `watcher.sh` because the marker file is gone.
1700
1701 ## 29. Incident Post-Mortem: Pre-Flight Check False Negatives due to VPN Firewall STUN Blocks (July 10, 2026)
1702
1703 ### Symptom
1704 Immediately after a cold boot or full restart of the gateway, the pre-flight connectivity check `[1/3] Gateway
reachable via Tailscale` failed, causing `toggle.sh` to skip the full exit-node configuration and fall back
to SOCKS5/local DNS proxy mode. However, manually running `tailscale ping 100.100.21.8` immediately after
startup succeeded in 1ms.
1705
1706 ### Root Cause
1707 An asynchronous Tailscale handshake delay caused by firewall restrictions. Inside the container network
namespace, the `warp` container (gluetun) implements a strict firewall that blocks all outbound UDP traffic
to the public internet except to the designated WireGuard endpoint. Because of this, the `tailscale`
container's STUN requests to public STUN servers are blocked, causing the local daemon to report `UDP is
blocked, trying HTTPS`.
1708 Without UDP STUN capability, Tailscale is forced to fall back to a slower DERP-assisted handshake (relayed via
HTTPS/TCP). This negotiation and direct-path punch-through takes roughly 30 to 35 seconds to establish on
cold boots (after a full `tailscale up --reset`). Since the pre-flight check only waited 15 seconds (15
attempts with 1-second sleeps), the check timed out and aborted before the path completed.
1709
1710 ### Fix (July 10, 2026)
1711 Increased the pre-flight ping retry limit from 15 to 45 attempts (45 seconds) in both `toggle.sh` and `scripts/
toggle-linux.sh`. This provides sufficient time for the DERP-assisted handshake to complete on cold boots,
while still passing immediately (in 1-2 seconds) on warm starts or once the path is established.
1712
1713 ## 30. Incident Post-Mortem: Docker Image Build DNS Failures on Immediate Restart (July 10, 2026)
1714
1715 ### Symptom
1716 When executing `toggle.sh --restart`, the script successfully stops the gateway but then immediately fails
during startup during `docker compose build` with DNS resolution errors:
1717 `failed to do request: Head "https://registry-1.docker.io/...": dial tcp: lookup registry-1.docker.io on
192.168.5.1:53: no such host`.
1718
1719 ### Root Cause
1720 A race condition between physical link negotiation and virtual machine startup. The STOP path of the restart
command invokes `cleanup_network_state` which power-cycles the host's physical Wi-Fi interface (`en0`) to
flush stale routing states. Immediately after, `toggle.sh` starts the gateway boot sequence. Because
`toggle.sh` did not verify the status of the physical interface, it proceeded to boot Colima and build
Docker containers while `en0` was still negotiating a DHCP lease. Consequently, the host (and by extension,
the VM bridge) had no active internet gateway/DNS path, causing Docker's registry check to fail.
1721
1722 ### Fix (July 10, 2026)

```

```

1723 1. Created a unified `wait_for_dhcp_settle` helper function in `scripts/common.sh` that polls `ipconfig
    getpacket` and `ifconfig` until a valid IP and router are assigned. To handle typical macOS Wi-Fi
    reassociation and DHCP negotiation latency (which commonly takes 15-20 seconds after a power-cycle), this
    loop was configured to poll up to 60 times (30 seconds total).
1724 2. Injected a call to `wait_for_dhcp_settle` at the beginning of the `START` action in `toggle.sh` (right
    before checking Colima status) to block container builds until the host's physical network is online.
1725 3. Refactored `recover.sh` to reuse the unified `wait_for_dhcp_settle` function.
1726
1727 ## 31. Incident Post-Mortem: PF Kill-Switch Bricks Host Internet in SOCKS5 Fallback Mode (July 10, 2026)
1728
1729 ### Symptom
1730 When the pre-flight checks fail (e.g. because host Tailscale was unauthenticated/logged out), the script falls
    back to SOCKS5 proxy mode but the host immediately loses all internet connectivity and tailscaled reports `
    You are logged out... connect: no route to host`. Even running `sudo tailscale up` to log in fails because
    the connection to the control plane times out.
1731
1732 ### Root Cause
1733 An architectural mismatch. The macOS Packet Filter (PF) Kill-Switch works at the IP level: it blocks all
    outbound traffic on the physical interface (`en0`) except to explicitly whitelisted VPN endpoints,
    expecting all other traffic to go through the virtual VPN interface (`utun*`).
1734 In SOCKS5 fallback mode, the exit-node (`utun*`) is NOT enabled; instead, only specific applications (like
    browsers) use the localhost SOCKS5 proxy, while all other host traffic (including the `tailscaled` daemon's
    UDP packets) continues to go through `en0`. Because the script still enabled the PF Kill-Switch during
    SOCKS5 fallback, it blocked all non-SOCKS5 traffic on `en0`, completely bricking the host's internet and
    locking the Tailscale daemon out of the control plane.
1735
1736 ### Fix (July 10, 2026)
1737 Modified `toggle.sh` to remove the `enable_killswitch` call from the SOCKS5 fallback path. The firewall kill-
    switch is now strictly reserved for exit-node VPN mode, ensuring SOCKS5 fallback leaves the host's direct
    internet route open for system daemons to authenticate.
1738
1739 ## 32. Incident Post-Mortem: Host Tailscale Logouts due to Aggressive reset Flags (July 10, 2026)
1740
1741 ### Symptom
1742 Every time the user runs `toggle.sh` or `recover.sh` which disconnects/reconnects the host client, the host
    client randomly gets logged out, forcing them to re-run `sudo tailscale up` to authenticate.
1743
1744 ### Root Cause
1745 An overly aggressive configuration reset. The host-side `tailscale up` calls (used in `
    disconnect_tailscale_host` and the startup/enable exit node paths) all passed the `--reset` flag. On macOS,
    `--reset` resets the entire configuration state and authentication token cache of the daemon. Since the
    script does not pass a `TS_AUTHKEY` to the host's commands (which are meant for the user's private
    interactive client), the `--reset` flag effectively logged the user out of their tailnet, requiring
    interactive re-login.
1746
1747 ### Fix (July 10, 2026)
1748 Removed the `--reset` flag from all host-side `tailscale up` invocations in `toggle.sh` and `scripts/common.sh
    `. This ensures that exit-node state transitions (enabling/disabling) are handled safely while preserving
    the host client's authenticated session.
1749
1750 ## 33. Incident Post-Mortem: Discarded Docker Exec Errors in logs (July 10, 2026)
1751
1752 ### Symptom
1753 When the `warp` container is stopped, dead, or failing to connect to its endpoints, the `toggle.sh` and `
    recover.sh` connectivity health checks fail, but no details are printed to `output.log`. The logs only show
    generic `FAIL` outputs, making it hard to see if the failure was a network timeout, Docker daemon error,
    or a missing binary.
1754
1755 ### Root Cause
1756 Overly broad error silencing. The `docker compose exec` commands in the health checks (Check 3 in `toggle.sh`
    and the traffic check in `recover.sh`) both redirected stderr to `/dev/null` (`&>/dev/null` and `2>/dev/
    null` respectively), completely throwing away any diagnostic error messages produced by Docker or `wget`.
1757
1758 ### Fix (July 10, 2026)
1759 Redirected stderr of the `docker compose exec` check commands to `output.log` (`2>> output.log`). This ensures
    that any command execution failures or connection timeout details are logged.
1760
1761 ## 34. Architectural Design: How nullexit Mimics Commercial VPNs during Roaming
1762
1763 The `nullexit` roaming/network recovery subsystem replicates the exact engineering mechanics used by commercial
    , premium VPN clients (like Mullvad or NordVPN) to transition across Wi-Fi networks safely and seamlessly.
1764
1765 ### The 4-Step Transition Cycle
1766 When your Mac disconnects from one Wi-Fi and associates with another, `nullexit` coordinates the following
    events:
1767
1768 1. Firewall Lock (Kill-Switch): The macOS Packet Filter (PF) anchor `com.apple/nullexit` remains locked on
    the physical interface (`en0`), blocking all direct outbound IPv4 and IPv6 traffic. This guarantees that
    your real ISP IP or unencrypted DNS requests never leak onto the new network while the connection is
    rebuilding.

```

```

1769 2. **System Event Interception**: A launchd LaunchAgent (`watcher.sh`) listens to the macOS System
    Configuration framework for the `State:/Network/Global/IPv4` key changes, spawning `recover.sh --post-wake`
    in under 500ms when a change is detected.
1770 3. **Bypass Routing**: The script dynamically resolves the new physical network's IP gateway (e.g.
    `192.168.137.1`), cleans up stale bypass routes from the previous network, and adds static bypass routes
    pointing to Cloudflare's WARP endpoints and the Tailscale control plane directly via the new gateway. This
    allows the VPN client to negotiate its handshake outside of the tunnel.
1771 4. **Hole-Punching / NAT Re-negotiation**: The script runs a target check on the `warp` container's tunnel. If
    the WireGuard connection is wedged by the network switch, it force-recreates the container to establish a
    fresh NAT binding to Cloudflare, restoring connection in ~15 seconds.

1772
1773 ## 35. Incident Post-Mortem: macOS Tailscale Flag Requirement Abort (July 10, 2026)
1774
1775 ### Symptom
1776 When the `toggle.sh` stop trap or the roaming watcher triggered, the gateway failed to shut down or disconnect
    properly. The `output.log` showed the error: `Error: changing settings via 'tailscale up' requires
    mentioning all non-default flags. To proceed, either re-run your command with --reset...`
1777
1778 ### Root Cause
1779 In a previous attempt to stop host-side logouts (Section 32), we removed the `--reset` flag from all `tailscale
    up` invocations. However, if the Tailscale daemon on macOS is already running with non-default flags (like
    `--ssh` or `--accept-routes`), invoking `tailscale up` with only a subset of flags (like `--exit-node=`)
    triggers a strict safety check in the Tailscale CLI that aborts the command completely. Because the command
    aborted, exit-node routing remained stuck.

1780
1781 ### Fix (July 10, 2026)
1782 Migrated all specific preference changes in `toggle.sh` and `scripts/common.sh` from `tailscale up` to a two-
    step sequence:
1783 1. `tailscale up` (with no flags) to safely ensure the interface is brought up using the current authenticated
    state.
1784 2. `tailscale set --exit-node=...` to modify the specific target preference cleanly without triggering the
    missing flags error or requiring `--reset`.

1785
1786 ## 36. Incident Post-Mortem: macOS SNAT Endpoint Poisoning & The P2P Blackhole (July 10, 2026)
1787
1788 ### Symptom
1789 When a Tailscale client (like an S24) connected to the exact same Wi-Fi network as the macOS gateway, it
    completely lost internet connectivity. However, if the S24 connected to a Windows PC's hotspot on the same
    network, the tunnel worked flawlessly.

1790
1791 ### Root Cause
1792 Claude's critique correctly dismantled the port collision theory: the gateway daemon wasn't failing to bind,
    and macOS wasn't load-balancing ports. The real culprit was **macOS SNAT Source Port Mangling poisoning
    Tailscale's path discovery**.

1793
1794 Here is the exact step-by-step of the "infinite loop" blackhole:
1795 1. **The P2P Handshake**: The S24 and the Mac are on the same Wi-Fi (`192.168.1.x`). The S24 discovers the Mac's
    local IP and sends a UDP hole-punch packet to `192.168.1.5:41641`. Colima successfully forwards this to
    the gateway container.
1796 2. **The Bypass Rule**: The gateway replies. Because `routing-fix.sh` forces all Tailscale UDP traffic out `eth0`
    to bypass WARP, the reply goes to the macOS host.
1797 3. **macOS SNAT Mangling**: macOS routes the reply out `en0` to the S24. Because the packet crosses from the
    Colima VM bridge to the Wi-Fi interface, macOS applies Source NAT (SNAT). Critically, macOS **randomizes
    the source port** (e.g., changes it from `41641` to `58291`).
1798 4. **Endpoint Poisoning**: The S24 receives the authenticated Tailscale packet from `192.168.1.5:58291`.
    Tailscale's `magicsock` is designed to handle NAT dynamically, so it assumes the Mac has roamed to port
    `58291`! The S24 updates its endpoint and starts sending all its encrypted internet traffic to
    `192.168.1.5:58291`.
1799 5. **The Blackhole**: macOS has no port forward for `58291`. It drops 100% of the S24's outbound data packets.
    The connection is completely dead.

1800
1801 ### Why did the Hotspot work?
1802 When the S24 connected to the PC hotspot, it was behind the PC's NAT. The S24 sent the packet, PC NATted it,
    and the Mac replied (mangled to `58291`). When the reply hit the PC, the PC's strict NAT dropped it because
    the source port (`58291`) didn't match the destination port it originally sent to (`41641`). Because the
    mangled packet was dropped, the S24 never poisoned its endpoint! It safely gave up on P2P, fell back to the
    100% reliable TCP DERP relay, and the internet worked perfectly.

1803
1804 ### Fix & Resolution (July 10, 2026)
1805
1806 **Phase 1 Port disambiguation**: Changed the gateway container's Tailscale listen port from the default
    `41641` to `41642` across `docker-compose.yml`, `post-rules.txt`, and `routing-fix.sh`. This prevents any
    bind conflict with the macOS host's native Tailscale daemon.
1807
1808 **Phase 2 RFC1918 DROP rules**: Updated `routing-fix.sh`'s `add_tailscale_p2p_bypass()` to explicitly drop all
    outgoing Tailscale UDP packets destined for private subnets (`192.168.0.0/16`, `10.0.0.0/8`,
    `172.16.0.0/12`) when `TAILSCALE_ALLOW_LAN_P2P=false`. This surgically kills the local P2P hole-punch
    attempt **before** it can trigger the macOS gvproxy SNAT mangling. The S24 receives a clean failure and
    immediately locks onto the public DERP relay with 0% packet loss.
1809

```

```

1810 **Phase 3 Automatic network detection:** Because `TAILSCALE_ALLOW_LAN_P2P=false` breaks P2P on trusted
      hotspots that genuinely don't have AP isolation, a `.env` toggle and a full auto-detection system were
      implemented:
1811
1812 - **`.env` toggle:** `TAILSCALE_ALLOW_LAN_P2P=true/false` static fallback, used only if auto-detection cannot
      run.
1813 - **`watcher.sh` auto-detection (`detect_lan_p2p_mode`):** On every network change, `watcher.sh` runs two
      checks:
1814 1. **WPA2-Enterprise detection:** `airport -I` is used to read the current Wi-Fi `link auth` field. If the
      auth type does not contain `-psk` (e.g., it is plain `wpa2` = 802.1x), the network is classified as
      enterprise and P2P is forced off.
1815 2. **AP Isolation probe:** For WPA2-Personal networks, a short `ping` is sent to the default gateway. If the
      gateway responds (it always should on a real home/hotspot network), P2P is allowed. If not, AP isolation is
      inferred and P2P is forced off.
1816 - The result (`true` or `false`) is written to `.lan_p2p_detected` in the repo root.
1817 - **`routing-fix.sh` dynamic reading:** `add_tailscale_p2p_bypass()` re-reads `.lan_p2p_detected` on every 30-
      second loop iteration and atomically adds or removes the RFC1918 DROP rules. **No container restart is
      needed when switching networks.**
1818
1819 ### Known Unknowns (Pending Verification)
1820 - The gvpnproxy SNAT source port mangling theory is mechanistically sound and backed by observed behavior (
      gvpnproxy's UDP proxy changelog has documented edge cases in this exact code path), but has not yet been
      confirmed via `tcpdump -i en0 udp portrange 40000-60000` while triggering a P2P handshake from the phone. A
      packet capture cross-referenced with `tailscale ping` endpoint reports on the phone side would confirm the
      theory definitively.
1821 - Claude's recommendation: run `tcpdump -i en0 udp port 41642 or portrange 40000-60000` on the Mac while
      triggering the handshake from the phone to see whether the reply leaves with a different source port than
      `41642`.
1822 - **Client-Side VPN Cycling (Roaming Recovery):** An observation (July 10, 2026) suggests that when a hung DNS
      probe state is triggered by roaming between Access Points (APs) on a WPA2-Enterprise network, simply
      toggling the VPN connection (Tailscale) off and on locally on the client phone immediately resolves the
      issue. You do not need to cycle the phone's physical Wi-Fi. This reinforces the theory that roaming on
      Enterprise networks leaves a stale/poisoned P2P endpoint in the phone's Tailscale magicsock state, which
      gets instantly flushed upon a local VPN restart. (Status: Possible but not formally verified).
1823
1824 ## 37. TODO
1825
1826 * **Verify Wi-Fi Roaming:** Investigate and test whether switching Wi-Fi networks now successfully and smoothly
      recovers the gateway end-to-end without any tailscale flag errors or dead tunnels. (Pending user
      verification).
1827 * **Fix Diagnostic Script Check 5/8:** ~Investigate why `diagnose-host-leak.sh` check 5/8 (`Host default route
      `) sometimes reports "default route goes via physical Wi-Fi Tailscale route assertion failed" on macOS
      even though the `warp=on` egress checks confirm that traffic is successfully tunneling.~ **Closed** fixed
      by switching check 5 from `netstat -rn | grep default` to `route -n get 1.1.1.1`. macOS Tailscale uses
      interface-scoped routes and does not replace the DHCP default route entry. See 36 Known Unknowns.
1828 * **Expose Tor Control Port (9051):** Consider mapping the Tor Control port to localhost and adding `nyx` (the
      Tor terminal status monitor) to allow users to inspect their entry, middle, and exit node IPs in real-time.
      Currently, only the exit node IP is discoverable by curling an external API.
1829 * **Host-Only Tor Mode (Pluggable Egress):** Implement an `EGRESS_TYPE=tor_host_only` mode for users in heavily
      censored (DPI-restricted) environments. This mode would skip booting `warp` and `tailscale`, boot `
      adguardhome` and `tor` (using Telegram `obfs4` bridges), set AdGuard's upstream to Tor's `DNSPort`, and use
      the `pf` kill-switch to force all host Mac traffic strictly through the Tor network. This provides a 100%
      free, DPI-resistant evasion path without requiring an external VPS.
1830 * **The Technical Realities (What You Need to Watch Out For):**
1831 * **The UDP Problem:** Tor only transports TCP traffic. It does not support UDP (aside from DNS requests
      passed directly to its DNSPort). macOS is incredibly chatty over UDP (FaceTime, mDNS, QUIC protocols). Your
      pf rules defined in `scripts/pf.conf` must be configured to aggressively and silently DROP all host UDP
      traffic (except for local DNS queries to AdGuard). If you don't drop it cleanly, macOS services might hang
      indefinitely while waiting for timeouts.
1832 * **The "Daily Driver" Friction:** Routing a whole host OS through Tor is brutal on performance. Background
      daemons (iCloud sync, macOS software updates, Spotlight indexing) will blindly attempt to download
      gigabytes of data over Tor, which could saturate the circuits or time out completely.
1833 * **Bridge Rot:** State firewalls actively hunt and block obfs4 bridges. When a user's bridges burn out,
      they will lose all internet access due to the pf kill-switch. You will need to ensure the user has a
      frictionless way to update the `tor-bridges.txt` file and restart the Tor container without exposing their
      IP.
1834 * **Exit Node Discrimination:** Because all traffic will exit from known Tor nodes, the user will face an
      onslaught of Cloudflare CAPTCHAs, and many banking or streaming services will flat-out reject the
      connections.
1835
1836 ## 38. Observation: AdGuard Returns Two Different Sinkhole IPs (Unverified)
1837
1838 > **Status: Observed but not formally verified. Needs a deeper audit of the rule-compiler sources to confirm.**
1839
1840 ### Observation (July 10, 2026)
1841 When querying AdGuard Home for blocked ad domains, some domains resolve to `0.0.0.0` and others resolve to
      `127.0.0.1`. Both are blocked, both cause connection failure, but the different responses were unexpected.
1842
1843 ```
1844 doubleclick.net      0.0.0.0      (blocked)
1845 ads.google.com       127.0.0.1    (blocked)

```

```

1846 pagead2.googlesyndication.com 0.0.0.0 (blocked)
1847 scorecardresearch.com 127.0.0.1 (blocked)
1848 ```
1849
1850 ### Likely Explanation (Unverified)
1851 There are three historical schools of thought on the "correct" DNS sinkhole response, and AdGuard faithfully
    preserves whichever the source blocklist used:
1852
1853 | Sinkhole IP | Convention | Origin |
1854 |---|---|---|
1855 | `0.0.0.0` | AdBlock-style lists (`\|\|domain`) | Modern DNS-level blocking; AdGuard's native format. Fails
    fast OS doesn't attempt a TCP connection. Works for both IPv4 and IPv6 without a separate rule. |
1856 | `127.0.0.1` | Hosts-file-style lists (`127.0.0.1 domain`) | 90s-era `/etc/hosts` tradition. Steven Black's
    hosts list (500k+ domains) still uses this. |
1857 | `NXDOMAIN` | Purist / Pi-hole default | Returns "domain doesn't exist." Semantically most accurate but
    requires browsers to handle NXDOMAIN gracefully. |
1858
1859 The dual-response behavior in this project is because `rule-compiler` pulls from both AdBlock-format sources (
    Hagezi, uBlock origin lists) and hosts-format sources (Spamhaus DROP, Steven Black, Feodo Tracker), and
    AdGuard returns whatever sinkhole IP the matching rule specifies.
1860
1861 ### To Verify
1862 Run `docker compose exec tailscale cat /etc/hosts` to confirm no hosts-file rules are being injected at the
    container level, and check which specific AdGuard rule matched each domain via the AdGuard query log at `
    http://100.100.21.8:3000/#logs`.
1863
1864 ## 39. Observation: AGY CLI Appears to Generate Native macOS Notification Pop-ups (Unverified)
1865
1866 > **Status: Observed but source unverified. macOS security logs ruled out system-level origin.**
1867
1868 ### Observation (July 10, 2026)
1869 While running a DNS blocklist verification via the Antigravity CLI (`agy`), a command was proposed that
    included `malware.wicar.org` (a known-safe DNS blocklist test domain) in a `dig` loop. Shortly after, a
    macOS-style notification popup appeared describing a "malicious" or "blocked" event. The AGY CLI terminal
    session then closed.
1870
1871 ### Investigation
1872 - **macOS XProtect / Gatekeeper logs:** Empty. Zero events in the 30-minute window around the incident.
1873 - **macOS Endpoint Security / System Policy logs:** Empty.
1874 - **Broad `log show` search for "malicious", "malware", "blocked":** Empty.
1875 - **Conclusion:** macOS itself did not generate the notification. No system security subsystem fired.
1876
1877 ### Likely Explanation (Unverified)
1878 The notification appears to have originated from **AGY CLI's own internal safety system**. AGY CLI is a native
    macOS application (not just a terminal process) and has access to the macOS Notification Center API (`
    UNUserNotificationCenter`). When its guardrails detected a domain containing the word "malware" (`malware.
    wicar.org`) in a proposed shell command, it likely:
1879 1. Blocked the command from executing
1880 2. Posted a native macOS-style notification via the Notification Center
1881
1882 This is surprising behavior for a CLI tool most terminal programs don't send GUI notifications but AGY CLI
    appears to bridge both worlds: it runs in a terminal but is packaged as a native app with full system API
    access. The notification would be indistinguishable from a system security alert at a glance.
1883
1884 ### Notes
1885 - `malware.wicar.org` is the DNS/AV equivalent of the EICAR test file a deliberately benign domain that
    triggers blocklist/AV systems to verify they work. It was included in the test set to confirm the AdGuard
    blocklist was catching it. AGY's safety system is not aware of this distinction.
1886 - For future DNS blocklist testing, use only ad/tracking domains (e.g., `doubleclick.net`, `adnxs.com`) to
    avoid triggering AGY's guardrails.
1887
1888 ## 40. Apple Silently Removed the `airport` Binary in macOS Sonoma 14.4 (March 2024)
1889
1890 ### What happened
1891 `watcher.sh`'s `detect_lan_p2p_mode()` was written to use the `airport` CLI tool at:
1892 ```
1893 /System/Library/PrivateFrameworks/Apple80211.framework/Versions/Current/Resources/airport
1894 ```
1895 This path returned `exit 127: no such file or directory`. The function silently fell back to `reason="no-wifi"`
    even while the Mac was actively connected to WPA2 Enterprise Wi-Fi.
1896
1897 ### Why Apple removed it
1898 `airport` was never a real public binary it lived inside a **private framework** and was always technically
    unsupported. Apple deprecated it quietly years ago and finally removed it in **macOS Sonoma 14.4 (March
    2024)**. The removal is part of a broader privacy clampdown on Wi-Fi metadata:
1899 - Starting in **macOS 14.5**, BSSIDs and other Wi-Fi association details are **redacted** (`<redacted>`) in
    most tools, including the official replacement `wdutil`
1900 - Apple's official recommendation is to use the **CoreWLAN framework** (Swift/Objective-C) with proper
    entitlements useless for a bash script
1901 - The official CLI replacement `wdutil` requires `sudo` for most useful output also useless for a background
    LaunchAgent

```

```

1902
1903 ### Fix
1904 Replaced `airport -I | awk '/link auth/'` with:
1905 ```bash
1906 system_profiler SPAirPortDataType | awk '/Current Network Information:/{found=1} found && /Security:/{print $NF
1907 ; exit}'
1908 ...
1909 This returns human-readable strings like `WPA2 Enterprise`, `WPA2 Personal`, or `None` for the currently
1910 associated network without `sudo`, and without any removed private binary.
1911
1912 **Confirmed working output on this machine:**
1913 ```
1914 P2P detect: security='Enterprise' allow=false (reason: 802.1x-enterprise)
1915 ```
1916
1917 ### Risk: `system_profiler` may not survive forever either
1918 `system_profiler SPAirPortDataType` still works as of macOS Sequoia (15.x) without sudo and without redaction.
1919 However, given Apple's trajectory on Wi-Fi privacy, this could be locked down in a future release. If `
1920 security` comes back empty on a future macOS version while the Mac is actively on Wi-Fi, the function will
1921 silently fall back to `allow=false` (safe default), but the detection will be broken again. The long-term
1922 fix would be a small Swift helper using CoreWLAN that we compile once and bundle in the repo.
1923
1924 ## 41. Fixed: Watcher.sh Suppressed Exit Code Bug
1925
1926 ### Observation
1927 The `scripts/watcher.sh` script is a long-running daemon that invokes `recover.sh --post-wake` when it detects
1928 system wake or network roam events. Previously, it contained the following bug:
1929
1930 ```bash
1931 bash "$RECOVER" --post-wake
1932 local exit_code=$?
1933 ...
1934 return $exit_code || true
1935 ```
1936
1937 The `|| true` mistakenly swallowed the actual `$exit_code` returned by `recover.sh`, causing `run_recover()` to
1938 always return `0` (success), masking any underlying failures in the wake recovery process.
1939
1940 ### Fix
1941 The `|| true` statement was removed from the return line:
1942 ```bash
1943 return $exit_code
1944 ```
1945
1946 Since `watcher.sh` explicitly omits `set -e` (to prevent pipe breakages from killing the daemon), removing `||
1947 true` safely allows the underlying exit code to propagate without risking daemon termination. The watcher
1948 daemon was then reloaded (`launchctl kickstart -k`) to pick up the script changes in memory.
1949
1950 ## 42. Network Readiness Race Condition on `--restart` (Fixed)
1951
1952 **Problem:** When running `./toggle.sh --restart`, the script tears down the gateway and immediately begins the
1953 startup sequence. On systems with Wi-Fi (or any wireless interface), the OS takes a moment to re-associate
1954 with the network after the teardown's DNS/routing changes. If the startup sequence ran before the OS had a
1955 default route, `get_active_service` would return nothing, routing bypass targets would be wrong, and DNS/
1956 WARP setup would silently fail resulting in a partially configured gateway that self-heals only once the
1957 `watcher.sh` re-runs.
1958
1959 **Root cause:** The original code had no gate before the first network-dependent call (`ACTIVE_SERVICE=$(
1960 get_active_service)` at the top of the start branch). This was a pure timing race.
1961
1962 **First fix (Wi-Fi specific):** A polling loop on `ipconfig getifaddr en0` was added to wait for an IPv4
1963 address on `en0` before proceeding. This worked but was brittle it only handled Wi-Fi and would fail for
1964 Ethernet, USB-C adapters, or iPhone USB tethering.
1965
1966 **Generalized fix:** Replaced the `en0` check with `route get default | grep 'gateway:'`. This detects a valid
1967 default gateway on *any* interface, covering all connection types. The loop polls every 2 seconds with a
1968 60-second hard timeout (then proceeds with a warning rather than hanging forever). The output prints the
1969 detected gateway IP so failures are easy to diagnose in logs.
1970
1971 ```bash
1972 # In toggle.sh, before ACTIVE_SERVICE=$(get_active_service)
1973 while true; do
1974     if route get default 2>/dev/null | grep -q 'gateway: '; then
1975         _gw=$(route get default 2>/dev/null | awk '/gateway:/{print $2}')
1976         echo " Network ready (default gateway: $_gw)."
1977         break
1978     fi
1979     ...
1980 done
1981 ```
1982
1983 **Why not ping?** Pinging an IP (e.g. `1.1.1.1`) during restart is unreliable because DNS may still be hijacked
1984 from the previous session, and the exit node routing may not yet be cleared, causing the ping to loop back
1985 through the tunnel. `route get default` is a pure local kernel query no packets sent.
1986
1987

```

```

1960 ## 43. Low Gateway Throughput (DERP Relay & MTU Fragmentation) Fixed
1961
1962 **Problem:** The host Mac's internet throughput through the gateway dropped significantly (e.g., from 34Mbps
1963 raw to 2.1Mbps via gateway). This was caused by a combination of two bottlenecks:
1964
1965 **1. Tailscale DERP Relay Bottleneck:**
1966 Because the host Mac and the Docker container operate behind the same public IP, standard hole-punching for
1967 Tailscale P2P failed, forcing traffic through Tailscale's NYC DERP relay. Normally, `routing-fix.sh`
explicitly drops container Tailscale UDP packets destined for local subnets (when `TAILSCALE_ALLOW_LAN_P2P`=
false` in `.env` or auto-detected). We nuke these P2P connections on purpose to prevent the severe macOS `
gvproxy` SNAT endpoint poisoning issue we faced earlier (see `devref.md 36`). However, this strict policy
unintentionally blocked direct communication between the container and the Mac host itself via the Docker
bridge network.
1968
1969 **Fix:** Tailscale's control plane registers the Mac host using its physical IP (e.g., `172.17.52.223`), not
the Docker bridge IP. If this physical IP falls within the dropped local subnets (like `172.16.0.0/12`),
the P2P handshake is dropped. To solve this natively, `scripts/common.sh` now features a `write_host_ips()`
function that actively grabs all physical IPv4 addresses on the Mac and writes them to `.host_ips`.
`routing-fix.sh` dynamically reads this file every 30 seconds and injects priority `RETURN` rules for those
exact physical IPs. This guarantees direct container-to-host P2P over the local bridge (bypassing DERP)
while continuing to block external LAN devices (like phones) to prevent the 36 SNAT poisoning bug.
1970
1971 **2. MTU Double-Encapsulation Fragmentation:**
1972 The host's Tailscale interface (`utun5`) defaulted to an MTU of 1280. The WARP tunnel inside the container also
uses an MTU of 1280. When a full 1280-byte Tailscale packet from the host entered the container and was
encapsulated by WARP (adding ~60 bytes of overhead), the resulting packet exceeded 1280 bytes, leading to
severe fragmentation and performance degradation.
1973
1974 **Fix:** In `toggle.sh`'s `setup_exit_node_routing` function, the host's Tailscale `utun` interface MTU is
explicitly lowered to `1200` (configurable via `HOST_MTU` in `.env`). This ensures that host-originated
packets can comfortably fit inside the container's WARP tunnel encapsulation without fragmenting.
1975
1976 ## 44. Architectural Limitation: Chrome Remote Desktop Performance (UDP NAT)
1977
1978 ### Observation (July 11, 2026)
1979 When the user connects to their host Mac using Chrome Remote Desktop (CRD) while `nullexit` is active, the
connection suffers from massive latency, lag, and degraded video quality. AdGuard Home's `blocked.log`
shows zero blocked DNS requests for Google, WebRTC, or `chromoting` endpoints, indicating this is not a DNS
sinkhole issue.
1980
1981 ### Root Cause
1982 This is a fundamental limitation of tunneling all host traffic through a strict commercial NAT (Cloudflare WARP
).
1983
1984 1. Chrome Remote Desktop attempts to use **STUN (UDP hole-punching)** to establish a lightning-fast, direct
peer-to-peer WebRTC connection between the client device and the host Mac.
1985
1986 2. Because all non-Tailscale traffic is forcefully routed through the `warp` container, the STUN packets exit
the network from a Cloudflare IP.
1987
1988 3. Cloudflare's enterprise NAT infrastructure strictly drops unsolicited inbound UDP packets. The UDP hole-
punch fails.
1989
1990 4. Because a direct P2P connection cannot be established, CRD falls back to **Relay Mode** (TURN), bouncing all
video data back and forth through Google's cloud servers instead of sending it directly over the local or
mesh network. This relaying introduces massive latency.
1991
1992 ### Workaround / Fix
1993 This lag is the accepted "tax" for maintaining absolute cryptographic anonymity; the only way to restore CRD
speed would be to leak its UDP traffic outside the tunnel (violating zero-trust).
1994
1995 To achieve low-latency remote access, users should bypass CRD entirely and use macOS's built-in **Screen
Sharing (VNC)** directly over the Tailscale IP. Tailscale's sophisticated custom DERP/WireGuard
architecture successfully UDP hole-punches through strict NATs, providing a direct, encrypted, high-speed
connection without relying on external cloud relays.
1996
1997 ## 45. Architectural Decision: Dynamic IP Fetch vs. MagicDNS
1998
1999 ### Observation
The `nullexit` gateway uses a dynamically fetched IP address (cached in `.gateway_ip`) to configure host
routing and the `pf` kill-switch, rather than utilizing Tailscale's user-friendly MagicDNS hostname (e.g.,
`nullexit-gateway`).
2000
2001 ### Why MagicDNS is Not Used
2002 1. **The "Chicken and Egg" Bootstrapping Problem:**
2003 macOS's Packet Filter (`pf`) and low-level routing commands (`route add`) operate strictly at Layer 3 and
require raw IP addresses. If `toggle.sh` or `recover.sh` attempted to use `nullexit-gateway`, the host Mac
would need to perform a DNS lookup to resolve it. However, the very first step of the gateway's
initialization is to completely hijack the host's DNS and point it **at the gateway itself**. The Mac cannot
resolve the gateway's MagicDNS name if it needs the gateway's IP to know where to send the DNS query!
2004
2005 2. **Dynamic Self-Healing (Avoiding Stale Cache):**
2006 To solve the bootstrapping problem without causing bugs if the node is deleted and re-authenticated on the
Tailscale Admin Console, `toggle.sh` explicitly runs `docker compose exec ... tailscale ip -4` during boot
to fetch the absolute freshest IP dynamically. It saves this to `.gateway_ip` so that fast-roaming scripts
like `recover.sh` can instantly retrieve the routing target without incurring the latency of querying
Docker.

```

5 diagrams.md

```
1 # nullexit Flow Diagrams & Architecture
2
3 ---
4
5 ## 1. System Architecture What's Running & Where
6
7 ```mermaid
8 graph TB
9     subgraph INTERNET[" Internet / Cloudflare Edge"]
10         CF["Cloudflare WARP\n(Exit Point)\nwarp=on"]
11     end
12
13     subgraph TAILNET[" Tailscale Mesh (100.x.x.x)"]
14         PHONE[" Phone / Laptop\n(Tailnet Client)"]
15     end
16
17     subgraph MACHOST[" macOS Host"]
18         direction TB
19         PFFIREWALL["macOS pf Firewall\n(Kill-Switch & TCP MSS Clamping)"]
20         TSDAEMON["tailscaled\n(brew service)\nhost Tailscale"]
21         HOSTDNS["Host DNS\n Gateway IP\n(hijacked by toggle.sh)"]
22         HOSTSOCKS["Host SOCKS5\n127.0.0.1:1080\n(fallback only)"]
23
24         subgraph MONITORS[" Background Daemons (toggle.sh children)"]
25             WARPWATCH["WARP Watcher\n(every 30s, via docker exec)\n output.log"]
26             LEAKPROBE["Host Leak Probe\n(every 300ms, via curl from HOST)\n output.log"]
27             DNSWATCHER["DNS Watcher\n(re-hijacks if drift detected)\n output.log"]
28             CAFFEINATE["caffeinate -i\n(sleep prevention)"]
29         end
30     end
31
32     subgraph COLIMAVM[" Colima VM (Linux, 600MB RAM)"]
33         subgraph NETNS["warp container network namespace (shared by ALL containers)"]
34             GLUETUN["warp / Gluetun\nWireGuard tun0\n(owns the namespace)"]
35             TS["tailscale container\nAdvertises exit node\ntailscale0 interface"]
36             SOCKS["tailscale\nSOCKS5\nport 1080"]
37             ADGUARD["adguardhome\nDNS sinkhole\nport 5335"]
38             ROUTINGFIX["routing-fix sidecar\nGeo-IP Blocking & Go Logger\n(writes blocked.log)"]
39         end
40     end
41
42     subgraph LAUNCHD[" launchd (always-on)"]
43         WATCHER["scripts/watcher.sh\n(sleep/wake + Wi-Fi roam\n+ LAN P2P auto-detection)"]
44     end
45
46     subgraph RECOVERY[" Recovery"]
47         RECOVER["recover.sh\n(nuclear reset)"]
48         RECOVERPOST["recover.sh --post-wake\n(light refresh)"]
49     end
50
51     %% Traffic flow
52     PHONE -->|"WireGuard / Tailscale mesh"| TSDAEMON
53     TSDAEMON -->|"exit-node route utun*"| TS
54     TS -->|"FORWARD tun0 (MASQUERADE)"| GLUETUN
55     GLUETUN -->|"WireGuard UDP (macOS Host Egress)"| PFFIREWALL
56     PFFIREWALL -->|"TCP MSS Clamped / Kill-Switch check"| CF
57
58     %% DNS
59     HOSTDNS -->|"UDP:53 TCP:5354"| ADGUARD
60     ADGUARD -->|"DNS upstream via tun0"| CF
61
62     %% Monitoring
63     WARPWATCH -->|"docker exec warp wget cdn-cgi/trace"| GLUETUN
64     LEAKPROBE -->|"curl cdn-cgi/trace (HOST NIC)"| CF
65     DNSWATCHER -->|"networksetup -getdnsservers"| HOSTDNS
66
67     %% Logging
68     ROUTINGFIX -->|"writes"| BLOCKEDLOG["blocked.log\n(Host-side)"]
69
70     %% Recovery triggers
71     WARPWATCH -->|"6 consecutive warp=off"| RECOVER
72     WATCHER -->|"sleep/wake / roam"| RECOVERPOST
73     RECOVERPOST -->|"if gateway unhealthy"| RECOVER
```

```

74     WATCHER -->|"writes .lan_p2p_detected"| ROUTINGFIX
75
76     style MONITORS fill:#1a1a2e,color:#e0e0ff
77     style COLIMAVM fill:#0d2137,color:#e0e0ff
78     style NETNS fill:#0a1628,color:#e0e0ff
79     style RECOVERY fill:#2d0a0a,color:#ffe0e0
80     style INTERNET fill:#0a2d1a,color:#e0ffe0
81     style TAILNET fill:#1a2d0a,color:#e8ffe0
82     style LAUNCHD fill:#2d1a2d,color:#ffe0ff
83     ---
84
85     ---
86
87     ## 2. toggle.sh Full START Flow
88
89     ```mermaid
90     flowchart TD
91         START(["./toggle.sh"])
92         CHECK{"is_gateway_active?\n(containers running OR\nDNS 1.1.1.1)"}
93
94         START --> CHECK
95         CHECK -->|"YES already ON"| STOPFLOW[" See STOP Flow"]
96         CHECK -->|"NO start it"| S1
97
98         S1["1. Reset DNS 1.1.1.1\n(prevent deadlocks)"]
99         S2["2. tailscale down\n(prevent exit-node routing during boot)"]
100        S3["3. Check Colima VM\n(colima start --memory 0.6 --vm-type vz --network-address --network-mode bridged)"]
101        S4["4. Configure VM swap\n(400MB, prevent OOM)"]
102        S5["5. Clean corrupted AdGuard config\n(prevent container crash loop)"]
103        S6["6. docker compose up -d --build\n(compile Go threat rules & boot containers)"]
104        S7["7. Poll tailscale status\n(wait for 'offers exit node'\nup to 60s)"]
105        TSREADY{"container\nTailscale ready?"}
106        ABORT(["ABORT\n cleanup_handler"])
107        S8["8. Resolve gateway Tailscale IP\n(.gateway_ip or docker exec)"]
108        S9["9. Verify tailscaled running\n(auto-start if needed)"]
109        S10["10. tailscale up --reset\n--exit-node=\n--accept-routes=true\n--ssh=true --accept-dns=false"]
110
111        PF1{"Pre-flight 1/3\n tailscale ping gateway"}
112        PF2{"Pre-flight 2/3\n dig +tcp AdGuard DNS"}
113        PF3{"Pre-flight 3/3\n WARP internet check"}
114        ALLPASS{"All 3 pass?"}
115
116        EXITPATH[" EXIT NODE PATH\n tailscale up --exit-node=\n force_dns_to_gateway (3 retry)\n SOCKS5 disabled"]
117        SOCKSPATH[" SOCKS5 FALLBACK PATH\n macOS SOCKS5 127.0.0.1:1080\n DNS proxy 127.0.0.1:53\n No exit node (\n SKIP_EXIT_NODE=true)"]
118
119        DAEMONS["Start background daemons\n caffeinate -i (sleep prevention)\n DNS Watcher\n WARP Watcher\n Host\n Leak Probe\n(all write output.log)"]
120
121        RULECOMPILE["Rule compiler status\n(log rule/IP counts)"]
122        WRITEMARKER["write_gateway_active_marker\n(/tmp/nullexit-gateway-active.marker)"]
123        DONE([" Gateway UP"])
124
125        S1 --> S2 --> S3 --> S4 --> S5 --> S6 --> S7
126        S7 --> TSREADY
127        TSREADY -->|"NO (40 NoState)"| ABORT
128        TSREADY -->|"YES"| S8
129        S8 --> S9 --> S10
130        S10 --> PF1 --> PF2 --> PF3
131        PF1 & PF2 & PF3 --> ALLPASS
132        ALLPASS -->|"YES"| EXITPATH
133        ALLPASS -->|"NO"| SOCKSPATH
134        EXITPATH --> DAEMONS
135        SOCKSPATH --> DAEMONS
136        DAEMONS --> RULECOMPILE --> WRITEMARKER --> DONE
137
138        style ABORT fill:#5c1010,color:#fff
139        style EXITPATH fill:#0a3d1a,color:#c0ffc0
140        style SOCKSPATH fill:#3d2d00,color:#ffe0a0
141        style DAEMONS fill:#0a1f3d,color:#c0d8ff
142        style DONE fill:#0a3d1a,color:#fff
143    ---
144
145    ---
146
147    ## 3. toggle.sh STOP Flow
148
149    ```mermaid
150    flowchart TD
151        STOP(["./toggle.sh\n(gateway already ON)"])
152

```

```

153 ST1["1. tailscale down\n(disconnect from mesh)"]
154 ST2["2. docker compose down -t 5\n(stop all containers)"]
155 ST3["3. Reset DNS 1.1.1.1\n(networksetup on all services)"]
156 ST4["4. stop_local_dns_proxy\n(kill Python UDP proxy)"]
157 ST5["5. stop_pidfile_daemon\n(kill caffeinate, rm PID)"]
158 ST6["6. stop_pidfile_daemon\n(kill DNS Watcher, rm PID)"]
159 ST7["7. stop_pidfile_daemon\n(kill WARP Watcher, rm PID)"]
160 ST8["8. stop_pidfile_daemon\n(kill Leak Probe, rm PID)"]
161 ST9["9. clear_gateway_active_marker\n(rm /tmp/nullexit-gateway-active.marker)"]
162 ST10["10. cleanup_network_state\n\n disable_all_proxies\n Flush DNS cache (dscacheutil)\n Flush stale routes\n\n Power-cycle Wi-Fi (offon)"]
163
164 DONE([" Gateway DOWN\nInternet restored"])
165
166 STOP --> ST1 --> ST2 --> ST3 --> ST4
167 ST4 --> ST5 --> ST6 --> ST7 --> ST8 --> ST9 --> ST10 --> DONE
168
169 style DONE fill:#0a3d1a,color:#fff
170
171 ---
172
173
174 ## 4. Monitoring Layer The 4 Background Daemons
175
176 ```mermaid
177 graph LR
178     subgraph DAEMONS["Background Daemons (all start with gateway, stop on teardown)"]
179         direction TB
180
181         subgraph D1[" Sleep Prevention"]
182             CAFF["caffeinate -i\nPID: /tmp/nullexit-caffeinate.pid\n\nKeeps Mac awake while\ngateway is active"]
183         end
184
185         subgraph D2[" DNS Watcher"]
186             DNSW["Polls networksetup every -30s\nPID: /tmp/nullexit-dns-watcher.pid\n\nIf DNS drifts from gateway IP\nre-hijacks automatically\n output.log"]
187         end
188
189         subgraph D3[" WARP Watcher"]
190             WARPW["Polls every 30s via:\ndocker compose exec warp wget cdn-cgi/trace\nPID: /tmp/nullexit-warp-watcher.pid\n\nCounts consecutive warp=off\n WARP_FAIL_THRESHOLD (default 6=30s)\n runs recover.sh (nuclear)\n output.log"]
191         end
192
193         subgraph D4[" Host Leak Probe"]
194             LEAKP["Polls every 300ms via:\ncurl cdn-cgi/trace (HOST NIC directly)\nPID: /tmp/nullexit-host-leak-probe.pid\n\nLogs ONLY on state change:\n LEAK / ROTATE / HOST-PROBE failed\nAll errors output.log\n(HOST_LEAK_PROBE=true in .env)"]
195         end
196     end
197
198     subgraph BLIND["What each monitor can/can't see"]
199         B1["WARP Watcher sees:\n Container WARP tunnel state\n Host NIC traffic\n Flash leaks < 5s"]
200         B2["Host Leak Probe sees:\n Host NIC egress (browser path)\n Sub-second transitions\n Container internals"]
201     end
202
203     D3 -->|"covers"| B1
204     D4 -->|"covers"| B2
205
206     style D1 fill:#1a2d0a
207     style D2 fill:#0a1a2d
208     style D3 fill:#2d0a2d
209     style D4 fill:#2d1a00
210     style BLIND fill:#1a1a1a,color:#aaa
211
212 ---
213
214
215 ## 5. Traffic Flow Data Path (Normal Operation)
216
217 ```mermaid
218 sequenceDiagram
219     participant PHONE as Phone<br/>(Tailnet client)
220     participant TS_HOST as tailscaled<br/>(macOS host)
221     participant TS_CONTAINER as tailscale<br/>(container)
222     participant GLUETUN as Gluetun/warp<br/>(container)
223     participant CF as Cloudflare<br/>WARP Edge
224     participant INTERNET as Internet
225

```

```

226 Note over PHONE,INTERNET: Normal EXIT NODE path (all 3 pre-flights passed)
227
228 PHONE-->TS_HOST: WireGuard packet<br/>(encrypted, UDP 41641)
229 TS_HOST-->TS_CONTAINER: route via utun* tailscale0
230 TS_CONTAINER-->GLUETUN: FORWARD chain<br/>(iptables MASQUERADE on tun0)
231 GLUETUN-->TS_HOST: WireGuard tunnel (tun0) egresses macOS Host
232 Note over TS_HOST: macOS pf Firewall applies<br/>Kill-Switch & TCP MSS Clamping
233 TS_HOST-->CF: re-encrypted UDP packet (MSS 1160)
234 CF-->INTERNET: Plain HTTPS from<br/>Cloudflare IP
235
236 Note over PHONE,INTERNET: DNS Resolution path
237 PHONE-->TS_HOST: DNS query
238 TS_HOST-->TS_CONTAINER: UDP:53 TCP:5354 AdGuard :5335
239 TS_CONTAINER-->GLUETUN: AdGuard upstream DNS via tun0
240 GLUETUN-->CF: Encrypted DNS query
241 CF-->GLUETUN: DNS response
242 GLUETUN-->TS_CONTAINER: response
243 TS_CONTAINER-->TS_HOST: response
244 TS_HOST-->PHONE: DNS answer
245 ...
246
247 ---
248
249 ## 6. recover.sh Decision Tree
250
251 ```mermaid
252 flowchart TD
253     INVOKE(["recover.sh called"])
254     MODE{"--post-wake\nflag?"}
255
256     subgraph POSTWAKE["--post-wake (light refresh, gateway stays UP)"]
257         PW1["Re-hijack DNS gateway IP"]
258         PW2["Flush DNS cache"]
259         PW3["Refresh tailscale exit-node"]
260         PW4["Force-recreate warp container\n(if Gluetun healthcheck failed)"]
261         PW5["Leave: containers, Wi-Fi,\ncaffeinate, DNS watcher,\nHost Leak Probe untouched"]
262     end
263
264     subgraph NUCLEAR["Default (nuclear gateway torn down)"]
265         N0["Sudo keep-alive loop (background)"]
266         N1["1. tailscale down\n(disconnect from mesh)"]
267         N2["2. Reset DNS 1.1.1.1\n(ALL network services)"]
268         N3["3. disable_all_proxies\n(all interfaces)"]
269         N4["4. Flush DNS cache\n(dscacheutil -flushcache)"]
270         N5["5. Flush stale routes\n(WARP endpoint routes)"]
271         N6a["6a. docker compose down -t 30\n(stop all containers)"]
272         N6b["6b. stop_pidfile_daemon\n(kill caffeinate)"]
273         N6c["6c. stop_pidfile_daemon\n(kill DNS Watcher)"]
274         N6d["6d. stop_pidfile_daemon\n(kill Leak Probe)"]
275         N7["7. Power-cycle Wi-Fi\n(networksetup offsleep 2on)"]
276         N8["8. Verify internet working\n(curl ifconfig.io)"]
277     end
278
279     DONE_PW(["Gateway refreshed\n(still running)"])
280     DONE_NUC(["Internet restored\n(gateway fully stopped)"])
281
282     INVOKE --> MODE
283     MODE -->|"yes"| PW1
284     PW1 --> PW2 --> PW3 --> PW4 --> PW5 --> DONE_PW
285     MODE -->|"no"| N0
286     N0 --> N1 --> N2 --> N3 --> N4 --> N5
287     N5 --> N6a --> N6b --> N6c --> N6d --> N7 --> N8 --> DONE_NUC
288
289     style POSTWAKE fill:#0a2d1a,color:#c0ffc0
290     style NUCLEAR fill:#2d0505,color:#fffc0c
291     style DONE_PW fill:#0a3d1a,color:#fff
292     style DONE_NUC fill:#3d1a00,color:#ffe0c0
293 ...
294
295 ---
296
297 ## 7. Failure Paths & Self-Healing
298
299 ```mermaid
300 flowchart LR
301     subgraph EVENTS["Failure / Event"]
302         E1["WARP tunnel drops\n(warp=off)"]
303         E2["Mac wakes from sleep\nor Wi-Fi roams"]
304         E3["DNS drifts from\ngateway IP"]
305         E4["toggle.sh crashes /\nCtrl-C / SIGTERM"]
306         E5["Host-side flash leak\nundetected (HOST NIC)"]

```

```

307     E6["Silent NAT timeout\n(ping stops working)"]
308 end
309
310 subgraph DETECTORS["Detector"]
311     D1["WARP Watcher\n(30s poll, docker exec)"]
312     D2["scripts/watcher.sh\n(launchd, always on)"]
313     D3["DNS Watcher\n(30s poll, networksetup)"]
314     D4["cleanup_handler\n(ERR/INT/TERM/HUP trap)"]
315     D5["Host Leak Probe\n(300ms poll, HOST curl)"]
316     D6["routing-fix.sh\n(30s curl over tun0)"]
317 end
318
319 subgraph ACTIONS["Response"]
320     A1[" threshold consecutive off\n recover.sh (nuclear)"]
321     A2["recover.sh --post-wake\n(gentle refresh)"]
322     A3["Re-hijack DNS\n(networksetup setdnsservers)"]
323     A4["Stop all daemons\nReset DNS 1.1.1.1\nTear down Tailscale"]
324     A5["Log LEAK to output.log"]
325     A6["Blackhole internet traffic\n+ Drop TUNNEL_FAILED_CLOSED.marker"]
326     A7["Push macOS Desktop Notification\n(via watcher.sh listener)"]
327 end
328
329 E1 --> D1 --> A1
330 E2 --> D2 --> A2
331 E3 --> D3 --> A3
332 E4 --> D4 --> A4
333 E5 --> D5 --> A5
334 E6 --> D6 --> A6
335 A6 -. -> |"marker detected"| D2
336 D2 --> A7
337
338 style EVENTS fill:#2d1010,color:#ffcccc
339 style DETECTORS fill:#0a1a2d,color:#cce0ff
340 style ACTIONS fill:#0a2d0a,color:#ccffcc
341 ...
342
343 ---
344
345 ## 8. output.log Who Writes What
346
347 All events converge into a single `output.log` at the repo root.
348
349 | Writer | Event Types | Format |
350 |-----|-----|-----|
351 | `toggle.sh` | All startup/shutdown steps, errors, pre-flight results | Plain text with timestamps |
352 | `recover.sh` | Every recovery step (nuclear or post-wake) | Plain text |
353 | **WARP Watcher** | `WARP DOWN`, `WARP RECOVERED`, `WARP SHUTDOWN` | `[UTC timestamp]` prefix |
354 | **Host Leak Probe** | `LEAK`, `ROTATE`, `HOST-PROBE failed/timeout` | `[HH:MM:SS.mmm]` prefix |
355 | **DNS Watcher** | DNS re-hijack events | Appended inline |
356 | `docker compose` | Container logs on failure (last 100 lines of warp) | Dumped on ERR path |
357 | `routing-fix.sh` | (via `nullexit-logger` inside container) | Structured |
358
359 **Grep cheatsheet:**
360 ```bash
361 grep LEAK output.log # Host-side warp=off events (real leaks)
362 grep ROTATE output.log # Cloudflare edge IP rotations (harmless)
363 grep 'HOST-PROBE' output.log # curl failures from probe
364 grep 'WARP DOWN' output.log # Container-side tunnel drops
365 grep 'WARP SHUTDOWN' output.log # Auto-recovery triggered
366 grep 'WARP RECOVERED' output.log # Self-healed before threshold
367 ```

```

6 toggle.sh

```

1 #!/bin/bash
2 # toggle.sh nullexit gateway toggle script for macOS
3 # Automatically handles Docker containers, Colima, DNS hijacking, and Tailscale exit node routing.
4
5 set -e
6
7 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
8
9 # Enforce Cryptographic Script Integrity
10 if [ -f "scripts/crypto.sh" ]; then
11     if ! bash scripts/crypto.sh --verify; then
12         exit 1

```

```

13 fi
14 fi
15
16 # Define log file
17 LOG_FILE="$PWD/output.log"
18
19 # Rotate log file if it exceeds 50MB (52428800 bytes)
20 if [ -f "$LOG_FILE" ]; then
21     log_size=$(stat -f%z "$LOG_FILE" 2>/dev/null || stat -c%s "$LOG_FILE" 2>/dev/null || echo 0)
22     if [ "$log_size" -gt 52428800 ]; then
23         mv "$LOG_FILE" "${LOG_FILE}.old" 2>/dev/null || rm -f "$LOG_FILE"
24         echo "[$(date '+%Y-%m-%d %H:%M:%S')] Log file exceeded 50MB; rotated to output.log.old" > "$LOG_FILE"
25     fi
26 fi
27
28 # --- MAIN EXECUTION LOGGING ---
29 echo -e "\n===== " >> "$LOG_FILE"
30 echo "[$(date '+%Y-%m-%d %H:%M:%S')] toggle.sh invoked" >> "$LOG_FILE"
31 echo "===== " >> "$LOG_FILE"
32
33 # (Previously this script chmod 600'd .env at start to "unlock" it for docker compose,
34 # and chmod 000'd it on exit. That pattern broke docker compose on macOS/Colima when
35 # the umask produced 0600 the script removed its own ability to read .env before it could even proceed.)
36
37 # --- PROCEDURAL PORT GENERATION ---
38 # To prevent static port fingerprinting by malware, we randomize exposed ports
39 # on every boot and persist them to a hidden environment file.
40 PORTS_FILE="/tmp/nullexit-ports.env"
41 if [[ "$1" == "" || "$1" == "--restart" ]]; then
42
43     # Collision-avoidance function
44     GENERATED_PORTS=""
45     get_free_port() {
46         local port
47         while true; do
48             port=$(jot -r 1 10000 65000)
49             # 1. Prevent self-collision within this loop
50             if [[ "$GENERATED_PORTS" == *"$port"* ]]; then
51                 continue
52             fi
53             # 2. Prevent collision with ANY process on the Mac (root daemons, terminal apps, etc)
54             # We use netstat instead of lsof because netstat reads the kernel socket table
55             # directly and doesn't require sudo to see ports bound by other users/root.
56             if ! netstat -an -p tcp | awk '{print $4}' | grep -q "\.$port$"; then
57                 echo "$port"
58                 return
59             fi
60         done
61     }
62
63     export TOR SOCKS_PORT=$(get_free_port)
64     GENERATED_PORTS="$GENERATED_PORTS $TOR SOCKS_PORT"
65
66     export SOCKS_PROXY_PORT=$(get_free_port)
67     GENERATED_PORTS="$GENERATED_PORTS $SOCKS_PROXY_PORT"
68
69     export DNS_PROXY_PORT=$(get_free_port)
70
71     echo "export TOR SOCKS_PORT=$TOR SOCKS_PORT" > "$PORTS_FILE"
72     echo "export SOCKS_PROXY_PORT=$SOCKS_PROXY_PORT" >> "$PORTS_FILE"
73     echo "export DNS_PROXY_PORT=$DNS_PROXY_PORT" >> "$PORTS_FILE"
74 elif [ -f "$PORTS_FILE" ]; then
75     # On STOP, source the existing ports so docker-compose down matches
76     source "$PORTS_FILE"
77 else
78     # Fallbacks if file is lost
79     export TOR SOCKS_PORT=9050
80     export SOCKS_PROXY_PORT=1080
81     export DNS_PROXY_PORT=5354
82 fi
83
84 # Start execution timer
85 TOGGLE_START_TIME=$SECONDS
86
87 if [[ "$1" == "--restart" ]]; then
88     echo "Executing Gateway Restart Sequence..."
89     # We must source common.sh here temporarily to check is_gateway_active before we proceed
90     source "$SCRIPT_DIR/scripts/common.sh"
91     if is_gateway_active; then
92         echo "Gateway is currently running. Stopping it first..."
93         bash "$0"

```

```

94     echo "Gateway stopped. Now starting it..."
95 else
96     echo "Gateway is already stopped. Starting it..."
97 fi
98 bash "$0"
99 exit 0
100 fi
101
102 # Global flag to track if the script completed successfully
103 SUCCESS_RUN=false
104
105 # Global variable to track the currently running background command PID
106 CURRENT_BG_PID=""
107
108 # Source common bash functions
109 source "$SCRIPT_DIR/scripts/common.sh"
110
111 # Prevent concurrent execution of lifecycle scripts (toggle.sh / recover.sh)
112 LOCK_FILE="/tmp/nullexit-toggle.lock"
113 if [ -f "$LOCK_FILE" ]; then
114     LOCK_PID=$(cat "$LOCK_FILE" 2>/dev/null || echo "")
115     if [ -n "$LOCK_PID" ] && [ "$LOCK_PID" != "$$" ] && kill -0 "$LOCK_PID" 2>/dev/null; then
116         if ps -p "$LOCK_PID" -o args= 2>/dev/null | grep -q -E "toggle\.sh|recover\.sh"; then
117             die "Another instance of toggle.sh or recover.sh (PID $LOCK_PID) is already running."
118         fi
119     fi
120 fi
121 echo "$$" > "$LOCK_FILE"
122
123 # Helper function to run a GUI command as the logged-in console user
124 # (Prevents permission failures if the script is run with sudo)
125 run_gui_cmd() {
126     local console_user
127     console_user=$(stat -f '%Su' /dev/console 2>> output.log || echo "$SUDO_USER")
128     if [ -z "$console_user" ] || [ "$console_user" = "root" ]; then
129         console_user=$(logname 2>> output.log || echo "$USER")
130     fi
131
132     if [ -n "$console_user" ] && [ "$console_user" != "root" ] && [ "$EUID" -eq 0 ]; then
133         sudo -u "$console_user" "$@"
134     else
135         "$@"
136     fi
137 }
138
139 # Active-state marker: written at end of START path so external watchers
140 # (see launchd/com.nullexit.wake-recovery.plist + scripts/watcher.sh +
141 # devref 10.29) know the gateway is currently up. They only fire
142 # recover.sh --post-wake when this file is present. Cleared at the top
143 # of the STOP path and inside cleanup_handler on any error/signal path
144 # so a half-broken gateway never re-triggers post-wake refreshes.
145 write_gateway_active_marker() {
146     date -u +%FT%TZ > /tmp/nullexit-gateway-active.marker
147     # Set the watcher debounce timestamp to now to prevent a redundant post-startup recovery run.
148     date +%s > /tmp/nullexit-watcher.last-recovery 2>/dev/null || true
149 }
150
151 clear_gateway_active_marker() {
152     rm -f /tmp/nullexit-gateway-active.marker
153     rm -f "$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/TUNNEL_FAILED_CLOSED.marker"
154 }
155
156 # Defensive: clear any stale marker from a prior crashed/aborted run. If a
157 # previous toggle.sh wrote the marker but was OOM-killed in the middle of the
158 # START block (before the END-of-START write) or if the user rebooted mid
159 # session this prevents the watcher from firing post-wake against a
160 # non-running gateway on every wake event. Placed AFTER the function
161 # definitions so bash resolves the call at parse time.
162 clear_gateway_active_marker
163
164 PID_FILE="$PID_CAFFEINATE"
165
166 # Start macOS caffeine to prevent sleep while gateway is running
167 start_sleep_prevention() {
168     if ! command -v caffeine >/dev/null 2>&1; then
169         echo " [Warning] caffeine tool not found. Sleep prevention unavailable."
170         return 1
171     fi
172
173     # Stop any existing sleep prevention process first to avoid duplicates
174     stop_pidfile_daemon "/tmp/nullexit-caffeinate.pid" "system sleep prevention"

```

```

175
176 echo " Enabling system sleep prevention (caffeinate)..."
177 # Run caffeinate wrapped in a bash trap to prevent idle system sleep.
178 # Critically, this traps macOS shutdown signals (SIGTERM) and automatically
179 # flushes hijacked DNS back to normal right before the Mac powers off.
180 # We do not use recover.sh here because it is a heavy recovery script (managing
181 # containers, restarting tailscaled, etc.) which takes too long and gets
182 # terminated by macOS before it can reset the DNS. Instead, we surgically and
183 # instantly restore normal DNS and proxy settings here.
184 nohup bash -c "
185     trap '
186         echo \"[Shutdown Trap] Reverting network settings...\\" >> \"\$PWD/output.log\" 2>&1
187         kill \$! 2>/dev/null || true
188         sudo -n networksetup -setdnsservers \"\$ACTIVE_SERVICE\" 1.1.1.1 >> \"\$PWD/output.log\" 2>&1 || true
189         sudo -n networksetup -setdnsservers \"\$ENO_SERVICE\" 1.1.1.1 >> \"\$PWD/output.log\" 2>&1 || true
190         sudo -n networksetup -setsearchdomains \"\$ACTIVE_SERVICE\" \"Empty\" >> \"\$PWD/output.log\" 2>&1 || true
191         sudo -n networksetup -setsearchdomains \"\$ENO_SERVICE\" \"Empty\" >> \"\$PWD/output.log\" 2>&1 || true
192         sudo -n networksetup -setsocksfirewallproxystate \"\$ACTIVE_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 ||
193         true
194         sudo -n networksetup -setsocksfirewallproxystate \"\$ENO_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
195         sudo -n networksetup -setwebproxystate \"\$ACTIVE_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
196         sudo -n networksetup -setwebproxystate \"\$ENO_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
197         sudo -n networksetup -setsecurewebproxystate \"\$ACTIVE_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
198         sudo -n networksetup -setsecurewebproxystate \"\$ENO_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
199         if command -v tailscale >/dev/null 2>&1; then
200             tailscale up >> \"\$PWD/output.log\" 2>&1 || true
201             tailscale set --accept-dns=false --exit-node= >> \"\$PWD/output.log\" 2>&1 || true
202             TS_DOWN_ARGS=\"\"
203             if [ \"\$KILL_SWITCH\" = \"true\" ]; then TS_DOWN_ARGS=\"--accept-risk=lose-ssh\"; fi
204             tailscale down \$TS_DOWN_ARGS >> \"\$PWD/output.log\" 2>&1 &
205         fi
206         sleep 1
207         echo \"[Shutdown Trap] Cleanup complete.\" >> \"\$PWD/output.log\" 2>&1
208         exit 0
209     ' SIGTERM SIGINT SIGHUP
210     caffeinate -i &
211     wait \$!
212     \" >> output.log 2>&1 &
213     local caffe_pid=$!
214     echo \"\$caffe_pid\" > \"\$PID_FILE\"
215     echo \" Sleep prevention active (PID \$caffe_pid). Your Mac won't sleep while the gateway is running.\"
216 }
217
218
219 DNS_WATCHER_PID_FILE=\"$PID_DNS_WATCHER\"
220 WARP_WATCHER_PID_FILE=\"/tmp/nullexit-warp-watcher.pid\"
221
222 start_dns_watcher() {
223     local target_ip=$1
224     stop_pidfile_daemon "/tmp/nullexit-dns-watcher.pid" "background DNS Watcher"
225     echo " Starting background DNS Watcher for seamless Wi-Fi roaming..."
226     nohup bash -c "
227         source \"\$SCRIPT_DIR/scripts/common.sh\"
228         trap 'exit 0' SIGTERM SIGINT SIGHUP
229         while true; do
230             ACTIVE_IF=$(get_active_service)
231             if [ -n \"\$ACTIVE_IF\" ]; then
232                 CURRENT_DNS=$(networksetup -getdnsservers \"\$ACTIVE_IF\" 2>/dev/null)
233                 if [ \"\$CURRENT_DNS\" != \"\$target_ip\" ]; then
234                     networksetup -setdnsservers \"\$ACTIVE_IF\" \"\$target_ip\" >/dev/null 2>&1
235                 fi
236             fi
237             sleep 30
238         done
239         \" >> output.log 2>&1 &
240     echo $! > \"\$DNS_WATCHER_PID_FILE\"
241 }
242
243
244
245 # Background WARP tunnel liveness monitor. Polls cdn-cgi/trace every 30s
246 # while healthy, but accelerates to polling every 5s if a failure is detected.
247 # Always logs state transitions to output.log. When warp=off persists for
248 # WARP_FAIL_THRESHOLD consecutive polls (default 6 = 30s of downtime), the watcher
249 # triggers a nuclear recovery: runs recover.sh to tear down the entire
250 # gateway disconnecting Tailscale, stopping containers, resetting DNS,
251 # and power-cycling Wi-Fi. Note: This minimizes exposure window, but
252 # only the pf kill switch guarantees absolutely zero leakage on drop.
253 #
254 # The threshold (default 6) is statistically chosen: even at an

```

```

255 # unrealistically high 10% false-positive rate per poll, P(6 consecutive
256 # false positives) = 0.1^6 0.0001%. Real-world false-positive rates
257 # are far lower, making 30s of continuous downtime overwhelmingly likely
258 # to be a genuine outage.
259 #
260 # Configurable via .env: WARP_FAIL_THRESHOLD=3 (15s, more aggressive)
261 start_warp_watcher() {
262     stop_warp_watcher
263     # Parse threshold from .env (same pattern as GATEWAY_MSS, GATEWAY_BYPASS_PING, etc.)
264     local threshold
265     threshold=$(read_env_var WARP_FAIL_THRESHOLD)
266     if [ -z "$threshold" ]; then
267         threshold="{WARP_FAIL_THRESHOLD:-6}"
268     fi
269     local trigger_file="/tmp/nullexit-warp-shutdown-triggered"
270     rm -f "$trigger_file"
271     echo " Starting background WARP Watcher (polling every 30s [accelerating to 5s on failure], shutdown after $
        {threshold} consecutive failures)..."
272     nohup bash -c "
273         trap 'exit 0' SIGTERM SIGINT SIGHUP
274         last_state='on'
275         consec_off=0
276         threshold='$threshold'
277         trigger_file='$trigger_file'
278         out_log='$PWD/output.log'
279         recover_bin='$PWD/recover.sh'
280         inhibit_file='/tmp/nullexit-warp-inhibit.marker'
281         while true; do
282             # Inhibit check
283             # recover.sh --post-wake writes this marker while force-recreating the
284             # warp container. During that window, the container is intentionally
285             # down don't count it as a failure or we'll fire nuclear recover.sh
286             # and kill a gateway that was in the process of healing itself.
287             if [ -f "\$inhibit_file" ]; then
288                 consec_off=0
289                 sleep 5
290                 continue
291             fi
292
293             state='off'
294             if docker compose exec -T warp wget -qO- --timeout=5 \
295                 https://www.cloudflare.com/cdn-cgi/trace 2>/dev/null \
296                 | grep -q 'warp=on'; then
297                 state='on'
298             fi
299
300             # State-transition logging
301             if [ "\$state" != "\$last_state" ]; then
302                 if [ "\$state" = 'off' ]; then
303                     echo "\[$(date -u +%FT%TZ)] WARP DOWN cdn-cgi/trace reports warp=off" >> "\$out_log"
304                 fi
305                 last_state="\$state"
306             fi
307
308             # Consecutive-failure tracking + auto-shutdown
309             if [ "\$state" = 'off' ]; then
310                 consec_off=$((consec_off + 1))
311                 if [ "\$consec_off" -ge "\$threshold" ] && [ ! -f "\$trigger_file" ]; then
312                     touch "\$trigger_file"
313                     echo "\[$(date -u +%FT%TZ)] WARP SHUTDOWN \$consec_off consecutive failures (threshold=\$threshold)
. Running recover.sh to kill the gateway and restore normal internet. Your IP is NO LONGER Cloudflare.\" >>
        \"\$out_log\"
314                     # Nuclear recovery: tear down the entire gateway.
315                     # recover.sh resets DNS 1.1.1.1, disconnects Tailscale,
316                     # stops Docker containers, flushes routes, power-cycles Wi-Fi.
317                     bash \"\$recover_bin\" --auto >> \"\$out_log\" 2>&1 || true
318                     echo "\[$(date -u +%FT%TZ)] WARP SHUTDOWN recover.sh completed, watcher exiting. Your IP is now
your real ISP IP.\" >> \"\$out_log\"
319                     exit 0
320                 fi
321             else
322                 if [ "\$consec_off" -gt 0 ]; then
323                     echo "\[$(date -u +%FT%TZ)] WARP RECOVERED warp=on after \$consec_off consecutive off readings (
threshold was \$threshold)\>> \"\$out_log\"
324                 fi
325                 consec_off=0
326             fi
327
328             if [ "\$state" = 'off' ]; then
329                 sleep 5
330             else

```

```

331     sleep 30
332 fi
333 done
334 " >> output.log 2>&1 &
335 echo $! > "$WARP_WATCHER_PID_FILE"
336 }
337
338 stop_warp_watcher() {
339     if [ -f "$WARP_WATCHER_PID_FILE" ]; then
340         local wp
341         wp=$(cat "$WARP_WATCHER_PID_FILE")
342         if [ -n "$wp" ] && kill -0 "$wp" 2>/dev/null; then
343             echo " Stopping background WARP Watcher (PID $wp)..."
344             kill "$wp" 2>/dev/null || true
345         fi
346         rm -f "$WARP_WATCHER_PID_FILE"
347     fi
348 }
349
350
351
352
353
354 # Cleanup handler to restore DNS to 1.1.1.1 on error or user interrupt (Ctrl+C / SIGTERM / SIGHUP)
355 cleanup_handler() {
356     local exit_code=$?
357     local trigger_type="$1"
358     local line_no="$2"
359
360     if [ "$SUCCESS_RUN" = "false" ]; then
361         echo -e "\n[Self-Correction] Script interrupted or failed ($trigger_type). Restoring host DNS to 1.1.1.1..."
362
363         if [ -n "$ACTIVE_SERVICE" ]; then
364             reset_dns
365         fi
366
367         if [ -n "$CURRENT_BG_PID" ]; then
368             echo "Terminating active command (PID $CURRENT_BG_PID)..."
369             kill -15 "$CURRENT_BG_PID" 2>> output.log || true
370         fi
371
372         # Disconnect host Tailscale and close the GUI application on failure
373         disconnect_tailscale_host
374
375         # Stop local DNS proxy if running
376         stop_local_dns_proxy
377
378         # Stop sleep prevention
379         stop_pidfile_daemon "/tmp/nullexit-cafeinate.pid" "system sleep prevention"
380         stop_pidfile_daemon "/tmp/nullexit-dns-watcher.pid" "background DNS Watcher"
381         stop_warp_watcher
382         clear_gateway_active_marker
383
384         # Capture warp logs on failure for debugging before teardown
385         if [ "$trigger_type" = "ERR" ]; then
386             echo -e "\n--- WARP FAILURE LOGS ---" >> output.log
387             docker compose logs warp --tail=100 >> output.log 2>&1 || true
388             echo "-----" >> output.log
389         fi
390
391         # Best-effort container teardown on failure
392         docker compose down --remove-orphans -t 30 2>> output.log || true
393
394         # Nuke leftover network state (proxies, routes, DNS cache, Wi-Fi)
395         if [ -n "$ACTIVE_SERVICE" ]; then
396             cleanup_network_state
397         fi
398
399         if [ "$trigger_type" = "ERR" ]; then
400             echo -e "\n=====
401             echo "ERROR: Script failed at line $line_no with exit code $exit_code."
402             echo "=====
403             echo "Troubleshooting steps:"
404             echo "1. Run 'colima status' to verify the VM is active."
405             echo "2. Run 'docker compose ps' to check container health."
406             echo "3. Run 'docker compose logs' to view application logs."
407             echo "4. Run 'tailscale status' to check host Tailscale status."
408             echo "=====
409         fi
410         rm -f "${LOCK_FILE:-/tmp/nullexit-toggle.lock}"

```

```

411 fi
412 }
413
414 trap 'cleanup_handler ERR $LINENO' ERR
415 trap 'cleanup_handler INT' INT
416 trap 'cleanup_handler TERM' TERM
417 trap 'cleanup_handler HUP' HUP
418
419 # NOPASSWD sudo
420 # The user has NOPASSWD in /etc/sudoers.d/nullexit for the specific commands
421 # needed (networksetup, dnsmasq, dscacheutil, killall, pkill, socat).
422 # All privileged calls below use `sudo -n` which will run without prompting.
423 # No credential caching loop is needed.
424
425 # Add Homebrew and standard paths
426 setup_standard_path
427
428 # Check for local DNS proxy tools (used when tailnet data plane is unavailable)
429 # Uses a Python DNS proxy (handles TCP wire format correctly).
430 # Python3 is built into macOS no external dependencies.
431 DNS_PROXY_BIN=""
432 if command -v python3 >> output.log 2>&1; then
433     DNS_PROXY_BIN="python3 -I"
434 elif command -v python >> output.log 2>&1; then
435     DNS_PROXY_BIN="python -I"
436 fi
437
438 # Path to the DNS proxy script (sibling of this script)
439 DNS_PROXY_SCRIPT="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/scripts/dns-proxy.py"
440
441 # Resolve Tailscale CLI on host. We rely on the standalone Homebrew formula
442 # (`brew install tailscale` + `brew services start tailscale`) NOT the
443 # Tailscale.app GUI so there is no .app bundle or Network Extension sandbox
444 # to launch, no menu-bar click to perform, and no scutil Network Service to
445 # start manually. tailscaled runs as a per-user LaunchAgent (or LaunchDaemon
446 # if you ran the install with sudo) and the control socket is always available.
447 TS_BIN=""
448 if command -v tailscale >> output.log 2>&1; then
449     TS_BIN="tailscale"
450 fi
451
452 # Baseline: disable Tailscale DNS management at the daemon level.
453 # `tailscale set` writes a persistent preference. However, `tailscale up --reset`
454 # (used in steps 7 and 9) RESETS all unspecified flags to defaults, which would
455 # re-enable `--accept-dns=true` and nullify this `set`.
456 # Therefore, EVERY `tailscale up --reset` call in this file MUST also pass
457 # `--accept-dns=false` explicitly. See steps 7 and 9 for the fix.
458 #
459 # This initial `set` is still useful as a baseline for non-reset operations
460 # (e.g. if the user runs `tailscale up` manually without --reset) and covers
461 # the brief window between this line and the first `--reset` call.
462 #
463 # Runs once at script start while tailscaled is still connected (if it was
464 # left running from a previous toggle), so the pref takes effect before any
465 # disconnection or reconnection logic.
466 if [ -n "$TS_BIN" ]; then
467     $TS_BIN set --accept-dns=false >> output.log 2>&1 || true
468 fi
469
470 # Disconnect host Tailscale from the mesh. With the standalone daemon, this is the
471 # *entire* teardown no scutil tunnel stop, no Tailscale.app GUI quit, no pkill.
472 # tailscaled keeps running (as a LaunchAgent / LaunchDaemon), which is intentional:
473 # the next `tailscale up` then reconnects in seconds rather than waiting for a
474 # slow .app launch + Network Extension activation. The `tailscale down` call also
475 # retains the user's auth state, so we don't force a browser re-login every cycle.
476 #
477 # Defined early (before the if/else branches and referenced by cleanup_handler's
478 # ERR trap) so that any failure before the main logic can still tear down Tailscale.
479
480
481 # Wait for Network Readiness
482 # On --restart, the gateway tears down while the host network interface may
483 # still be reconnecting (Wi-Fi, Ethernet, USB tethering, etc.). If we proceed
484 # too early, get_active_service returns nothing, routing targets the wrong
485 # interface, and DNS/WARP setup silently fails.
486 #
487 # We use `route get default` to detect a valid default gateway rather than
488 # checking a specific interface (e.g. en0) this generalizes across all
489 # connection types: Wi-Fi, Ethernet, USB-C adapters, iPhone tethering, etc.
490 # The moment the OS has any routable default gateway, we proceed.
491 _net_wait_secs=0

```

```

492 _net_timeout=60
493 echo "Waiting for network connectivity before starting..."
494 while true; do
495     if route get default 2>/dev/null | grep -q 'gateway: '; then
496         _gw=$(route get default 2>/dev/null | awk '/gateway:/{print $2}')
497         echo "    Network ready (default gateway: $_gw)."
498         break
499     fi
500     if [ "$_net_wait_secs" -ge "$_net_timeout" ]; then
501         echo "    [Warning] No default gateway after ${_net_timeout}s proceeding anyway."
502         break
503     fi
504     sleep 2
505     _net_wait_secs=$(($_net_wait_secs + 2))
506 done
507
508 ACTIVE_SERVICE=$(get_active_service)
509
510 # Resolve the service name for en0 (usually "Wi-Fi") macOS scutil DNS resolver
511 # is commonly scoped to en0, so per-service DNS changes on other interfaces are
512 # ignored unless en0's service is also updated.
513 ENO_SERVICE=$(get_en0_service)
514
515 # Helper to add host-side bypass routes for the WARP WireGuard endpoints.
516 # This prevents an infinite recursive routing loop (tunnel loop) where
517 # the container's WireGuard packets exit the VM, reach the host, match the
518 # default route (the exit node), and get routed back into the tunnel.
519 # We route via the gateway IP (not interface) to ensure packets are routed
520 # properly even when the host's default route changes to the Tailscale interface.
521
522 # Helper to manually override the host's default route to point through the
523 # Tailscale utun* interface. Standalone tailscaled on macOS (Homebrew version)
524 # lacks the platform-specific integration to override the default gateway
525 # automatically when --exit-node is used.
526 setup_exit_node_routing() {
527     local ts_iface
528     ts_iface=$(ifconfig | awk '/^[a-z0-9]+:/{iface=$1} /inet 100\./{print iface; exit}' | tr -d ':')
529
530     if [ -n "$ts_iface" ]; then
531         echo "Re-routing default gateway to $ts_iface using 0.0.0.0/1 split..."
532         local host_mtu
533         host_mtu=$(read_env_var HOST_MTU)
534         host_mtu=${host_mtu:-1200}
535
536         # Lower the host Tailscale MTU (defaults to 1200) to prevent fragmentation when passing
537         # through the container's WARP tunnel (which also has an MTU of 1280).
538         sudo -n ifconfig "$ts_iface" mtu "$host_mtu" >> output.log 2>&1 || true
539
540         # DO NOT delete the physical default route! It crashes Colima.
541         # Use the /1 trick to mathematically override the default route.
542         sudo -n route delete -net 0.0.0.0/1 >> output.log 2>&1 || true
543         sudo -n route delete -net 128.0.0.0/1 >> output.log 2>&1 || true
544         sudo -n route add -net 0.0.0.0/1 -interface "$ts_iface" >> output.log 2>&1 || true
545         sudo -n route add -net 128.0.0.0/1 -interface "$ts_iface" >> output.log 2>&1 || true
546
547         if netstat -nr | grep -E -q "^(0\.\0\.\0/1|0/1)[[:space:]]+.*$ts_iface"; then
548             echo "    [] Default route successfully overridden via $ts_iface."
549         else
550             echo "    [!] Failed to verify exit node routing on $ts_iface."
551         fi
552     else
553         echo "[Warning] Could not detect Tailscale utun interface for host routing."
554     fi
555 }
556
557 # Helper to restore host DNS to a clean state (used on script start, on failure,
558 # and after a successful STOP). Without this, a successful toggle-off leaves the
559 # host's resolver pinned to a now-dead gateway IP every lookup stalls for macOS's
560 # DNS timeout (~5s) before falling through to whatever next server is configured.
561 # With it, we always leave the host in '1.1.1.1 / no search domain'.
562 #
563 # We set DNS on BOTH the active service AND the en0 service because macOS's scutil DNS
564 # resolver is commonly scoped to en0 (Wi-Fi). A per-service change on e.g.
565 # "USB 10/100 LAN" is ignored by the system resolver unless the en0 service is also updated.
566 reset_dns() {
567     if [ -n "$ACTIVE_SERVICE" ]; then
568         for dns_svc in "$ACTIVE_SERVICE" "$ENO_SERVICE"; do
569             networksetup -setsearchdomains "$dns_svc" "Empty" 2>> output.log || true
570             networksetup -setdnsservers "$dns_svc" 1.1.1.1 2>> output.log || true
571         done
572     fi

```

```

573 }
574
575 # Helper to nuke leftover network state after teardown (proxy settings, routing
576 # table, DNS cache, Wi-Fi). Prevents the "had to write a whole recovery script"
577 # problem where state state kills internet after the gateway stops.
578 cleanup_network_state() {
579     echo -e "\nCleaning up network state (proxies, routes, DNS cache, Wi-Fi)..."
580
581     # 1. Disable any leftover proxy settings (needs root on many macOS versions)
582     for svc in "$ACTIVE_SERVICE" "$ENO_SERVICE"; do
583         disable_all_proxies "$svc"
584     done
585     echo " Proxies disabled."
586
587     # 2. Flush DNS cache
588     if command -v dscacheutil >> output.log 2>&1; then
589         sudo -n dscacheutil -flushcache 2>> output.log || true
590     fi
591     sudo -n killall -HUP mDNSResponder 2>> output.log || true
592     echo " DNS cache flushed."
593
594     # 3. Flush stale routing table entries
595     if command -v route >> output.log 2>&1; then
596         sudo -n route delete -net 0.0.0.0/1 >> output.log 2>&1 || true
597         sudo -n route delete -net 128.0.0.0/1 >> output.log 2>&1 || true
598     fi
599     remove_warp_bypass_routes
600     disable_killswitch
601     echo " Routing table and firewall flushed."
602
603     # 4. Power-cycle Wi-Fi to clear any lingering interface state
604     WIFI_PORT=$(networksetup -listallhardwareports 2>> output.log | awk '/Hardware Port: Wi-Fi/{getline; print $2
605     }')
606     if [ -n "$WIFI_PORT" ]; then
607         sudo -n networksetup -setairportpower "$WIFI_PORT" off 2>> output.log || true
608         sleep 2
609         sudo -n networksetup -setairportpower "$WIFI_PORT" on 2>> output.log || true
610         echo " Wi-Fi power-cycled (interface $WIFI_PORT)."
611         sleep 3
612     else
613         echo " Wi-Fi interface not detected; bouncing en0..."
614         sudo -n ifconfig en0 down 2>> output.log || true
615         sudo -n ifconfig en0 up 2>> output.log || true
616     fi
617
618     # Force restart sharing services to prevent AirDrop / AirPlay discovery freezes
619     # after network interface/DNS changes.
620     echo " Resetting macOS sharing services (AirDrop/AirPlay)..."
621     reset_sharing_services
622
623     echo "Network state cleanup complete."
624 }
625
626 # SOCKS5 Proxy (WARP Tunnel)
627 # When Tailscale's exit node can't route through WARP (due to userspace
628 # forwarding), we use a SOCKS5 proxy running inside the warp container's
629 # network namespace. Connections created by this proxy go through the
630 # kernel routing table (table 200 -> tun0 -> WARP), unlike Tailscale's
631 # userspace exit node which bypasses it.
632 #
633 # The SOCKS5 proxy is exposed on localhost:${SOCKS_PROXY_PORT} via Docker port mapping.
634 # `networksetup -setsocksfirewallproxy` on macOS routes system TCP traffic
635 # through the proxy, which then goes through WARP.
636 #
637 # IMPORTANT: CLI tools like curl do NOT respect macOS system proxy settings.
638 # They must use --socks5-hostname or the ALL_PROXY env variable explicitly.
639
640 SOCKS_PROXY_PORT=${SOCKS_PROXY_PORT}
641
642 enable_socks_proxy() {
643     local svc="$1"
644     if [ -z "$svc" ]; then
645         svc="$ACTIVE_SERVICE"
646     fi
647
648     echo -n " Enabling system-wide SOCKS5 proxy on $svc (localhost:${SOCKS_PROXY_PORT})... "
649     sudo -n networksetup -setsocksfirewallproxy "$svc" 127.0.0.1 $SOCKS_PROXY_PORT 2>> output.log || true
650     sudo -n networksetup -setsocksfirewallproxystate "$svc" on 2>> output.log || true
651
652     # Also set on en0 service for macOS resolver consistency
653     if [ "$svc" != "$ENO_SERVICE" ]; then

```

```

653     sudo -n networksetup -setsocksfirewallproxy "$ENO_SERVICE" 127.0.0.1 $SOCKS_PROXY_PORT 2>> output.log ||
654         true
655     sudo -n networksetup -setsocksfirewallproxystate "$ENO_SERVICE" on 2>> output.log || true
656 fi
657 # IMPORTANT: Exclude local LAN and mDNS traffic from the proxy so P2P works natively
658 # without source-IP masking from the Colima VM bridge.
659 sudo -n networksetup -setproxybypassdomains "$svc" 127.0.0.1 localhost 192.168.0.0/16 10.0.0.0/8
660     172.16.0.0/12 "*.local" 2>> output.log || true
661 if [ "$svc" != "$ENO_SERVICE" ]; then
662     sudo -n networksetup -setproxybypassdomains "$ENO_SERVICE" 127.0.0.1 localhost 192.168.0.0/16 10.0.0.0/8
663     172.16.0.0/12 "*.local" 2>> output.log || true
664 fi
665
666 echo "done."
667 echo " All TCP traffic now routed through gateway -> WARP tunnel -> internet."
668 echo " (CLI tools: export ALL_PROXY=socks5://127.0.0.1:$SOCKS_PROXY_PORT)"
669 }
670
671 disable_socks_proxy() {
672     for svc in "$ACTIVE_SERVICE" "$ENO_SERVICE"; do
673         sudo -n networksetup -setsocksfirewallproxystate "$svc" off 2>> output.log || true
674     done
675     echo " SOCKS5 proxy disabled."
676 }
677
678 # Local DNS Proxy (Python)
679 # When the tailnet data plane cannot establish, we use a Python DNS proxy.
680 # It listens on UDP:53, receives DNS queries, forwards them over TCP to
681 # AdGuard at 127.0.0.1:${DNS_PROXY_PORT} (Docker port mapping), and returns the response.
682 # The Python script handles the DNS-over-TCP wire format (2-byte length prefix)
683 # correctly unlike socat which just forwards raw bytes.
684 #
685 # Python3 is built into macOS no external dependencies.
686 DNS_PROXY_PID=""
687
688 # Kill any leftover DNS proxy process
689 stop_local_dns_proxy() {
690     if [ -n "$DNS_PROXY_PID" ]; then
691         kill "$DNS_PROXY_PID" 2>/dev/null || true
692         wait "$DNS_PROXY_PID" 2>/dev/null || true
693         DNS_PROXY_PID=""
694     fi
695     # Clean up any stale Python DNS proxy processes
696     sudo -n pkill -f "dns-proxy.py" 2>/dev/null || true
697 }
698
699 # Start local DNS proxy (Python)
700 start_local_dns_proxy() {
701     if [ -z "$DNS_PROXY_BIN" ]; then
702         echo " Python not found. Python3 is built into macOS this shouldn't happen."
703         return 1
704     fi
705     if [ ! -f "$DNS_PROXY_SCRIPT" ]; then
706         echo " DNS proxy script not found at $DNS_PROXY_SCRIPT"
707         return 1
708     fi
709     # Kill any leftover process first
710     stop_local_dns_proxy
711
712     echo -n " Starting local DNS proxy via Python (UDP:53 TCP:${DNS_PROXY_PORT})... "
713     sudo -n bash -c "TARGET_PORT=$DNS_PROXY_PORT \"\$DNS_PROXY_BIN\" \"\$DNS_PROXY_SCRIPT\" &
714     DNS_PROXY_PID=$!
715     disown \"$DNS_PROXY_PID\" 2>/dev/null || true
716
717     # Give it a moment to bind
718     sleep 0.5
719
720     if kill -0 "$DNS_PROXY_PID" 2>/dev/null; then
721         echo "started (PID $DNS_PROXY_PID)."
722
723         # Hijack host DNS to localhost
724         echo -n " Hijacking host DNS to 127.0.0.1 for ad-blocking... "
725         networksetup -setsearchdomains "$ACTIVE_SERVICE" "Empty" 2>/dev/null || true
726         networksetup -setsearchdomains "$ENO_SERVICE" "Empty" 2>/dev/null || true
727         if ! networksetup -setdnsservers "$ACTIVE_SERVICE" 127.0.0.1; then
728             echo "FAILED (networksetup error check permissions)."
729             stop_local_dns_proxy
730             return 1
731         fi
732     fi
733 }

```

```

731 fi
732 networksetup -setdnsservers "$ENO_SERVICE" 127.0.0.1 2>/dev/null || true
733 sudo -n dscacheutil -flushcache 2>/dev/null || true
734 sudo -n killall -HUP mDNSResponder 2>/dev/null || true
735
736 # Verify DNS actually works through the proxy
737 echo -n " Verifying DNS resolution... "
738 if dig @127.0.0.1 google.com +short +timeout=5 &>/dev/null; then
739     echo "ok."
740     return 0
741 else
742     echo "FAILED (DNS queries not reaching AdGuard)."

```

```

810 # would prepend it to every lookup and most public DNS queries would resolve to NXDOMAIN.
811
812
813 echo "Initializing DNS to 1.1.1.1 to ensure reliable internet access..."
814 reset_dns
815
816
817
818
819 echo "Checking Gateway Status..."
820 HIJACK_HOST=$(read_env_var GATEWAY_HIJACK_HOST | tr '[:upper:]' '[:lower:]')
821
822 if is_gateway_active; then
823     echo -e "\n=====
824     echo "Gateway is RUNNING. Stopping it now..."
825     echo -e "=====\\n"
826
827     # 1. Disconnect host Tailscale cleanly first before stopping the exit-node container
828     if [[ "$HIJACK_HOST" != "false" ]]; then
829         disconnect_tailscale_host
830         echo ""
831     fi
832
833     echo "Stopping Docker containers..."
834     docker compose down --remove-orphans -t 30
835
836     # Only stop Colima if the user explicitly opted in (false by default) to prevent breaking their other Docker
837     # dev projects
838     STOP_COLIMA=$(read_env_var STOP_COLIMA_ON_EXIT | tr '[:upper:]' '[:lower:]')
839     if [[ "$STOP_COLIMA" == "true" ]]; then
840         echo -e "\nStopping Colima VM to free up host RAM and battery..."
841         run_with_timeout 30 colima stop >> output.log 2>&1 || echo "Warning: Failed to stop Colima gracefully."
842     else
843         echo -e "\nLeaving Colima running. (Set STOP_COLIMA_ON_EXIT=true in .env to change this)"
844     fi
845
846     if [[ "$HIJACK_HOST" != "false" ]]; then
847         # The host's DNS was hijacked to the gateway IP during ENABLE; now that the
848         # gateway is down, restore DNS to 1.1.1.1 immediately so subsequent lookups
849         # don't stall for the macOS DNS timeout before the 1.1.1.1 fallback engages.
850         echo -e "\nRestoring host DNS to 1.1.1.1 (gateway is gone)... "
851         reset_dns
852
853         # 3. Stop local DNS proxy if running
854         stop_local_dns_proxy
855
856         # Stop background daemons
857         stop_pidfile_daemon "/tmp/nullexit-cafeinate.pid" "system sleep prevention"
858         stop_pidfile_daemon "/tmp/nullexit-dns-watcher.pid" "background DNS Watcher"
859         stop_warp_watcher
860         clear_gateway_active_marker
861
862         # 4. Nuke leftover network state so internet actually works after teardown
863         cleanup_network_state
864     fi
865
866     ELAPSED=$(( SECONDS - TOGGLE_START_TIME ))
867     echo -e "\nGateway has been successfully STOPPED in ${ELAPSED} seconds."
868 else
869     START_GATEWAY=true
870     echo "[$(date '+%Y-%m-%d %H:%M:%S')] Action: STARTUP GATEWAY" >> "$LOG_FILE"
871     echo -e "\n=====
872     echo "Gateway is STOPPED. Starting it now..."
873     echo -e "=====\\n"
874
875     # 1. Prevent host exit-node deadlock during VM / Container startup
876     if [[ "$HIJACK_HOST" != "false" ]]; then
877         disconnect_tailscale_host
878     fi
879
880     # 2. Wait for physical DHCP lease to settle (crucial for restarts/wake-up/roaming)
881     wait_for_dhcp_settle
882
883     # 3. Boot Colima VM if it is not already running
884     echo -e "\nChecking Colima VM status..."
885     if ! run_with_timeout 15 colima status >> output.log 2>&1; then
886         colima_network_mode="bridged"
887         security_type=""
888         if [[ "$OSTYPE" == "darwin"* ]]; then
889             security_type=$(system_profiler SPAirPortDataType 2>/dev/null | awk '/Current Network Information:/{found
            =1} found && /Security:/{print $NF; exit}')

```

```

889     if echo "$security_type" | grep -qi "Enterprise"; then
890         echo -e "\n [!] WPA2-Enterprise Wi-Fi detected! Bridged mode will cause MAC-spoofing deauthentication.
"
891         echo "          Falling back to '--network-mode shared' for Colima."
892         colima_network_mode="shared"
893     fi
894 fi
895 echo "Colima is not running. Starting Colima (600MB RAM allocation, vz VM, network address,
$colima_network_mode)..."
896 run_with_timeout 120 colima start --memory 0.6 --vm-type vz --network-address --network-mode "
$colima_network_mode"
897 echo "$colima_network_mode" > /tmp/nullexit_colima_mode.txt
898 else
899     echo "Colima is already running."
900 fi
901
902 # 3b. Auto-detect and fix Docker Subnet Collisions (e.g. if the user travels to a 172.17.x.x Wi-Fi)
903 if [ -f "scripts/fix-docker-bridge-collision.sh" ]; then
904     if bash scripts/fix-docker-bridge-collision.sh --dry | grep -q "collision detected"; then
905         echo -e "\n[!] WARNING: Docker Subnet Collision Detected with your current Wi-Fi!"
906         echo -e "          Auto-fixing Colima config to prevent internet jitter..."
907         bash scripts/fix-docker-bridge-collision.sh --skip-toggle || true
908     fi
909 fi
910
911 # 3c. Configure swap file inside the VM to prevent OOM on low-memory limits
912 # We also set vm.swappiness=10 so the Linux kernel strictly prefers physical RAM
913 # and avoids unnecessarily wearing out the SSD with proactive background swapping.
914 if ! run_with_timeout 15 colima ssh -- grep -q 'swapfile' /proc/swaps >> output.log 2>&1; then
915     echo "Configuring 400MB swap file inside the VM to prevent OOM..."
916     run_with_timeout 30 colima ssh -- sudo sh -c "if [ ! -f /swapfile ]; then dd if=/dev/zero of=/swapfile bs=1
M count=400 status=none && mkswap /swapfile; fi && swapon /swapfile && sysctl vm.swappiness=10" >> output.
log 2>&1 || echo "Warning: Failed to enable swap file inside the VM."
917 fi
918
919 # 4. Clean up corrupted AdGuardHome configurations
920 if [ -f "adguard/conf/AdGuardHome.yaml" ] && [ ! -s "adguard/conf/AdGuardHome.yaml" ]; then
921     rm -f "adguard/conf/AdGuardHome.yaml"
922     echo "Removed corrupted empty AdGuardHome.yaml to prevent container crash loop."
923 fi
924
925 # 4b. Auto-heal stale Docker socket from hard reboots
926 if ! run_with_timeout 15 docker ps >/dev/null 2>&1; then
927     echo "Docker daemon is unresponsive (likely a stale socket from a crash). Auto-healing Colima..."
928     run_with_timeout 120 colima restart >> output.log 2>&1
929 fi
930
931 # 4c. Inject TCP MSS clamp from .env into post-rules.txt before starting
932 if [ -f .env ] && grep -q "^GATEWAY_MSS=" .env; then
933     GATEWAY_MSS=$(read_env_var GATEWAY_MSS)
934     if [ "$GATEWAY_MSS" =~ ^[0-9]+$ ]; then
935         # macOS sed syntax
936         sed -i '' "s/--set-mss [0-9]*/--set-mss ${GATEWAY_MSS}/g" post-rules.txt 2>/dev/null || \
937         # Linux sed fallback
938         sed -i "s/--set-mss [0-9]*/--set-mss ${GATEWAY_MSS}/g" post-rules.txt 2>/dev/null
939     fi
940 fi
941
942 # 5. Start compose services
943 write_host_ips
944
945 # Procedurally generated ports have already been exported at the top of the script.
946 echo " [Security] Procedurally randomized proxy ports generated to evade fingerprinting."
947
948 # --- HONEY-PORT TRIPWIRE ---
949 # Generate a random trap port. If any malware does a sequential port scan on localhost,
950 # it will inevitably hit this port. When it does, we instantly fire a macOS alert.
951 HONEY_PORT=$(get_free_port)
952 (
953     if nc -l localhost "$HONEY_PORT" >/dev/null 2>&1; then
954         echo "[$(date +%Y-%m-%d %H:%M:%S)] [CRITICAL SECURITY ALERT] PORT SCAN DETECTED! An unknown application
just pinged the local Honey-Port ($HONEY_PORT)!" >> "$LOG_FILE"
955         osascript -e 'display notification "An unknown application is aggressively scanning your local ports.
Malware port-scan suspected." with title "nullexit OPSEC Alert" sound name "Basso"' 2>/dev/null || true
956     fi
957 ) &
958 disown %1 2>/dev/null || true
959 echo " [Security] Honey-Port Tripwire armed on random port to detect malware port-scans."
960
961 echo -e "\nStarting Docker containers..."
962 docker compose up -d --build

```

```

963
964 # Log the output of the rule compiler for debugging before removing it
965 docker compose logs rule-compiler >> output.log 2>&1
966 # Clean up the one-off rule compiler container so it doesn't clutter the Docker UI
967 docker compose rm -s -f rule-compiler >> output.log 2>&1
968
969 RULE_COUNT=$(grep "! Total Block Rules:" adguard/work/userfilters/compiled_rules.txt 2>/dev/null | awk -F': ' '{print $2}' || echo "0")
970 NATIVE_COUNT=$(grep "! Native AdGuard Rules:" adguard/work/userfilters/compiled_rules.txt 2>/dev/null | awk -F': ' '{print $2}' || echo "0")
971
972 if [ "$RULE_COUNT" != "0" ] && [ "$RULE_COUNT" != "" ]; then
973     if [ "$NATIVE_COUNT" != "0" ] && [ "$NATIVE_COUNT" != "" ]; then
974         TOTAL_COUNT=$((RULE_COUNT + NATIVE_COUNT))
975         # Format with commas for readability (e.g. 441,578)
976         if command -v printf &> /dev/null; then
977             FORMATTED_RULE=$(printf "%'d" "$RULE_COUNT")
978             FORMATTED_TOTAL=$(printf "%'d" "$TOTAL_COUNT")
979             echo " DNS: Compiled $FORMATTED_RULE unique custom rules (Total active in AdGuard: ~$FORMATTED_TOTAL rules)."
980         else
981             echo " DNS: Compiled $RULE_COUNT unique custom rules (Total active in AdGuard: ~$TOTAL_COUNT rules)."
982         fi
983     else
984         echo " DNS: Compiled and loaded $RULE_COUNT optimized rules."
985     fi
986 fi
987
988 # Report IP blacklist compilation results
989 IP_COUNT=$(grep "^# Entries:" adguard/work/userfilters/ip_blocklist.ipset 2>/dev/null | awk -F': ' '{print $2}' || echo "0")
990 if [ "$IP_COUNT" != "0" ] && [ "$IP_COUNT" != "" ]; then
991     if command -v printf &> /dev/null; then
992         FORMATTED_IP=$(printf "%'d" "$IP_COUNT")
993         echo " IP: Loaded $FORMATTED_IP threat intelligence IPs/CIDRs into kernel firewall."
994     else
995         echo " IP: Loaded $IP_COUNT threat intelligence IPs/CIDRs into kernel firewall."
996     fi
997 fi
998
999 # 6. Wait for the gateway container's Tailscale connection to be ready
1000 BYPASS_PING=$(read_env_var GATEWAY_BYPASS_PING | tr '[:upper:]' '[:lower:]')
1001 USE_EXIT_NODE=$(read_env_var GATEWAY_USE_EXIT_NODE | tr '[:upper:]' '[:lower:]')
1002
1003 echo "Waiting for gateway container's Tailscale to connect to the tailnet..."
1004 echo " (This can take 30-60s Tailscale goes through: NoState Starting Running Online)"
1005 connected=false
1006 consecutive_failures=0
1007 for i in {1..60}; do
1008     # Run tailscale status and capture its output so the user can see what's happening.
1009     # Previously this was hidden behind 2>> output.log, making it impossible to debug hangs.
1010     ts_output=$(run_with_timeout 5 docker compose exec -T tailscale tailscale status 2>&1 || true)
1011
1012     if echo "$ts_output" | grep -q "offers exit node"; then
1013         connected=true
1014         break
1015     fi
1016
1017     # Track consecutive failures to detect a stuck state.
1018     # If we see 40+ consecutive 'NoState' responses, give up early
1019     # the daemon is alive but can't connect to the coordination server.
1020     if echo "$ts_output" | grep -q "NoState"; then
1021         consecutive_failures=$((consecutive_failures + 1))
1022         if [ "$consecutive_failures" -ge 40 ]; then
1023             echo ""
1024             echo " [attempt $i/60] Stuck in 'NoState' for $consecutive_failures checks."
1025             echo " Tailscale daemon is running but can't reach the coordination server."
1026             break
1027         fi
1028     else
1029         consecutive_failures=0
1030     fi
1031
1032     # Print a condensed status every 10 seconds so the user can see progress
1033     if [ $((i % 10)) -eq 0 ]; then
1034         ts_short=$(echo "$ts_output" | head -1 | tr -d '\r\n')
1035         echo " [attempt $i/60] $ts_short"
1036     elif [ $((i % 5)) -eq 0 ]; then
1037         echo -n "$[i]"
1038     else
1039         echo -n "."

```

```

1040     fi
1041     sleep 1
1042 done
1043 echo ""
1044
1045 if [ "$connected" = "true" ]; then
1046     echo "Gateway container is online on the Tailscale mesh."
1047 elif [ "$BYPASS_PING" = "true" ]; then
1048     echo "[Warning] Gateway container check timed out. Proceeding anyway (GATEWAY_BYPASS_PING is true)..."
1049 else
1050     echo "ERROR: Gateway container failed to initialize Tailscale (timed out after 60s)." >&2
1051     echo " The container was seen in the following states during the wait:" >&2
1052     echo "$ts_output" | sed 's/~/  /' >&2
1053     echo "" >&2
1054     echo " This could mean:" >&2
1055     echo "   1. Tailscale auth key has expired check TS_AUTHKEY in .env" >&2
1056     echo "   2. Network issue inside the container check: docker compose logs tailscale" >&2
1057     echo "   3. gluetun VPN tunnel not connecting check: docker compose logs warp | tail -20" >&2
1058     echo "" >&2
1059     echo " Diagnose manually:" >&2
1060     echo "     docker compose exec -T tailscale tailscale status" >&2
1061     echo "     docker compose exec -T tailscale tailscale netcheck" >&2
1062 cleanup_handler ERR $LINENO
1063 exit 1
1064 fi
1065
1066 # 6b. Resolve the gateway's Tailscale IP.
1067 # Dynamically query the container for the IP (handles IP changes after re-auth)
1068 # and cache it in .gateway_ip for fast retrieval by recover.sh during Wi-Fi roaming.
1069 #
1070 # CRITICAL: `docker compose exec -T` on macOS/Colima injects carriage
1071 # returns (\r) into captured output. If $TS_IP ends with \r, then
1072 # `networksetup -setdnsservers` silently rejects the malformed IP and
1073 # DNS stays at 1.1.1.1 the primary bug this project was hitting.
1074 # `tr -d '\r'` strips the CR; `awk 'NR==1{print $1; exit}'` takes only
1075 # the first field of the first line.
1076 TS_IP=""
1077 if [ "$connected" = "true" ]; then
1078     TS_IP=$(run_with_timeout 10 docker compose exec -T tailscale tailscale ip -4 2>> output.log | tr -d '\r' |
1079         awk 'NR==1{print $1; exit}' || true)
1080     if [ -n "$TS_IP" ]; then
1081         echo "Resolved gateway Tailscale IP dynamically: $TS_IP"
1082         echo "$TS_IP" > .gateway_ip
1083     else
1084         echo " [!] Failed to resolve gateway IP dynamically. Trying fallback cache..." >&2
1085         TS_IP=$(read_adguard_ip || true)
1086     fi
1087 else
1088     TS_IP=$(read_adguard_ip || true)
1089 fi
1090
1091 # 7. Connect host Mac to Tailscale mesh.
1092 # With the standalone tailscaled daemon (registered as a per-user LaunchAgent or
1093 # system LaunchDaemon via 'brew services start tailscale'), the previous five-
1094 # phase flow collapses into two trivial CLI calls. There is no .app GUI to launch,
1095 # no menu bar item to click, and no Accessibility permission required.
1096 #
1097 # CRITICAL: If tailscaled is not running, `tailscale up` will silently fail.
1098 # We auto-start it before proceeding, and SKIP the rest of the Tailscale setup
1099 # if it can't be started (DNS hijack to a 100.x.x.x IP and exit-node routing
1100 # are impossible without the host on the mesh).
1101 #
1102 # We connect in two phases:
1103 #   Phase A Join mesh WITHOUT exit node (so pre-flight checks can run)
1104 #   Phase B Set exit node only after pre-flight checks confirm the gateway works
1105 HOST_ON_MESH=false
1106 SKIP_EXIT_NODE=false
1107 if [ "$HIJACK_HOST" = "false" ]; then
1108     echo -e "\n[Info] HIJACK_HOST is false (Headless Mode). Host networking will remain untouched."
1109     SKIP_EXIT_NODE=true
1110 elif [ -n "$TS_BIN" ]; then
1111     echo -n "Verifying tailscaled is reachable"
1112     daemon_ready=false
1113     status_exit=0
1114     for i in {1..10}; do
1115         # Test if the daemon responds to status check.
1116         # If status returns non-zero, check if it was a normal "stopped/disconnected" state
1117         # (exit code 1 is normal for disconnected). If the command exited quickly (not killed by timeout),
1118         # and the error is just "Tailscale is stopped", then the daemon is alive and ready.
1119         # But if it timed out (exit code 143) or is completely unresponsive, it's wedged.
1120         status_exit=0

```

```

1120     if run_with_timeout 5 $TS_BIN status >> output.log 2>&1; then
1121         daemon_ready=true
1122         break
1123     else
1124         status_exit=$?
1125         if [ "$status_exit" -ne 143 ] && [ -S /var/run/tailscaled.socket ]; then
1126             daemon_ready=true
1127             break
1128         fi
1129     fi
1130     echo -n "."
1131     sleep 1
1132 done
1133 echo ""
1134
1135 # If the daemon was unresponsive or socket check failed, attempt auto-restart
1136 if [ "$daemon_ready" != "true" ]; then
1137     if [ "$status_exit" -eq 143 ]; then
1138         warn "tailscaled daemon is wedged/unresponsive. Restarting it..."
1139     else
1140         warn "tailscaled daemon is not running. Attempting to start it..."
1141     fi
1142     restart_tailscaled_daemon
1143
1144 # Re-verify
1145 if run_with_timeout 5 $TS_BIN status >> output.log 2>&1 || [ -S /var/run/tailscaled.socket ]; then
1146     daemon_ready=true
1147 fi
1148 fi
1149
1150 if [ "$daemon_ready" = "true" ]; then
1151     echo "tailscaled is responsive (running as a system service)."
1152
1153 # Phase A: Join mesh without exit node
1154 echo "Connecting host to Tailscale mesh (no exit node yet)..."
1155 if $TS_BIN up; then
1156     $TS_BIN set --ssh=true --accept-dns=false --accept-routes=true --exit-node= || true
1157     HOST_ON_MESH=true
1158     echo "Host is on Tailscale mesh."
1159 else
1160     warn "tailscale up failed even though the daemon is running."
1161     echo " This usually means the host hasn't authenticated with your tailnet yet."
1162     echo " Run this command in a terminal to authenticate:"
1163     echo ""
1164     echo "         sudo tailscale up"
1165     echo ""
1166     echo " A browser will open. Log in to your Tailscale account, then re-run toggle.sh."
1167 fi
1168 else
1169     fail "tailscaled could NOT be started automatically."
1170     echo " Run this in a terminal to start and authenticate Tailscale:"
1171     echo ""
1172     echo "         sudo brew services start tailscale"
1173     echo "         sudo tailscale up"
1174     echo ""
1175     echo " Then re-run toggle.sh."
1176 fi
1177 else
1178     echo -e "\n[Warning] tailscale CLI not found on PATH. Skipping host Tailscale configuration..."
1179 fi
1180
1181 # Phase B: Pre-flight checks + enable exit node only if safe
1182 if [ "$HOST_ON_MESH" = "true" ] && [ "$USE_EXIT_NODE" != "false" ] && [ -n "$TS_IP" ]; then
1183     echo -e "\n"
1184     echo "Pre-flight connectivity check"
1185     echo ""
1186
1187 # Check 1: Can we reach the gateway via Tailscale (uses tailscale ping, not ICMP)?
1188 # NOTE: tailscale ping exits 1 when the connection goes through a DERP relay
1189 # ("direct connection not established") even though the pong was received.
1190 # We check for 'pong' in the output (stderr discarded) to accept relayed connections.
1191 echo -n " [1/3] Gateway reachable via Tailscale... "
1192 ping_ok=false
1193 # The host tailscaled needs a few seconds to propagate the new network map and
1194 # establish a connection route after 'tailscale up --reset'. We retry up to 45 times.
1195 for attempt in {1..45}; do
1196     if $TS_BIN ping --until-direct=false -c 2 --timeout 2s "$TS_IP" 2>/dev/null | grep -q "pong"; then
1197         ping_ok=true
1198         break
1199     fi
1200     sleep 1

```

```

1201 done
1202 if [ "$ping_ok" = "true" ]; then
1203     echo "PASS"
1204 else
1205     echo "FAIL"
1206     SKIP_EXIT_NODE=true
1207 fi
1208
1209 # Check 2: Can we reach AdGuard via localhost (exposed port ${DNS_PROXY_PORT})?
1210 # Colima's SSH tunnel only forwards TCP (not UDP), so we use +tcp.
1211 echo -n " [2/3] AdGuard DNS via localhost... "
1212 if command -v dig &>/dev/null; then
1213     if dig +tcp @127.0.0.1 -p ${DNS_PROXY_PORT} google.com +short +timeout=5 &>/dev/null; then
1214         echo "PASS"
1215     else
1216         echo "FAIL"
1217         SKIP_EXIT_NODE=true
1218     fi
1219 elif command -v nslookup &>/dev/null; then
1220     if nslookup -port=${DNS_PROXY_PORT} google.com 127.0.0.1 &>/dev/null; then
1221         echo "PASS"
1222     else
1223         echo "FAIL"
1224         SKIP_EXIT_NODE=true
1225     fi
1226 else
1227     echo "SKIP (dig/nslookup not available)"
1228 fi
1229
1230 # Check 3: Does the WARP container have external internet?
1231 echo -n " [3/3] WARP container internet... "
1232 tmp_err=$(mktemp)
1233 if docker compose exec -T warp wget -qO- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace >/dev/null 2>
1234 "$tmp_err"; then
1235     echo "PASS"
1236 else
1237     echo "FAIL"
1238     if [ -s "$tmp_err" ]; then
1239         err_msg=$(cat "$tmp_err" | tr -d '\r')
1240         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Check 3] WARP container check failed: $err_msg" >> output.log
1241     fi
1242     SKIP_EXIT_NODE=true
1243 fi
1244 rm -f "$tmp_err"
1245
1246 # Act based on check results
1247 if [ "$SKIP_EXIT_NODE" != "true" ]; then
1248     echo -e "\nAll checks passed. Enabling exit node $TS_IP..."
1249     $TS_BIN up || true
1250     if $TS_BIN set --ssh=true --accept-dns=false --accept-routes=true --exit-node="$TS_IP" --exit-node-allow-
1251 lan-access=true; then
1252         echo "Exit node enabled."
1253         add_warp_bypass_routes
1254         setup_exit_node_routing
1255         enable_killswitch
1256     else
1257         echo "[Warning] Failed to set exit node."
1258         SKIP_EXIT_NODE=true
1259     fi
1260 fi
1261
1262 if [ "$SKIP_EXIT_NODE" = "true" ]; then
1263     warn "Pre-flight checks failed EXIT NODE + DNS HIJACK SKIPPED."
1264     echo " Host stays on Tailscale mesh but your internet routes through your normal connection."
1265     echo " Troubleshoot with:"
1266     echo "     docker compose logs warp | tail -20"
1267     echo "     docker compose exec -T warp wget -qO- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace"
1268     echo " Falling back to SOCKS5 proxy for traffic routing..."
1269 fi
1270
1271 elif [ "$HOST_ON_MESH" = "true" ] && [ "$USE_EXIT_NODE" = "false" ]; then
1272     echo "USE_EXIT_NODE is false. Skipping exit node and DNS hijack."
1273     SKIP_EXIT_NODE=true
1274 fi
1275
1276 # 8. Apply DNS Hijacking.
1277 # If exit node is enabled: hijack DNS to gateway's Tailscale IP (routes through tailnet).
1278 # Otherwise, leave DNS at 1.1.1.1 (already set by reset at the top).
1279 # AdGuard is also available at localhost:${DNS_PROXY_PORT} for manual configuration in apps.
1280 if [ -n "$TS_IP" ] && [ "$HOST_ON_MESH" = "true" ] && [ "$SKIP_EXIT_NODE" != "true" ]; then
1281     echo -e "\nHijacking host DNS to point ONLY at AdGuard Home ($TS_IP)..."

```

```

1280 echo "Setting MagicDNS search domain 'ts.net' so tailnet hostnames resolve..."
1281 if force_dns_to_gateway "$TS_IP"; then
1282     echo "DNS hijack verified: single server = $TS_IP."
1283 else
1284     echo "[Warning] DNS hijack could not be verified."
1285     echo "    If DNS is broken, run manually:"
1286     echo "        networksetup -setdnsservers \"\$ACTIVE_SERVICE\" \"$TS_IP"
1287     echo "        networksetup -setsearchdomains \"\$ACTIVE_SERVICE\" ts.net"
1288 fi
1289 elif [ -n "$TS_IP" ] && [ "$HIJACK_HOST" != "false" ]; then
1290     echo -e "\n[Info] Exit node not enabled (pre-flight checks failed)."
1291     echo "    Trying local DNS proxy for ad-blocking..."
1292     if start_local_dns_proxy; then
1293         echo "    Ad-blocking active via local DNS proxy through AdGuard."
1294     else
1295         echo "    Local DNS proxy unavailable. DNS stays at 1.1.1.1."
1296         echo "    AdGuard available at http://localhost:80 (port ${DNS_PROXY_PORT} for DNS via TCP)."
1297     fi
1298
1299 # Enable SOCKS5 proxy for traffic routing through WARP
1300 # (enabled whenever exit node is skipped, regardless of HOST_ON_MESH)
1301 echo -e "\n    Enabling SOCKS5 proxy for TCP traffic routing through gateway WARP tunnel..."
1302 echo "    (SOCKS5 proxy runs inside gateway container, routes through tun0 -> WARP)"
1303 enable_socks_proxy "$ACTIVE_SERVICE"
1304
1305 # Verify the SOCKS5 proxy works through WARP
1306 echo -n "    Verifying SOCKS5 proxy routes through WARP... "
1307 if curl --socks5-hostname 127.0.0.1:${SOCKS_PROXY_PORT} --max-time 10 -s https://www.cloudflare.com/cdn-cgi/
1308     trace 2>/dev/null | grep -q "warp=on"; then
1309     echo "ok (warp=on)."
1310     echo "    All TCP traffic now encrypted through Cloudflare WARP tunnel."
1311 else
1312     echo "FAILED (proxy not routing through WARP)."
1313     echo "    Disabling SOCKS5 proxy internet stays on direct connection."
1314     disable_socks_proxy
1315     echo "    SOCKS proxy available at localhost:${SOCKS_PROXY_PORT} for manual configuration."
1316 fi
1317 else
1318     echo -e "\n[Warning] Could not resolve gateway Tailscale IP."
1319 fi
1320
1321 # 9. Verify host connectivity
1322 if [ "$HOST_ON_MESH" = "true" ] && [ -n "$TS_BIN" ]; then
1323     if run_with_timeout 5 $TS_BIN status >> output.log 2>&1; then
1324         echo "Host is online on the Tailscale mesh."
1325     else
1326         echo "[Warning] Host is not responding on Tailscale."
1327     fi
1328 fi
1329
1330 ELAPSED=$(( SECONDS - TOGGLE_START_TIME ))
1331 echo -e "\nGateway has been successfully STARTED in ${ELAPSED} seconds."
1332
1333 # Prevent system from going to sleep while gateway is running
1334 start_sleep_prevention
1335
1336 if [ [ "$HIJACK_HOST" != "false" ] ]; then
1337     if [ -n "$TS_IP" ] && [ "$TS_IP" != "1.1.1.1" ]; then
1338         start_dns_watcher "$TS_IP"
1339     fi
1340
1341     # Start WARP liveness monitor (logs to output.log on warp flip)
1342     start_warp_watcher
1343 fi
1344
1345 # Reset sharing services after network configuration to prevent AirDrop freezes
1346 echo "Resetting macOS sharing services (AirDrop/AirPlay)..."
1347 reset_sharing_services
1348
1349 # Tell external watchers (post-wake / network-change) the gateway is up.
1350 write_gateway_active_marker
1351
1352 # Local Network Surveillance Check
1353 echo -e "\n"
1354 echo "Local Network Surveillance Check"
1355 echo ""
1356 # Count populated ARP cache entries to detect network scanning or high congestion
1357 # Using '-an' avoids hanging on DNS reverse resolution when the network state is in transition.
1358 ARP_COUNT=$(arp -an 2>/dev/null | grep -iv 'incomplete' | wc -l | tr -d ' ')
1359 if [ "$ARP_COUNT" -gt 15 ]; then
1360     warn "High ARP activity detected ($ARP_COUNT devices in cache)."
```

```

1360     warn "This Wi-Fi network may be heavily congested or actively scanned (e.g., arp-scan)."
1361     echo -e " [] Your traffic remains fully encrypted and invisible.\n"
1362 else
1363     echo -e " [] Local network looks quiet ($ARP_COUNT devices). No aggressive scanning detected.\n"
1364 fi
1365 fi
1366
1367 SUCCESS_RUN=true
1368
1369 # Final DNS state summary
1370 if [ -n "$ACTIVE_SERVICE" ]; then
1371     FINAL_DNS=$(networksetup -getdnsservers "$ACTIVE_SERVICE" 2>> output.log || true)
1372     echo -e "\n"
1373     echo -e "DNS STATE: $FINAL_DNS"
1374     echo -e ""
1375 fi
1376
1377 if [ "$START_GATEWAY" = "true" ] && command -v docker >/dev/null 2>&1; then
1378     if docker compose logs rule-compiler 2>/dev/null | grep -q "Warning: Failed to fetch"; then
1379         echo -e "\n[Warning] One or more blocklist URLs failed to download (404/Offline)."
1380         echo " The gateway gracefully fell back to yesterday's cached blocklist."
1381         echo " (Run 'docker logs rule-compiler' later to investigate which link died.)"
1382     fi
1383 fi
1384
1385 rm -f "${LOCK_FILE:-/tmp/nullexit-toggle.lock}"
1386 if [ -t 0 ]; then
1387     read -rp "Press [r] and Enter to instantly reverse state, or just press Enter to exit: " USER_CHOICE
1388     if [[ "${USER_CHOICE}" == "r" || "${USER_CHOICE}" == "R" ]]; then
1389         echo "Reversing gateway state..."
1390         exec bash "$0"
1391     fi
1392 fi
1393
1394 echo -e "\nYou can close this terminal window now."

```

7 recover.sh

```

1 #!/bin/bash
2 # recover.sh nullexit KILL-SWITCH recovery script for macOS
3 # Fixes internet when toggle.sh leaves you stranded.
4 #
5 # This is a NUCLEAR option: it undoes EVERYTHING toggle.sh does by:
6 # 1. Disconnecting host Tailscale from the mesh (exit-node off)
7 # 2. Resetting DNS to default on ALL network services
8 # 3. Disabling rogue proxy settings (SOCKS, web, secure web)
9 # 4. Flushing macOS DNS cache
10 # 5. Flushing stale routing table entries
11 # 6. Stopping gateway Docker containers
12 # 7. Power-cycling Wi-Fi
13 # 8. Verifying internet is working
14 #
15 # Usage: double-click Recover-Gateway.applescript
16 #         OR ./recover.sh
17 #         OR ./recover.sh --post-wake (lighter-touch refresh, keeps gateway live)
18
19 set -e
20
21 # Resolve script directory (used for path-relative refs)
22 # recover.sh lives at the repo root; SCRIPT_DIR lets us launch toggle.sh
23 # (and reference any other repo-root file) from the same directory no
24 # matter where the user invoked recover.sh from.
25 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
26 cd "$SCRIPT_DIR" || exit 1
27
28 # Enforce Cryptographic Script Integrity
29 if [ -f "scripts/crypto.sh" ]; then
30     if ! bash scripts/crypto.sh --verify; then
31         exit 1
32     fi
33 fi
34
35 # Argument parsing
36 --post-wake: lightweight refresh after sleep/wake or Wi-Fi roam.
37 #           Keeps the gateway live while HUPping DNS caches, refreshing the
38 #           Tailscale exit-node leak, re-hijacking host DNS, and force-

```

```

39 # recreating the warp container if its gluetun healthcheck failed.
40 # Does NOT tear down Docker, Tailscale, sleep prevention, or Wi-Fi.
41 # Called from scripts/watcher.sh on every wake / network-change event
42 # while /tmp/nullexit-gateway-active.marker is present (i.e. the
43 # gateway is currently up). See devref.md 10.29 for the why.
44 POST_WAKE=false
45 AUTO_MODE=false
46 for arg in "$@"; do
47     if [ "$arg" = "--post-wake" ]; then
48         POST_WAKE=true
49     elif [ "$arg" = "--auto" ] || [ "$arg" = "--non-interactive" ]; then
50         AUTO_MODE=true
51     fi
52 done
53
54 rm -f "$SCRIPT_DIR/TUNNEL_FAILED_CLOSED.marker"
55
56 # Prevent concurrent execution of lifecycle scripts (toggle.sh / recover.sh)
57 LOCK_FILE="/tmp/nullexit-toggle.lock"
58 if [ -f "$LOCK_FILE" ]; then
59     LOCK_PID=$(cat "$LOCK_FILE" 2>/dev/null || echo "")
60     if [ -n "$LOCK_PID" ] && [ "$LOCK_PID" != "$$" ] && kill -0 "$LOCK_PID" 2>/dev/null; then
61         if ps -p "$LOCK_PID" -o args= 2>/dev/null | grep -q -E "toggle\.sh|recover\.sh"; then
62             if [ "$POST_WAKE" = "true" ]; then
63                 echo "[$(date -u +%FT%TZ)] POST-WAKE skipped: another lifecycle script (PID $LOCK_PID) is running." >>
64                 output.log
65                 exit 0
66             else
67                 echo "Error: Another instance of toggle.sh or recover.sh (PID $LOCK_PID) is already running." >&2
68                 exit 1
69             fi
70         fi
71     fi
72     echo "$$" > "$LOCK_FILE"
73
74 # Clear the active-state marker in nuclear recovery mode since the gateway is being
75 # completely torn down. This prevents scripts/watcher.sh from triggering post-wake
76 # recoveries in response to network changes caused by the teardown itself.
77 if [ "$POST_WAKE" = "false" ]; then
78     rm -f "/tmp/nullexit-gateway-active.marker"
79 fi
80
81 # WARP Watcher inhibit marker: written immediately in --post-wake mode so the
82 # background WARP Watcher (in toggle.sh) skips failure counting while we are
83 # intentionally tearing down / recreating the warp container. Without this, the
84 # watcher accumulates 6 consecutive warp=off readings during force-recreate (~35s)
85 # and fires nuclear recover.sh, killing a gateway that would have been healthy.
86 WARP_INHIBIT_FILE="/tmp/nullexit-warp-inhibit.marker"
87 if [ "$POST_WAKE" = "true" ]; then
88     touch "$WARP_INHIBIT_FILE"
89     echo "[$(date -u +%FT%TZ)] POST-WAKE started WARP Watcher inhibited to prevent spurious shutdown during warp
90     recreate." >> output.log
91 fi
92
93 # Ensure the inhibit marker and lock file are always removed when the script exits (success or error)
94 cleanup_recover() {
95     rm -f "$WARP_INHIBIT_FILE"
96     if [ -f "$LOCK_FILE" ] && [ "$(cat "$LOCK_FILE" 2>/dev/null)" = "$$" ]; then
97         rm -f "$LOCK_FILE"
98     fi
99 }
100 trap cleanup_recover EXIT
101
102 # Source common formatting and helper functions
103 source "$SCRIPT_DIR/scripts/common.sh"
104
105 # Always add Homebrew path
106 setup_standard_path
107
108 echo ""
109 if [ "$POST_WAKE" = "true" ]; then
110     echo -e "${YELLOW}${BOLD}${NC}"
111     echo -e "${YELLOW}${BOLD} Nullexit POST-WAKE / POST-ROAM LIGHT ${NC}"
112     echo -e "${YELLOW}${BOLD} (keeps the gateway live, refreshes) ${NC}"
113     echo -e "${YELLOW}${BOLD}${NC}"
114     echo ""
115     echo "Re-hijacking DNS, refreshing Tailscale exit-node, force-recreating"
116     echo "warp if unhealthy, and resetting sharing services. The gateway stays"
117     echo "up. docker compose / Wi-Fi / caffeinate are NOT touched."
118     echo ""

```

```

118 else
119     echo -e "${RED}${BOLD}${NC}"
120     echo -e "${RED}${BOLD} Nullexit Gateway RECOVERY MODE ${NC}"
121     echo -e "${RED}${BOLD}${NC}"
122     echo ""
123     echo "This script will tear down the gateway and restore normal internet."
124     echo ""
125 fi
126
127 # Cache sudo credentials upfront (default mode ONLY)
128 # Skip in --post-wake: the LaunchAgent-launched watcher has no TTY, so `sudo -v`
129 # would block forever waiting for a password. All post-wake commands use `sudo -n`
130 # which is non-blocking and relies on /etc/sudoers.d/nullexit NOPASSWD entries.
131 SUDO_KEEPER_PID=""
132 if [ "$POST_WAKE" = "false" ] && [ "$AUTO_MODE" = "false" ] && [ -t 0 ]; then
133     echo -e "${YELLOW}${BOLD}Authentication required for network recovery...${NC}"
134     sudo -v
135     (
136         while true; do
137             sudo -n true 2>/dev/null
138             sleep 60
139         done
140     ) &
141     SUDO_KEEPER_PID=$!
142     trap 'kill $SUDO_KEEPER_PID 2>/dev/null' EXIT
143 fi
144
145 # Resolve physical default interface and gateway upfront
146 PHYSICAL_IFACE=$(networksetup -listallhardwareports 2>> output.log | awk '/Hardware Port: (Wi-Fi|Ethernet)/{
    getline; print $2; exit}')
147 [ -z "$PHYSICAL_IFACE" ] && PHYSICAL_IFACE="en0"
148
149 if [ "$POST_WAKE" = "true" ]; then
150     wait_for_dhcp_settle
151 else
152     # Quick resolution in non-post-wake mode (non-blocking)
153     PHYSICAL_GW=$(ipconfig getpacket "$PHYSICAL_IFACE" 2>> output.log | awk -F'[{}]' '/router/{print $2}')
154 fi
155
156 # 1. Disconnect host Tailscale (default) / Refresh exit-node (post-wake)
157
158 if [ "$POST_WAKE" = "true" ]; then
159     step "Refreshing host Tailscale exit-node routing (post-wake)"
160     if command -v tailscale >> output.log 2>&1; then
161         # Re-issue the SAME exit-node up command toggle.sh START uses. This refreshes
162         # the DERP relay mapping and re-asserts the exit-node preference without
163         # dropping the host's mesh connection (which `tailscale down` would).
164         TS_IP=""
165         for src in .gateway_ip; do
166             [ -f "$src" ] || continue
167             TS_IP=$(cat "$src" 2>> output.log | tr -d '\r' | awk 'NR==1{print $1; exit}' || true)
168             [ -n "$TS_IP" ] && break
169         done
170
171         if [ -n "$TS_IP" ]; then
172             step "Re-establishing exit node $TS_IP via 'tailscale set --exit-node'"
173             # `tailscale set` only mutates the named preference (exit-node here).
174             # Crucially it does NOT call --reset, which would re-apply ALL flags and
175             # trigger a sub-second route-table re-evaluation cycle each post-wake.
176             # On already-connected nodes, the lighter `set` keeps the existing tunnel
177             # alive while only changing the exit-node preference.
178             if run_with_timeout 15 tailscale set --exit-node="$TS_IP" \
179                 --exit-node-allow-lan-access=true \
180                 >> output.log 2>&1; then
181                 ok "Exit-node $TS_IP re-asserted via 'tailscale set'"
182             else
183                 warn "tailscale set --exit-node=$TS_IP didn't respond; falling back to mesh-only"
184                 run_with_timeout 10 tailscale set --exit-node= \
185                     >> output.log 2>&1 || true
186             fi
187         else
188             warn ".gateway_ip missing cannot re-assert exit node"
189             run_with_timeout 10 tailscale up --reset --ssh=true --accept-dns=false --exit-node= \
190                 >> output.log 2>&1 || true
191         fi
192     else
193         warn "tailscale CLI not found"
194     fi
195 else
196     step "Disconnecting host Tailscale from mesh"
197     disconnect_tailscale_host "tailscale"

```

```

198 fi
199
200 # 2. Reset DNS to default on ALL network services (default) / Re-hijack gateway DNS (post-wake)
201 if [ "$POST_WAKE" = "true" ]; then
202     step "Re-hijacking host DNS to gateway IP (post-wake / post-roam)"
203     TS_IP=""
204     for src in .gateway_ip; do
205         [ -f "$src" ] || continue
206         TS_IP=$(cat "$src" 2>> output.log | tr -d '\r' | awk 'NR==1{print $1; exit}' || true)
207         [ -n "$TS_IP" ] && break
208     done
209
210     if [ -n "$TS_IP" ]; then
211         # Detect the active network service (using helper in common.sh)
212         ACTIVE_SVC=$(get_active_service)
213         ENO_SVC=$(get_eno_service)
214
215         networksetup -setsearchdomains "$ACTIVE_SVC" "ts.net" 2>> output.log || true
216         networksetup -setdnsservers "$ACTIVE_SVC" "$TS_IP" 2>> output.log || true
217         networksetup -setsearchdomains "$ENO_SVC" "ts.net" 2>> output.log || true
218         networksetup -setdnsservers "$ENO_SVC" "$TS_IP" 2>> output.log || true
219
220         HITS=$(networksetup -getdnsservers "$ACTIVE_SVC" 2>> output.log | grep -Fx "$TS_IP" || true)
221         if [ -n "$HITS" ]; then
222             ok "DNS re-hijacked to $TS_IP on services: $ACTIVE_SVC, $ENO_SVC"
223         else
224             warn "networksetup accepted but DNS read-back didn't include $TS_IP"
225         fi
226     else
227         warn ".gateway_ip missing cannot re-hijack DNS (user must run toggle.sh START)"
228     fi
229 else
230     step "Resetting DNS to default on all network services (empty = DHCP)"
231
232     # Get ALL network services (handles spaces in names, skips disabled ones)
233     SERVICES=$(networksetup -listallnetworkservices 2>> output.log | grep -v "^An asterisk" | grep -v "^$" || true)
234
235     if [ -z "$SERVICES" ]; then
236         # Fallback to common names
237         SERVICES="Wi-Fi Ethernet Thunderbolt Ethernet USB 10/100 LAN"
238     fi
239
240     RESET_COUNT=0
241     while IFS= read -r service; do
242         # Strip leading asterisk (disabled services)
243         service_clean=$(echo "$service" | sed 's/^\*//')
244         [ -z "$service_clean" ] && continue
245
246         # Check if this service has a hardware port (skip VPN/service-only entries)
247         if networksetup -listallhardwareports 2>> output.log | grep -A1 "Port: $service_clean$" | grep -q "Device:"
248             || [ "$service_clean" = "Wi-Fi" ]; then
249             (
250                 networksetup -setsearchdomains "$service_clean" "Empty" 2>> output.log
251                 # Set to "empty" so macOS uses DHCP-assigned DNS (not hardcoded 1.1.1.1)
252                 sudo networksetup -setdnsservers "$service_clean" "empty" 2>> output.log
253             ) && RESET_COUNT=$((RESET_COUNT + 1)) || true
254         fi
255     done <<< "$SERVICES"
256
257     ok "DNS reset on $RESET_COUNT service(s)"
258 fi
259
260 # 3. Disable rogue proxy settings (default only gateway uses proxy in non-exit-node path)
261 if [ "$POST_WAKE" = "false" ]; then
262     step "Disabling proxy settings (SOCKS, web, secure web)"
263     for svc in $(networksetup -listallnetworkservices 2>> output.log | grep -v "^An asterisk" | grep -v "^$" ||
264         echo "Wi-Fi"); do
265         svc_clean=$(echo "$svc" | sed 's/^\*//')
266         [ -z "$svc_clean" ] && continue
267         disable_all_proxies "$svc_clean"
268     done
269     ok "Proxy settings cleared"
270 else
271     step "Proxy settings: leaving untouched (post-wake keeps gateway SOCKS wiring alive)"
272 fi
273
274 # 4. Flush macOS DNS cache (always safe and useful in both modes)
275 step "Flushing DNS cache"
276 if command -v dscacheutil >> output.log 2>&1; then
277     flush_dns_cache

```

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276 ok "DNS cache flushed (mDNSResponder HUP)"
277 else
278 warn "dscacheutil not found"
279 fi
280
281 # 5. Clear stale routing table (always safe in both modes)
282 step "Flushing stale routing table entries"
283 # Resolve physical default gateway IP before flushing
284 # We cannot trust `route get default` here because Tailscale may have already hijacked
285 # the default route to utunX. We must read the actual DHCP router assignment from the hardware.
286 # PHYSICAL_IFACE and PHYSICAL_GW resolved upfront
287
288 # Instead of full route -n flush (which destroys macOS loopback/multicast and wedges the OS until reboot),
289 # we cleanly delete the Tailscale overrides and Cloudflare WARP bypass routes in ALL modes.
290 # This ensures tailscaled isn't blocked from reaching its control plane when waking from sleep.
291 sudo -n route delete -net 0.0.0.0/1 >> output.log 2>&1 || true
292 sudo -n route delete -net 128.0.0.0/1 >> output.log 2>&1 || true
293 remove_warp_bypass_routes
294 ok "Routing overrides cleared"
295
296 if [ "$POST_WAKE" = "false" ]; then
297     disable_killswitch
298     ok "Routing table flush completed via targeted deletes and firewall disabled"
299 else
300     ok "Routing overrides cleared (post-wake mode to preserve Colima bridge100)"
301 fi
302
303 # In post-wake mode we don't bounce Wi-Fi, so we MUST manually restore the physical default route
304 if [ "$POST_WAKE" = "true" ] && [ -n "$PHYSICAL_GW" ]; then
305     # Ensure physical gateway is set just in case it was dropped
306     sudo -n route add default "$PHYSICAL_GW" >> output.log 2>&1 || true
307 fi
308
309 # CONDITIONAL BYPASS ROUTE RESTORATION:
310 # If this was a post-wake recovery and the exit node is active, the route flush
311 # just wiped the static bypass routes for the WARP endpoints. We must re-add
312 # them immediately to prevent a routing loop when Tailscale starts sending traffic.
313 if [ "$POST_WAKE" = "true" ]; then
314     if [ -z "${ACTIVE_IF:-}" ]; then
315         ACTIVE_IF=$(route get default 2>> output.log | awk '/interface:/{print $2; exit}')
316         if [ -z "$ACTIVE_IF" ] || [[ "$ACTIVE_IF" =~ ^utun ]] || [[ "$ACTIVE_IF" =~ ^tun ]]; then
317             for i in en0 en1 en2 en3; do
318                 if ifconfig "$i" 2>> output.log | grep -q 'status: active'; then
319                     ACTIVE_IF="$i"; break
320                 fi
321             done
322         fi
323         [ -z "$ACTIVE_IF" ] && ACTIVE_IF="en0"
324     fi
325
326     echo "Re-adding host bypass routes for Cloudflare WARP endpoints..."
327     remove_warp_bypass_routes
328     if [ -n "${PHYSICAL_GW:-}" ]; then
329         add_warp_bypass_routes "$PHYSICAL_GW"
330     else
331         add_warp_bypass_routes "$ACTIVE_IF"
332     fi
333
334     # Also re-apply the default route pointing to the Tailscale interface
335     ts_iface=$(ifconfig 2>> output.log | grep -B4 "inet 100." | grep -E '^[a-z0-9]+' | cut -d: -f1 | head -n 1)
336     if [ -n "$ts_iface" ]; then
337         echo "Re-routing default gateway to $ts_iface using 0.0.0.0/1 split..."
338         # DO NOT delete the physical default route! It crashes Colima.
339         # Use the /1 trick to mathematically override the default route.
340         sudo -n route delete -net 0.0.0.0/1 >> output.log 2>&1 || true
341         sudo -n route delete -net 128.0.0.0/1 >> output.log 2>&1 || true
342         sudo -n route add -net 0.0.0.0/1 -interface "$ts_iface" >> output.log 2>&1 || true
343         sudo -n route add -net 128.0.0.0/1 -interface "$ts_iface" >> output.log 2>&1 || true
344         enable_killswitch
345     fi
346 fi
347
348 # 6. Stop gateway Docker containers (default) / Force-recreate warp if unhealthy (post-wake)
349 if [ "$POST_WAKE" = "true" ]; then
350     # 6a. Check for Roaming Network Mismatch (Bridged vs Enterprise Wi-Fi)
351     if [ -f "/tmp/nullexit_colima_mode.txt" ] && [ -f "$SCRIPT_DIR/../../lan_p2p_detected" ]; then
352         current_colima_mode=$(cat /tmp/nullexit_colima_mode.txt)
353         p2p_allowed=$(cat "$SCRIPT_DIR/../../lan_p2p_detected")
354         required_colima_mode="bridged"
355         [ "$p2p_allowed" == "false" ] && required_colima_mode="shared"
356     fi

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```

357     if [ "$current_colima_mode" != "$required_colima_mode" ]; then
358         warn "Network roaming mismatch detected!"
359         warn "Colima is running in '$current_colima_mode' mode but new network requires '$required_colima_mode'."
360         warn "Executing full gateway restart to protect against MAC-spoofing deauthentication..."
361         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Roam] Mismatch ($current_colima_mode vs $required_colima_mode).
Triggering toggle.sh --restart." >> output.log
362         nohup bash "$SCRIPT_DIR/./toggle.sh" --restart >/dev/null 2>&1 &
363         exit 0
364     fi
365 fi
366
367 step "Checking Colima VM state"
368 colima_status=$(colima status 2>&1)
369 if echo "$colima_status" | grep -qi "running"; then
370     ok "Colima VM is running"
371     echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Colima status check passed." >> output.log
372 else
373     warn "Colima VM is NOT running or unresponsive"
374     echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Colima is unhealthy or dead! Output: $colima_status"
>> output.log
375     echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Attempting to restart Colima..." >> output.log
376     colima restart >> output.log 2>&1 || warn "Failed to restart Colima"
377 fi
378
379 step "Checking warp container health (gluetun UDP tunnel)"
380 echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Inspecting warp container state..." >> output.log
381 # The warp container is the most failure-prone link in the chain at wake/roam:
382 # its UDP NAT binding to Cloudflare's edge (162.159.192.1:2408) silently dies
383 # during a Wi-Fi roam. We verify the tunnel is actually passing traffic via curl.
384 WARP_STATE=$(docker inspect --format '{{.State.Health.Status}}' warp 2>> output.log || echo "missing")
385 echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container status: $WARP_STATE" >> output.log
386
387 if [ "$WARP_STATE" = "missing" ]; then
388     warn "warp container is missing leaving gateway containers alone (full relaunch requires \"$SCRIPT_DIR/
toggle.sh\")"
389     echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container missing from Docker engine. Requires
full toggle.sh launch." >> output.log
390 else
391     tmp_err=$(mktemp)
392     if docker compose exec -T warp wget -q0- --timeout=3 https://www.cloudflare.com/cdn-cgi/trace 2>"$tmp_err"
| grep -q "warp=on"; then
393         ok "warp container is healthy and actively tunneling traffic (no recreate needed)"
394         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container passed live traffic healthcheck (
Cloudflare trace successful)." >> output.log
395     else
396         warn "warp container tunnel is dead (Docker status: $WARP_STATE) force-recreating all gateway containers
to restore network namespace"
397         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container is DEAD or STUCK. Forcing recreation
of gateway stack..." >> output.log
398         if [ -s "$tmp_err" ]; then
399             err_msg=$(cat "$tmp_err" | tr -d '\r')
400             echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Healthcheck failure details: $err_msg" >> output.
log
401         fi
402         docker compose up -d --force-recreate >> output.log 2>&1 || warn "force-recreate failed"
403
404         NEW_STATE=$(docker inspect --format '{{.State.Health.Status}}' warp 2>> output.log || echo "missing")
405         ok "warp container health after recreate: $NEW_STATE"
406         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container health after recreation: $NEW_STATE"
>> output.log
407     fi
408     rm -f "$tmp_err"
409 fi
410 else
411     step "Stopping gateway Docker containers"
412     if command -v docker >> output.log 2>&1 && docker info >> output.log 2>&1; then
413         if [ -f "docker-compose.yml" ]; then
414             docker compose down -t 5 2>> output.log && ok "Containers stopped" || warn "No containers were running"
415         else
416             warn "docker-compose.yml not found in current directory"
417         fi
418     else
419         warn "Docker not available skipping"
420     fi
421 fi
422
423 # 6b. Stopping sleep prevention (caffeinate) default only
424 if [ "$POST_WAKE" = "false" ]; then
425     step "Stopping sleep prevention"
426     stop_pidfile_daemon "$PID_CAFFEINATE" "system sleep prevention"
427 else

```

```

428     step "Sleep prevention: leaving untouched (caffeinate already re-acquired by parent shell)"
429 fi
430
431 # 6c. Stopping DNS Watcher default only
432 if [ "$POST_WAKE" = "false" ]; then
433     step "Stopping DNS Watcher"
434     stop_pidfile_daemon "$PID_DNS_WATCHER" "background DNS Watcher"
435 else
436     step "DNS Watcher: leaving untouched (it keeps re-hijacking DNS on every roam)"
437 fi
438
439 # 7. Power-cycle Wi-Fi default only
440 if [ "$POST_WAKE" = "false" ]; then
441     step "Power-cycling Wi-Fi"
442     bounce_wifi_interfaces
443 else
444     step "Wi-Fi: leaving untouched (post-wake keeps the current Wi-Fi association)"
445 fi
446
447 # 8. Verify (default checks direct internet; post-wake verifies the gateway)
448 if [ "$POST_WAKE" = "true" ]; then
449     step "Verifying gateway is healthy after post-wake refresh"
450
451     # Wait a moment for tailscale + warp healthcheck to settle
452     sleep 3
453
454     GATEWAY_OK=false
455     if command -v dig >> output.log 2>&1; then
456         # Source procedural ports if available
457         if [ -f "/tmp/nullexit-ports.env" ]; then
458             source "/tmp/nullexit-ports.env"
459         fi
460         DNS_PORT=${DNS_PROXY_PORT:-5354}
461
462         # AdGuard is exposed at localhost:${DNS_PORT}. If this resolves, the warp tunnel
463         # is functioning properly because AdGuard routes upstream to warp.
464         if dig +tcp @127.0.0.1 -p "$DNS_PORT" google.com +short +timeout=5 >> output.log 2>&1; then
465             ok "Gateway DNS (AdGuard via localhost:${DNS_PORT}) works"
466             GATEWAY_OK=true
467         else
468             warn "Gateway DNS not responding yet"
469         fi
470     fi
471
472     # Direct host gateway check via Tailscale ping (relayed pong counts as success)
473     if command -v tailscale >> output.log 2>&1; then
474         TS_IP=""
475         [ -f .gateway_ip ] && TS_IP=$(read_adguard_ip || true)
476         if [ -n "$TS_IP" ] && tailscale ping --until-direct=false -c 1 --timeout 4s "$TS_IP" 2>> output.log | grep
477             -q pong; then
478             ok "Gateway $TS_IP reachable via Tailscale"
479             GATEWAY_OK=true
480         else
481             warn "Tailscale ping to gateway did not pong exit-node may still be re-establishing"
482         fi
483     fi
484
485     if [ "$GATEWAY_OK" = "true" ]; then
486         echo -e " ${GREEN}${BOLD} Gateway is healthy after post-wake refresh.${NC}"
487     else
488         echo -e " ${YELLOW}${BOLD} Gateway recovery is in-progress.${NC}"
489         echo "    The route may need another 30s before queries succeed."
490         echo "    If it stays broken for >60s, run: bash \"${SCRIPT_DIR}/toggle.sh\""
491     fi
492 else
493     step "Verifying internet connectivity"
494
495     # Wait a moment for the network to settle
496     sleep 2
497
498     INTERNET_OK=false
499
500     # Try DNS resolution first
501     if host -W 3 google.com 1.1.1.1 >> output.log 2>&1; then
502         ok "DNS resolution works (google.com via 1.1.1.1)"
503         INTERNET_OK=true
504     elif nslookup google.com 1.1.1.1 >> output.log 2>&1; then
505         ok "DNS resolution works (nslookup google.com via 1.1.1.1)"
506         INTERNET_OK=true
507     else
508         warn "DNS resolution check failed will still try ping/curl"

```

```

508 fi
509
510 # Try actual HTTP connectivity
511 if command -v curl >> output.log 2>&1; then
512     if curl -sf --max-time 5 https://1.1.1.1 >> output.log 2>&1; then
513         ok "Internet reachable via HTTP"
514         INTERNET_OK=true
515     elif curl -sf --max-time 5 http://1.1.1.1 >> output.log 2>&1; then
516         ok "Internet reachable via HTTP (plain)"
517         INTERNET_OK=true
518     else
519         warn "HTTP check failed"
520     fi
521 elif command -v ping >> output.log 2>&1; then
522     if ping -c 1 -W 3 1.1.1.1 >> output.log 2>&1; then
523         ok "Internet reachable via ping"
524         INTERNET_OK=true
525     else
526         warn "Ping check failed"
527     fi
528 fi
529 fi
530
531 # Summary
532 echo ""
533 echo -e "${BOLD}${NC}"
534 if [ "$POST_WAKE" = "true" ]; then
535     echo -e "${BOLD}POST-WAKE SUMMARY${NC}"
536 else
537     echo -e "${BOLD}RECOVERY SUMMARY${NC}"
538 fi
539 echo -e "${BOLD}${NC}"
540
541 # Show final DNS state
542 FINAL_DNS=$(networksetup -getdnsservers "Wi-Fi" 2>/dev/null || echo "N/A")
543 echo " DNS (Wi-Fi): $FINAL_DNS"
544
545 FINAL_DNS2=$(networksetup -getdnsservers "Ethernet" 2>/dev/null || true)
546 if [[ -z "$FINAL_DNS2" || "$FINAL_DNS2" == *"not a recognized"* || "$FINAL_DNS2" == *"Error"* ]]; then
547     FINAL_DNS2="N/A (Not configured)"
548 fi
549 echo " DNS (Ethernet): $FINAL_DNS2"
550
551 echo ""
552
553 if [ "$POST_WAKE" = "false" ]; then
554     if [ "$INTERNET_OK" = "true" ]; then
555         echo -e " ${GREEN}${BOLD} Internet is working.${NC}"
556     else
557         echo -e " ${YELLOW}${BOLD} Could not verify internet connectivity.${NC}"
558         echo " This might mean:"
559         echo " - Your Wi-Fi is disconnected from the router"
560         echo " - Tailscale is still interfering (try restarting tailscale:"
561         echo "   brew services restart tailscale)"
562         echo " - You need to re-connect to Wi-Fi in the menu bar"
563         echo " - Try toggling Wi-Fi off/on in the menu bar"
564         echo ""
565         echo " Or try manually:"
566         echo "   networksetup -setdnsservers Wi-Fi empty"
567         echo "   tailscale down"
568         echo "   sudo route delete -net 0.0.0.0/1"
569         echo "   sudo route delete -net 128.0.0.0/1"
570         echo "   brew services restart tailscale"
571     fi
572 fi
573
574 # Reset sharing services to prevent AirDrop freezes after interface changes
575 # This is the SAME step in BOTH modes (post-wake inherits the previous fix from 10.27).
576 echo -e " Resetting macOS sharing services (AirDrop/AirPlay)..."
577 reset_sharing_services
578
579 echo ""
580 if [ "$POST_WAKE" = "true" ]; then
581     echo -e "${YELLOW}${BOLD}Post-wake refresh complete.${NC}"
582     # Remove the WARP Watcher inhibit marker now that the gateway is fully healthy.
583     # The watcher will resume normal liveness monitoring on its next 5s poll.
584     rm -f "$WARP_INHIBIT_FILE"
585     echo "[$(date -u +%FT%TZ)] POST-WAKE complete WARP Watcher inhibit released." >> output.log
586 else
587     echo -e "${GREEN}${BOLD}Recovery complete.${NC}"
588 fi

```

```
589 echo ""
```

8 setup.sh

```
1 #!/bin/bash
2 # setup.sh nullexit one-shot setup script (macOS)
3 # Configures Cloudflare WARP + Tailscale + AdGuard Home from scratch.
4 set -euo pipefail
5
6 # Source common formatting and helper functions
7 source "${dirname "$0"}/scripts/common.sh"
8
9 # Run common installation logic
10 source "${dirname "$0"}/scripts/setup-common.sh"
11
12 # Compile macOS Shortcuts
13 echo -e "\n${BOLD}Compiling macOS Desktop Shortcuts...${NC}"
14 if command -v osacompile &> /dev/null; then
15     osacompile -o "Toggle Gateway.app" "Toggle-Gateway.applescript" >/dev/null 2>&1
16     osacompile -o "Recover Gateway.app" "Recover-Gateway.applescript" >/dev/null 2>&1
17     echo "    Created 'Toggle Gateway.app' and 'Recover Gateway.app'"
18 fi
19
20 # Done
21 echo ""
22 echo -e "${GREEN}${BOLD}${NC}"
23 echo -e "${GREEN}${BOLD}                Setup complete!                ${NC}"
24 echo -e "${GREEN}${BOLD}${NC}"
25 echo ""
26 echo -e "${BOLD}One manual step remaining  approve the exit node:${NC}"
27 echo "    https://login.tailscale.com/admin/machines"
28 echo "    Find your gateway '...' 'Edit route settings'  enable Exit Node"
29 echo ""
30
31 echo -e "${YELLOW}${BOLD}Sudoers / Background Execution Requirements:${NC}"
32 echo "    To allow silent, non-interactive execution of the firewall and network routes (especially during sleep/
wake recovery), you must configure passwordless sudo."
33 echo "    Run this exact command in your terminal:"
34 echo "        echo \"\$USER ALL=(root) NOPASSWD: /sbin/pfctl, /usr/sbin/networksetup, /usr/bin/dsccacheutil, /usr/
bin/killall, /usr/bin/pkill, /bin/kill, /sbin/route, /sbin/ifconfig, /usr/bin/true, /opt/homebrew/bin/brew,
/usr/local/bin/brew, /usr/bin/python3 -I \$PWD/scripts/dns-proxy.py, /opt/homebrew/bin/python3 -I \$PWD/
scripts/dns-proxy.py, /usr/local/bin/python3 -I \$PWD/scripts/dns-proxy.py\" | sudo tee /etc/sudoers.d/
nullexit"
35 echo ""
36
37 if [[ -n "${TS_IP:-}" ]]; then
38     echo -e "${BOLD}AdGuard Home dashboard:${NC}    http://${TS_IP}:3000    (username: admin)"
39     echo ""
40 fi
41
42 echo -e "${BOLD}To toggle the gateway on/off:${NC}"
43 echo "    macOS:    double-click the 'Toggle Gateway' app icon!"
44 echo "    Windows: double-click Toggle-Gateway.bat"
45 echo ""
46
47 if [[ "$OSTYPE" == "darwin*" ]]; then
48     # LuLu localhost filtering check
49     if [ -d "/Applications/LuLu.app" ] || systemextensionsctl list 2>/dev/null | grep -q "com.objective-see.
lulu"; then
50         echo -e "${YELLOW}${BOLD}LuLu Firewall Detected:${NC}"
51         echo "    LuLu's localhost (loopback) filtering is OFF by default. To properly protect the"
52         echo "    local nullexit proxy ports from malicious local processes, you must manually enable"
53         echo "    'Allow Loopback' (or block it selectively) in LuLu's Preferences -> Rules."
54         echo ""
55     fi
56
57     echo -e "${YELLOW}${BOLD}Note on macOS Tailscale Sandboxing:${NC}"
58     echo "    The Tailscale GUI app (tailscale-app) installed on macOS is sandboxed."
59     echo "    While you cannot run a Tailscale SSH server on this Mac host, you can"
60     echo "    still SSH outbound to other devices on the mesh using their MagicDNS"
61     echo "    addresses directly (e.g. ssh user@device.tailnet-name.ts.net)."
62     echo ""
63 fi
```

9 scripts/toggle-linux.sh

```
1 #!/bin/bash
2 # toggle-linux.sh nullexit gateway toggle script for Linux
3 # Automatically handles Docker containers, Colima, DNS hijacking, and Tailscale exit node routing.
4
5 set -e
6
7 # --- PROCEDURAL PORT GENERATION ---
8 # To prevent static port fingerprinting by malware, we randomize exposed ports
9 # on every boot and persist them to a hidden environment file.
10 PORTS_FILE="/tmp/nullexit-ports.env"
11 if [[ "$1" == "" || "$1" == "--restart" || "$1" == "restart" ]]; then
12
13     # Collision-avoidance function
14     GENERATED_PORTS=""
15     get_free_port() {
16         local port
17         while true; do
18             # Linux uses shuf or $RANDOM instead of jot
19             port=$(shuf -i 10000-65000 -n 1 2>/dev/null || echo $((10000 + RANDOM % 55000)))
20             # 1. Prevent self-collision within this loop
21             if [[ "$GENERATED_PORTS" == *"$port"* ]]; then
22                 continue
23             fi
24             # 2. Prevent collision with ANY process on the machine (root daemons, terminal apps, etc)
25             # We use ss to read the kernel socket table directly.
26             if ! ss -lnt | awk '{print $4}' | grep -q ":$port$"; then
27                 echo "$port"
28                 return
29             fi
30         done
31     }
32
33     export TOR SOCKS_PORT=$(get_free_port)
34     GENERATED_PORTS="$GENERATED_PORTS $TOR SOCKS_PORT"
35
36     export SOCKS_PROXY_PORT=$(get_free_port)
37     GENERATED_PORTS="$GENERATED_PORTS $SOCKS_PROXY_PORT"
38
39     export DNS_PROXY_PORT=$(get_free_port)
40
41     echo "export TOR SOCKS_PORT=$TOR SOCKS_PORT" > "$PORTS_FILE"
42     echo "export SOCKS_PROXY_PORT=$SOCKS_PROXY_PORT" >> "$PORTS_FILE"
43     echo "export DNS_PROXY_PORT=$DNS_PROXY_PORT" >> "$PORTS_FILE"
44 elif [ -f "$PORTS_FILE" ]; then
45     # On STOP, source the existing ports so docker-compose down matches
46     source "$PORTS_FILE"
47 else
48     # Fallbacks if file is lost
49     export TOR SOCKS_PORT=9050
50     export SOCKS_PROXY_PORT=1080
51     export DNS_PROXY_PORT=5354
52 fi
53
54 # Start execution timer
55 TOGGLE_START_TIME=$SECONDS
56
57 if [[ "$1" == "restart" ]] || [[ "$1" == "--restart" ]]; then
58     echo "Executing Gateway Restart Sequence..."
59     # We must source common.sh here temporarily to check is_gateway_active before we proceed
60     source "$(dirname "${BASH_SOURCE[0]}")/common.sh"
61     if is_gateway_active; then
62         echo "Gateway is currently running. Stopping it first..."
63         bash "$0"
64         echo "Gateway stopped. Now starting it..."
65     else
66         echo "Gateway is already stopped. Starting it..."
67     fi
68     bash "$0"
69     exit 0
70 fi
71
72 # Global flag to track if the script completed successfully
73 SUCCESS_RUN=false
74
75 # Global variable to track the currently running background command PID
76 CURRENT_BG_PID=""
77
```

```

78 # Source common bash functions
79 source "${dirname "${BASH_SOURCE[0]}"}"/common.sh"
80
81
82 PID_FILE="/tmp/nullexit-cafeinate.pid"
83
84 # Start macOS caffeinate to prevent sleep while gateway is running
85 start_sleep_prevention() {
86     if ! command -v systemd-inhibit >/dev/null 2>&1; then
87         echo " [Warning] systemd-inhibit tool not found. Sleep prevention unavailable."
88         return 1
89     fi
90
91     # Stop any existing sleep prevention process first to avoid duplicates
92     stop_sleep_prevention
93
94     echo " Enabling system sleep prevention (systemd-inhibit)..."
95     # Run systemd-inhibit wrapped in a bash trap to prevent idle system sleep.
96     # Critically, this traps shutdown signals (SIGTERM) and automatically
97     # flushes hijacked DNS back to normal right before the PC powers off.
98     # We do not use recover-linux.sh here because it is a heavy recovery script
99     # which takes too long and gets terminated before it can reset the DNS. Instead,
100    # we surgically and instantly restore normal DNS settings here.
101    nohup bash -c "
102        trap '
103            echo \"[Shutdown Trap] Reverting network settings...\" >> \"\$PWD/output.log\" 2>&1
104            kill \$! 2>/dev/null || true
105            sudo -n resolvectl domain \"\$ACTIVE_SERVICE\" \"\" >> \"\$PWD/output.log\" 2>&1 || true
106            sudo -n resolvectl revert \"\$ACTIVE_SERVICE\" >> \"\$PWD/output.log\" 2>&1 || true
107            resolvectl domain \"\$ACTIVE_SERVICE\" \"\" >> \"\$PWD/output.log\" 2>&1 || true
108            resolvectl revert \"\$ACTIVE_SERVICE\" >> \"\$PWD/output.log\" 2>&1 || true
109            if command -v tailscale >/dev/null 2>&1; then
110                tailscale up --accept-dns=false --exit-node= >> \"\$PWD/output.log\" 2>&1 &
111                TS_DOWN_ARGS=\"\"
112                if [ \"\$KILL_SWITCH\" = \"true\" ]; then TS_DOWN_ARGS=\"--accept-risk=lose-ssh\"; fi
113                tailscale down \$TS_DOWN_ARGS >> \"\$PWD/output.log\" 2>&1 &
114            fi
115            sleep 1
116            echo \"[Shutdown Trap] Cleanup complete.\" >> \"\$PWD/output.log\" 2>&1
117            exit 0
118        ' SIGTERM SIGINT SIGHUP
119        systemd-inhibit --what=sleep --who=nullexit --why=\"gateway running\" --mode=block sleep infinity &
120        wait \$!
121        \" >> output.log 2>&1 &
122        local caffe_pid=$!
123        echo \"\$caffe_pid\" > \"\$PID_FILE\"
124        echo \" Sleep prevention active (PID \$caffe_pid). Your PC won't sleep while the gateway is running.\"
125    }
126
127 # Stop sleep prevention when gateway is stopped
128 stop_sleep_prevention() {
129     if [ -f \"\$PID_FILE\" ]; then
130         local caffe_pid
131         caffe_pid=$(cat \"\$PID_FILE\")
132         if [ -n \"\$caffe_pid\" ] && kill -0 \"\$caffe_pid\" 2>/dev/null && ps -p \"\$caffe_pid\" -o command= 2>/dev/null |
133             grep -q -E \"systemd-inhibit|recover-linux.sh\"; then
134             echo \" Stopping system sleep prevention (systemd-inhibit wrapper PID \$caffe_pid)..."
135             kill \"\$caffe_pid\" 2>/dev/null || true
136         fi
137         rm -f \"\$PID_FILE\"
138     fi
139 }
140
141 DNS_WATCHER_PID_FILE="/tmp/nullexit-dns-watcher.pid"
142
143 start_dns_watcher() {
144     local target_ip=$1
145     stop_dns_watcher
146     echo \" Starting background DNS Watcher for seamless Wi-Fi roaming..."
147     nohup bash -c "
148         trap 'exit 0' SIGTERM SIGINT SIGHUP
149         while true; do
150             if [ -n \"\$ACTIVE_SERVICE\" ]; then
151                 CURRENT_DNS=$(resolvectl dns \"\$ACTIVE_SERVICE\" 2>/dev/null | grep -oE
152                 '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | head -1)
153                 if [ \"\$CURRENT_DNS\" != \"\$target_ip\" ]; then
154                     resolvectl dns \"\$ACTIVE_SERVICE\" \"\$target_ip\" >/dev/null 2>&1 || true
155                 fi
156                 sleep 30
157             done

```

```

157 " >> output.log 2>&1 &
158 echo $! > "$DNS_WATCHER_PID_FILE"
159 }
160
161 stop_dns_watcher() {
162 if [ -f "$DNS_WATCHER_PID_FILE" ]; then
163     local watcher_pid
164     watcher_pid=$(cat "$DNS_WATCHER_PID_FILE")
165     if [ -n "$watcher_pid" ] && kill -0 "$watcher_pid" 2>/dev/null; then
166         echo " Stopping background DNS Watcher (PID $watcher_pid)..."
167         kill -9 "$watcher_pid" 2>/dev/null || true
168     fi
169     rm -f "$DNS_WATCHER_PID_FILE"
170 fi
171 }
172
173 # Cleanup handler to restore DNS to 1.1.1.1 on error or user interrupt (Ctrl+C / SIGTERM / SIGHUP)
174 cleanup_handler() {
175     local exit_code=$?
176     local trigger_type="$1"
177     local line_no="$2"
178
179     if [ "$SUCCESS_RUN" = "false" ]; then
180         echo -e "\n[Self-Correction] Script interrupted or failed ($trigger_type). Restoring host DNS to 1.1.1.1..."
181
182         if [ -n "$ACTIVE_SERVICE" ]; then
183             reset_dns
184         fi
185
186         if [ -n "$CURRENT_BG_PID" ]; then
187             echo "Terminating active command (PID $CURRENT_BG_PID)..."
188             kill -15 "$CURRENT_BG_PID" 2>> output.log || true
189         fi
190
191         # Disconnect host Tailscale and close the GUI application on failure
192         disconnect_tailscale_host
193
194         # Stop local DNS proxy if running
195         stop_local_dns_proxy
196
197         # Stop sleep prevention
198         stop_sleep_prevention
199         stop_dns_watcher
200
201         # Nuke leftover network state (proxies, routes, DNS cache, Wi-Fi)
202         if [ -n "$ACTIVE_SERVICE" ]; then
203             cleanup_network_state
204         fi
205
206         if [ "$trigger_type" = "ERR" ]; then
207             echo -e "\n=====
208             echo "ERROR: Script failed at line $line_no with exit code $exit_code."
209             echo "=====
210             echo "Troubleshooting steps:"
211             echo "1. Run 'colima status' to verify the VM is active."
212             echo "2. Run 'docker compose ps' to check container health."
213             echo "3. Run 'docker compose logs' to view application logs."
214             echo "4. Run 'tailscale status' to check host Tailscale status."
215             echo "=====
216         fi
217     fi
218 }
219
220 trap 'cleanup_handler ERR $LINENO' ERR
221 trap 'cleanup_handler INT' INT
222 trap 'cleanup_handler TERM' TERM
223 trap 'cleanup_handler HUP' HUP
224
225 # NOPASSWD sudo
226 # The user has NOPASSWD in /etc/sudoers.d/nullexit for the specific commands
227 # needed (networksetup, dnsmasq, dscacheutil, killall, pkill, socat).
228 # All privileged calls below use `sudo -n` which will run without prompting.
229 # No credential caching loop is needed.
230
231 # Add Homebrew and standard paths
232 export PATH="/opt/homebrew/bin:/opt/homebrew/sbin:/usr/local/bin:$PATH"
233
234 # Check for local DNS proxy tools (used when tailnet data plane is unavailable)
235 # Uses a Python DNS proxy (handles TCP wire format correctly).
236 # Python3 is built into macOS no external dependencies.

```

```

237 DNS_PROXY_BIN=""
238 if command -v python3 >/dev/null 2>&1; then
239     DNS_PROXY_BIN="python3 -I"
240 elif command -v python >/dev/null 2>&1; then
241     DNS_PROXY_BIN="python -I"
242 fi
243
244 # Path to the DNS proxy script (sibling of this script)
245 DNS_PROXY_SCRIPT="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/dns-proxy.py"
246
247 # Resolve Tailscale CLI on host. We rely on the standalone Homebrew formula
248 # (`brew install tailscale` + `systemctl start tailscaled`) NOT the
249 # Tailscale.app GUI so there is no .app bundle or Network Extension sandbox
250 # to launch, no menu-bar click to perform, and no scutil Network Service to
251 # start manually. tailscaled runs as a per-user LaunchAgent (or LaunchDaemon
252 # if you ran the install with sudo) and the control socket is always available.
253 TS_BIN=""
254 if command -v tailscale >> output.log 2>&1; then
255     TS_BIN="tailscale"
256 fi
257
258 # Baseline: disable Tailscale DNS management at the daemon level.
259 # `tailscale set` writes a persistent preference. However, `tailscale up --reset`
260 # (used in steps 7 and 9) RESETS all unspecified flags to defaults, which would
261 # re-enable `--accept-dns=true` and nullify this `set`.
262 # Therefore, EVERY `tailscale up --reset` call in this file MUST also pass
263 # `--accept-dns=false` explicitly. See steps 7 and 9 for the fix.
264 #
265 # This initial `set` is still useful as a baseline for non-reset operations
266 # (e.g. if the user runs `tailscale up` manually without --reset) and covers
267 # the brief window between this line and the first `--reset` call.
268 #
269 # Runs once at script start while tailscaled is still connected (if it was
270 # left running from a previous toggle), so the pref takes effect before any
271 # disconnection or reconnection logic.
272 if [ -n "$TS_BIN" ]; then
273     $TS_BIN set --accept-dns=false >> output.log 2>&1 || true
274 fi
275
276 # Disconnect host Tailscale from the mesh. With the standalone daemon, this is the
277 # *entire* teardown no need to mess with system network managers.
278 # tailscaled keeps running (as a systemd service), which is intentional:
279 # the next `tailscale up` then reconnects in seconds rather than waiting for a
280 # slow daemon launch. The `tailscale down` call also retains the user's
281 # auth state, so we don't force a browser re-login every cycle.
282 #
283 # Defined early (before the if/else branches and referenced by cleanup_handler's
284 # ERR trap) so that any failure before the main logic can still tear down Tailscale.
285 restart_tailscaled_daemon() {
286     echo " [Tailscale Recovery] tailscaled daemon appears to be wedged/unresponsive."
287     echo " [Tailscale Recovery] Attempting to restart tailscaled service via systemctl..."
288     if run_with_timeout 15 sudo -n systemctl restart tailscaled >> output.log 2>&1; then
289         echo " [Tailscale Recovery] Successfully restarted tailscaled service."
290     else
291         echo " [Tailscale Recovery] WARNING: Failed to restart tailscaled. Manual intervention may be needed."
292     fi
293     sleep 3
294 }
295 disconnect_tailscale_host() {
296     if [ -n "$TS_BIN" ]; then
297         echo "Disconnecting host Tailscale from mesh (tailscaled stays running as system service)..."
298         local ts_args=""
299         if is_kill_switch_enabled; then ts_args="--accept-risk=lose-ssh"; fi
300         if ! run_with_timeout 10 $TS_BIN down $ts_args >> output.log 2>&1; then
301             restart_tailscaled_daemon
302             run_with_timeout 10 $TS_BIN down $ts_args >> output.log 2>&1 || true
303         fi
304     fi
305 }
306
307 # Function to get the active network service name (e.g., "Wi-Fi" or "USB 10/100 LAN")
308 get_active_service() {
309     # Get the interface used for default internet routing
310     local iface
311     iface=$(ip route get 1.1.1.1 2>/dev/null | awk '{print $5; exit}')
312     if [ -z "$iface" ]; then
313         echo ""
314         return 1
315     fi
316     echo "$iface"
317 }

```

```

318
319 ACTIVE_SERVICE=$(get_active_service)
320
321 # Resolve the service name for en0 (usually "Wi-Fi") macOS scutil DNS resolver
322 # is commonly scoped to en0, so per-service DNS changes on other interfaces are
323 # ignored unless en0's service is also updated.
324 # Helper to restore host DNS to a clean state (used on script start, on failure,
325 # and after a successful STOP). Without this, a successful toggle-off leaves the
326 # host's resolver pinned to a now-dead gateway IP every lookup stalls for macOS's
327 # DNS timeout (~5s) before falling through to whatever next server is configured.
328 # With it, we always leave the host in `1.1.1.1 / no search domain`.
329 reset_dns() {
330     if [ -n "$ACTIVE_SERVICE" ]; then
331         resolvectl domain "$ACTIVE_SERVICE" "" 2>/dev/null || true
332         resolvectl revert "$ACTIVE_SERVICE" 2>/dev/null || true
333     fi
334 }
335
336 # Helper to nuke leftover network state after teardown (proxy settings, routing
337 # table, DNS cache, Wi-Fi). Prevents the "had to write a whole recovery script"
338 # problem where stale state kills internet after the gateway stops.
339 cleanup_network_state() {
340     echo -e "\nCleaning up network state (proxies, routes, DNS cache, Wi-Fi)..."
341
342     # 1. Disable any leftover proxy settings (needs root on many macOS versions)
343     # (Skipped for Linux since we don't set global SOCKS proxy)
344     echo " Proxies disabled."
345
346     # 2. Flush DNS cache
347     if command -v resolvectl >> output.log 2>&1; then
348         sudo -n resolvectl flush-caches 2>> output.log || true
349     elif command -v systemd-resolve >> output.log 2>&1; then
350         sudo -n systemd-resolve --flush-caches 2>> output.log || true
351     fi
352     echo " DNS cache flushed."
353
354     # 3. Flush stale routing table entries
355     if command -v ip >> output.log 2>&1; then
356         sudo -n ip route flush cache >> output.log 2>&1 || true
357     fi
358     echo " Routing table flushed."
359
360     # 4. Power-cycle Wi-Fi to clear any lingering interface state
361     echo " Restarting network interface..."
362     sudo -n ip link set dev "$ACTIVE_SERVICE" down 2>> output.log || true
363     sleep 1
364     sudo -n ip link set dev "$ACTIVE_SERVICE" up 2>> output.log || true
365     echo " Interface power-cycled ($ACTIVE_SERVICE)."
366     sleep 3
367
368     echo "Network state cleanup complete."
369 }
370
371 # SOCKS5 Proxy (WARP Tunnel)
372 # When Tailscale's exit node can't route through WARP (due to userspace
373 # forwarding), we use a SOCKS5 proxy running inside the warp container's
374 # network namespace. Connections created by this proxy go through the
375 # kernel routing table (table 200 -> tun0 -> WARP), unlike Tailscale's
376 # userspace exit node which bypasses it.
377 #
378 # The SOCKS5 proxy is exposed on localhost:${SOCKS_PROXY_PORT} via Docker port mapping.
379 # We route system TCP traffic through the proxy, which then goes through WARP.
380 #
381 # IMPORTANT: CLI tools like curl do NOT respect macOS system proxy settings.
382 # They must use --socks5-hostname or the ALL_PROXY env variable explicitly.
383
384 enable_socks_proxy() {
385     local svc="$1"
386     echo " (Linux global SOCKS proxy configuration is desktop-environment specific, skipping.)"
387     echo " SOCKS5 proxy is active and listening on 127.0.0.1:${SOCKS_PROXY_PORT}"
388 }
389
390 disable_socks_proxy() {
391     local svc="$1"
392     echo " (Linux global SOCKS proxy configuration skipped.)"
393 }
394
395 # Local DNS Proxy (Python)
396 # When the tailnet data plane cannot establish, we use a Python DNS proxy.
397 # It listens on UDP:53, receives DNS queries, forwards them over TCP to
398 # AdGuard at 127.0.0.1:${DNS_PROXY_PORT} (Docker port mapping), and returns the response.

```

```

399 # The Python script handles the DNS-over-TCP wire format (2-byte length prefix)
400 # correctly unlike socat which just forwards raw bytes.
401 #
402 # Python3 is built into macOS no external dependencies.
403 DNS_PROXY_PID=""
404
405 # Kill any leftover DNS proxy process
406 stop_local_dns_proxy() {
407     if [ -n "$DNS_PROXY_PID" ]; then
408         kill "$DNS_PROXY_PID" 2>/dev/null || true
409         wait "$DNS_PROXY_PID" 2>/dev/null || true
410         DNS_PROXY_PID=""
411     fi
412     # Clean up any stale Python DNS proxy processes
413     sudo -n pkill -f "dns-proxy.py" 2>/dev/null || true
414 }
415
416 # Start local DNS proxy (Python)
417 start_local_dns_proxy() {
418     if [ -z "$DNS_PROXY_BIN" ]; then
419         echo " Python not found. Python3 is built into macOS this shouldn't happen."
420         return 1
421     fi
422
423     if [ ! -f "$DNS_PROXY_SCRIPT" ]; then
424         echo " DNS proxy script not found at $DNS_PROXY_SCRIPT"
425         return 1
426     fi
427
428     # Kill any leftover process first
429     stop_local_dns_proxy
430
431     echo -n " Starting local DNS proxy via Python (UDP:53 TCP:${DNS_PROXY_PORT})... "
432     sudo -n bash -c "TARGET_PORT=$DNS_PROXY_PORT \"$DNS_PROXY_BIN\" \"$DNS_PROXY_SCRIPT\" \" &
433     DNS_PROXY_PID=$!
434     disown "$DNS_PROXY_PID" 2>/dev/null || true
435
436     # Give it a moment to bind
437     sleep 0.5
438
439     if kill -0 "$DNS_PROXY_PID" 2>/dev/null; then
440         echo "started (PID $DNS_PROXY_PID).\"
441
442         # Hijack host DNS to localhost
443         echo -n " Hijacking host DNS to 127.0.0.1 for ad-blocking... \"
444         resolvectl domain \"$ACTIVE_SERVICE\" \" 2>/dev/null || true
445         resolvectl domain \"$ACTIVE_SERVICE\" \" 2>/dev/null || true
446         if ! sudo -n resolvectl dns \"$ACTIVE_SERVICE\" 127.0.0.1; then
447             echo "FAILED (resolvectl error check permissions).\"
448             stop_local_dns_proxy
449             return 1
450         fi
451         resolvectl dns \"$ACTIVE_SERVICE\" 127.0.0.1 2>/dev/null || true
452         sudo -n resolvectl flush-caches 2>/dev/null || true
453
454         # Verify DNS actually works through the proxy
455         echo -n " Verifying DNS resolution... \"
456         if dig @127.0.0.1 google.com +short +timeout=5 &>/dev/null; then
457             echo "ok.\"
458             return 0
459         else
460             echo "FAILED (DNS queries not reaching AdGuard).\"
461             echo " Restoring DNS to 1.1.1.1...\"
462             reset_dns
463             stop_local_dns_proxy
464             return 1
465         fi
466     else
467         echo "FAILED (could not bind port 53 is it in use?).\"
468         DNS_PROXY_PID=""
469         return 1
470     fi
471 }
472
473 # Helper to FORCIBLY hijack host DNS to a single server (the gateway's AdGuard IP)
474 # and VERIFY via `networksetup -getdnsservers` that the only name server is exactly
475 # $1. Returns 0 iff the live state matches; non-zero if it can't be confirmed.
476 #
477 # This deliberately does NOT append a 1.1.1.1 fallback: macOS's resolver queries
478 # the list in order and falls back to the next entry on timeout, so on a
479 # misbehaving AdGuard (filter compile crash, OOM kill, etc.) the resolver still

```

```

480 # leaks to 1.1.1.1 and silently bypasses ad-blocking. With no fallback, a broken
481 # gateway manifests as a *visible* DNS outage that prompts the user to
482 # investigate which is the correct failure mode.
483 force_dns_to_gateway() {
484     local target_ip="$1"
485
486     for attempt in {1..3}; do
487         echo -n " [force_dns] Attempt $attempt/3: setting DNS to $target_ip on $ACTIVE_SERVICE... "
488         sudo -n resolvectl domain "$ACTIVE_SERVICE" "-ts.net" 2>/dev/null || true
489         if ! sudo -n resolvectl dns "$ACTIVE_SERVICE" "$target_ip"; then
490             echo "FAILED (resolvectl error)"
491         else
492             # Verify
493             entries=$(resolvectl dns "$ACTIVE_SERVICE" 2>> output.log | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+')
494             if [ "$entries" == "$target_ip" ]; then
495                 echo "VERIFIED"
496                 return 0
497             else
498                 echo "MISMATCH (Resolver refused lock)"
499             fi
500         fi
501         sleep 2
502     done
503     return 1
504 }
505
506 # Ensure DNS is set to 1.1.1.1 immediately at script start to prevent any DNS deadlocks/hangs
507 # during status checks, container startup, VM boot, or teardown. We *also* clear any leftover
508 # `ts.net` search domain from a previous `tailscale up --accept-dns=true` run, otherwise macOS
509 # would prepend it to every lookup and most public DNS queries would resolve to NXDOMAIN.
510 # Capture current DNS before resetting so we can check if it was hijacked
511 INITIAL_DNS=$(resolvectl dns "$ACTIVE_SERVICE" 2>/dev/null | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | head
512 -1 || true)
513
514 echo "Initializing DNS to 1.1.1.1 to ensure reliable internet access..."
515 reset_dns
516
517 # Local Network Surveillance Check
518 echo -e "\n"
519 echo "Local Network Surveillance Check"
520 echo ""
521 # Count populated ARP cache entries to detect network scanning or high congestion
522 if command -v ip >/dev/null 2>&1; then
523     ARP_COUNT=$(ip neigh | grep -iv 'incomplete' | wc -l | tr -d ' ')
524 else
525     # Using '-an' avoids hanging on DNS reverse resolution when the network state is in transition.
526     ARP_COUNT=$(arp -an 2>/dev/null | grep -iv 'incomplete' | wc -l | tr -d ' ')
527 fi
528 if [ "$ARP_COUNT" -gt 15 ]; then
529     echo -e " \033[1;33m[!] WARNING: High ARP activity detected ($ARP_COUNT devices in cache).\033[0m"
530     echo -e " \033[1;33m[!] This Wi-Fi network may be heavily congested or actively scanned (e.g., arp-scan)
531     .\033[0m"
532     echo -e " [] Your traffic remains fully encrypted and invisible.\n"
533 else
534     echo -e " [] Local network looks quiet ($ARP_COUNT devices). No aggressive scanning detected.\n"
535 fi
536
537 echo "Checking Gateway Status..."
538 if is_gateway_active; then
539     echo -e "\n=====
540     echo "Gateway is RUNNING. Stopping it now..."
541     echo -e "=====\\n"
542
543 # 1. Disconnect host Tailscale cleanly first before stopping the exit-node container
544 disconnect_tailscale_host
545 echo ""
546
547 echo "Stopping Docker containers..."
548 docker compose down -t 5
549
550 echo -e "\n\\nDocker daemon handles container teardown automatically."
551
552 # The host's DNS was hijacked to the gateway IP during ENABLE; now that the
553 # gateway is down, restore DNS to 1.1.1.1 immediately so subsequent lookups
554 # don't stall for the macOS DNS timeout before the 1.1.1.1 fallback engages.
555 echo -e "\n\\nRestoring host DNS to 1.1.1.1 (gateway is gone)... "
556 reset_dns
557
558 # 3. Stop local DNS proxy if running
559 stop_local_dns_proxy

```

```

559
560 # Stop sleep prevention
561 stop_sleep_prevention
562 stop_dns_watcher
563
564 # 4. Nuke leftover network state so internet actually works after teardown
565 cleanup_network_state
566
567 ELAPSED=$(( SECONDS - TOGGLE_START_TIME ))
568 echo -e "\nGateway has been successfully STOPPED in ${ELAPSED} seconds."
569 else
570     echo -e "\n=====
571     echo "Gateway is STOPPED. Starting it now..."
572     echo -e "=====\\n"
573
574 # 1. Prevent host exit-node deadlock during VM / Container startup
575 disconnect_tailscale_host
576
577 echo -e "\nUsing native Linux Docker..."
578
579 # 4. Clean up corrupted AdGuardHome configurations
580 if [ -f "adguard/conf/AdGuardHome.yaml" ] && [ ! -s "adguard/conf/AdGuardHome.yaml" ]; then
581     rm -f "adguard/conf/AdGuardHome.yaml"
582     echo "Removed corrupted empty AdGuardHome.yaml to prevent container crash loop."
583 fi
584
585
586 # 5. Start compose services
587 write_host_ips
588 # Procedurally generated ports have already been exported at the top of the script.
589 echo " [Security] Procedurally randomized proxy ports generated to evade fingerprinting."
590
591 # --- HONEY-PORT TRIPWIRE ---
592 # Generate a random trap port. If any malware does a sequential port scan on localhost,
593 # it will inevitably hit this port. When it does, we instantly fire a desktop alert.
594 HONEY_PORT=$(get_free_port)
595 (
596     if nc -l -p "$HONEY_PORT" 127.0.0.1 >/dev/null 2>&1 || nc -l localhost "$HONEY_PORT" >/dev/null 2>&1; then
597         echo "[$(date +%Y-%m-%d %H:%M:%S)] [CRITICAL SECURITY ALERT] PORT SCAN DETECTED! An unknown application
598             just pinged the local Honey-Port ($HONEY_PORT)!" >> "$PWD/output.log"
599         if command -v notify-send >/dev/null 2>&1; then
600             notify-send -u critical "nullexit OPSEC Alert" "An unknown application is aggressively scanning your
601             local ports. Malware port-scan suspected."
602         fi
603     ) &
604     disown %1 2>/dev/null || true
605     echo " [Security] Honey-Port Tripwire armed on random port to detect malware port-scans."
606
607     echo -e "\nStarting Docker containers..."
608     docker compose up -d
609
610 # Clean up the one-off rule compiler container so it doesn't clutter the Docker UI
611 docker compose rm -s -f rule-compiler >> output.log 2>&1
612
613 # 6. Wait for the gateway container's Tailscale connection to be ready
614 BYPASS_PING=$(read_env_var GATEWAY_BYPASS_PING | tr '[:upper:]' '[:lower:]')
615 USE_EXIT_NODE=$(read_env_var GATEWAY_USE_EXIT_NODE | tr '[:upper:]' '[:lower:]')
616
617 echo "Waiting for gateway container's Tailscale to connect to the tailnet..."
618 echo " (This can take 30-60s Tailscale goes through: NoState Starting Running Online)"
619 connected=false
620 consecutive_failures=0
621 for i in {1..60}; do
622     # Run tailscale status and capture its output so the user can see what's happening.
623     # Previously this was hidden behind 2>> output.log, making it impossible to debug hangs.
624     ts_output=$(run_with_timeout 5 docker compose exec -T tailscale tailscale status 2>&1 || true)
625
626     if echo "$ts_output" | grep -q "offers exit node"; then
627         connected=true
628         break
629     fi
630
631 # Track consecutive failures to detect a stuck state.
632 # If we see 40+ consecutive 'NoState' responses, give up early
633 # the daemon is alive but can't connect to the coordination server.
634 if echo "$ts_output" | grep -q "NoState"; then
635     consecutive_failures=$((consecutive_failures + 1))
636     if [ "$consecutive_failures" -ge 40 ]; then
637         echo ""

```

```

638     echo " [attempt $i/60] Stuck in 'NoState' for $consecutive_failures checks."
639     echo "     Tailscale daemon is running but can't reach the coordination server."
640     break
641 fi
642 else
643     consecutive_failures=0
644 fi
645
646 # Print a condensed status every 10 seconds so the user can see progress
647 if [ $((i % 10)) -eq 0 ]; then
648     ts_short=$(echo "$ts_output" | head -1 | tr -d '\r\n')
649     echo " [attempt $i/60] $ts_short"
650 elif [ $((i % 5)) -eq 0 ]; then
651     echo -n "$[i]"
652 else
653     echo -n "."
654 fi
655 sleep 1
656 done
657 echo ""
658
659 if [ "$connected" = "true" ]; then
660     echo "Gateway container is online on the Tailscale mesh."
661 elif [ "$BYPASS_PING" = "true" ]; then
662     echo "[Warning] Gateway container check timed out. Proceeding anyway (GATEWAY_BYPASS_PING is true)..."
663 else
664     echo "ERROR: Gateway container failed to initialize Tailscale (timed out after 60s)." >&2
665     echo " The container was seen in the following states during the wait:" >&2
666     echo "$ts_output" | sed 's/^/ /' >&2
667     echo "" >&2
668     echo " This could mean:" >&2
669     echo " 1. Tailscale auth key has expired check TS_AUTHKEY in .env" >&2
670     echo " 2. Network issue inside the container check: docker compose logs tailscale" >&2
671     echo " 3. gluetun VPN tunnel not connecting check: docker compose logs warp | tail -20" >&2
672     echo "" >&2
673     echo " Diagnose manually:" >&2
674     echo " docker compose exec -T tailscale tailscale status" >&2
675     echo " docker compose exec -T tailscale tailscale netcheck" >&2
676 cleanup_handler ERR $LINENO
677 exit 1
678 fi
679
680 # 6b. Resolve the gateway's Tailscale IP.
681 # Dynamically query the container for the IP (handles IP changes after re-auth)
682 # and cache it in .gateway_ip for fast retrieval by recover.sh during Wi-Fi roaming.
683 #
684 # CRITICAL: `docker compose exec -T` on macOS/Colima injects carriage
685 # returns (\r) into captured output. If $TS_IP ends with \r, then
686 # `networksetup -setdnsservers` silently rejects the malformed IP and
687 # DNS stays at 1.1.1.1 the primary bug this project was hitting.
688 # `tr -d '\r'` strips the CR; `awk 'NR==1{print $1; exit}'` takes only
689 # the first field of the first line.
690 TS_IP=""
691 if [ "$connected" = "true" ]; then
692     TS_IP=$(run_with_timeout 10 docker compose exec -T tailscale tailscale ip -4 2>> output.log | tr -d '\r' |
693     awk 'NR==1{print $1; exit}' || true)
694     if [ -n "$TS_IP" ]; then
695         echo "Resolved gateway Tailscale IP dynamically: $TS_IP"
696         echo "$TS_IP" > .gateway_ip
697     else
698         echo " [!] Failed to resolve gateway IP dynamically. Trying fallback cache..." >&2
699         TS_IP=$(read_adguard_ip || true)
700     fi
701 else
702     TS_IP=$(read_adguard_ip || true)
703 fi
704
705 # 7. Connect host Mac to Tailscale mesh.
706 # With the standalone tailscaled daemon (registered as a per-user LaunchAgent or
707 # system LaunchDaemon via 'systemctl start tailscaled'), the previous five-
708 # phase flow collapses into two trivial CLI calls. There is no .app GUI to launch,
709 # no menu bar item to click, and no Accessibility permission required.
710 #
711 # CRITICAL: If tailscaled is not running, `tailscale up` will silently fail.
712 # We auto-start it before proceeding, and SKIP the rest of the Tailscale setup
713 # if it can't be started (DNS hijack to a 100.x.x.x IP and exit-node routing
714 # are impossible without the host on the mesh).
715 #
716 # We connect in two phases:
717 # Phase A Join mesh WITHOUT exit node (so pre-flight checks can run)
718 # Phase B Set exit node only after pre-flight checks confirm the gateway works

```

```

718 HOST_ON_MESH=false
719 SKIP_EXIT_NODE=false
720 if [ -n "$TS_BIN" ]; then
721     echo -n "Verifying tailscaled is reachable"
722     daemon_ready=false
723     for i in {1..10}; do
724         # `tailscale status` returns exit code 1 when the daemon is alive but
725         # disconnected ("Tailscale is stopped."). We check the daemon socket
726         # directly as a fallback if the socket exists, tailscaled is running
727         # even if the host isn't connected to the mesh yet.
728         if run_with_timeout 5 $TS_BIN status >> output.log 2>&1 || [ -S /var/run/tailscaled.socket ]; then
729             daemon_ready=true
730             break
731         fi
732         echo -n "."
733         sleep 1
734     done
735     echo ""
736
737     if [ "$daemon_ready" != "true" ]; then
738         echo -e "\033[0;33m[!] tailscaled daemon is not running. Attempting to start it...\033[0m"
739
740         # Try system-wide daemon (with timeout sudo may prompt for password)
741         if [ "$daemon_ready" != "true" ]; then
742             if run_with_timeout 10 sudo systemctl start tailscaled 2>> output.log; then
743                 echo "Started tailscaled as system LaunchDaemon."
744                 for i in {1..15}; do
745                     if run_with_timeout 5 $TS_BIN status >> output.log 2>&1; then
746                         daemon_ready=true
747                         break
748                     fi
749                     sleep 1
750                 done
751             fi
752         fi
753     fi
754
755     if [ "$daemon_ready" = "true" ]; then
756         echo "tailscaled is responsive (running as a system service)."
757
758         # Phase A: Join mesh without exit node
759         echo "Connecting host to Tailscale mesh (no exit node yet)..."
760         if $TS_BIN up --reset --ssh=true --accept-dns=false --exit-node=; then
761             HOST_ON_MESH=true
762             echo "Host is on Tailscale mesh."
763         else
764             echo -e "\033[0;33m[!] tailscale up failed even though the daemon is running.\033[0m"
765             echo "This usually means the host hasn't authenticated with your tailnet yet."
766             echo "Run this command in a terminal to authenticate:"
767             echo ""
768             echo "        sudo tailscale up"
769             echo ""
770             echo "A browser will open. Log in to your Tailscale account, then re-run toggle-linux.sh."
771         fi
772     else
773         echo -e "\033[0;31m[!] tailscaled could NOT be started automatically.\033[0m"
774         echo "Run this in a terminal to start and authenticate Tailscale:"
775         echo ""
776         echo "        sudo systemctl start tailscaled"
777         echo "        sudo tailscale up"
778         echo ""
779         echo "Then re-run toggle-linux.sh."
780     fi
781 else
782     echo -e "\n[Warning] tailscale CLI not found on PATH. Skipping host Tailscale configuration..."
783 fi
784
785 # Phase B: Pre-flight checks + enable exit node only if safe
786 if [ "$HOST_ON_MESH" = "true" ] && [ "$USE_EXIT_NODE" != "false" ] && [ -n "$TS_IP" ]; then
787     echo -e "\n"
788     echo "Pre-flight connectivity check"
789     echo ""
790
791     # Check 1: Can we reach the gateway via Tailscale (uses tailscale ping, not ICMP)?
792     # NOTE: tailscale ping exits 1 when the connection goes through a DERP relay
793     # ("direct connection not established") even though the pong was received.
794     # We check for 'pong' in the output (stderr discarded) to accept relayed connections.
795     echo -n " [1/3] Gateway reachable via Tailscale... "
796     ping_ok=false
797     # The host tailscaled needs a few seconds to propagate the new network map and
798     # establish a connection route after 'tailscale up --reset'. We retry up to 45 times.

```

```

799     for attempt in {1..45}; do
800         if $TS_BIN ping --until-direct=false -c 2 --timeout 2s "$TS_IP" 2>/dev/null | grep -q "pong"; then
801             ping_ok=true
802             break
803         fi
804         sleep 1
805     done
806     if [ "$ping_ok" = "true" ]; then
807         echo "PASS"
808     else
809         echo "FAIL"
810         SKIP_EXIT_NODE=true
811     fi
812
813     # Check 2: Can we reach AdGuard via localhost (exposed port ${DNS_PROXY_PORT})?
814     echo -n " [2/3] AdGuard DNS via localhost... "
815     if command -v dig >/dev/null 2>&1; then
816         if dig +tcp @127.0.0.1 -p ${DNS_PROXY_PORT} google.com +short +timeout=5 >/dev/null 2>&1; then
817             echo "PASS"
818         else
819             echo "FAIL"
820             SKIP_EXIT_NODE=true
821         fi
822     elif command -v nslookup >/dev/null 2>&1; then
823         if nslookup -port=${DNS_PROXY_PORT} google.com 127.0.0.1 >/dev/null 2>&1; then
824             echo "PASS"
825         else
826             echo "FAIL"
827             SKIP_EXIT_NODE=true
828         fi
829     else
830         echo "SKIP (dig/nslookup not available)"
831     fi
832
833     # Check 3: Does the WARP container have external internet?
834     echo -n " [3/3] WARP container internet... "
835     if docker compose exec -T warp wget -qO- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace &>/dev/null;
836     then
837         echo "PASS"
838     else
839         echo "FAIL"
840         SKIP_EXIT_NODE=true
841     fi
842
843     # Act based on check results
844     if [ "$SKIP_EXIT_NODE" != "true" ]; then
845         echo -e "\nAll checks passed. Enabling exit node $TS_IP..."
846         if $TS_BIN up --reset --ssh=true --accept-dns=false --exit-node="$TS_IP" --exit-node-allow-lan-access=
847         true; then
848             echo "Exit node enabled."
849         else
850             echo "[Warning] Failed to set exit node."
851             SKIP_EXIT_NODE=true
852         fi
853     fi
854
855     if [ "$SKIP_EXIT_NODE" = "true" ]; then
856         echo -e "\n\033[0;33m[!] Pre-flight checks failed EXIT NODE + DNS HIJACK SKIPPED.\033[0m"
857         echo " Host stays on Tailscale mesh but your internet routes through your normal connection."
858         echo " Troubleshoot with:"
859         echo "     docker compose logs warp | tail -20"
860         echo "     docker compose exec -T warp wget -qO- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace"
861         echo ""
862         echo " Falling back to SOCKS5 proxy for traffic routing..."
863     fi
864
865     elif [ "$HOST_ON_MESH" = "true" ] && [ "$USE_EXIT_NODE" = "false" ]; then
866         echo "USE_EXIT_NODE is false. Skipping exit node and DNS hijack."
867         SKIP_EXIT_NODE=true
868     fi
869
870     # 8. Apply DNS Hijacking.
871     # If exit node is enabled: hijack DNS to gateway's Tailscale IP (routes through tailnet).
872     # Otherwise, leave DNS at 1.1.1.1 (already set by reset at the top).
873     # AdGuard is also available at localhost:${DNS_PROXY_PORT} for manual configuration in apps.
874     if [ -n "$TS_IP" ] && [ "$HOST_ON_MESH" = "true" ] && [ "$SKIP_EXIT_NODE" != "true" ]; then
875         echo -e "\nHijacking host DNS to point ONLY at AdGuard Home ($TS_IP)..."
876         echo "Setting MagicDNS search domain 'ts.net' so tailnet hostnames resolve..."
877         if force_dns_to_gateway "$TS_IP"; then
878             echo "DNS hijack verified: single server = $TS_IP."
879             start_dns_watcher "$TS_IP"
880         else

```

```

878     echo "[Warning] DNS hijack could not be verified."
879     echo "    If DNS is broken, run manually:"
880     echo "        networksetup -setdnsservers \"\$ACTIVE_SERVICE\" \$TS_IP"
881     echo "        networksetup -setsearchdomains \"\$ACTIVE_SERVICE\" ts.net"
882 fi
883 elif [ -n "$TS_IP" ]; then
884     echo -e "\n[Info] Exit node not enabled (pre-flight checks failed)."
885     echo "    Trying local DNS proxy for ad-blocking..."
886     if start_local_dns_proxy; then
887         echo "    Ad-blocking active via local DNS proxy through AdGuard."
888     else
889         echo "    Local DNS proxy unavailable. DNS stays at 1.1.1.1."
890         echo "    AdGuard available at http://localhost:80 (port ${DNS_PROXY_PORT} for DNS via TCP)."
891     fi
892
893     # Enable SOCKS5 proxy for traffic routing through WARP
894     # (enabled whenever exit node is skipped, regardless of HOST_ON_MESH)
895     echo -e "\n    Enabling SOCKS5 proxy for TCP traffic routing through gateway WARP tunnel..."
896     echo "    (SOCKS5 proxy runs inside gateway container, routes through tun0 -> WARP)"
897     enable_socks_proxy "$ACTIVE_SERVICE"
898
899     # Verify the SOCKS5 proxy works through WARP
900     echo -n "    Verifying SOCKS5 proxy routes through WARP... "
901     if curl --socks5-hostname 127.0.0.1:${SOCKS_PROXY_PORT} --max-time 10 -s https://www.cloudflare.com/cdn-cgi/
902         trace 2>/dev/null | grep -q "warp=on"; then
903         echo "ok (warp=on)."
904         echo "    All TCP traffic now encrypted through Cloudflare WARP tunnel."
905     else
906         echo "FAILED (proxy not routing through WARP)."
907         echo "    Disabling SOCKS5 proxy internet stays on direct connection."
908         disable_socks_proxy
909         echo "    SOCKS proxy available at localhost:${SOCKS_PROXY_PORT} for manual configuration."
910     fi
911 else
912     echo -e "\n[Warning] Could not resolve gateway Tailscale IP."
913 fi
914
915 # 9. Verify host connectivity
916 if [ "$HOST_ON_MESH" = "true" ] && [ -n "$TS_BIN" ]; then
917     if run_with_timeout 5 $TS_BIN status >> output.log 2>&1; then
918         echo "Host is online on the Tailscale mesh."
919     else
920         echo "[Warning] Host is not responding on Tailscale."
921     fi
922 fi
923
924 ELAPSED=$(( SECONDS - TOGGLE_START_TIME ))
925 echo -e "\nGateway has been successfully STARTED in ${ELAPSED} seconds."
926
927 # Prevent system from going to sleep while gateway is running
928 start_sleep_prevention
929 fi
930
931 SUCCESS_RUN=true
932
933 # Final DNS state summary
934 if [ -n "$ACTIVE_SERVICE" ]; then
935     FINAL_DNS=$(resolvectl dns "$ACTIVE_SERVICE" 2>/dev/null | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | tr '
936 ' ' || true)
937     echo -e "\n"
938     echo -e "DNS STATE: $FINAL_DNS"
939     echo -e ""
940 fi
941
942 if [ "$START_GATEWAY" = "true" ] && command -v docker >/dev/null 2>&1; then
943     if docker compose logs rule-compiler 2>/dev/null | grep -q "Warning: Failed to fetch"; then
944         echo -e "\n[Warning] One or more blocklist URLs failed to download (404/Offline)."
945         echo "    The gateway gracefully fell back to yesterday's cached blocklist."
946         echo "    (Run 'docker logs rule-compiler' later to investigate which link died.)"
947     fi
948 fi
949
950 echo -e "\n"
951 read -rp "Press [r] and Enter to instantly reverse state, or just press Enter to exit: " USER_CHOICE
952 if [[ "${USER_CHOICE}" == "r" || "${USER_CHOICE}" == "R" ]]; then
953     echo "Reversing gateway state..."
954     exec bash "$0"
955 fi
956
957 echo -e "\nYou can close this terminal window now."

```

10 scripts/recover-linux.sh

```
1 #!/bin/bash
2 # recover.sh nullexit KILL-SWITCH recovery script for Linux
3 # Fixes internet when toggle.sh leaves you stranded.
4 #
5 # This is a NUCLEAR option: it undoes EVERYTHING toggle.sh does by:
6 # 1. Disconnecting host Tailscale from the mesh (exit-node off)
7 # 2. Resetting DNS to default on ALL network services
8 # 3. Disabling rogue proxy settings (SOCKS, web, secure web)
9 # 4. Flushing macOS DNS cache
10 # 5. Flushing stale routing table entries
11 # 6. Stopping gateway Docker containers
12 # 7. Power-cycling Wi-Fi
13 # 8. Verifying internet is working
14 #
15 # Usage: double-click Recover-Gateway.applescript
16 # OR ./recover.sh
17
18 set -e
19
20 # Source common formatting and helper functions
21 source "$(dirname "$0")/common.sh"
22
23 # Always add Homebrew path
24 export PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:$PATH"
25
26 echo ""
27 echo -e "${RED}${BOLD}${NC}"
28 echo -e "${RED}${BOLD} Nullexit Gateway RECOVERY MODE ${NC}"
29 echo -e "${RED}${BOLD}${NC}"
30 echo ""
31 echo "This script will tear down the gateway and restore normal internet."
32 echo ""
33
34 # Cache sudo credentials upfront
35 # Prevents the script from hanging halfway through when it needs to run
36 # privileged commands (route flush, Wi-Fi power cycle, etc.).
37 echo -e "${YELLOW}${BOLD}Authentication required for network recovery...${NC}"
38 sudo -v
39 (
40     while true; do
41         sudo -n true 2>/dev/null
42         sleep 60
43     done
44 ) &
45 SUDO_KEEPER_PID=$!
46 trap 'kill $SUDO_KEEPER_PID 2>/dev/null' EXIT
47
48 # 1. Disconnect host Tailscale
49 step "Disconnecting host Tailscale from mesh"
50 if command -v tailscale >> output.log 2>&1; then
51     # Check if actually connected first (with timeout tailscaled could be wedged)
52     if run_with_timeout 5 tailscale status >> output.log 2>&1; then
53         # Also explicitly reset any exit-node so it doesn't linger.
54         if run_with_timeout 10 tailscale up --reset --ssh=true --accept-dns=false --exit-node= >> output.log 2>&1;
55         then
56             ok "Exit-node preference cleared"
57         else
58             warn "tailscale up --reset didn't respond (tailscaled may be wedged)"
59         fi
60         ts_args=""
61         if is_kill_switch_enabled; then ts_args="--accept-risk=lose-ssh"; fi
62         if run_with_timeout 10 tailscale down $ts_args >> output.log 2>&1; then
63             ok "Tailscale disconnected"
64         else
65             warn "tailscale down didn't respond"
66         fi
67     else
68         warn "tailscaled not reachable skipping (timeout after 5s)"
69     fi
70 else
71     warn "tailscale CLI not found skipping"
72 fi
73
74 # 2. Reset DNS to default on ALL network services
75 step "Resetting DNS to default on active interface"
76 ACTIVE_SERVICE=$(ip route get 1.1.1.1 2>/dev/null | awk '{print $5; exit}')
```

```

77 if [ -n "$ACTIVE_SERVICE" ]; then
78     sudo resolvectl revert "$ACTIVE_SERVICE" 2>> output.log || true
79     ok "DNS reset on $ACTIVE_SERVICE"
80 else
81     warn "Could not determine active network interface"
82 fi
83
84 # 3. Disable rogue proxy settings
85 step "Disabling proxy settings (SOCKS, web, secure web)"
86 echo " (Linux global SOCKS proxy configuration skipped.)"
87 ok "Proxy settings cleared"
88
89 # 4. Flush macOS DNS cache
90 step "Flushing DNS cache"
91 if command -v resolvectl >> output.log 2>&1; then
92     sudo resolvectl flush-caches 2>> output.log || true
93     ok "DNS cache flushed"
94 else
95     warn "resolvectl not found"
96 fi
97
98 # 5. Clear stale routing table
99 step "Flushing stale routing table entries"
100 sudo ip route flush cache >> output.log 2>&1 || true
101 ok "Routing table flushed"
102
103 # 6. Stop gateway Docker containers
104 step "Stopping gateway Docker containers"
105 if command -v docker >> output.log 2>&1 && docker info >> output.log 2>&1; then
106     if [ -f "docker-compose.yml" ]; then
107         docker compose down -t 5 2>> output.log && ok "Containers stopped" || warn "No containers were running"
108     else
109         warn "docker-compose.yml not found in current directory"
110     fi
111 else
112     warn "Docker not available skipping"
113 fi
114
115 # 6b. Stopping sleep prevention (systemd-inhibit)
116 step "Stopping sleep prevention"
117 PID_FILE="/tmp/nullexit-cafeinate.pid"
118 if [ -f "$PID_FILE" ]; then
119     INHIBIT_PID=$(cat "$PID_FILE")
120     if [ -n "$INHIBIT_PID" ] && kill -0 "$INHIBIT_PID" 2>/dev/null; then
121         kill "$INHIBIT_PID" 2>/dev/null || true
122         ok "Sleep prevention stopped (PID $INHIBIT_PID)"
123     else
124         warn "Sleep prevention process not running, cleaning up stale PID file"
125     fi
126     rm -f "$PID_FILE"
127 else
128     ok "Sleep prevention was not active"
129 fi
130
131 # 7. Power-cycle Wi-Fi
132 step "Power-cycling Wi-Fi"
133 if command -v nmcli &>/dev/null; then
134     sudo nmcli radio wifi off 2>> output.log || true
135     sleep 2
136     sudo nmcli radio wifi on 2>> output.log || true
137     ok "Wi-Fi bounced via nmcli"
138     sleep 3
139 else
140     warn "nmcli not found. Falling back to ip link..."
141     WIFI_IFACE=$(ip link | grep -E '[0-9]+: (wl[a-zA-Z0-9]+|wlan[0-9]+):' | awk -F': ' '{print $2}')
142     if [ -n "$WIFI_IFACE" ]; then
143         sudo ip link set "$WIFI_IFACE" down 2>> output.log || true
144         sleep 2
145         sudo ip link set "$WIFI_IFACE" up 2>> output.log || true
146         ok "Wi-Fi bounced via ip link (interface $WIFI_IFACE)"
147         sleep 3
148     else
149         warn "No Wi-Fi interface detected."
150     fi
151 fi
152
153 # 8. Verify internet connectivity
154 step "Verifying internet connectivity"
155
156 # Wait a moment for the network to settle

```

```

158 sleep 2
159
160 INTERNET_OK=false
161
162 # Try DNS resolution first
163 if host -W 3 google.com 1.1.1.1 >> output.log 2>&1; then
164     ok "DNS resolution works (google.com via 1.1.1.1)"
165     INTERNET_OK=true
166 elif nslookup google.com 1.1.1.1 >> output.log 2>&1; then
167     ok "DNS resolution works (nslookup google.com via 1.1.1.1)"
168     INTERNET_OK=true
169 else
170     warn "DNS resolution check failed will still try ping/curl"
171 fi
172
173 # Try actual HTTP connectivity
174 if command -v curl >> output.log 2>&1; then
175     if curl -sf --max-time 5 https://1.1.1.1 >> output.log 2>&1; then
176         ok "Internet reachable via HTTP"
177         INTERNET_OK=true
178     elif curl -sf --max-time 5 http://1.1.1.1 >> output.log 2>&1; then
179         ok "Internet reachable via HTTP (plain)"
180         INTERNET_OK=true
181     else
182         warn "HTTP check failed"
183     fi
184 elif command -v ping >> output.log 2>&1; then
185     if ping -c 1 -W 3 1.1.1.1 >> output.log 2>&1; then
186         ok "Internet reachable via ping"
187         INTERNET_OK=true
188     else
189         warn "Ping check failed"
190     fi
191 fi
192
193 # Summary
194 echo ""
195 echo -e "${BOLD}${NC}"
196 echo -e "${BOLD}RECOVERY SUMMARY${NC}"
197 echo -e "${BOLD}${NC}"
198
199 # Show final DNS state
200 FINAL_DNS=$(resolvectl dns "$ACTIVE_SERVICE" 2>>> output.log | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | tr '
201 ' ' ' ' || echo "N/A")
202 echo "  DNS ($ACTIVE_SERVICE):  $FINAL_DNS"
203
204 echo ""
205
206 if [ "$INTERNET_OK" = "true" ]; then
207     echo -e "  ${GREEN}${BOLD} Internet is working.${NC}"
208 else
209     echo -e "  ${YELLOW}${BOLD} Could not verify internet connectivity.${NC}"
210     echo "    This might mean:"
211     echo "    - Your Wi-Fi is disconnected from the router"
212     echo "    - Tailscale is still interfering (try restarting tailscaled:"
213     echo "      systemctl restart tailscaled)"
214     echo "    - You need to re-connect to Wi-Fi in the menu bar"
215     echo "    - Try toggling Wi-Fi off/on in the menu bar"
216     echo ""
217     echo "    Or try manually:"
218     echo "      resolvectl revert $ACTIVE_SERVICE"
219     extra_args=""
220     if is_kill_switch_enabled; then extra_args="--accept-risk=lose-ssh"; fi
221     echo "      tailscale down $extra_args"
222     echo "      sudo ip route flush cache"
223     echo "      systemctl restart tailscaled"
224 fi
225
226 echo ""
227 echo -e "${GREEN}${BOLD}Recovery complete.${NC}"
228 echo ""

```

11 scripts/setup-linux.sh

```

1 #!/bin/bash
2 # scripts/setup-linux.sh nullexit one-shot setup script (Linux)

```

```

3 # Configures Cloudflare WARP + Tailscale + AdGuard Home from scratch.
4 set -euo pipefail
5
6 # Source common formatting and helper functions
7 source "${dirname "$0"}/common.sh"
8
9 # Run common installation logic
10 source "${dirname "$0"}/setup-common.sh"
11
12 # Compile Linux Desktop Shortcuts
13 echo -e "\n${BOLD}Creating Linux Desktop Shortcuts...${NC}"
14 DIR="$(pwd)"
15 cat <<EOF > "Toggle Gateway.desktop"
16 [Desktop Entry]
17 Version=1.0
18 Name=Toggle Gateway
19 Comment=Start or Stop Nullexit Gateway
20 Exec=bash -c "cd '$DIR' && ./scripts/toggle-linux.sh; read -p 'Press Enter to close...' dummy"
21 Icon=utilities-terminal
22 Terminal=true
23 Type=Application
24 Categories=Network;Security;
25 EOF
26
27 cat <<EOF > "Recover Gateway.desktop"
28 [Desktop Entry]
29 Version=1.0
30 Name=Recover Gateway
31 Comment=Emergency DNS and Network Recovery
32 Exec=bash -c "cd '$DIR' && ./scripts/recover-linux.sh; read -p 'Press Enter to close...' dummy"
33 Icon=utilities-terminal
34 Terminal=true
35 Type=Application
36 Categories=Network;Security;
37 EOF
38 echo "    Created 'Toggle Gateway.desktop' and 'Recover Gateway.desktop'"
39
40 # Done
41 echo ""
42 echo -e "${GREEN}${BOLD}${NC}"
43 echo -e "${GREEN}${BOLD}                Setup complete!                ${NC}"
44 echo -e "${GREEN}${BOLD}${NC}"
45 echo ""
46 echo -e "${BOLD}One manual step remaining approve the exit node:${NC}"
47 echo "    https://login.tailscale.com/admin/machines"
48 echo "    Find your gateway '...' 'Edit route settings' enable Exit Node"
49 echo ""
50
51 if [[ -n "${TS_IP:-}" ]]; then
52     echo -e "${BOLD}AdGuard Home dashboard:${NC} http://${TS_IP}:3000 (username: admin)"
53     echo ""
54 fi
55
56 echo -e "${BOLD}To toggle the gateway on/off:${NC}"
57 echo "    Linux: double-click the 'Toggle Gateway' desktop icon!"
58 echo "    (or run ./scripts/toggle-linux.sh in terminal)"
59 echo ""

```

12 docker-compose.yml

```

1 services:
2   warp:
3     image: qmcgaw/gluetun:v3.41.1
4     container_name: warp
5     hostname: nullexit-gateway
6     cap_add:
7       - NET_ADMIN
8     devices:
9       - /dev/net/tun:/dev/net/tun
10    ports:
11      - "127.0.0.1:${DNS_PROXY_PORT:-5354}:5335/udp" # AdGuard DNS (5335 is used by SSH on host, use
12        procedural port instead)
13      - "127.0.0.1:${DNS_PROXY_PORT:-5354}:5335/tcp" # AdGuard DNS (TCP)
14      - "41642:41642/udp" # Tailscale WireGuard (direct connection to host)
15      - "127.0.0.1:${SOCKS_PROXY_PORT:-1080}:1080/tcp" # SOCKS5 proxy (routed through WARP tunnel)
16      - "127.0.0.1:${TOR SOCKS_PORT:-9050}:9050/tcp" # Tor SOCKS5 proxy (procedurally generated)

```

```

16 environment:
17   - TZ=America/New_York
18   - VPN_SERVICE_PROVIDER=custom
19   - VPN_TYPE=wireguard
20   - WIREGUARD_PRIVATE_KEY=${WIREGUARD_PRIVATE_KEY}
21   - WIREGUARD_PUBLIC_KEY=${WIREGUARD_PUBLIC_KEY}
22   - WIREGUARD_ADDRESSES=${WIREGUARD_ADDRESSES}
23   - WIREGUARD_ENDPOINT_IP=${WARP_ENDPOINT_1:-162.159.192.1}
24   - WIREGUARD_ENDPOINT_PORT=2408
25   # Set to 25s (WireGuard default) to beat standard 30s hardware NAT/firewall UDP timeouts
26   - WIREGUARD_PERSISTENT_KEEPALIVE_INTERVAL=25s
27   # MTU Fix (Critical): Prevent packet fragmentation from double-encryption
28   - WIREGUARD_MTU=1280
29   # Give WARP more time to establish the connection before internal restart (default is 6s which is too
short)
30   - HEALTH_VPN_DURATION_INITIAL=30s
31   # Use plaintext DNS resolvers instead of DNS-over-TLS (DoT) to prevent startup failures on networks where
port 853 is blocked
32   - DNS_UPSTREAM_RESOLVER_TYPE=plain
33   - DNS_UPSTREAM_PLAIN_ADDRESSES=1.1.1.1:53,8.8.8.8:53
34   # Allow local network traffic to bypass WARP so Tailscale can establish fast, direct peer-to-peer
connections (no DERP relays) on the LAN
35   - FIREWALL_OUTBOUND_SUBNETS=192.168.0.0/16,172.16.0.0/12,10.0.0.0/8
36   # Built-in DNS blocking for ads, malicious domains, and tracking
37   # NOTE: This is the massive "Hagezi-style" built-in blocklist.
38   # You can turn these to 'on' for maximum security if you have spare RAM
39   # or are running this on a cloud exit node with adequate autonomous resources (requires approx 1-2GB of
RAM).
40   # Disabled intentionally: AdGuard Home provides the DNS filtering layer instead.
41   # If AdGuard fails, there is no fallback filtering, but the healthcheck
42   # dependency chain ensures Tailscale won't come up if AdGuard is down.
43   - BLOCK_MALICIOUS=off
44   - BLOCK_SURVEILLANCE=off
45   - BLOCK_ADS=off
46 dns:
47   - 1.1.1.1
48 sysctls:
49   - net.ipv4.ip_forward=1
50 volumes:
51   - ./post-rules.txt:/iptables/post-rules.txt:ro
52 restart: unless-stopped
53 healthcheck:
54   test: ["CMD-SHELL", "/gluetun-entrypoint healthcheck"]
55   interval: 2s
56   timeout: 5s
57   retries: 15
58   start_period: 60s
59 logging:
60   driver: "json-file"
61   options:
62     max-size: "10m"
63     max-file: "3"
64
65 tailscale:
66   image: tailscale/tailscale:v1.98.4
67   container_name: tailscale
68   network_mode: service:warp
69   depends_on:
70     warp:
71       condition: service_healthy
72     adguardhome:
73       condition: service_healthy
74   cap_add:
75     - NET_ADMIN
76     - NET_RAW
77     - SYS_MODULE
78   sysctls:
79     - net.ipv4.ip_forward=1
80     - net.ipv6.conf.all.forwarding=1
81 environment:
82   - TZ=America/New_York
83   - TS_EXTRA_ARGS=--advertise-exit-node
84   - PORT=41642
85   - TS_USERSPACE=false
86   - TS_AUTHKEY=${TS_AUTHKEY}
87   - TS_STATE_DIR=/var/lib/tailscale
88   # (Note: For headless/cloud deployments, you can bypass this footgun by
89   # using Tailscale Ephemeral Keys which auto-clean themselves).
90   - TS_AUTH_ONCE=true
91   # MTU Optimization: Force Tailscale packets to be smaller than WARP's 1280 MTU
92   # to prevent WireGuard-in-WireGuard fragmentation, completely eliminating bufferbloat

```

```

93     - TS_DEBUG_MTU=1200
94     # Enable Tailscale's built-in SOCKS5 proxy to route out of the WARP tunnel
95     - TS_SOCKS5_SERVER=:1080
96 devices:
97     - /dev/net/tun:/dev/net/tun
98 volumes:
99     - tailscale_state:/var/lib/tailscale
100 restart: unless-stopped
101 logging:
102     driver: "json-file"
103     options:
104         max-size: "10m"
105         max-file: "3"
106
107 routing-fix:
108     image: nullexit-routing-fix:v1.0.0
109     build:
110         context: ./scripts
111         dockerfile: Dockerfile.routing-fix
112     container_name: routing-fix
113     network_mode: service:warp
114     # NOTE: Deliberately NOT sharing tailscale's PID namespace the shared
115     # namespace caused routing-fix to be killed whenever tailscale restarted
116     # (e.g. during gluetun health-check flaps), which dropped all routing
117     # table changes and left traffic leaking through eth0 instead of tun0.
118     cap_add:
119         - NET_ADMIN
120         - SYSLOG
121     depends_on:
122         warp:
123             condition: service_healthy
124         rule-compiler:
125             condition: service_completed_successfully
126     environment:
127         - TZ=America/New_York
128         - WARP_ENDPOINT=${WARP_ENDPOINT_1:-162.159.192.1}
129         - BLOCKED_COUNTRIES=${BLOCKED_COUNTRIES}
130         - TAILSCALE_ALLOW_LAN_P2P=${TAILSCALE_ALLOW_LAN_P2P:-false}
131     volumes:
132         - ./scripts/routing-fix.sh:/routing-fix.sh:ro
133         # SECURITY: ./adguard/work/ is bind-mounted into this container.
134         # Never write to it from the host or another container. The
135         # single allowed writer is `rule-compiler` container manual
136         # edits corrupt the AdGuard cache + BoltDB session store.
137         - ./adguard/work/userfilters:/userfilters:ro
138         - ./adguard/work:/adguard_work:ro
139         - ./app
140     restart: unless-stopped
141     stop_grace_period: 1s
142     logging:
143         driver: "json-file"
144         options:
145             max-size: "1m"
146             max-file: "1"
147
148 adguardhome:
149     # WARNING: AdGuard is pre-configured with hardcoded credentials (admin/nullexit).
150     # This is fine for a local Mac running behind a NAT, but if you are deploying
151     # this on a public cloud VPS with port 80/3000 exposed, change this immediately!
152     image: adguard/adguardhome:v0.107.77
153     container_name: adguardhome
154     network_mode: service:warp
155     depends_on:
156         warp:
157             condition: service_healthy
158         rule-compiler:
159             condition: service_completed_successfully
160     environment:
161         - TZ=America/New_York
162     healthcheck:
163         test: ["CMD-SHELL", "nc -z 127.0.0.1 5335 || exit 1"]
164         interval: 2s
165         timeout: 5s
166         retries: 10
167     volumes:
168         # SECURITY: ./adguard/work/ is bind-mounted into this container.
169         # Never write to it from the host or another container. The
170         # single allowed writer is `rule-compiler` container manual
171         # edits corrupt the AdGuard cache + BoltDB session store.
172         - ./adguard/work:/opt/adguardhome/work
173         - ./adguard/conf:/opt/adguardhome/conf

```

```

174 restart: unless-stopped
175 stop_grace_period: 2s
176 logging:
177   driver: "json-file"
178   options:
179     max-size: "10m"
180     max-file: "3"
181
182 rule-compiler:
183   image: nullexit-rule-compiler:v1.0.0
184   build:
185     context: ./scripts
186     dockerfile: Dockerfile.rule-compiler
187   container_name: rule-compiler
188   volumes:
189     - ./app
190
191 tor:
192   build: ./docker/tor
193   image: nullexit-tor:v1.0.0
194   container_name: tor
195   network_mode: service:warp
196   depends_on:
197     warp:
198       condition: service_healthy
199   environment:
200     - TZ=America/New_York
201     - TOR_USE_BRIDGES=${TOR_USE_BRIDGES:-false}
202     - TOR_BRIDGE_LINES_FILE=${TOR_BRIDGE_LINES_FILE:-./tor-bridges.txt}
203   working_dir: /nullexit
204   volumes:
205     - ./:/nullexit:ro
206     - ./output.log:/output.log:rw
207   restart: unless-stopped
208   logging:
209     driver: "json-file"
210     options:
211       max-size: "10m"
212       max-file: "3"
213
214 volumes:
215   tailscale_state:
216
217 networks:
218   default:
219     ipam:
220       config:
221         - subnet: 10.200.1.0/24

```

13 scripts/common.sh

```

1 #!/bin/bash
2 # common.sh shared bash utilities for nullexit scripts
3
4 # Formatting
5 RED='\033[0;31m'
6 GREEN='\033[0;32m'
7 YELLOW='\033[1;33m'
8 BLUE='\033[0;34m'
9 BOLD='\033[1m'
10 NC='\033[0m'
11
12 # Constants & Environment
13 export MARKER_FILE="/tmp/nullexit-gateway-active.marker"
14 export PID_CAFFEINATE="/tmp/nullexit-caffeinate.pid"
15 export PID_DNS_WATCHER="/tmp/nullexit-dns-watcher.pid"
16 export DEFAULT_WARP_ENDPOINT_1="162.159.192.1"
17 export DEFAULT_WARP_ENDPOINT_2="162.159.193.1"
18
19 setup_standard_path() {
20   export PATH="/opt/homebrew/bin:/opt/homebrew/sbin:/usr/local/bin:$PATH"
21 }
22
23 step() { echo -e "\n[${date '+%Y-%m-%d %H:%M:%S'}] ${BLUE}${BOLD} ${*}${NC}"; }
24 ok() { echo -e "[${date '+%Y-%m-%d %H:%M:%S'}] ${GREEN} ${*}${NC}"; }
25 warn() { echo -e "[${date '+%Y-%m-%d %H:%M:%S'}] ${YELLOW} ${*}${NC}"; }

```

```

26 fail() { echo -e "[$(date '+%Y-%m-%d %H:%M:%S')] ${RED} ${NC}"; }
27 die() { echo -e "\n[$(date '+%Y-%m-%d %H:%M:%S')] ${RED} ${NC}\n"; exit 1; }
28
29 # Helper Functions
30
31 # Pure-bash timeout (no dependency on GNU coreutils' timeout)
32 # Runs a command with a safety cutoff so a wedged daemon can't hang the script.
33 run_with_timeout() {
34     local timeout_sec="$1"
35     shift
36     if [ $# -eq 0 ]; then return 1; fi
37
38     "$@" &
39     local cmd_pid=$!
40     CURRENT_BG_PID="$cmd_pid"
41
42     (
43         sleep "$timeout_sec"
44         if kill -0 "$cmd_pid" 2>> output.log; then
45             echo -e "\n[Timeout] Command '$*' exceeded $timeout_sec seconds. Terminating..." >&2
46             kill -15 "$cmd_pid" 2>> output.log || true
47             sleep 2
48             if kill -0 "$cmd_pid" 2>> output.log; then
49                 kill -9 "$cmd_pid" 2>> output.log || true
50             fi
51         fi
52     ) &
53     local watcher_pid=$!
54
55     # Disable set -e temporarily to capture exit status of wait
56     set +e
57     wait "$cmd_pid" 2>> output.log
58     local exit_status=$?
59     set -e
60
61     # Kill the watcher since the command finished
62     kill "$watcher_pid" 2>> output.log && wait "$watcher_pid" 2>> output.log || true
63
64     CURRENT_BG_PID=""
65     return "$exit_status"
66 }
67
68 # Check if the gateway is active (either containers are running, or host DNS is hijacked)
69 is_gateway_active() {
70     # 1. Check if containers are running (suppress stderr to avoid errors if docker is down)
71     if run_with_timeout 15 docker compose ps --status running 2>/dev/null | grep -q 'warp'; then
72         echo "[$(date '+%Y-%m-%d %H:%M:%S')] is_gateway_active: TRUE (warp container is running)" >> "$LOG_FILE"
73         return 0
74     fi
75
76     # 2. Check if host DNS was hijacked (not 1.1.1.1 and not default/empty)
77     local current_dns=""
78     local active_svc=""
79     if [[ "$OSTYPE" == "darwin"* ]]; then
80         active_svc=$(get_active_service)
81         current_dns=$(networksetup -getdnsservers "$active_svc" 2>> output.log || true)
82     else
83         current_dns=$(resolvectl dns 2>/dev/null | awk '/Global|Link/ {for(i=4;i<=NF;i++) print $i}' | head -n1)
84     fi
85
86     if [[ -n "$current_dns" && "$current_dns" != "1.1.1.1" && ! "$current_dns" =~ "There aren't any DNS Servers"
87         ]]; then
88         echo "[$(date '+%Y-%m-%d %H:%M:%S')] is_gateway_active: TRUE (DNS was hijacked: $current_dns)" >> "
89             $LOG_FILE"
90         return 0
91     fi
92
93     # 3. Check if SOCKS proxy is enabled (macOS only)
94     if [[ "$OSTYPE" == "darwin"* ]]; then
95         local socks_proxy
96         socks_proxy=$(networksetup -getsocksfirewallproxy "$active_svc" 2>> output.log || true)
97         if echo "$socks_proxy" | grep -q "Enabled: Yes"; then
98             echo "[$(date '+%Y-%m-%d %H:%M:%S')] is_gateway_active: TRUE (SOCKS proxy enabled)" >> "$LOG_FILE"
99             return 0
100         fi
101     fi
102
103     echo "[$(date '+%Y-%m-%d %H:%M:%S')] is_gateway_active: FALSE (Containers down, DNS clean, SOCKS disabled)"
104     >> "$LOG_FILE"
105     return 1
106 }

```

```

104
105 # Reads an environment variable from a file (default: .env), strips quotes and whitespace
106 read_env_var() {
107     local var_name="$1"
108     local env_file="${2:-.env}"
109     grep -E "${var_name}=" "$env_file" 2>/dev/null | cut -d '=' -f2- | tr -d "\"\`" | sed -e 's/^[[:space:]]*//'
110     -e 's/[[:space:]]*$//' || echo ""
111 }
112
113 # Load KILL_SWITCH variable from .env (defaults to false if not found/empty)
114 export KILL_SWITCH
115 KILL_SWITCH=$(read_env_var "KILL_SWITCH" 2>/dev/null | tr '[:upper:]' '[:lower:]')
116 if [ -z "$KILL_SWITCH" ]; then
117     KILL_SWITCH="false"
118 fi
119
120 # Function to get the active network service name (e.g., "Wi-Fi" or "USB 10/100 LAN")
121 get_active_service() {
122     local iface
123     iface=$(route get default 2>> output.log | awk '/interface:/ {print $2}')
124
125     # If the default interface is empty or a VPN tunnel (like utunX), fallback to the active physical interface
126     if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
127         for i in en0 en1 en2 en3; do
128             if ifconfig "$i" 2>> output.log | grep -q "status: active" && ifconfig "$i" 2>> output.log | grep -q "
129                 inet "; then
130                 iface="$i"
131                 break
132             fi
133         done
134     fi
135
136     # Default fallback if still empty or not enX
137     if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
138         iface="en0"
139     fi
140
141     # Map interface (e.g., en0) to service name (e.g., Wi-Fi)
142     local service
143     service=$(networksetup -listnetworkserviceorder 2>> output.log | grep -B 1 "Device: $iface" | head -n 1 | sed
144         -E 's/^\([0-9\*\]+\)/ /')
145
146     if [ -n "$service" ]; then
147         echo "$service"
148     else
149         echo "Wi-Fi"
150     fi
151 }
152
153 get_en0_service() {
154     local service
155     service=$(networksetup -listnetworkserviceorder 2>> output.log | grep -B1 "Device: en0" | head -1 | sed -E 's
156         /\([0-9\*\]+\)/ /' || true)
157     if [ -z "$service" ]; then
158         echo "Wi-Fi"
159     else
160         echo "$service"
161     fi
162 }
163
164 get_warp_endpoint_1() { read_env_var "WARP_ENDPOINT_1" ".env" | grep -Eo '[0-9\.]+' || echo "
165     $DEFAULT_WARP_ENDPOINT_1"; }
166 get_warp_endpoint_2() { read_env_var "WARP_ENDPOINT_2" ".env" | grep -Eo '[0-9\.]+' || echo "
167     $DEFAULT_WARP_ENDPOINT_2"; }
168
169 restart_tailscaled_daemon() {
170     echo " [Tailscale Recovery] tailscaled daemon appears to be wedged/unresponsive."
171     echo " [Tailscale Recovery] Attempting to restart tailscaled service..."
172
173     if run_with_timeout 15 brew services restart tailscale >> output.log 2>&1; then
174         echo " [Tailscale Recovery] Successfully restarted tailscaled (user service)."
175     elif run_with_timeout 15 sudo -n brew services restart tailscale >> output.log 2>&1; then
176         echo " [Tailscale Recovery] Successfully restarted tailscaled (system service)."
177     else
178         echo " [Tailscale Recovery] WARNING: Failed to restart tailscaled. Manual intervention may be needed."
179     fi
180     sleep 3
181 }
182
183 disconnect_tailscale_host() {
184     local ts_bin="${1:-tailscale}"

```

```

179 if command -v "$ts_bin" >> output.log 2>&1; then
180     echo " Disconnecting host Tailscale from mesh..."
181
182     # Check if actually connected first (with timeout tailscaled could be wedged)
183     local status_ok=false
184     if run_with_timeout 5 "$ts_bin" status >> output.log 2>&1; then
185         status_ok=true
186     else
187         local status_exit=$?
188         # If it was a quick exit code 1 (disconnected), it is still reachable
189         if [ "$status_exit" -ne 143 ] && [ -S /var/run/tailscaled.socket ]; then
190             status_ok=true
191         fi
192     fi
193
194     if [ "$status_ok" = "false" ]; then
195         restart_tailscaled_daemon
196     fi
197
198     # Explicitly reset any exit-node so it doesn't linger.
199     "$ts_bin" up >> output.log 2>&1 || true
200     if run_with_timeout 10 "$ts_bin" set --ssh=true --accept-dns=false --exit-node= >> output.log 2>&1; then
201         echo " [] Exit-node preference cleared."
202     else
203         echo " [!] tailscale up didn't respond (tailscaled may be wedged)"
204     fi
205
206     local ts_args=""
207     if is_kill_switch_enabled; then ts_args="--accept-risk=lose-ssh"; fi
208     if run_with_timeout 10 "$ts_bin" down $ts_args >> output.log 2>&1; then
209         echo " [] Tailscale disconnected."
210     else
211         echo " [!] tailscale down didn't respond"
212     fi
213 else
214     echo " [!] tailscale CLI not found skipping"
215 fi
216 }
217
218 stop_pidfile_daemon() {
219     local pid_file="$1"
220     local label="$2"
221     if [ -f "$pid_file" ]; then
222         local pid
223         pid=$(cat "$pid_file")
224         if kill -0 "$pid" 2>/dev/null; then
225             echo " Stopping $label (PID $pid)..."
226             sudo -n kill "$pid" 2>> output.log || kill "$pid" 2>> output.log || true
227             sleep 0.5
228             if kill -0 "$pid" 2>/dev/null; then
229                 sudo -n kill -9 "$pid" 2>> output.log || kill -9 "$pid" 2>> output.log || true
230             fi
231         fi
232         rm -f "$pid_file"
233     fi
234 }
235
236 disable_all_proxies() {
237     local svc="$1"
238     if [ -n "$svc" ]; then
239         sudo -n networksetup -setsocksfirewallproxystate "$svc" off 2>> output.log || true
240         sudo -n networksetup -setwebproxystate "$svc" off 2>> output.log || true
241         sudo -n networksetup -setsecurewebproxystate "$svc" off 2>> output.log || true
242     fi
243 }
244
245 flush_dns_cache() {
246     if [[ "$OSTYPE" == "darwin"* ]]; then
247         sudo -n dscacheutil -flushcache 2>> output.log || true
248         sudo -n killall -HUP mDNSResponder 2>> output.log || true
249     else
250         resolvectl flush-caches 2>> output.log || true
251     fi
252 }
253
254 wait_for_dhcp_settle() {
255     # Resolve physical interface (Wi-Fi or Ethernet)
256     local physical_iface
257     physical_iface=$(networksetup -listallhardwareports 2>> output.log | awk '/Hardware Port: (Wi-Fi|Ethernet)/{
258         getline; print $2; exit}')
259     [ -z "$physical_iface" ] && physical_iface="en0"

```

```

259 echo -n "Waiting for physical network interface ($physical_iface) DHCP lease to settle..."
260 local settled=false
261 for attempt in {1..60}; do
262     PHYSICAL_GW=$(ipconfig getpacket "$physical_iface" 2>> output.log | awk -F'[{ }]' '/router /{print $2}')
263     local local_ip
264     local_ip=$(ifconfig "$physical_iface" 2>/dev/null | awk '/inet /{print $2}')
265
266     if [ -n "$PHYSICAL_GW" ] && [[ "$PHYSICAL_GW" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]] && \
267         [ -n "$local_ip" ] && [[ "$local_ip" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]] && \
268         [[ "$local_ip" != "169.254.*" ]]; then
269         echo "settled (Interface IP: $local_ip, Router IP: $PHYSICAL_GW)."
270         settled=true
271         break
272     fi
273
274     if [ "$attempt" -gt 15 ] && [ -n "$local_ip" ] && [[ "$local_ip" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]] && \
275         [[ "$local_ip" != "169.254.*" ]]; then
276         local fallback_gw
277         fallback_gw=$(route get default 2>> output.log | awk '/gateway:/ {print $2}')
278         if [ -n "$fallback_gw" ] && [[ "$fallback_gw" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]]; then
279             PHYSICAL_GW="$fallback_gw"
280             fi
281         echo "static/settled detected (Interface IP: $local_ip, Gateway IP: ${PHYSICAL_GW:-unknown})."
282         settled=true
283         break
284     fi
285     echo -n "."
286     sleep 0.5
287 done
288 if [ "$settled" = "false" ]; then
289     echo "timed out or no active link. Proceeding anyway."
290 fi
291 }
292
293 reset_sharing_services() {
294     if [[ "$OSTYPE" == "darwin"* ]]; then
295         sudo -n killall sharingd rapportd 2>> output.log || true
296     fi
297 }
298
299 read_adguard_ip() {
300     if [ -f ".gateway_ip" ]; then
301         tr -d '\r' < ".gateway_ip" | awk 'NR==1{print $1;exit}'
302     fi
303 }
304
305 is_kill_switch_enabled() {
306     [ "$KILL_SWITCH" = "true" ]
307 }
308
309 add_warp_bypass_routes() {
310     local target="${1:-}"
311     local ep1
312     ep1=$(get_warp_endpoint_1)
313     local ep2
314     ep2=$(get_warp_endpoint_2)
315
316     if [[ "$OSTYPE" == "darwin"* ]]; then
317         local routing_arg=""
318         local msg_via=""
319         if [ -n "$target" ]; then
320             if [[ "$target" =~ ^en[0-9]+$ || "$target" == "bridge"* ]]; then
321                 routing_arg="-interface $target"
322                 msg_via="interface $target"
323             else
324                 routing_arg="$target"
325                 msg_via="gateway $target"
326             fi
327         else
328             local gateway_ip
329             gateway_ip=$(route get default 2>> output.log | awk '/gateway:/ {print $2}')
330             if [ -z "$gateway_ip" ]; then
331                 local iface
332                 iface=$(route get default 2>> output.log | awk '/interface:/ {print $2}')
333                 if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
334                     iface="en0"
335                 fi
336                 routing_arg="-interface $iface"
337                 msg_via="interface $iface"
338             else

```

```

339     routing_arg="$gateway_ip"
340     msg_via="gateway $gateway_ip"
341     fi
342 fi
343
344 echo -e "\nAdding host bypass routes for Cloudflare WARP endpoints via $msg_via..."
345 sudo -n route delete -host "$sep1" 2>/dev/null || true
346 sudo -n route delete -host "$sep2" 2>/dev/null || true
347 sudo -n route add -host "$sep1" $routing_arg >> output.log 2>&1 || true
348 sudo -n route add -host "$sep2" $routing_arg >> output.log 2>&1 || true
349
350 echo -e "Adding host bypass routes for Tailscale control plane via $msg_via..."
351 sudo -n route delete -net 192.200.0.0/24 2>/dev/null || true
352 sudo -n route add -net 192.200.0.0/24 $routing_arg >> output.log 2>&1 || true
353
354 echo -e "Adding host bypass routes for Tailscale DERP relays via $msg_via..."
355 local derp_ips
356 derp_ips=$(curl -s --connect-timeout 5 https://login.tailscale.com/derpmap/default | grep -oE '"IPv4"[[:
space:]]*:[[:space:]]*[0-9]+\.' | cut -d'.' -f4 | grep -E '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' || true)
357 if [ -n "$derp_ips" ]; then
358     echo "$derp_ips" > /tmp/nullexit-derp-ips.txt
359     for ip in $derp_ips; do
360         sudo -n route delete -host "$ip" 2>/dev/null || true
361         sudo -n route add -host "$ip" $routing_arg >> output.log 2>&1 || true
362     done
363 fi
364 else
365     local gateway_ip="${target}"
366     if [ -z "$gateway_ip" ]; then
367         gateway_ip=$(ip route show default 2>/dev/null | awk '/default/ {print $3}')
368     fi
369     if [ -n "$gateway_ip" ]; then
370         sudo ip route add "$sep1" via "$gateway_ip" >> output.log 2>&1 || true
371         sudo ip route add "$sep2" via "$gateway_ip" >> output.log 2>&1 || true
372     fi
373 fi
374 }
375
376 remove_warp_bypass_routes() {
377     local ep1
378     ep1=$(get_warp_endpoint_1)
379     local ep2
380     ep2=$(get_warp_endpoint_2)
381
382     if [[ "$OSTYPE" == "darwin"* ]]; then
383         echo -e "\nRemoving host bypass routes for Cloudflare WARP endpoints..."
384         sudo -n route delete -host "$sep1" >> output.log 2>&1 || true
385         sudo -n route delete -host "$sep2" >> output.log 2>&1 || true
386         echo -e "Removing host bypass routes for Tailscale control plane..."
387         sudo -n route delete -net 192.200.0.0/24 >> output.log 2>&1 || true
388         if [ -f /tmp/nullexit-derp-ips.txt ]; then
389             echo -e "Removing host bypass routes for Tailscale DERP relays..."
390             while read -r ip; do
391                 sudo -n route delete -host "$ip" >> output.log 2>&1 || true
392             done < /tmp/nullexit-derp-ips.txt
393             rm -f /tmp/nullexit-derp-ips.txt
394         fi
395     else
396         sudo ip route del "$sep1" >> output.log 2>&1 || true
397         sudo ip route del "$sep2" >> output.log 2>&1 || true
398     fi
399 }
400
401 bounce_wifi_interfaces() {
402     if [[ "$OSTYPE" == "darwin"* ]]; then
403         local wifi_port
404         wifi_port=$(networksetup -listallhardwareports 2>> output.log | awk '/Hardware Port: Wi-Fi/{getline; print
$2}')
405         if [ -n "$wifi_port" ]; then
406             sudo -n networksetup -setairportpower "$wifi_port" off 2>> output.log || true
407             sleep 2
408             sudo -n networksetup -setairportpower "$wifi_port" on 2>> output.log || true
409             echo "Wi-Fi bounced (interface $wifi_port)"
410             sleep 3
411         else
412             echo "Could not detect Wi-Fi interface. Bouncing fallback interfaces..."
413             set +e
414             for iface in en0 en1 en2 en3 en4 en5; do
415                 if ifconfig "$iface" >> output.log 2>&1; then
416                     sudo -n ifconfig "$iface" down 2>> output.log || true
417                     sudo -n ifconfig "$iface" up 2>> output.log || true

```

```

418         fi
419     done
420     set -e
421     echo "Fallback interfaces bounced"
422 fi
423 fi
424 }
425
426 get_active_interface() {
427     local iface
428     iface=$(route get default 2>> output.log | awk '/interface:/ {print $2}')
429
430     # If the default interface is empty or a VPN tunnel (like utunX), fallback to the active physical interface
431     if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
432         for i in en0 en1 en2 en3; do
433             if ifconfig "$i" 2>> output.log | grep -q "status: active" && ifconfig "$i" 2>> output.log | grep -q "
434                 inet "; then
435                 iface="$i"
436                 break
437             fi
438         done
439     fi
440
441     # Default fallback if still empty or not enX
442     if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
443         iface="en0"
444     fi
445
446     echo "$iface"
447 }
448
449 enable_killswitch() {
450     if [[ "$OSTYPE" == "darwin"* ]]; then
451         if ! is_kill_switch_enabled; then
452             return 0
453         fi
454
455         local ext_if
456         ext_if=$(get_active_interface)
457         local ep1
458         ep1=$(get_warp_endpoint_1)
459         local ep2
460         ep2=$(get_warp_endpoint_2)
461         local pf_conf_path
462         pf_conf_path="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/pf.conf"
463
464         echo -e "\nEnabling macOS Packet Filter (PF) Kill-Switch on $ext_if..."
465
466         # Enable PF and capture the reference token
467         local token_out
468         token_out=$(sudo -n pfctl -E 2>> output.log || true)
469         local token
470         token=$(echo "$token_out" | awk '/Token/{print $NF}')
471         if [ -n "$token" ]; then
472             echo "$token" > /tmp/nullexit-pf-token.txt
473             echo " [] PF enabled with token $token."
474         else
475             echo " [!] WARNING: Could not capture PF reference token (may already be enabled or sudo failed)."
476         fi
477
478         # Load the rules into the anchor
479         if sudo -n pfctl -D "ext_if=$ext_if" -a com.apple/nullexit -f "$pf_conf_path" >> output.log 2>&1; then
480             echo " [] Ruleset loaded into anchor com.apple/nullexit."
481         else
482             echo " [!] ERROR: Failed to load ruleset into anchor."
483         fi
484
485         # Populate tables in the anchor
486         sudo -n pfctl -a com.apple/nullexit -t vpn_endpoints -T flush >> output.log 2>&1 || true
487         sudo -n pfctl -a com.apple/nullexit -t vpn_endpoints -T add "$ep1" >> output.log 2>&1 || true
488         sudo -n pfctl -a com.apple/nullexit -t vpn_endpoints -T add "$ep2" >> output.log 2>&1 || true
489
490         if [ -f /tmp/nullexit-derp-ips.txt ]; then
491             local derp_ips
492             derp_ips=$(tr '\n' ' ' < /tmp/nullexit-derp-ips.txt)
493             if [ -n "$derp_ips" ]; then
494                 sudo -n pfctl -a com.apple/nullexit -t derp_relays -T flush >> output.log 2>&1 || true
495                 sudo -n pfctl -a com.apple/nullexit -t derp_relays -T add $derp_ips >> output.log 2>&1 || true
496             fi
497         fi
498
499         echo " [] Kill-switch tables populated."

```

```

498 fi
499 }
500
501 disable_killswitch() {
502     if [[ "$OSTYPE" == "darwin"* ]]; then
503         if ! is_kill_switch_enabled; then
504             return 0
505         fi
506         echo -e "\nDisabling macOS PF Kill-Switch..."
507
508         # Flush rules from anchor com.apple/nullexit
509         sudo -n pfctl -a com.apple/nullexit -F all >> output.log 2>&1 || true
510
511         # Release the enable reference if token is present
512         if [ -f /tmp/nullexit-pf-token.txt ]; then
513             local token
514             token=$(cat /tmp/nullexit-pf-token.txt)
515             if [ -n "$token" ]; then
516                 sudo -n pfctl -X "$token" >> output.log 2>&1 || true
517                 echo " [] Released PF enable reference token $token."
518             fi
519             rm -f /tmp/nullexit-pf-token.txt
520         else
521             echo " [] Rules flushed from anchor com.apple/nullexit."
522         fi
523     fi
524 }
525
526
527 write_host_ips() {
528     # Gather all active IPv4 addresses on the host and write to .host_ips
529     # routing-fix.sh will read this file to explicitly whitelist them for direct Tailscale P2P
530     local repo_dir
531     repo_dir="$(cd "$(dirname "${BASH_SOURCE[0]}")/.." && pwd)"
532
533     if [[ "$OSTYPE" == "darwin"* ]]; then
534         ifconfig | awk '/inet /{print $2}' | grep -v '127.0.0.1' > "$repo_dir/.host_ips" 2>/dev/null || true
535     else
536         ip -4 addr show 2>/dev/null | awk '/inet /{print $2}' | cut -d/ -f1 | grep -v '127.0.0.1' > "$repo_dir/.
537         host_ips" 2>/dev/null || true
538     fi
539 }

```

14 scripts/watcher.sh

```

1 #!/bin/bash
2 # scripts/watcher.sh nullexit post-wake / post-roam recovery watcher
3 #
4 # Long-running daemon. Launched by launchd as a LaunchAgent at
5 # ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
6 # (label com.nullexit.wake-recovery).
7 #
8 # Two independent listeners run as backgrounded pipelines:
9 #
10 # (1) WAKE listener `log stream` live-follows the macOS unified log for
11 # power-management events (lid closewake, manual wake, DarkWake from
12 # clamshell). Each event triggers run_recover (with a global debounce so
13 # a single wake that emits 2-3 log lines only causes ONE post-wake run).
14 #
15 # (2) NETWORK listener `scutil n.watch` on `State:/Network/Global/IPv4`
16 # fires on every Wi-Fi roam, ethernet swap, hotspot join, captive-portal
17 # re-bind, VPN up/down, etc. Same debounce.
18 #
19 # Both listeners call run_recover, which:
20 # - checks /tmp/nullexit-gateway-active.marker (only act when toggle.sh left
21 # the gateway up)
22 # - debounces (10s default) so multiple events don't pile up
23 # - shells out to `bash <repo>/recover.sh --post-wake` directly (no GUI auth prompt needed due to NOPASSWD
24 # sudoers)
25 #
26 # See devref.md 10.29 for the full Apple power management primer and the
27 # rationale behind using `log stream` + `scutil n.watch` from a shell daemon.
28 #
29 # NOTE: errexit (`set -e`) is intentionally NOT enabled. This is a long-running
30 # daemon, not a discrete-step script, and errexit actively breaks the reconnect
31 # loops below: any failed pipe (log stream on rotation, scutil on state churn)

```

```

31 # would kill the outer subshell before `echo "reconnecting..."; sleep 5;` runs,
32 # AND the parent's final `wait` would propagate a non-zero child exit and kill
33 # the entire watcher. Every error path is already wrapped in `|| true` /
34 # `2>/dev/null` fallbacks.
35 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
36 source "$SCRIPT_DIR/common.sh"
37 setup_standard_path
38
39 LOG="$SCRIPT_DIR/../output.log"
40 # Resolve our own directory; recover.sh lives one level up at the repo root.
41 # This makes the daemon install-agnostic no need to template the path
42 # in launchd, no need to keep a deploy-specific hardcoded location in sync.
43 RECOVER="$SCRIPT_DIR/../recover.sh"
44 MARKER="$MARKER_FILE"
45 DEBOUNCE_FILE="/tmp/nullexit-watcher.last-recovery"
46 DEBOUNCE_SECONDS="${NULLEXIT_DEBOUNCE_SECONDS:-10}"
47
48 exec >> "$LOG" 2>&1
49
50 # Single-instance lock
51 # Without this, repeated lid-closewake cycles under macOS Sonoma+ accumulate
52 # orphan watcher.sh processes: launchd suspends on sleep and SIGCONT-resumes
53 # on wake, and KeepAlive.Crashed=true caused relaunch on any non-zero exit;
54 # the listener grandchildren (log stream | while ... and (echo n.add ...;
55 # sleep 86400) | scutil | while ...) stay alive because pipe producers don't
56 # reliably respond to plain SIGTERM if the consumer is in a half-closed state.
57 #
58 # We solve it with a PID-file lock (POSIX-portable; macOS does NOT ship `flock`).
59 # The trap below removes the lock file on every exit so a stale lock never
60 # outlives a crashed watcher. On launch: read the existing PID, check it is
61 # alive AND is actually a `scripts/watcher.sh` process; if so, exit 0; else
62 # overwrite with our PID and proceed. The check-then-write window has a TOCTOU
63 # race but is microseconds wide and only matters for hand-launched duplicate
64 # test invocations; production is single-launch via launchctl load -w.
65 LOCK_FILE="/tmp/nullexit-watcher.lock"
66 LOCK_PID=""
67 if [ -e "$LOCK_FILE" ]; then
68     LOCK_PID=$(cat "$LOCK_FILE" 2>/dev/null || echo "")
69     if [ -n "$LOCK_PID" ] && kill -0 "$LOCK_PID" 2>/dev/null; then
70         # Verify the holder is actually a scripts/watcher.sh process (not some
71         # unrelated process that got the recycled PID). The exact-launchd pattern
72         # `bash <abs-path>/scripts/watcher.sh` is hard to accidentally match with
73         # a `grep scripts/watcher.sh` invocation because we anchor on `^bash`
74         # followed by whitespace and end-of-line on the script path.
75         if ps -p "$LOCK_PID" -o args= 2>/dev/null | grep -qE "(^[[[:space:]])bash[[[:space:]]+.*scripts/watcher\.sh$"; then
76             echo "[$(date -u +%FT%TZ)] watcher.sh: another instance holds $LOCK_FILE (pid=$LOCK_PID), exiting"
77             exit 0
78         fi
79     fi
80 fi
81 echo $$ > "$LOCK_FILE"
82
83 echo "[$(date -u +%FT%TZ)] watcher.sh start (pid=$$ ppid=$PPID user=$(id -un) lock=acquired)"
84
85 # Treat launchd-resume as a wake signal. Apple's launchd SUSPENDS LaunchAgents
86 # during system sleep and restores them on wake; during that suspended gap,
87 # macOS's `log stream` cannot catch the wake event that prompted the resume.
88 # The next thing that happens after wake IS our own startup, so we always
89 # fire one post-wake run unconditionally here. The marker check + debounce
90 # in run_recover() make this idempotent: if the gateway isn't up or we just
91 # debounced, it no-ops. Without this, the FIRST wake-after-install goes
92 # unseen and the watcher appears broken until the SECOND wake.
93 # Helper: run recover.sh --post-wake if the gateway is active
94 run_recover() {
95     local why="$1"
96     if [ ! -f "$MARKER" ]; then
97         echo "[$(date -u +%FT%TZ)] $why no $MARKER, skip"
98         return 0
99     fi
100     local now last
101     now=$(date +%s)
102     last=$(cat "$DEBOUNCE_FILE" 2>/dev/null || echo 0)
103     if [ "$(now - last)" -lt "$DEBOUNCE_SECONDS" ]; then
104         echo "[$(date -u +%FT%TZ)] $why debounced ($(now - last)s since last, threshold ${DEBOUNCE_SECONDS}s)"
105         return 0
106     fi
107     echo "$now" > "$DEBOUNCE_FILE"
108     echo "[$(date -u +%FT%TZ)] $why bash $RECOVER --post-wake"
109     bash "$RECOVER" --post-wake
110     local exit_code=$?

```

```

111 echo "$(date +%s)" > "$DEBOUNCE_FILE"
112 echo "[$(date -u +%FT%TZ)] $why exit=$exit_code (now=$(date +%s))"
113 return $exit_code
114 }
115
116 if [ -f "$MARKER" ]; then
117     run_recover "initial post-wake (launchd resume = wake signal)"
118 fi
119
120 # LAN P2P Auto-Detection
121 # Detects whether the current network allows safe Tailscale LAN P2P by:
122 # 1. Checking WPA2-Enterprise (802.1x) via airport -I always forces P2P=false
123 #    because enterprise networks almost universally have AP Client Isolation.
124 # 2. AP Isolation probe sends a broadcast ping and checks if ANY LAN host responds.
125 #    If no LAN hosts respond on a non-empty network, AP Isolation is active.
126 # Writes 'true' or 'false' to .lan_p2p_detected in the repo root.
127 # routing-fix.sh reads this file dynamically every 30s loop iteration.
128 P2P_OVERRIDE_FILE="$SCRIPT_DIR/../../lan_p2p_detected"
129
130
131 detect_lan_p2p_mode() {
132     local reason="unknown" allow="false"
133
134     # Step 1: Detect Wi-Fi security type via system_profiler (airport binary removed in macOS 14.4+)
135     # SPAirPortDataType lists all known networks; the first "Security:" line is the current association.
136     local security
137     security=$(system_profiler SPAirPortDataType 2>/dev/null \
138         | awk '/Current Network Information:/{found=1} found && /Security:/{print $NF; exit}')
139
140     if [ -z "$security" ]; then
141         # Not on Wi-Fi or system_profiler unavailable fail safe
142         reason="no-wifi" allow="false"
143     elif echo "$security" | grep -qi "Enterprise"; then
144         # WPA2/WPA3 Enterprise = 802.1x always AP isolated
145         reason="802.1x-enterprise" allow="false"
146     elif echo "$security" | grep -qi "Personal\\None\\Open"; then
147         # WPA2/WPA3 Personal or open probe for AP isolation
148         local gw
149         gw=$(route -n get 1.1.1.1 2>/dev/null | awk '/gateway:/{print $2}')
150         if [ -z "$gw" ]; then
151             reason="no-default-route" allow="false"
152         else
153             local responses
154             responses=$(ping -c 3 -t 1 -q "$gw" 2>/dev/null | awk '/packets transmitted/{print $4}')
155             if [ "${responses:-0}" -gt 0 ]; then
156                 reason="wpa-personal-lan-reachable" allow="true"
157             else
158                 reason="wpa-personal-ap-isolated" allow="false"
159             fi
160         fi
161     else
162         # Unknown security type fail safe
163         reason="unknown-security-type" allow="false"
164     fi
165
166     echo "[$(date -u +%FT%TZ)] P2P detect: security='${security:-none}' allow=${allow} (reason: ${reason})"
167     echo "$allow" > "$P2P_OVERRIDE_FILE"
168     write_host_ips
169 }
170
171 # Run once on startup
172 detect_lan_p2p_mode
173
174 # Listener 1: WAKE events via unified log
175 # Predicate picks up:
176 # * "didWake" regular kernel wake events (lid open, RTC alarm)
177 # * "Wake from Sleep" powerd-driven normal wake
178 # * "DarkWake" clamshell-display wake that didn't fully wake the
179 # Mac (relevant when lid opens during display sleep)
180 # * "Waking from" generic catch-all for variations in newer macOS
181 # `[c]` makes the match case-insensitive (the unified log uses mixed-case
182 # phrases). --info reduces noise by dropping debug/trace log levels.
183 # Subsystem-based predicate catches every powermanagement emission (willSleep,
184 # didWake, didDim, didUndim, DarkWake, etc.) regardless of message phrasing
185 # across kernel versions. Phrasing-based predicates ('didWake', 'Wake from Sleep')
186 # are fragile across macOS releases. We accept the extra log noise because the
187 # run_recover() debounce + marker check filter out anything that isn't a real
188 # gateway-relevant event.
189 (
190     # Reconnect loop: macOS rotates the unified log buffer periodically; when
191     # that happens, this `log stream` subscriber's pipe EOFs, the inner

```

```

192 # `while read` consumer terminates, and without this wrapper the listener
193 # would be silently dead for the rest of the watcher's lifetime.
194 while ;; do
195     log stream --predicate \
196         'subsystem == "com.apple.powermanagement"' \
197         --style compact --info 2>/dev/null \
198         | while IFS= read -r line; do
199             [ -z "$line" ] && continue
200             run_recover "WAKE: $(echo "$line" | head -c 100)"
201         done
202         echo "[$(date -u +%FT%TZ)] log stream subshell exited; reconnecting in 5s"
203         sleep 5
204     done
205 ) &
206
207 # Listener 2: NETWORK state changes via scutil
208 # `scutil n.watch` on a state key prints SCEventUpdate lines whenever the key
209 # changes. We watch Global/IPv4 because that's the umbrella that catches link,
210 # address, route, and DNS changes for ALL interfaces in one namespace.
211 # The pipe is kept alive by an `sleep 86400` loop so scutil never sees EOF
212 # from us (which would silently terminate the watch).
213 (
214     # Reconnect loop: scutil can exit noisily on certain state changes (rare,
215     # but observed). Without the wrapper, the network listener's pipe EOFs and
216     # every subsequent roam goes unseen until launchd restarts us.
217     while ;; do
218         (
219             # Watch both IPv4 and IPv6 umbrella state IPv6-only or dual-stack
220             # networks never fire on the IPv4 key alone, and the user's first roam
221             # in a hotel / airport / on cellular hotspot is often IPv6-first under
222             # NAT64.
223             echo "n.add State:/Network/Global/IPv4"
224             echo "n.add State:/Network/Global/IPv6"
225             echo "n.watch"
226             while true; do sleep 86400; done
227         ) | script -q /dev/null scutil 2>/dev/null \
228         | while IFS= read -r line; do
229             case "$line" in
230                 *changedKey*|*State:/Network/Global/IPv*|*n.state*|*SCEventUpdate*)
231                     detect_lan_p2p_mode
232                     run_recover "NET: $(echo "$line" | head -c 100)"
233                     ;;
234                 *) ;;
235             esac
236         done
237         echo "[$(date -u +%FT%TZ)] scutil pipe exited; reconnecting in 5s"
238         sleep 5
239     done
240 ) &
241 # Listener 3: TUNNEL_FAILED_CLOSED.marker Watcher
242 # Polls for the fail-closed marker dropped by the routing-fix container.
243 (
244     last_notified=0
245     while ;; do
246         if [ -f "$SCRIPT_DIR/./TUNNEL_FAILED_CLOSED.marker" ]; then
247             now=$(date +%s)
248             # Notify once every 5 minutes (300s) if it stays failed
249             if [ "$(now - last_notified)" -ge 300 ]; then
250                 osascript -e 'display notification "VPN Tunnel Health Check Failed. Internet traffic is blackholed to
251                 prevent IP leak." with title "NullExit Alert"' || true
252                 last_notified=$now
253             fi
254         else
255             last_notified=0
256         fi
257         sleep 3
258     done
259 ) &
260
261 # Lifetime
262 # `wait` blocks until both background pipelines exit (which they normally
263 # never do). launchctl `bootout`/SIGTERM triggers the trap, which kills the
264 # whole process group so the sleep-forever in the scutil pipe also dies.
265 # We catch TERM/INT/HUP for explicit signals and EXIT for any normal-exit path
266 # (so listeners always get cleaned up even if `exit 0` runs somewhere).
267 # `rm -f $LOCK_FILE` first so a stale lock can never block the next launch.
268 # `kill -TERM -P $$` then targets direct listener-children, then `kill 0`
269 # takes care of any backgrounded group-mates not reachable via the parent.
270 trap 'echo "[$(date -u +%FT%TZ)] watcher.sh terminating (signal/exit=$?)"; rm -f "$LOCK_FILE" 2>/dev/null ||
271 true; kill -TERM -P $$ 2>/dev/null || true; kill 0 2>/dev/null || true; exit 0' TERM INT HUP EXIT
272 wait

```

15 scripts/crypto.sh

```
1  #!/usr/bin/env bash
2  # crypto.sh - Cryptographic Integrity Check
3
4  set -euo pipefail
5  cd "$(dirname "$0")/.."
6
7  if [ "$#" -ne 1 ] || { [ "$1" != "--sign" ] && [ "$1" != "--verify" ]; }; then
8      echo "Usage: $0 --sign | --verify"
9      exit 1
10 fi
11
12 if [ ! -f .env ]; then
13     echo "Error: .env not found."
14     exit 1
15 fi
16
17 SEED=$(grep "^NULLEXIT_SEED=" .env | cut -d '=' -f2)
18 if [ -z "$SEED" ]; then
19     echo "Error: NULLEXIT_SEED not found in .env."
20     exit 1
21 fi
22
23 if [ "$1" == "--sign" ]; then
24     echo "Generating cryptographic signatures using seed..."
25     rm -f .signatures
26     for file in toggle.sh recover.sh scripts/common.sh scripts/routing-fix.sh scripts/toggle-linux.sh scripts/
27         recover-linux.sh scripts/diagnose-host-leak.sh scripts/watcher.sh scripts/pf.conf post-rules.txt docker-
28         compose.yml; do
29         if [ -f "$file" ]; then
30             hash=$(openssl dgst -sha256 -hmac "$SEED" "$file" | awk '{print $NF}')
31             echo "$file:$hash" >> .signatures
32             echo "    [Signed] $file"
33         else
34             echo "    [Skipped] $file not found"
35         fi
36     done
37     echo "Signatures saved to .signatures"
38     exit 0
39 fi
40
41 if [ "$1" == "--verify" ]; then
42     if [ ! -f .signatures ]; then
43         echo "Error: .signatures file missing. Run $0 --sign first."
44         exit 1
45     fi
46     FAIL=0
47     while IFS=: read -r file expected_hash; do
48         if [ -z "$file" ]; then continue; fi
49         if [ ! -f "$file" ]; then
50             echo "[FAIL] $file is missing!"
51             FAIL=1
52             continue
53         fi
54         actual_hash=$(openssl dgst -sha256 -hmac "$SEED" "$file" | awk '{print $NF}')
55         if [ "$actual_hash" != "$expected_hash" ]; then
56             echo "[FAIL] $file has been modified! Hash mismatch."
57             FAIL=1
58         fi
59     done < .signatures
60
61     if [ "$FAIL" -eq 1 ]; then
62         echo ""
63         echo "CRITICAL: Script integrity verification failed!"
64         echo "One or more core files have been modified without authorization."
65         echo "To authorize these changes, run: ./scripts/crypto.sh --sign"
66         echo ""
67         exit 1
68     fi
69     exit 0
70 fi
```

16 .aiignore

```
1 # Environment variables and secrets
2 .env
3 tor-bridges.txt
```

17 .gitignore

```
1 # Environment variables and secrets
2 .env
3
4 # wgcf generated files containing keys and account info
5 wgcf-account.toml
6 wgcf-profile.conf
7
8 # Tailscale persistent state containing node identity
9 tailscale-state/
10
11 # Binaries
12 wgcf
13
14 # macOS App bundles
15 *.app/
16 *.app
17
18 # AdGuard Home persistent state
19 adguard/
20 .gateway_ip
21 logs.txt
22
23 #
24 stability_incident_report.md
25 output.log
26 .DS_Store
27 *.desktop
28 blocked.log
29 wtf.md
30 host-leak-diagnostic-*.txt
31 nullexit_review.md
32 nullexit_unified.tex
33 nullexit_unified.txt
34
35 nullexit_unified.synctex.gz
36 tor-bridges.txt
37
38 # Runtime state files
39 .host_ips
40 .lan_p2p_detected
41 .signatures
42 TUNNEL_FAILED_CLOSED.marker
43
44 # Python cache
45 __pycache__/_
46 *.py[cod]
```

18 .host__ips

```
1 172.17.52.223
2 100.109.94.19
3 192.168.64.1
```

19 Recover-Gateway.applescript

```
1 tell application "Finder" to set scriptDir to POSIX path of (container of (path to me) as alias)
2 try
3     do shell script "true" with administrator privileges
```

```

4 on error
5     return
6 end try
7 tell application "Terminal"
8     activate
9     do script "cd " & quoted form of scriptDir & "; bash recover.sh"
10 end tell

```

20 Toggle-Gateway.applescript

```

1 tell application "Finder" to set scriptDir to POSIX path of (container of (path to me) as alias)
2 try
3     do shell script "true" with administrator privileges
4 on error
5     return
6 end try
7 tell application "Terminal"
8     activate
9     do script "cd " & quoted form of scriptDir & "; ./toggle.sh"
10 end tell

```

21 benchmarks/benchmark.sh

```

1 #!/usr/bin/env bash
2
3 # benchmark.sh
4 # Automates the full benchmarking suite (both Python download test and Lighthouse HTML test)
5 # Runs the suite with Gateway ON, toggles OFF, runs again, then toggles back ON.
6
7 set -e
8
9 # Change to project root so we can access toggle.sh reliably
10 cd "$(dirname "$0")/.."
11
12 # Ensure jq is installed
13 if ! command -v jq &> /dev/null; then
14     echo "Error: 'jq' is not installed. Please run 'brew install jq'"
15     exit 1
16 fi
17
18 run_lighthouse() {
19     local url=$1
20     local file_prefix=$2
21     echo " [Lighthouse] Testing $url ..."
22
23     # Run lighthouse headlessly, silencing standard output
24     npx -y lighthouse "$url" --chrome-flags="--headless" --only-categories=performance --output=json --output-
25     path="${file_prefix}.json" >/dev/null 2>&1
26
27     # Parse the results
28     local score=$(jq -r '.categories.performance.score' "${file_prefix}.json")
29     local tti=$(jq -r '.audits["interactive"].displayValue' "${file_prefix}.json")
30     local si=$(jq -r '.audits["speed-index"].displayValue' "${file_prefix}.json")
31
32     # Convert score to percentage
33     local score_pct=$(awk "BEGIN {print $score*100}")
34
35     echo "    -> Performance Score: ${score_pct}%"
36     echo "    -> Time to Interactive: $tti"
37     echo "    -> Speed Index: $si"
38
39     # Cleanup
40     rm -f "${file_prefix}.json"
41 }
42
43 echo "=====
44 echo " PHASE 1: Testing with Gateway ON"
45 echo "=====
46 # 1. Run raw Python download test
47 python3 benchmarks/test_load.py
48 # 2. Run Lighthouse HTML render test

```

```

49 echo ""
50 echo "Starting Lighthouse HTML Tests (this takes ~30-60s per site)..."
51 run_lighthouse "https://www.cnet.com/" "on_cnet"
52 run_lighthouse "https://www.independent.co.uk/" "on_ind"
53
54
55 echo ""
56 echo "=====
57 echo "    PHASE 2: Toggling Gateway OFF"
58 echo "=====
59 ./toggle.sh
60 echo "Waiting 10 seconds for macOS Wi-Fi to stabilize after DNS reset..."
61 sleep 10
62
63
64 echo ""
65 echo "=====
66 echo "    PHASE 3: Testing with Gateway OFF"
67 echo "=====
68 # 1. Run raw Python download test
69 python3 benchmarks/test_load.py
70
71 # 2. Run Lighthouse HTML render test
72 echo ""
73 echo "Starting Lighthouse HTML Tests (this takes ~30-60s per site)..."
74 run_lighthouse "https://www.cnet.com/" "off_cnet"
75 run_lighthouse "https://www.independent.co.uk/" "off_ind"
76
77
78 echo ""
79 echo "=====
80 echo "    PHASE 4: Restoring Gateway ON"
81 echo "=====
82 ./toggle.sh
83
84 echo ""
85 echo "=====
86 echo " BENCHMARKS COMPLETE!"
87 echo "You can scroll up to compare Phase 1 (AdGuard ON) vs Phase 3 (AdGuard OFF)."
88 echo "=====

```

22 benchmarks/test_load.py

```

1 import urllib.request
2 import urllib.error
3 import time
4 from html.parser import HTMLParser
5 import concurrent.futures
6
7 class ResourceParser(HTMLParser):
8     def __init__(self):
9         super().__init__()
10        self.urls = set()
11
12    def handle_starttag(self, tag, attrs):
13        attrs = dict(attrs)
14        url = None
15        if tag in ['img', 'script']:
16            url = attrs.get('src')
17        elif tag == 'link' and attrs.get('rel') == 'stylesheet':
18            url = attrs.get('href')
19
20        if url and url.startswith('http'):
21            self.urls.add(url)
22
23    def fetch_url(url):
24        try:
25            req = urllib.request.Request(url, headers={'User-Agent': 'Mozilla/5.0'})
26            urllib.request.urlopen(req, timeout=3)
27            return True
28        except Exception:
29            return False
30
31    def measure_url(target_url):
32        print(f"\nTesting {target_url} Load...")
33        start_total = time.time()

```

```

34
35 # Fetch main HTML
36 try:
37     req = urllib.request.Request(target_url, headers={'User-Agent': 'Mozilla/5.0 (Macintosh; Intel Mac OS X
10_15_7) AppleWebKit/537.36'})
38     response = urllib.request.urlopen(req, timeout=5)
39     html = response.read().decode('utf-8', errors='ignore')
40 except Exception as e:
41     print(f"Failed to fetch {target_url}: {e}")
42     return
43
44 parser = ResourceParser()
45 parser.feed(html)
46 urls = list(parser.urls)
47 print(f"Found {len(urls)} subresources to fetch (scripts, images, css)...")
48
49 # Concurrently fetch subresources
50 success_count = 0
51 fail_count = 0
52 max_threads = min(64, max(1, len(urls)))
53 with concurrent.futures.ThreadPoolExecutor(max_workers=max_threads) as executor:
54     results = list(executor.map(fetch_url, urls))
55
56 success_count = sum(1 for r in results if r)
57 fail_count = len(results) - success_count
58
59 total_time = time.time() - start_total
60 print(f"Time taken: {total_time:.2f} seconds")
61 print(f"Successfully fetched: {success_count}, Failed/Blocked: {fail_count}")
62
63 urls_to_test = [
64     "https://www.dailymail.co.uk/home/index.html",
65     "https://www.cnet.com/",
66     "https://www.independent.co.uk/"
67 ]
68
69 for u in urls_to_test:
70     measure_url(u)

```

23 black_list.txt

```

1 ad.doubleclick.net
2 adclick.g.doubleclick.net
3 adeventtracker.spotify.com
4 adfstat.yandex.ru
5 ads-api.tiktok.com
6 ads-api.twitter.com
7 ads-fa.spotify.com
8 ads-sg.tiktok.com
9 ads.tiktok.com
10 adserver.unityads.unity3d.com
11 adsfs.oppomobile.com
12 an.facebook.com$important
13 connect.facebook.net$important
14 pixel.facebook.com$important
15 analytics.query.yahoo.com
16 analytics.s3.amazonaws.com
17 analytics.spotify.com
18 analyticsengine.s3.amazonaws.com
19 appmetrica.yandex.ru
20 audio2.spotify.com
21 bdapi-ads.realmemobile.com
22 bdapi-in-ads.realmemobile.com
23 bounceexchange.com
24 bs.serving-sys.com
25 business-api.tiktok.com
26 ck.ads.oppomobile.com
27 crashdump.spotify.com
28 data.ads.oppomobile.com
29 data.mistat.india.xiaomi.com
30 data.mistat.rus.xiaomi.com
31 data.mistat.xiaomi.com
32 desktop.spotify.com
33 doubleclick.net
34 ds.metric.spotify.com
35 gemini.yahoo.com

```

```

36 googleads.g.doubleclick.net
37 googleadservices.com
38 googlesyndication.com
39 grs.hicloud.com
40 gtm.spotify.com
41 iot-eu-logser.realme.com
42 iot-logser.realme.com
43 log.byteoversea.com
44 log.fc.yahoo.com
45 log.spotify.com
46 m.doubleclick.net
47 mediavisor.doubleclick.net
48 metrics.spotify.com
49 metrika.yandex.ru
50 omaze.com
51 pagead2.googlesyndication.com
52 securepubads.g.doubleclick.net
53 static.doubleclick.net
54 stats.g.doubleclick.net
55 tracking.rus.miui.com
56 udc.m.yahoo.com
57 v.jwpcdn.com
58 weblb-wg.gslb.spotify.com
59 webview.unityads.unity3d.com
60 diagassets.apple.com
61 iphonesubmissions.apple.com
62 radarsubmissions.apple.com
63 metrics.apple.com
64 xp.apple.com
65 heads-fa.spotify.com
66 heads4.spotify.com
67 pixel.spotify.com
68 segs.spotify.com
69 audio-fa.spotify.com
70 events.data.microsoft.com
71 datarouter-prod.ak.epicgames.com
72 nel.cloudflare.com
73 eu-mobile.events.data.microsoft.com
74 stocks-analytics-events.apple.com

```

24 cloud-init.yaml

```

1 #cloud-config
2 # Generic Cloud-Init Bootstrap Script for Remote Exit Nodes (Ubuntu/Debian)
3 # Paste this into the "User Data" or "Initialization Script" section when creating a VPS.
4
5 package_update: true
6 package_upgrade: true
7
8 packages:
9   - apt-transport-https
10   - ca-certificates
11   - curl
12   - gnupg
13   - lsb-release
14   - git
15   - python3
16   - python3-pip
17
18 runcmd:
19   # 1. Install Docker natively
20   - curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /usr/share/keyrings/docker-
     archive-keyring.gpg
21   - echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/docker-archive-keyring.gpg]
     https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/
     docker.list > /dev/null
22   - sudo apt-get update
23   - sudo apt-get install -y docker-ce docker-ce-cli containerd.io docker-compose-plugin
24
25   # 2. Add default user (ubuntu) to docker group
26   - usermod -aG docker ubuntu || true
27   - usermod -aG docker debian || true
28
29   # 3. Create the installation directory
30   - mkdir -p /opt/nullexit
31   - chown -R ubuntu:ubuntu /opt/nullexit || chown -R debian:debian /opt/nullexit

```

```

32
33 # 4. Enable IPv4 and IPv6 forwarding for VPN routing
34 - echo "net.ipv4.ip_forward=1" >> /etc/sysctl.d/99-tailscale.conf
35 - echo "net.ipv6.conf.all.forwarding=1" >> /etc/sysctl.d/99-tailscale.conf
36 - sysctl -p /etc/sysctl.d/99-tailscale.conf
37
38 final_message: "The remote exit node has been fully bootstrapped. SSH in, clone the repo to /opt/nullexit, add
    your .env, and run ./setup-linux.sh!"

```

25 docker/tor/Dockerfile

```

1 FROM debian:bookworm-slim
2
3 RUN apt-get update && \
4     apt-get install -y tor obfs4proxy gosu bash && \
5     rm -rf /var/lib/apt/lists/*
6
7 COPY entrypoint.sh /entrypoint.sh
8 RUN chmod +x /entrypoint.sh
9
10 ENTRYPOINT ["/entrypoint.sh"]

```

26 docker/tor/entrypoint.sh

```

1 #!/bin/bash
2 set -e
3
4 TORRC="/etc/tor/torrc"
5 mkdir -p /etc/tor /var/lib/tor
6
7 # Base configuration (SOCKS proxy on 9050, strictly localhost/container boundaries)
8 cat <<EOF > $TORRC
9 SocksPort 0.0.0.0:9050
10 Log notice stdout
11 DataDirectory /var/lib/tor
12 EOF
13
14 if [ "${TOR_USE_BRIDGES:-false}" = "true" ]; then
15     BRIDGE_FILE="${TOR_BRIDGE_LINES_FILE:-/tor-bridges.txt}"
16
17     if ! grep -vE "^[[:space:]]*#" "$BRIDGE_FILE" | grep -q '^[[:space:]]'; then
18         MSG="[$(date +%Y-%m-%d %H:%M:%S)] [CRITICAL] [Tor Proxy] TOR_USE_BRIDGES is enabled, but no bridge
19             lines found at $BRIDGE_FILE. Proxy startup ABORTED. Get bridges from https://bridges.torproject.org."
20         echo "$MSG"
21         if [ -w "/output.log" ]; then
22             echo "$MSG" >> /output.log
23         fi
24         # Sleep indefinitely to prevent docker-compose 'unless-stopped' from triggering a crash loop
25         exec sleep infinity
26     fi
27
28     echo "UseBridges 1" >> $TORRC
29     echo "ClientTransportPlugin obfs4 exec /usr/bin/obfs4proxy" >> $TORRC
30
31     while IFS= read -r line; do
32         # Ignore comments and empty lines
33         [[ -z "$line" || "$line" == \#* ]] && continue
34         echo "Bridge $line" >> $TORRC
35     done < "$BRIDGE_FILE"
36 fi
37
38 # Debian tor package uses 'debian-tor' user
39 chown -R debian-tor:debian-tor /var/lib/tor /etc/tor
40
41 exec gosu debian-tor /usr/bin/tor -f $TORRC

```

27 launchd/com.nullexit.wake-recovery.plist

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
3 <plist version="1.0">
4 <dict>
5   <key>Label</key>
6   <string>com.nullexit.wake-recovery</string>
7
8   <!-- WorkingDirectory + relative program arg keeps this plist install-agnostic.
9        The source-of-truth for the install path is __NULLEXIT_HOME__, which
10       setup.sh or the user must substitute with their absolute nullexit
11       install root at install time. Program arg is `scripts/watcher.sh`
12       RELATIVE to WorkingDirectory, so launchd's evaluated cwd IS the
13       install root and BASH_SOURCE inside scripts/watcher.sh still
14       resolves to the real <install>/scripts/watcher.sh, preserving the
15       SCRIPT_DIR pattern. After install-time `sed -i ' ' 's|__NULLEXIT_HOME__|<abs_path>|'`,
16       the plist is portable to any clone location. -->
17   <key>WorkingDirectory</key>
18   <string>__NULLEXIT_HOME__</string>
19
20   <key>ProgramArguments</key>
21   <array>
22     <string>/bin/bash</string>
23     <string>scripts/watcher.sh</string>
24   </array>
25
26   <!-- Start the daemon immediately on launchctl load (instead of waiting
27        for the next user login). Combined with keepAlive below this means
28        once installed: load once, runs forever across logouts/reboots. -->
29   <key>RunAtLoad</key>
30   <true/>
31
32   <!-- Restart policy:
33        SuccessfulExit=false    SIGTERM / clean exit does NOT relaunch (so
34                                `launchctl bootout` cleanly stops, not loops).
35        Crashed=false          HARD crashes ALSO do NOT relaunch. We've seen
36                                that repeated sleep/wake cycles on macOS
37                                Sonoma+ accumulate orphan watcher.sh PIDs
38                                because SIGCONT/resume doesn't reliably kill
39                                the prior instance. The watcher.sh script
40                                now enforces single-instance via a manual
41                                PID-file and process-identity check at the
42                                top, so a re-launch on crash is
43                                actively dangerous (it would skip the lock
44                                and exit; better to surface the crash in
45                                logs than to silently 0-process). -->
46   <key>KeepAlive</key>
47   <dict>
48     <key>SuccessfulExit</key>
49     <false/>
50     <key>Crashed</key>
51     <false/>
52   </dict>
53
54   <!-- Background hint to App Nap so it doesn't get suspended when
55        the user is inactive. We're the entire reason this LaunchAgent
56        needs to NOT nap. -->
57   <key>ProcessType</key>
58   <string>Background</string>
59
60   <!-- launchd-level stdout/stderr (the bash launcher shell). The script
61        itself writes to /tmp/nullexit-watcher.log via exec >>. These
62        separate channels help us tell "launchd crashed" from "the bash
63        wrapper piped to recover.sh errored". -->
64   <key>StandardOutPath</key>
65   <string>/tmp/nullexit-watcher.out.log</string>
66   <key>StandardErrorPath</key>
67   <string>/tmp/nullexit-watcher.err.log</string>
68 </dict>
69 </plist>

```

28 post-rules.txt

```

1 iptables -A FORWARD -i tailscale0 -o tun0 -j ACCEPT
2 iptables -A FORWARD -i tun0 -o tailscale0 -j ACCEPT
3 iptables -A INPUT -i tailscale0 -j ACCEPT
4 iptables -A INPUT -i eth0 -p udp --dport 41642 -j ACCEPT

```

```

5 iptables -t nat -A POSTROUTING -o tun0 -j MASQUERADE
6 # Clamp MSS to 1120. With Tailscale overhead on top of WARP (each WireGuard encapsulation adds ~60 bytes),
7 # the safe MSS for double-tunneled traffic is 1120 to prevent strict-MSS endpoints from stalling.
8 iptables -t mangle -A POSTROUTING -p tcp --tcp-flags SYN,RST SYN -j TCPMSS --set-mss 1120
9
10 iptables -t nat -A PREROUTING -i tailscale0 -p udp --dport 53 -j REDIRECT --to-port 5335
11 iptables -t nat -A PREROUTING -i tailscale0 -p tcp --dport 53 -j REDIRECT --to-port 5335
12
13 # Also redirect DNS from Docker bridge interface so the host can reach AdGuard via localhost:53
14 iptables -t nat -A PREROUTING -i eth0 -p udp --dport 53 -j REDIRECT --to-port 5335
15 iptables -t nat -A PREROUTING -i eth0 -p tcp --dport 53 -j REDIRECT --to-port 5335
16
17 # Block IPv6 exit-node traffic to prevent leakage (Gluetun/WARP is IPv4-only)
18 ip6tables -A FORWARD -i tailscale0 -j DROP
19
20 # Block DNS-over-TLS (Port 853) to force Android Private DNS to fallback to Port 53
21 iptables -A FORWARD -i tailscale0 -p tcp --dport 853 -j REJECT --reject-with tcp-reset
22 iptables -A FORWARD -i tailscale0 -p udp --dport 853 -j REJECT
23
24 # Block QUIC (UDP 443) to force Facebook/Google to fallback to TCP.
25 # UDP bypasses TCP MSS clamping, causing fragmentation. Over weak Wi-Fi (far from gateway),
26 # dropping a single fragment destroys the entire packet, stalling image downloads.
27 iptables -A FORWARD -i tailscale0 -p udp --dport 443 -j REJECT

```

29 scripts/Dockerfile.routing-fix

```

1 FROM golang:1.22-alpine AS builder
2 WORKDIR /app
3 COPY logger/ ./logger/
4 RUN cd logger && go build -o logger main.go
5
6 FROM alpine:3.20
7 RUN apk add --no-cache iptables iproute2 curl ipset
8 COPY --from=builder /app/logger/logger /usr/local/bin/nullexit-logger
9 COPY routing-fix.sh /routing-fix.sh
10 CMD ["sh", "/routing-fix.sh"]

```

30 scripts/Dockerfile.rule-compiler

```

1 FROM golang:1.22-alpine AS builder
2 WORKDIR /app
3 COPY rule-compiler/ ./rule-compiler/
4 RUN cd rule-compiler && go build -o sync-rules main.go
5
6 FROM alpine:3.20
7 COPY --from=builder /app/rule-compiler/sync-rules /usr/local/bin/sync-rules
8 WORKDIR /app
9 CMD ["sync-rules"]

```

31 scripts/Toggle-Gateway.bat

```

1 <# :
2 @echo off
3 powershell -NoProfile -ExecutionPolicy Bypass -Command "Invoke-Expression ([System.IO.File]::ReadAllText('%-f0
4 ')) -replace '^(?s).*?#\s*>\s*\r?\n','')
5 exit /b
6 #>
7 # Toggle-Gateway.bat (Hybrid Batch-PowerShell Toggler) nullexit Gateway Toggler for Windows
8 # Mirrors the sophisticated error handling, checks, and automation of toggle.sh
9 # 1. Administrator Check & Elevation
10 $isAdmin = ([Security.Principal.WindowsPrincipal][Security.Principal.WindowsIdentity]::GetCurrent()).IsInRole([
11 Security.Principal.WindowsBuiltInRole]::Administrator)
12 if (-not $isAdmin) {
13     Write-Host "Requesting Administrator privileges..." -ForegroundColor Yellow
14     Start-Process PowerShell -ArgumentList "-NoProfile -ExecutionPolicy Bypass -Command `\"Invoke-Expression ([
15 System.IO.File]::ReadAllText('$PSCommandPath') -replace '^(?s).*?#\s*>\s*\r?\n','')`\" -Verb RunAs
16 Exit

```

```

15 }
16
17 # Set console output encoding to UTF8 for clean symbols
18 [Console]::OutputEncoding = [System.Text.Encoding]::UTF8
19 Clear-Host
20
21 Write-Host "" -ForegroundColor Green
22 Write-Host "          nullexit Windows Toggler          " -ForegroundColor Green
23 Write-Host "" -ForegroundColor Green
24 Write-Host ""
25
26 # 2. Helpers
27 function Reset-HostDNS {
28     Write-Host "Restoring default DNS (DHCP/DHCP-assigned DNS)..." -ForegroundColor Cyan
29     $ActiveAdapters = Get-NetAdapter | Where-Object { $_.Status -eq 'Up' }
30     foreach ($adapter in $ActiveAdapters) {
31         Write-Host "    -> Resetting DNS on adapter: $($adapter.Name)" -ForegroundColor Gray
32         Set-DnsClientServerAddress -InterfaceIndex $adapter.InterfaceIndex -ResetServerAddresses -ErrorAction
33         SilentlyContinue
34     }
35 }
36
37 function Hijack-HostDNS ($ip) {
38     Write-Host "Hijacking DNS to route through AdGuard ($ip)..." -ForegroundColor Yellow
39     $ActiveAdapters = Get-NetAdapter | Where-Object { $_.Status -eq 'Up' }
40     foreach ($adapter in $ActiveAdapters) {
41         Write-Host "    -> Pointing DNS to $ip on adapter: $($adapter.Name)" -ForegroundColor Gray
42         Set-DnsClientServerAddress -InterfaceIndex $adapter.InterfaceIndex -ServerAddresses $ip -ErrorAction
43         SilentlyContinue
44     }
45 }
46
47 function Get-HostTailscalePath {
48     $paths = @(
49         "$env:ProgramFiles\Tailscale\tailscale.exe",
50         "$env:ProgramFiles (x86)\Tailscale\tailscale.exe",
51         "tailscale.exe"
52     )
53     foreach ($path in $paths) {
54         if (Test-Path $path) { return $path }
55     }
56     return $null
57 }
58
59 # 3. Detect State
60 # Prevent deadlocks by resetting DNS to default temporarily during checks
61 Reset-HostDNS
62
63 Write-Host "Checking Docker daemon status..." -ForegroundColor Cyan
64 & docker info >$null 2>&1
65 if ($LASTEXITCODE -ne 0) {
66     Write-Host "Docker is not running. Starting Docker Desktop..." -ForegroundColor Yellow
67     $dockerPath = "C:\Program Files\Docker\Docker\Docker Desktop.exe"
68     if (Test-Path $dockerPath) {
69         Start-Process $dockerPath
70         Write-Host "Waiting for Docker daemon to become responsive..." -ForegroundColor Gray
71         $timeout = 60
72         while ($timeout -gt 0) {
73             Start-Sleep -Seconds 2
74             & docker info >$null 2>&1
75             if ($LASTEXITCODE -eq 0) {
76                 Write-Host "Docker is ready!" -ForegroundColor Green
77                 break
78             }
79             $timeout -= 2
80         }
81         if ($timeout -le 0) {
82             Write-Error "Docker failed to start in time. Aborting."
83             Exit 1
84         }
85     } else {
86         Write-Error "Docker Desktop executable not found at standard path. Please start Docker manually."
87         Exit 1
88     }
89 }
90
91 # Detect if gateway is already running
92 $runningWarp = docker compose ps --status running --format json | ConvertFrom-Json -ErrorAction
93     SilentlyContinue | Where-Object { $_.Service -eq "warp" }
94 $isGatewayActive = ($null -ne $runningWarp)

```

```

93 # 4. Execution
94 if ($isGatewayActive) {
95     # STOP PATH
96     Write-Host "Gateway is currently RUNNING. Disabling now..." -ForegroundColor Yellow
97     Write-Host ""
98
99     # Disconnect host Tailscale from exit node if installed
100     $tsCli = Get-HostTailscalePath
101     if ($null -ne $tsCli) {
102         Write-Host "Disconnecting host Tailscale from exit node..." -ForegroundColor Cyan
103         & $tsCli up --exit-node= --accept-dns=true >$null 2>&1
104     }
105
106     Write-Host "Stopping Docker containers..." -ForegroundColor Cyan
107     docker compose down -t 5
108
109     Reset-HostDNS
110     Write-Host "Gateway has been successfully STOPPED." -ForegroundColor Green
111 } else {
112     # START PATH
113     Write-Host "Gateway is currently STOPPED. Enabling now..." -ForegroundColor Yellow
114     Write-Host ""
115
116     # Clean up corrupted AdGuardHome configurations
117     $confFile = "adguard\conf\AdGuardHome.yaml"
118     if (Test-Path $confFile) {
119         $fileSize = (Get-Item $confFile).Length
120         if ($fileSize -eq 0) {
121             Remove-Item $confFile -Force
122             Write-Host "Removed corrupted empty AdGuardHome.yaml to prevent container crash loop." -
123                 ForegroundColor Yellow
124         }
125     }
126
127     Write-Host "Starting Docker containers..." -ForegroundColor Cyan
128     docker compose up -d
129
130     Write-Host "Waiting for container's Tailscale connection to offer Exit Node..." -ForegroundColor Cyan
131     Write-Host " (This can take up to 60 seconds..." -ForegroundColor Gray
132
133     $elapsed = 0
134     $maxWait = 60
135     $resolvedIP = $null
136
137     while ($elapsed -lt $maxWait) {
138         $statusJson = docker compose exec -T tailscale tailscale status --json 2>$null
139         if ($null -ne $statusJson) {
140             $status = $statusJson | ConvertFrom-Json -ErrorAction SilentlyContinue
141             if ($null -ne $status -and $null -ne $status.Self) {
142                 # Check if it has a valid Tailscale IP and is running
143                 if ($status.Self.Online -and $status.Self.TailscaleIPs.Count -gt 0) {
144                     $resolvedIP = $status.Self.TailscaleIPs[0]
145                     break
146                 }
147             }
148             Start-Sleep -Seconds 2
149             $elapsed += 2
150         }
151     }
152
153     if ($null -ne $resolvedIP) {
154         Write-Host "Tailscale IP resolved: $resolvedIP" -ForegroundColor Green
155         Write-Host ""
156
157         # Configure host Tailscale client to route through this exit node
158         $tsCli = Get-HostTailscalePath
159         if ($null -ne $tsCli) {
160             Write-Host "Configuring host Tailscale to use Exit Node ($resolvedIP)..." -ForegroundColor Cyan
161             & $tsCli up --exit-node=$resolvedIP --accept-dns=false >$null 2>&1
162         }
163
164         # Hijack DNS
165         Hijack-HostDNS $resolvedIP
166         Write-Host ""
167         Write-Host "Gateway has been successfully STARTED." -ForegroundColor Green
168     } else {
169         Write-Host "Warning: Could not resolve gateway Tailscale IP. DNS hijacking skipped." -ForegroundColor
170         Red
171
172         # Fallback to .gateway_ip if it exists
173         if (Test-Path ".gateway_ip") {
174             $fallbackIP = Get-Content ".gateway_ip" -TotalCount 1
175         }
176     }
177 }

```

```

172         if (-not [string]::IsNullOrEmpty($fallbackIP)) {
173             Write-Host "[Fallback] Using static IP from .gateway_ip: $fallbackIP" -ForegroundColor Yellow
174             Hijack-HostDNS $fallbackIP
175             Write-Host "Gateway STARTED (with static fallback IP)." -ForegroundColor Green
176         }
177     } else {
178         Write-Host "Gateway started in offline/unauthenticated state. Run setup.sh if this is a fresh setup"
179         . " -ForegroundColor Yellow
180     }
181 }
182
183 Write-Host ""
184 Write-Host "Press any key to exit..."
185 [void]$Host.UI.RawUI.ReadKey("NoEcho,IncludeKeyDown")

```

32 scripts/diagnose-host-leak.sh

```

1  #!/bin/bash
2  # scripts/diagnose-host-leak.sh  nullexit host-routing diagnostic
3  #
4  # Symptom this addresses: tests like https://www.whatismyip.com on the
5  # *HOST MAC* return the underlying physical ISP/ASN
6  # local ISP / campus network") even though the nullexit gateway is active and remote
7  # tailnet clients correctly egress via Cloudflare WARP. The container and
8  # the gateway itself are healthy only the host's own routing is leaking.
9  #
10 # This script runs 7 checks in sequence, classifies the result into ONE
11 # of the three known leak scenarios, writes the full report to a
12 # timestamped file in the project root, and prints a verdict + a
13 # ready-to-run fix command on stdout. Pass --fix to apply the matching
14 # remediation in the same invocation and re-verify egress afterwards.
15 # See devref.md 10.30 for the deep-dive rationale.
16 #
17 # Usage:
18 #   bash scripts/diagnose-host-leak.sh           # diagnose + write report
19 #   bash scripts/diagnose-host-leak.sh --fix      # diagnose + fix + re-verify
20 #   bash scripts/diagnose-host-leak.sh --watch   # full baseline then loop checks every 60s
21 #   bash scripts/diagnose-host-leak.sh --watch 30 # same, with custom interval (seconds)
22 #   bash scripts/diagnose-host-leak.sh --help    # this message
23 #
24 # Multiple runs produce timestamped files (one per run) so historical
25 # reports can be diffed. The newest file is the most recent.
26
27 set -uo pipefail
28 # NOTE: errexit (`set -e`) is intentionally NOT enabled. Several checks
29 # below are EXPECTED to fail and that's diagnostic data (e.g. `tailscale
30 # status` returns non-zero when the daemon is wedged). Every command
31 # pipes to a helper that records the exit code without killing the script.
32
33 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
34 PROJECT_ROOT="$(cd "$SCRIPT_DIR/.." && pwd)"
35 TIMESTAMP="$(date -u +%Y%m%dT%H%M%SZ)"
36 source "$SCRIPT_DIR/common.sh"
37 LOG_FILE="$PROJECT_ROOT/output.log"
38 REPORT_FILE="$PROJECT_ROOT/host-leak-diagnostic-$TIMESTAMP.txt"
39
40 # Argument parsing
41 DO_FIX=false
42 DO_WATCH=false
43 WATCH_INTERVAL=60
44 WATCH_LOG="$PROJECT_ROOT/host-leak-watch.log"
45 case "${1:-}" in
46     --fix) DO_FIX=true ;;
47     --watch)
48         DO_WATCH=true
49         # Optional second argument: custom interval in seconds
50         if [ -n "${2:-}" ] && [[ "$2" =~ ^[0-9]+$ ]]; then
51             WATCH_INTERVAL="$2"
52         fi
53     ;;
54     --help|-h)
55         sed -n '2,31p' "$0"
56         exit 0
57     ;;
58     *)

```

```

59 : # default: diagnose only
60 ;;
61 *)
62     echo "Unknown argument: $1" >&2
63     echo "Run with --help for usage." >&2
64     exit 2
65 ;;
66 esac
67
68 # Colours (auto-disabled when stdout is not a tty, e.g. piped)
69 if [ -t 1 ]; then
70     RED='\033[0;31m'; GREEN='\033[0;32m'; YELLOW='\033[1;33m'
71     BOLD='\033[1m'; NC='\033[0m'
72 else
73     RED=''; GREEN=''; YELLOW=''; BOLD=''; NC=''
74 fi
75
76 # Output helpers write to BOTH stdout and the report file
77 exec 3>> "$REPORT_FILE" || {
78     printf '\n[FATAL] cannot open %s for write\n' "$REPORT_FILE" >&2
79     exit 1
80 }
81 section() {
82     local title="$1"
83     printf '\n%b %s %b\n' "$BOLD" "$title" "$NC" >&3
84     printf '\n%b %s %b\n' "$BOLD" "$title" "$NC"
85 }
86 line() { printf ' %b\n' "$1" >&3; printf ' %b\n' "$1"; }
87 ok() { line "${GREEN}${NC} $1"; }
88 warn() { line "${YELLOW}${NC} $1"; }
89 fail() { line "${RED}${NC} $1"; }
90 note() { line "(no ${1})"; }
91
92 # Watch-mode header (only shown when entering watch loop)
93 if [ "$DO_WATCH" = "true" ]; then
94     echo ""
95     echo ""
96     echo " n u l l e x i t   w a t c h   m o d e "
97     echo ""
98     echo " Interval:          ${WATCH_INTERVAL}s"
99     echo " Watch log:          $WATCH_LOG"
100     echo ""
101 else
102     echo ""
103     echo " n u l l e x i t   h o s t - r o u t i n g   d i a g n o s t i c "
104     echo ""
105     echo " Timestamp (UTC): $TIMESTAMP"
106     echo " Project root:    $PROJECT_ROOT"
107     echo " Report file:     $REPORT_FILE"
108     echo " Mode:            ${([ "$DO_FIX" = true ] && echo "diagnose + apply fix" || echo "diagnose only (pass --
109         fix to remediate))}"
109     echo ""
110     echo "" >&3
111     echo "" >&3
112     echo " n u l l e x i t   h o s t - r o u t i n g   d i a g n o s t i c " >&3
113     echo "" >&3
114     echo " Timestamp (UTC): $TIMESTAMP" >&3
115     echo " Mode:            ${([ "$DO_FIX" = true ] && echo "diagnose + apply fix" || echo "diagnose only")}" >&3
116 fi
117
118 # Always add Homebrew to PATH (same as toggle.sh / recover.sh)
119 export PATH="/opt/homebrew/bin:/usr/local/bin:$PATH"
120
121 # Pure-bash timeout (macOS lacks GNU `timeout`)
122 run_with_timeout() {
123     local timeout_sec="$1"
124     shift
125     if [ "$#" -eq 0 ]; then return 1; fi
126     "$@" &
127     local cmd_pid=$!
128     (
129         sleep "$timeout_sec"
130         if kill -0 "$cmd_pid" 2>> "$LOG_FILE"; then
131             kill -9 "$cmd_pid" 2>> "$LOG_FILE" || true
132         fi
133     ) >> "$LOG_FILE" 2>&1 &
134     local watcher_pid=$!
135     set +e
136     wait "$cmd_pid"
137     local exit_status=$?
138     set -uo pipefail

```

```

139 kill "$watcher_pid" 2>> "$LOG_FILE" || true
140 return "$exit_status"
141 }
142
143 # Resolve the active network service (same heuristic as toggle.sh)
144 resolve_active_service() {
145     get_active_service
146 }
147
148 #
149 # Quick-check functions (used by --watch loop; return simple status strings)
150 # Each echoes its result so the caller can capture and compare.
151 #
152
153 quick_warp() {
154     # Returns: "on", "off", "unknown"
155     local out
156     out=$(run_with_timeout 8 curl -s --max-time 6 \
157         https://www.cloudflare.com/cdn-cgi/trace 2>/dev/null || echo "")
158     local warp
159     warp=$(printf '%s' "$out" | awk -F=' ' /^warp={print $2; exit}')
160     printf '%s' "${warp:-unknown}"
161 }
162
163 quick_ipv6() {
164     # Returns: the IPv6 address if leaking, or empty string if clean
165     local out
166     out=$(run_with_timeout 7 curl -6 -s --max-time 5 ifconfig.co 2>/dev/null || echo "")
167     if [ -n "$out" ] && printf '%s' "$out" | grep -qE '[0-9a-fA-F:]+$'; then
168         printf '%s' "$out"
169     fi
170 }
171
172 quick_default_route() {
173     # Returns: "utun" if default route is via Tailscale utun*,
174     #           "wifi" if via physical Wi-Fi (192.168.x.x),
175     #           "other" otherwise
176     local route
177     route=$(netstat -rn 2>/dev/null | awk '/^default/ {print $0; exit}')
178     if [ -z "$route" ]; then
179         printf '%s' "none"
180     elif printf '%s' "$route" | grep -qE 'utun[0-9]+'; then
181         printf '%s' "utun"
182     elif printf '%s' "$route" | grep -qE '(default|0\.0\.0\.0).*\b192\.168\.'; then
183         printf '%s' "wifi"
184     else
185         printf '%s' "other"
186     fi
187 }
188
189 # Watch loop (runs after baseline diagnostic when --watch is passed)
190 run_watch_loop() {
191     local baseline_warp="$1"
192     local baseline_default="$2"
193     local baseline_ipv6_leak="$3"
194     local interval="$4"
195
196     echo ""
197     echo ""
198     echo " Baseline established. Monitoring every ${interval}s..."
199     echo "   warp:      $baseline_warp"
200     echo "   default:    $baseline_default"
201     echo ""
202
203     # Safety: warn if interval is very short (avoids hammering APIs)
204     if [ "$interval" -lt 30 ]; then
205         printf '%b' Short interval (%ss) rate-limiting risk on external APIs%b\n' "$YELLOW" "$interval" "$NC"
206         printf ' Consider 30s or longer for production use.\n'
207     fi
208     printf '[%s] WATCH START warp=%s default=%s interval=%ss\n' \
209         "$(date -u +%Y-%m-%dT%H:%M:%SZ)" "$baseline_warp" "$baseline_default" "$interval" \
210         >> "$WATCH_LOG"
211
212     local iteration=0
213     local last_warp="$baseline_warp"
214     local last_default="$baseline_default"
215     local last_ipv6_ok=$(( [ "$baseline_ipv6_leak" = "false" ] && echo true || echo false ))
216
217     trap 'printf "\n%bWatch stopped by user. Log at: %s%b\n" "$YELLOW" "$WATCH_LOG" "$NC"; exit 0' INT TERM
218
219     while true; do

```

```

220     iteration=$((iteration + 1))
221
222     local warp_now ipv6_now default_now
223     warp_now=$(quick_warp)
224     ipv6_now=$(quick_ipv6)
225     default_now=$(quick_default_route)
226
227     local ts
228     ts=$(date -u +%H:%M:%S)
229
230     # WARP check
231     if [ "$warp_now" != "$last_warp" ]; then
232         if [ "$warp_now" = "off" ] || [ "$warp_now" = "unknown" ]; then
233             printf '\n\b ALERT %b\n' "$RED$BOLD" "$NC"
234             printf '%b[%s] %bWARP FLIPPED: %s %s\b YOUR IP IS LEAKING!\n' \
235                 "$RED" "$ts" "$BOLD" "$last_warp" "$warp_now" "$NC"
236             printf '[%s] ALERT warp flipped %s%s\n' "$ts" "$last_warp" "$warp_now" >> "$WATCH_LOG"
237         else
238             printf '\n\b[%s] warp recovered: %s %s\b\n' \
239                 "$GREEN" "$ts" "$last_warp" "$warp_now" "$NC"
240             printf '[%s] OK warp recovered %s%s\n' "$ts" "$last_warp" "$warp_now" >> "$WATCH_LOG"
241         fi
242         last_warp="$warp_now"
243     fi
244
245     # IPv6 check
246     if [ -n "$ipv6_now" ]; then
247         if [ "$last_ipv6_ok" = true ]; then
248             printf '\n\b ALERT %b\n' "$RED$BOLD" "$NC"
249             printf '%b[%s] %bIPv6 LEAK DETECTED %s\b\n' \
250                 "$RED" "$ts" "$BOLD" "$ipv6_now" "$NC"
251             printf '[%s] ALERT IPv6 leak %s\n' "$ts" "$ipv6_now" >> "$WATCH_LOG"
252             last_ipv6_ok=false
253         fi
254     else
255         if [ "$last_ipv6_ok" = false ]; then
256             printf '%b[%s] IPv6 leak resolved\b\n' "$GREEN" "$ts" "$NC"
257             printf '[%s] OK IPv6 resolved\n' "$ts" >> "$WATCH_LOG"
258         fi
259         last_ipv6_ok=true
260     fi
261
262     # Default route check
263     if [ "$default_now" != "$last_default" ]; then
264         if [ "$default_now" != "utun" ]; then
265             printf '\n\b ALERT %b\n' "$RED$BOLD" "$NC"
266             printf '%b[%s] %bDEFAULT ROUTE CHANGED: %s %s\b traffic may bypass WARP!\n' \
267                 "$RED" "$ts" "$BOLD" "$last_default" "$default_now" "$NC"
268             printf '[%s] ALERT route changed %s%s\n' "$ts" "$last_default" "$default_now" >> "$WATCH_LOG"
269         else
270             printf '%b[%s] default route recovered: %s %s\b\n' \
271                 "$GREEN" "$ts" "$last_default" "$default_now" "$NC"
272             printf '[%s] OK route recovered %s%s\n' "$ts" "$last_default" "$default_now" >> "$WATCH_LOG"
273         fi
274         last_default="$default_now"
275     fi
276
277     # Heartbeat (every 10th iteration or if all OK)
278     if [ $((iteration % 10)) -eq 0 ]; then
279         printf '[%s] #d warp=%s ipv6=%s route=%s\n' \
280             "$ts" "$iteration" "$warp_now" "${ipv6_now:-none}" "$default_now" >> "$WATCH_LOG"
281         printf '\r\b[%s] #d warp=%s route=%s (Ctrl+C to stop)%b' \
282             "$GREEN" "$ts" "$iteration" "$warp_now" "$default_now" "$NC"
283     fi
284
285     sleep "$interval"
286 done
287 }
288
289 #
290 # Begin main diagnostic (skipped in non-watch modes if we're just watching)
291 #
292
293 ACTIVE_SERVICE=$(resolve_active_service)
294 ENO_SERVICE=$(networksetup -listnetworkserviceorder 2>> "$LOG_FILE" \
295     | grep -B 1 "Device: en0" | head -1 \
296     | sed -E 's/^([0-9\*]+)\s+//' | true)
297 [ -z "$ENO_SERVICE" ] && ENO_SERVICE="Wi-Fi"
298
299 # Resolve the gateway's Tailscale IP.
300 GATEWAY_TS_IP=""

```

```

301 if [ -f "$PROJECT_ROOT/.gateway_ip" ]; then
302     GATEWAY_TS_IP=$(cat "$PROJECT_ROOT/.gateway_ip" 2>> "$LOG_FILE" \
303         | tr -d '\r' | awk 'NR==1{print $1; exit}')
304 fi
305 if [ -z "$GATEWAY_TS_IP" ] && command -v docker >> "$LOG_FILE" 2>&1 \
306     && docker compose ps --status running 2>/dev/null | grep -q 'tailscale'; then
307     GATEWAY_TS_IP=$(run_with_timeout 10 docker compose exec -T tailscale \
308         tailscale ip -4 2>> "$LOG_FILE" \
309         | tr -d '\r' | awk 'NR==1{print $1; exit}')
310 fi
311
312 # Scenario classification state
313 SCENARIO=""
314 WARP_STATUS=""
315 SOCKS5_ENABLED=false
316 TS_ON_MESH=false
317 DEFAULT_VIA_UTUN=false
318 IPV6_LEAK=false
319
320 #
321 section "1/8 Host Tailscale mesh status"
322 #
323 if ! command -v tailscale >> "$LOG_FILE" 2>&1; then
324     fail "tailscale CLI not on PATH install via 'brew install tailscale' then re-run"
325     echo " (a Tailscale daemon is required for the exit-node path to work)"
326 else
327     set +e
328     TS_OUT=$(run_with_timeout 5 tailscale status 2>&1)
329     TS_EXIT=$?
330     set -uo pipefail
331     TS_HEAD=$(printf '%s\n' "$TS_OUT" | head -5 | tr -d '\r')
332     if [ "$TS_EXIT" -eq 137 ]; then
333         fail "tailscaled daemon is wedged/unresponsive (5s timeout)"
334         echo " Fix path: 'sudo brew services restart tailscale'"
335     elif printf '%s' "$TS_OUT" | grep -qE 'Logged out|not logged in|expired'; then
336         fail "tailscale is logged out / auth expired on this host"
337         echo " Fix path: 'sudo tailscale up' (browser auth once)"
338     elif printf '%s' "$TS_OUT" | grep -q '100\.'; then
339         ok "Host is on tailnet (100.x.x.x visible in status)"
340         TS_ON_MESH=true
341         if printf '%s' "$TS_OUT" | grep -q "$GATEWAY_TS_IP"; then
342             ok "Gateway $GATEWAY_TS_IP visible in host's peer list"
343         elif [ -n "$GATEWAY_TS_IP" ]; then
344             warn "Gateway $GATEWAY_TS_IP NOT in host's peer list possible auth/key mismatch"
345         fi
346         line " first 5 lines of 'tailscale status' "
347         line "$(printf '%s' "$TS_HEAD" | awk '{print " " $0}')"
348     else
349         fail "tailscale output unrecognized:"
350         line "$(printf '%s' "$TS_OUT" | head -3 | awk '{print " " $0}')"
351     fi
352 fi
353
354 #
355 section "2/8 Active SOCKS5 proxy on Wi-Fi"
356 #
357 SOCKS_OUT=$(networksetup -getsocksfirewallproxy "$ACTIVE_SERVICE" 2>> "$LOG_FILE" \
358     || networksetup -getsocksfirewallproxy Wi-Fi 2>> "$LOG_FILE" \
359     || echo "Could not query SOCKS5 proxy state")
360 if printf '%s' "$SOCKS_OUT" | grep -q "Enabled: Yes"; then
361     fail "SOCKS5 proxy IS enabled on $ACTIVE_SERVICE toggle.sh fell back to SOCKS5 mode last run"
362     SOCKS5_ENABLED=true
363     line " Full state:"
364     line "$(printf '%s' "$SOCKS_OUT" | awk '{print " " $0}')"
365     line " Browse on macOS ignores system SOCKS5 by default traffic bypasses WARP."
366 else
367     ok "SOCKS5 proxy is OFF on $ACTIVE_SERVICE exit-node path should be active"
368 fi
369
370 #
371 section "3/8 Egress IP + WARP tunnel verification (definitive test)"
372 #
373 CDN_OUT=$(run_with_timeout 8 curl -s --max-time 6 \
374     https://www.cloudflare.com/cdn-cgi/trace 2>&1 || echo "(curl failed)")
375 CDN_WARP=$(printf '%s' "$CDN_OUT" | awk -F=' ' /^warp={print $2; exit}')
376 CDN_IP=$(printf '%s' "$CDN_OUT" | awk -F=' ' /^ip={print $2; exit}')
377 CDN_COLO=$(printf '%s' "$CDN_OUT" | awk -F=' ' /^colo={print $2; exit}')
378 WARP_STATUS="$CDN_WARP"
379 if [ "$CDN_WARP" = "on" ]; then
380     ok "warp=on tunnel is hot, host traffic IS going through Cloudflare WARP"
381     line " Public IP: $CDN_IP"

```

```

382 line " Cloudflare:  ${CDN_COLO:-unknown colo}"
383 line "  whatismyip.com's 'local ISP' result is its ISP-database error, not yours."
384 elif [ "$CDN_WARP" = "off" ]; then
385 fail "warp=off host traffic is NOT going through WARP"
386 line " Public IP:  $CDN_IP (NOT Cloudflare)"
387 warn "This is the actual leak."
388 else
389 warn "could not determine warp status (curl failed or returned unexpected shape)"
390 line " Raw output: $(printf '%s' "$CDN_OUT" | head -3 | tr '\n' '|')"
391 fi
392
393 #
394 section "4/8 IPv6 leak probe (bypasses IPv4 exit-node on campus networks)"
395 #
396 IPV6_OUT=$(run_with_timeout 7 curl -6 -s --max-time 5 ifconfig.co 2>&1 || echo "")
397 if [ -n "$IPV6_OUT" ] && printf '%s' "$IPV6_OUT" | grep -qE '[0-9a-fA-F:]+$'; then
398 fail "host has live IPv6 egress $IPV6_OUT (A6 record bypasses WARP)"
399 IPV6_LEAK=true
400 line " Tailscale exit-node advertisement is IPv4-only, so IPv6 traffic skips it."
401 line " Quick-fix: 'sudo networksetup -setv6off $ACTIVE_SERVICE'"
402 else
403 ok "no IPv6 egress detected (good IPv6 won't bypass the tunnel)"
404 fi
405
406 #
407 section "5/8 Host egress route for real traffic (route -n get 1.1.1.1)"
408 #
409 # WHY NOT 'netstat -rn | grep default':
410 # On macOS, Tailscale does NOT replace the physical default route installed by
411 # DHCP. Instead it adds a second interface-scoped route bound to the utun
412 # interface that the kernel prefers for real traffic forwarding. The literal
413 # "default" entry (192.168.x.x via en0) is left untouched in the table as a
414 # BSD routing quirk it's not lying, it's just answering a different question.
415 # The correct question is: "where does a real destination actually resolve?"
416 # `route -n get 1.1.1.1` asks the kernel's forwarding logic directly.
417 ROUTE_GET=$(route -n get 1.1.1.1 2>/dev/null)
418 ROUTE_IFACE=$(echo "$ROUTE_GET" | awk '/interface:/{print $2}')
419 ROUTE_GATEWAY=$(echo "$ROUTE_GET" | awk '/gateway:/{print $2}')
420 if [ -z "$ROUTE_IFACE" ]; then
421 warn "route -n get 1.1.1.1 returned no interface routing table may be empty"
422 else
423 line " interface: $ROUTE_IFACE gateway: ${ROUTE_GATEWAY:-n/a}"
424 if echo "$ROUTE_IFACE" | grep -qE '^utun[0-9]+'; then
425 ok "1.1.1.1 resolves via $ROUTE_IFACE Tailscale interface-scoped route is winning"
426 DEFAULT_VIA_UTUN=true
427 elif echo "$ROUTE_IFACE" | grep -qE '^en[0-9]+'; then
428 fail "1.1.1.1 resolves via $ROUTE_IFACE (physical Wi-Fi) exit-node interface route not active"
429 else
430 warn "1.1.1.1 resolves via unexpected interface: $ROUTE_IFACE"
431 fi
432 fi
433
434 #
435 section "6/8 Last toggle.sh START-relevant lines in output.log (if any)"
436 #
437 if [ ! -f "$LOG_FILE" ]; then
438 note "output.log"
439 line " toggle.sh was never run from this directory. Run './toggle.sh' first,"
440 line " then re-run this diagnostic."
441 else
442 HITS=$(grep -E 'Exit node enabled|SKIP_EXIT_NODE|Pre-flight|Falling back to SOCKS|Starting|Successfully' \
443 "$LOG_FILE" 2>/dev/null | tail -8 || true)
444 if [ -z "$HITS" ]; then
445 note "matching lines in output.log"
446 else
447 line "$(printf '%s\n' "$HITS" | awk '{print " " "$0}')"
448 fi
449 fi
450
451 #
452 section "7/8 Gateway IP we should be pointing at"
453 #
454 if [ -n "$GATEWAY_TS_IP" ]; then
455 ok "gateway Tailscale IP: $GATEWAY_TS_IP"
456 else
457 warn "could not resolve gateway Tailscale IP"
458 line " (looked in .gateway_ip and tried docker compose exec tailscale ip -4)"
459 fi
460
461 #
462 section "8/8 Tailscale System Extension (App Store conflict check)"

```

```

463 #
464 SE_PENDING_UNINSTALL=false
465 SE_ACTIVE=false
466 if command -v systemextensionsctl >/dev/null 2>&1; then
467     SE_OUT=$(systemextensionsctl list 2>/dev/null | grep -iE '(io|com)\.tailscale')
468     if [ -n "$SE_OUT" ]; then
469         line " Tailscale System Extension(s) detected:"
470         line "$(printf '%s' "$SE_OUT" | awk '{print "    "$0}')"
471         if printf '%s' "$SE_OUT" | grep -q 'waiting to uninstall'; then
472             SE_PENDING_UNINSTALL=true
473             warn "System Extension is pending uninstall on next reboot."
474         else
475             SE_ACTIVE=true
476             warn "Active Tailscale System Extension detected. This will conflict with Homebrew Tailscale."
477         fi
478     else
479         ok "no Tailscale System Extension detected"
480     fi
481 else
482     note "systemextensionsctl not available (not on macOS or permission restricted)"
483 fi
484
485 #
486 section "CLASSIFICATION    applies ONE of the four known scenarios"
487 #
488
489 if [ "$WARP_STATUS" = "on" ] && [ "$IPV6_LEAK" = "false" ]; then
490     SCENARIO="OK"
491 elif [ "$SE_PENDING_UNINSTALL" = "true" ]; then
492     SCENARIO="SE_PENDING"
493 elif [ "$SOCKS5_ENABLED" = "true" ]; then
494     SCENARIO="A"
495 elif [ "$IPV6_LEAK" = "true" ]; then
496     SCENARIO="B"
497 elif [ "$DEFAULT_VIA_UTUN" = "false" ] && [ "$TS_ON_MESH" = "true" ]; then
498     SCENARIO="C"
499 else
500     SCENARIO="UNKNOWN"
501 fi
502
503 case "$SCENARIO" in
504     SE_PENDING)
505         echo ""
506         echo -e " ${RED}${BOLD}VERDICT: Scenario SE_PENDING Tailscale System Extension Pending Uninstall${NC}"
507         echo " A prior App Store Tailscale System Extension is pending uninstall."
508         echo " Since System Integrity Protection (SIP) is enabled, macOS blocks manual"
509         echo " uninstallation of terminated system extensions until the next reboot."
510         echo " Brew-managed tailscaled cannot engage the Network Extension slot until this is resolved."
511         echo ""
512         echo "${BOLD}Recommended fix:${NC}"
513         echo " sudo shutdown -r now"
514         echo " # After reboot, run: ./toggle.sh"
515         ;;
516     A)
517         echo ""
518         echo -e " ${RED}${BOLD}VERDICT: Scenario A SOCKS5 fallback lane${NC}"
519         echo " toggle.sh's pre-flight checks failed last run. The script activated"
520         echo " SOCKS5 + local DNS proxy as a fallback. DNS gets hijacked but"
521         echo " browsers on macOS ignore the system SOCKS5 traffic egresses"
522         echo " direct through the local network."
523         echo ""
524         echo "${BOLD}Recommended fix:${NC}"
525         echo " echo 'GATEWAY_BYPASS_PING=true' >> $PROJECT_ROOT/.env"
526         echo " sudo tailscale up --reset --ssh=true --accept-dns=false --accept-routes=true \\"
527         if [ -n "$GATEWAY_TS_IP" ]; then
528             echo " --exit-node=$GATEWAY_TS_IP --exit-node-allow-lan-access=true"
529         else
530             echo " --exit-node=$GATEWAY_TS_IP --exit-node-allow-lan-access=true # fill in $GATEWAY_TS_IP"
531         fi
532         echo " sudo dscacheutil -flushcache && sudo killall -HUP mDNSResponder"
533         echo " ./toggle.sh"
534         ;;
535     B)
536         echo ""
537         echo -e " ${RED}${BOLD}VERDICT: Scenario B IPv6 leak over dual-stack campus/local IPv6${NC}"
538         echo " Tailscale exit-node advertises only IPv4. macOS sends AAAA queries"
539         echo " straight out en0, which is dual-stack on most campus APs IPv6"
540         echo " traffic bypasses the WARP tunnel entirely."
541         echo ""
542         echo "${BOLD}Recommended fix:${NC}"
543         echo " sudo networksetup -setv6off $ACTIVE_SERVICE"

```

```

544     echo "      sudo dscacheutil -flushcache && sudo killall -HUP mDNSResponder"
545     echo "      # Re-enable IPv6 later with: sudo networksetup -setv6automatic $ACTIVE_SERVICE"
546     ;;
547 C)
548     echo ""
549     echo -e " ${RED}${BOLD}VERDICT: Scenario C Tailscale route-freeze${NC}"
550     echo " Host's default route points at the Wi-Fi gateway instead of the"
551     echo " Tailscale utun* interface. The exit-node preference is set but the"
552     echo " routing-table assertion did not take effect (devref 10.26)."

```

```

624 ;;
625 A)
626 if [ -n "$GATEWAY_TS_IP" ]; then
627     line " writing GATEWAY_BYPASS_PING=true to .env"
628     ENV="$PROJECT_ROOT/.env"
629     [ -f "$ENV" ] || touch "$ENV"
630     if grep -q '^GATEWAY_BYPASS_PING=' "$ENV" 2>/dev/null; then
631         sed -i 's/^GATEWAY_BYPASS_PING=.*GATEWAY_BYPASS_PING=true/' "$ENV"
632     else
633         printf '\nGATEWAY_BYPASS_PING=true\n' >> "$ENV"
634     fi
635     line " re-asserting tailscale exit-node (with --reset)"
636     sudo -n tailscale up --reset --ssh=true --accept-dns=false --accept-routes=true \
637         --exit-node="$GATEWAY_TS_IP" --exit-node-allow-lan-access=true \
638         >> "$LOG_FILE" 2>&1 \
639         && ok "tailscale exit-node re-asserted for $GATEWAY_TS_IP" \
640         || warn "tailscale up did not return success (check $LOG_FILE)"
641 else
642     warn "no gateway Tailscale IP resolved skipping tailscale up"
643     warn "fix manually: 'sudo tailscale up --reset ... --exit-node=<TS_IP>'"
644 fi
645 ;;
646 B)
647 line " disabling IPv6 on $ACTIVE_SERVICE while gateway is up"
648 sudo -n networksetup -setv6off "$ACTIVE_SERVICE" >> "$LOG_FILE" 2>&1 \
649     && ok "IPv6 disabled on $ACTIVE_SERVICE (re-enable with 'setv6automatic')\" \
650     || warn "networksetup -setv6off failed (check $LOG_FILE)"
651 ;;
652 C)
653 line " restarting tailscaled (route-table fix)"
654 run_with_timeout 15 brew services restart tailscale >> "$LOG_FILE" 2>&1 \
655     || run_with_timeout 15 sudo -n brew services restart tailscale >> "$LOG_FILE" 2>&1
656 sleep 3
657 line " flushing stale host routes + DNS cache"
658 sudo -n route delete -net 0.0.0.0/1 >> "$LOG_FILE" 2>&1 || true
659 sudo -n route delete -net 128.0.0.0/1 >> "$LOG_FILE" 2>&1 || true
660 sudo -n dscacheutil -flushcache >> "$LOG_FILE" 2>&1 || true
661 sudo -n killall -HUP mDNSResponder >> "$LOG_FILE" 2>&1 || true
662 if [ -n "$GATEWAY_TS_IP" ]; then
663     line " re-asserting exit-node after restart"
664     sudo -n tailscale up --reset --ssh=true --accept-dns=false --accept-routes=true \
665         --exit-node="$GATEWAY_TS_IP" --exit-node-allow-lan-access=true \
666         >> "$LOG_FILE" 2>&1 \
667         && ok "tailscale exit-node re-asserted for $GATEWAY_TS_IP" \
668         || warn "tailscale up did not return success (check $LOG_FILE)"
669 fi
670 ;;
671 esac
672
673 echo ""
674 echo "Re-verifying after fix..."
675 sleep 3
676 CDN_WARP_AFTER=$(run_with_timeout 8 curl -s --max-time 6 \
677     https://www.cloudflare.com/cdn-cgi/trace 2>&1 \
678     | awk -F'=' '{print $2; exit}')
679 IPV6_AFTER=$(run_with_timeout 7 curl -6 -s --max-time 5 ifconfig.co 2>&1 || echo "")
680 printf '\n[AFTER FIX]\n' >&3
681 printf ' warp status:    %s\n' "$CDN_WARP_AFTER" >&3
682 printf ' IPv6 egress:    %s\n' "${IPV6_AFTER:-none}" >&3
683 line " warp status:    ${CDN_WARP_AFTER:-unknown}"
684 line " IPv6 egress:    ${IPV6_AFTER:-none}"
685 if [ "$CDN_WARP_AFTER" = "on" ] && [ -z "$IPV6_AFTER" ]; then
686     echo ""
687     ok "Fix verified host is now routing through WARP. Remote clients unaffected."
688 else
689     echo ""
690     warn "Fix did not fully take effect on first try. Likely causes:"
691     line " tailscaled needed a longer settling window (re-run diagnostic in 30s)"
692     line " The classified scenario was wrong; paste this report for triage."
693     line " Tailscale needs a full auth refresh: 'sudo tailscale up'"
694 fi
695 fi
696
697 echo ""
698 echo ""
699 echo " Full report written to:"
700 echo " $REPORT_FILE"
701 echo ""
702 echo ""
703
704 #

```

```

705 # --watch mode: enter the continuous monitoring loop after baseline diagnostic
706 #
707 if [ "$DO_WATCH" = "true" ]; then
708     # Build baseline from the full diagnostic just completed
709     BASELINE_WARP="$WARP_STATUS"
710     BASELINE_DEFAULT=$( [ "$DEFAULT_VIA_UTUN" = "true" ] && echo "utun" || echo "wifi/other")
711
712     if [ "$SCENARIO" != "OK" ]; then
713         warn "Baseline diagnostic found issues (scenario: $SCENARIO).\"
714         line \" Watch loop will still run alerts fire on any STATE CHANGE from current baseline.\"
715         line \" Fix the issue first? Re-run with: bash scripts/diagnose-host-leak.sh --fix\"
716         echo \"
717     fi
718
719     run_watch_loop \"$BASELINE_WARP\" \"$BASELINE_DEFAULT\" \"$IPV6_LEAK\" \"$WATCH_INTERVAL\"
720 fi
721
722 exit 0

```

33 scripts/dns-proxy.py

```

1  #!/usr/bin/env python3
2  """
3  dns-proxy.py Local DNS proxy for nullexit gateway.
4
5  Listens on UDP port 53, receives DNS queries, forwards them over TCP
6  to AdGuard at 127.0.0.1:5354 (the Docker port mapping), and returns
7  the response via UDP. Handles the DNS-over-TCP wire format (2-byte
8  length prefix) correctly unlike socat which just forwards raw bytes.
9  """
10
11 import socket
12 import struct
13 import sys
14 import os
15 from concurrent.futures import ThreadPoolExecutor
16
17 LISTEN_ADDR = os.environ.get("LISTEN_ADDR", "127.0.0.1")
18 LISTEN_PORT = int(os.environ.get("LISTEN_PORT", 53))
19 TARGET_HOST = os.environ.get("TARGET_HOST", "127.0.0.1")
20 TARGET_PORT = int(os.environ.get("TARGET_PORT", 5354))
21
22
23 def handle_query(data: bytes, client_addr: tuple, sock: socket.socket) -> None:
24     """Forward a single DNS query over TCP and send the response back via UDP."""
25     try:
26         tcp = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
27         tcp.settimeout(5)
28         tcp.connect((TARGET_HOST, TARGET_PORT))
29
30         # DNS over TCP: prepend 2-byte length (big-endian)
31         tcp.sendall(struct.pack("!H", len(data)) + data)
32
33         # Read the 2-byte response length
34         raw_len = tcp.recv(2)
35         if len(raw_len) < 2:
36             tcp.close()
37             return
38         resp_len = struct.unpack("!H", raw_len)[0]
39
40         # Read the full response
41         response = b""
42         while len(response) < resp_len:
43             chunk = tcp.recv(resp_len - len(response))
44             if not chunk:
45                 break
46             response += chunk
47
48         tcp.close()
49
50         # Send the response back over UDP (strip TCP length prefix)
51         if response:
52             sock.sendto(response, client_addr)
53     except Exception:
54         # Silently drop failed queries (malformed, timeout, etc.)
55         pass

```

```

56
57
58 def main() -> None:
59     # Create UDP socket
60     sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
61     sock.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
62
63     try:
64         sock.bind((LISTEN_ADDR, LISTEN_PORT))
65     except PermissionError:
66         print(f"dns-proxy: ERROR need root to bind port {LISTEN_PORT}", flush=True)
67         sys.exit(1)
68     except OSError as e:
69         print(f"dns-proxy: ERROR {e}", flush=True)
70         sys.exit(1)
71
72     print(f"dns-proxy: listening on UDP {LISTEN_ADDR}:{LISTEN_PORT} TCP {TARGET_HOST}:{TARGET_PORT}", flush=True)
73
74     # Use a thread pool to handle concurrent DNS queries efficiently without thread exhaustion
75     executor = ThreadPoolExecutor(max_workers=64)
76
77     while True:
78         try:
79             data, client_addr = sock.recvfrom(4096) # Support EDNSO (up to 4096 bytes) to prevent truncation
80             executor.submit(handle_query, data, client_addr, sock)
81         except KeyboardInterrupt:
82             break
83         except Exception:
84             pass
85
86     executor.shutdown(wait=False)
87     sock.close()
88
89
90 if __name__ == "__main__":
91     main()

```

34 scripts/fix-docker-bridge-collision.sh

```

1 #!/bin/bash
2 # scripts/fix-docker-bridge-collision.sh nullexit fix for devref 9.13
3 #
4 # Symptom: host default route points at Docker's bridge gateway (172.17.0.1)
5 # instead of the real Wi-Fi gateway. ALL host internet egress fails. Remote
6 # tailnet clients route through the gateway correctly (their tunnel is
7 # independent of the host's broken route table), so they look healthy while
8 # the host itself cannot reach anything.
9 #
10 # Root cause: Docker's default bridge claims 172.17.0.0/16 by default. When the
11 # host's physical Wi-Fi also lands in the same /16 extremely common on
12 # university networks which hand out 172.16.0.0/12 the kernel
13 # installs Docker's bridge as the default route, and every packet bound for
14 # the internet is silently absorbed by docker0 instead of egressing.
15 #
16 # This script:
17 # 1. Detects the actual collision (Docker's 172.17.0.0/16 vs the host's
18 # Wi-Fi subnet, queried live via the ACTIVE interface not hardcoded en0)
19 # 2. Picks a non-colliding Docker bip using /8-family routing (works for
20 # 172.16.0.0/12 campus networks; naive prefix-match would pick 172.26/24
21 # which still lives INSIDE a /12 of 172.16.0.0/12)
22 # 3. Backs up ~/.colima/default/colima.yaml with an ISO-8601 timestamp
23 # 4. Edits the file in place, idempotent (replace existing docker.bip /
24 # append new docker.bip without disturbing other keys; skip YAML comments)
25 # 5. Restarts Colima so the VM rebuilds with the new bridge subnet
26 # 6. Rebinds Wi-Fi DHCP so the host re-learns its real Wi-Fi gateway
27 # 7. Verifies the new default route is no longer the Docker bridge
28 # 8. Re-runs toggle.sh to bring the gateway back up cleanly
29 # 9. Re-runs scripts/diagnose-host-leak.sh and prints the new verdict
30 #
31 # Usage:
32 # bash scripts/fix-docker-bridge-collision.sh # apply the fix
33 # bash scripts/fix-docker-bridge-collision.sh --dry # show what would change, then exit
34 # bash scripts/fix-docker-bridge-collision.sh --skip-toggle # fix but don't restart gateway
35 # bash scripts/fix-docker-bridge-collision.sh --help # usage
36

```

```

37 set -uo pipefail
38 # NOTE: errexit (`set -e`) is intentionally NOT enabled so intermediate
39 # `command || warn` patterns can continue past non-fatal failures.
40 # Every critical failure (colima restart, sed, cp backup, toggle.sh)
41 # uses an explicit `|| die` so a half-applied state never escapes.
42
43 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
44 PROJECT_ROOT="$(cd "$SCRIPT_DIR/.." && pwd)"
45 LOG_FILE="$PROJECT_ROOT/output.log"
46 source "$SCRIPT_DIR/common.sh"
47 TIMESTAMP="$(date -u +%Y%m%dT%H%M%SZ)"
48 COLIMA_YAML="$HOME/.colima/default/colima.yaml"
49 PLATFORM="$(uname -s)"
50
51 # Argument parsing
52 DRY_RUN=false
53 SKIP_TOGGLE=false
54 case "${1:-}" in
55   --dry|-n)      DRY_RUN=true ;;
56   --skip-toggle) SKIP_TOGGLE=true ;;
57   --help|-h)
58     sed -n '28,37p' "$0"
59     exit 0
60   ;;
61   *)
62     : # default: apply
63   ;;
64 *)
65   printf 'Unknown argument: %s\n' "$1" >&2
66   exit 2
67   ;;
68 esac
69
70
71
72 # Helper: print "[dry-run] $cmd" or actually run $cmd. Wraps the redirect
73 # INSIDE the function so outer `>> $FILE` statements never accidentally
74 # append dry-run noise to the real file. (Earlier draft didn't and the
75 # dry-run polluted $COLIMA_YAML.)
76 run() {
77   if [ "$DRY_RUN" = "true" ]; then
78     printf '    [dry-run] %s\n' "$*"
79     return 0
80   fi
81   printf '    $ %s\n' "$*"
82   "$@"
83 }
84
85 # Platform-aware sed in-place. macOS BSD sed needs an empty argument;
86 # GNU sed (Linux) does not. Earlier draft used `-i` everywhere and
87 # would have failed on Linux silently.
88 sed_inplace() {
89   local expr="$1"
90   local file="$2"
91   case "$PLATFORM" in
92     Darwin) run sed -i '' "$expr" "$file" ;;
93     *)      run sed -i "$expr" "$file" ;;
94   esac
95 }
96
97 # 0. Resolve the ACTIVE physical interface once (used everywhere)
98 # Replaces the earlier `en0` hardcoding. Same algorithm recover.sh uses:
99 # `route get default` if utun/tun, walk en0..en5 first with `status: active`
100 # + `inet` fall back to en0. This means the script also works on hosts
101 # whose primary NIC is en1/en2 (Thunderbolt Ethernet, USB-LAN, etc.).
102 resolve_active_iface() {
103   local iface
104   iface=$(route get default 2>> "$LOG_FILE" | awk '/interface:/{print $2; exit}')
105   if [ -z "$iface" ] || [ "${iface#tun}" != "$iface" ] || [ "${iface#tun}" != "$iface" ]; then
106     for i in en0 en1 en2 en3 en4 en5; do
107       if ifconfig "$i" 2>> "$LOG_FILE" | grep -q "status: active" \
108         && ifconfig "$i" 2>> "$LOG_FILE" | grep -q "inet "; then
109         iface="$i"; break
110       fi
111     done
112   fi
113   [ -z "$iface" ] && iface="en0"
114   printf '%s\n' "$iface"
115 }
116 ACTIVE_IFACE=$(resolve_active_iface)
117

```

```

118 # 1. Detect collision
119 step "1/9 Detecting Docker-bridge vs Wi-Fi subnet collision"
120
121 # Host subnet on the ACTIVE interface (not hardcoded en0).
122 HOST_SUBNET=$(ifconfig "$ACTIVE_IFACE" 2>> "$LOG_FILE" \
123 | awk '/inet /{print $2; exit}')
124 HOST_GATEWAY=$(route get default 2>> "$LOG_FILE" \
125 | awk '/gateway:/{print $2; exit}')
126 DEFAULT_DEV=$(route get default 2>> "$LOG_FILE" \
127 | awk '/interface:/{print $2; exit}')
128
129 printf ' active iface: %s\n' "$ACTIVE_IFACE"
130 printf ' iface inet: %s\n' "${HOST_SUBNET:-none}"
131 printf ' default via: %s\n' "${HOST_GATEWAY:-none}"
132 printf ' default dev: %s\n' "${DEFAULT_DEV:-none}"
133
134 # Collision fires when the host's default-route gateway is 172.17.0.1 (Docker's
135 # bridge) OR when the host subnet is in 172.17.0.0/16. The full-bridge-collision
136 # case (gateway is literally the docker bridge IP) is the smoking gun.
137 COLLISION=false
138 if [ -n "$HOST_GATEWAY" ] && [ "${HOST_GATEWAY#172.17.}" != "$HOST_GATEWAY" ]; then
139 COLLISION=true
140 fail "default gateway $HOST_GATEWAY collides with Docker's default bridge (172.17.0.0/16)"
141 elif [ -n "$HOST_SUBNET" ] && [ "${HOST_SUBNET#172.17.}" != "$HOST_SUBNET" ]; then
142 COLLISION=true
143 fail "host subnet $HOST_SUBNET overlaps Docker's 172.17.0.0/16 bridge"
144 else
145 ok "no collision detected between host Wi-Fi and Docker's default bridge"
146 fi
147
148 if [ "$COLLISION" = "false" ]; then
149 echo ""
150 echo "Nothing to fix. If you still see egress issues, run:"
151 echo "    bash scripts/diagnose-host-leak.sh"
152 exit 0
153 fi
154
155 # 2. Pick non-colliding bip via /8-family routing
156 step "2/9 Picking a non-colliding Docker bip for ~/.colima/default/colima.yaml"
157
158 # Earlier draft tried naive prefix-match: "if candidate's 3-octet prefix is
159 # not a literal prefix of host gateway, pick it". That breaks on /12 campus
160 # networks (172.16.0.0/12 covers 172.16-172.31): a candidate like
161 # 172.26.0.1/24 LIVES INSIDE the host's /12 even though the prefix test
162 # passes. Fix: classify the host's address family FIRST, then choose ALL
163 # candidates from a DIFFERENT RFC1918 family. /8 boundaries are coarse
164 # enough that no /24 picked this way can ever collide.
165 case "$HOST_GATEWAY" in
166 172.16.*|172.17.*|172.18.*|172.19.*|172.20.*|172.21.*|172.22.*|172.23.*|\
167 172.24.*|172.25.*|172.26.*|172.27.*|172.28.*|172.29.*|172.30.*|172.31.*)
168 HOST_FAMILY="172"
169 # Host is in 172.x jump past 172.31 entirely. Pick from 10.x / 192.168.x.
170 CANDIDATES="10.200.0.1/24 10.201.0.1/24 192.168.99.1/24"
171 ;;
172 10.*)
173 HOST_FAMILY="10"
174 # Host is in 10.x pick from 172.x (well outside any commonly-deployed
175 # 10.0.0.0/8 slice) / 192.168.x.
176 CANDIDATES="172.26.0.1/24 172.28.0.1/24 192.168.99.1/24"
177 ;;
178 192.168.*)
179 HOST_FAMILY="192"
180 CANDIDATES="172.26.0.1/24 10.200.0.1/24 10.201.0.1/24"
181 ;;
182 *)
183 HOST_FAMILY="unknown"
184 warn "host gateway $HOST_GATEWAY is not in any RFC1918 range (/8 family)"
185 warn "falling back to abstract candidate set; verify collision-free manually"
186 CANDIDATES="172.26.0.1/24 10.200.0.1/24 192.168.99.1/24"
187 ;;
188 esac
189
190 printf ' host address family: %s\n' "$HOST_FAMILY"
191 printf ' candidate pool: %s\n' "$CANDIDATES"
192
193 # Sanity-check the chosen pool has at least one entry that doesn't literally
194 # share a /24 with the host gateway. Naive check (just /24 prefix) is fine
195 # here because we already picked across families.
196 CHOSEN=""
197 for cand in $CANDIDATES; do
198 PREFIX3=$(printf '%s' "$cand" | cut -d. -f1-3)

```

```

199 if [ -n "$HOST_GATEWAY" ] && [ "${HOST_GATEWAY#"$PREFIX3".}" != "$HOST_GATEWAY" ]; then
200     warn "skip $cand (literal /24 collision with host gateway $HOST_GATEWAY)"
201     continue
202 fi
203 CHOSEN="$cand"
204 ok "chose bip=$CHOSEN (cross-family, no /24 literal collision)"
205 break
206 done
207
208 if [ -z "$CHOSEN" ]; then
209     CHOSEN="172.26.0.1/24"
210     warn "could not find a clean candidate; defaulting to $CHOSEN"
211     warn "verify with 'route get default' after restart"
212 fi
213
214 # 3. Backup colima.yaml
215 step "3/9 Backing up $COLIMA_YAML"
216 mkdir -p "$(dirname "$COLIMA_YAML")"
217 BACKUP="$HOME/.colima/default/colima.yaml.bak.$TIMESTAMP"
218 if [ -f "$COLIMA_YAML" ]; then
219     # `cp` is safe in dry-run because run() suppresses the inner command
220     # entirely no chance of accidentally writing a real backup.
221     run cp -p "$COLIMA_YAML" "$BACKUP" || die "backup failed"
222     if [ "$DRY_RUN" = "false" ]; then ok "backup: $BACKUP"; fi
223 else
224     warn "no existing $COLIMA_YAML (will create fresh)"
225 fi
226
227 # 4. Edit colima.yaml in place (idempotent)
228 step "4/9 Setting docker.bip=$CHOSEN in colima.yaml (in place)"
229
230 # Resolve the existing docker.bip (if any) without picking up commented-out
231 # YAML keys. awk-based extraction; the negated-comment check (! /^[[:space:]]*#`)
232 # ensures `# bip: 1.2.3.4/24` is treated as a comment, not the active value.
233 EXISTING_BIP=""
234 if [ -f "$COLIMA_YAML" ]; then
235     EXISTING_BIP=$(awk '
236         /^[[:space:]]*#/{next}
237         /^[[:space:]]*bip:[[:space:]]*/{print $2; exit}
238     ' "$COLIMA_YAML" 2>> "$LOG_FILE" || true)
239 fi
240
241 if [ -n "$EXISTING_BIP" ] && [ "$EXISTING_BIP" = "$CHOSEN" ]; then
242     ok "colima.yaml already has docker.bip=$CHOSEN no edit needed"
243 elif [ -n "$EXISTING_BIP" ]; then
244     printf ' (replacing existing docker.bip=%s with %s)\n' "$EXISTING_BIP" "$CHOSEN"
245     # Use sed_inplace so Linux / macOS get the right syntax. Skip YAML comment
246     # lines so existing `# bip:` comments aren't rewired.
247     sed_inplace "s|^\([[:space:]]*bip:[[:space:]]*\)\).*\${1}$CHOSEN|" "$COLIMA_YAML" \
248     || die "sed replacement failed (colima.yaml malformed?)"
249     ok "colima.yaml updated"
250 else
251     # No existing `bip:` key. Branch:
252     # - file has `docker:` block append ` bip:` under it
253     # - file has no `docker:` block append new docker: section
254     # - file does not exist create fresh
255     # Each branch is dry-run-aware so no branch can accidentally leak noise
256     # into $COLIMA_YAML during a preview.
257     HAS_DOCKER_BLOCK=false
258     if [ -f "$COLIMA_YAML" ] && grep -q '^docker:' "$COLIMA_YAML"; then
259         HAS_DOCKER_BLOCK=true
260     fi
261
262     if [ "$HAS_DOCKER_BLOCK" = "true" ]; then
263         if [ "$DRY_RUN" = "true" ]; then
264             printf ' [dry-run] would append to %s:\n bip: %s\n' "$COLIMA_YAML" "$CHOSEN"
265         else
266             printf '\n bip: %s\n' "$CHOSEN" >> "$COLIMA_YAML" \
267             || die "append failed"
268             ok "appended bip under existing docker: block"
269         fi
270     elif [ -f "$COLIMA_YAML" ]; then
271         if [ "$DRY_RUN" = "true" ]; then
272             printf ' [dry-run] would append to %s:\ndocker:\n bip: %s\n' "$COLIMA_YAML" "$CHOSEN"
273         else
274             printf '\ndocker:\n bip: %s\n' "$CHOSEN" >> "$COLIMA_YAML" \
275             || die "append failed"
276             ok "appended new docker: block at end of file"
277         fi
278     else
279         if [ "$DRY_RUN" = "true" ]; then

```

```

280     printf '    [dry-run] would create %s with:\\ndocker:\\n  bip: %s\\n' "$COLIMA_YAML" "$CHOSEN"
281   else
282     printf 'docker:\\n  bip: %s\\n' "$CHOSEN" > "$COLIMA_YAML" \
283     || die "create failed"
284     ok "created fresh $COLIMA_YAML with docker.bip=$CHOSEN"
285   fi
286 fi
287 fi
288
289 # Always show the final colima.yaml docker section for visibility
290 printf '\\n\\b  colima.yaml `docker:` section now reads:%b\\n' "$BOLD" "$NC"
291 awk '
292   /^docker:/{p=1}
293   p && /[^\n]/ && !/^docker:/{exit}
294   p
295   ' "$COLIMA_YAML" 2>/dev/null | sed 's/^/  /' || true
296
297 if [ "$DRY_RUN" = "true" ]; then
298   printf '\\n  %b(dry-run: exiting before colima restart / dhcp rebind / toggle.sh)%b\\n' "$YELLOW" "$NC"
299   exit 0
300 fi
301
302 # 5. Restart Colima
303 step "5/9 Restarting Colima VM with new bip"
304 if ! command -v colima >> "$LOG_FILE" 2>&1; then
305   die "colima not on PATH install with 'brew install colima' first"
306 fi
307 # No timeout wrapper (would mask actual VM startup time). If colima hangs,
308 # the user kills with Ctrl+C and can manually `colima restart` from a fresh
309 # terminal the rest of this script is idempotent so re-running works.
310 colima restart >> "$LOG_FILE" 2>&1 || die "colima restart failed; check $LOG_FILE"
311 ok "colima restarted"
312
313 # 6. Rebind DHCP on the ACTIVE interface (not hardcoded en0)
314 step "6/9 Rebinding DHCP on $ACTIVE_IFACE so host re-learns its real gateway"
315
316 # Map ifname macOS network service name (Wi-Fi, USB 10/100 LAN, etc.).
317 # Same logic as the diagnostic + toggle.sh.
318 ACTIVE_SVC=""
319 if [ "$PLATFORM" = "Darwin" ]; then
320   ACTIVE_SVC=$(get_active_service)
321 fi
322
323 case "$PLATFORM" in
324   Darwin)
325     sudo -n networksetup -setdhcp "$ACTIVE_SVC" renew >> "$LOG_FILE" 2>&1 \
326     && ok "DHCP rebind issued on $ACTIVE_SVC" \
327     || warn "setdhcp renew failed; try 'sudo ipconfig set $ACTIVE_IFACE DHCP' manually"
328     ;;
329   Linux)
330     sudo -n dhclient -r "$ACTIVE_IFACE" >> "$LOG_FILE" 2>&1 || true
331     sudo -n dhclient "$ACTIVE_IFACE" >> "$LOG_FILE" 2>&1 \
332     && ok "dhclient rebind on $ACTIVE_IFACE" \
333     || warn "dhclient rebind failed"
334     ;;
335   *)
336     warn "unsupported platform $PLATFORM; skip DHCP rebind"
337     ;;
338 esac
339
340 # 7. Verify new default route
341 step "7/9 Verifying the new default route is NOT the Docker bridge"
342 sleep 2 # give DHCP + kernel a beat
343 NEW_DEFAULT=$(netstat -rn 2>> "$LOG_FILE" | awk '/^default/ {print $0; exit}')
344 printf '  %s\\n' "${NEW_DEFAULT:-no default route detected}"
345 NEW_GATEWAY=$(printf '%s' "$NEW_DEFAULT" | awk '{print $2; exit}')
346
347 if [ -n "$NEW_GATEWAY" ] && [ "${NEW_GATEWAY#172.17.}" != "$NEW_GATEWAY" ]; then
348   fail "default route STILL points at 172.17.x.x DHCP rebind may need manual touch"
349   warn "try: sudo ipconfig set $ACTIVE_IFACE DHCP (or reconnect Wi-Fi in the menu bar)"
350 elif [ -n "$NEW_GATEWAY" ]; then
351   ok "default route is now via $NEW_GATEWAY (no longer Docker's bridge)"
352 else
353   warn "could not determine new gateway; verify with 'route get default'"
354 fi
355
356 # 8. Re-run toggle.sh to bring the gateway back up
357 if [ "$SKIP_TOGGLE" = "true" ]; then
358   step "8/9 Skipping toggle.sh (--skip-toggle)"
359   warn "gateway is currently down run ./toggle.sh manually when ready"
360 else

```

```

361 step "8/9 Re-running toggle.sh to bring the gateway back up"
362 if bash "$PROJECT_ROOT/toggle.sh" 2>&1 | tee -a "$LOG_FILE"; then
363     ok "toggle.sh completed"
364 else
365     warn "toggle.sh returned non-zero see $LOG_FILE for details"
366 fi
367 fi
368
369 # 9. Re-run the diagnostic + show verdict
370 step "9/9 Re-running host-leak diagnostic to confirm fix"
371 if bash "$SCRIPT_DIR/diagnose-host-leak.sh" 2>&1 | tee -a "$LOG_FILE"; then
372     LATEST_REPORT=$(ls -t "$PROJECT_ROOT"/host-leak-diagnostic-*.txt 2>/dev/null | head -1)
373     if [ -n "$LATEST_REPORT" ]; then
374         VERDICT=$(awk '/^ Scenario:/ {print $2; exit}' "$LATEST_REPORT" 2>/dev/null)
375         WARP=$(awk -F': *' '/^ WARP egress:/ {print $2; exit}' "$LATEST_REPORT" 2>/dev/null)
376         printf '\n Most recent verdict: Scenario=%s WARP=%s\n' \
377             "${VERDICT:-?}" "${WARP:-?}"
378         if [ "$VERDICT" = "OK" ]; then
379             ok "fix confirmed by diagnostic host traffic is going through WARP"
380         fi
381     fi
382 else
383     warn "diagnostic exited non-zero; verify manually"
384 fi
385
386 printf '\n%b%b\n' "$BOLD" "$NC"
387 printf '%bDone.%b Full transcript at %s\n' "$BOLD" "$NC" "$LOG_FILE"
388 printf '%b%b\n' "$BOLD" "$NC"
389 exit 0

```

35 scripts/fix-ssh-delay.sh

```

1 #!/bin/bash
2 # scripts/fix-ssh-delay.sh Fix ~20s SSH connection delay caused by UseDNS
3 #
4 # macOS OpenSSH defaults UseDNS to "yes". When Tailscale SSH connects from
5 # a peer (phone/laptop), sshd performs a reverse-DNS (PTR) lookup on the
6 # client's 100.x.x.x Tailscale IP. Since DNS is hijacked to the nullexit
7 # gateway AdGuard Cloudflare WARP, and 100.64.0.0/10 is CGNAT private
8 # space, the PTR query times out (~15-20s) before SSH proceeds.
9 #
10 # This script uncomments "UseDNS no" in /etc/ssh/sshd_config and reloads
11 # sshd. Existing SSH sessions are NOT dropped.
12 #
13 # Usage:
14 #   sudo bash scripts/fix-ssh-delay.sh
15
16 set -euo pipefail
17
18 SSHD_CONFIG="/etc/ssh/sshd_config"
19
20 # Check we're root
21 if [ "$(id -u)" -ne 0 ]; then
22     echo "ERROR: This script must be run with sudo (it edits $SSHD_CONFIG)"
23     echo "Usage: sudo bash scripts/fix-ssh-delay.sh"
24     exit 1
25 fi
26
27 # Apply config (idempotent)
28 if grep -qE '^UseDNS no' "$SSHD_CONFIG" 2>/dev/null; then
29     echo "UseDNS is already set to 'no' in $SSHD_CONFIG"
30 elif grep -qE '^#UseDNS' "$SSHD_CONFIG" 2>/dev/null; then
31     echo "Uncommenting 'UseDNS no' in $SSHD_CONFIG"
32     sed -i 's/^#UseDNS no/UseDNS no/' "$SSHD_CONFIG"
33 elif grep -qEi '^UseDNS' "$SSHD_CONFIG" 2>/dev/null; then
34     echo "Changing existing UseDNS line to 'UseDNS no'"
35     sed -i 's/^UseDNS.*/UseDNS no/' "$SSHD_CONFIG"
36 else
37     echo "Appending 'UseDNS no' to $SSHD_CONFIG"
38     echo "" >> "$SSHD_CONFIG"
39     echo "UseDNS no" >> "$SSHD_CONFIG"
40 fi
41
42 # Verify
43 if ! grep -qE '^UseDNS no' "$SSHD_CONFIG"; then
44     echo "Failed to apply UseDNS no check $SSHD_CONFIG manually"

```

```

45     exit 1
46 fi
47
48 # Reload sshd (does NOT drop existing connections)
49 echo " Reloading sshd..."
50
51 SSHD_PID=$(pgrep -f /usr/sbin/sshd 2>/dev/null | head -1 || true)
52 if [ -n "$SSHD_PID" ]; then
53     kill -HUP "$SSHD_PID"
54     echo " Sent SIGHUP to sshd (PID $SSHD_PID)  config reloaded"
55 else
56     echo " No running sshd found. macOS starts sshd on-demand per connection."
57     echo " The new 'UseDNS no' config will take effect on the next SSH connection."
58 fi
59
60 echo ""
61 echo " Done. SSH connections should now connect instantly."
62 echo " Existing sessions were not interrupted."

```

36 scripts/generate__tex.py

```

1  #!/usr/bin/env python3
2  import os
3
4  PROJECT_ROOT = os.path.abspath(os.path.join(os.path.dirname(__file__), '..'))
5
6  IGNORE_DIRS = {'.git', '.github', 'adguard', '__pycache__', 'Recover Gateway.app', 'Toggle Gateway.app'}
7  IGNORE_FILES = {'.env', 'nullexit_unified.tex', 'nullexit_unified.pdf', 'output.log', 'blocked.log', '.DS_Store',
8  '.lan_p2p_detected', '.signatures', '.gateway_ip', 'TUNNEL_FAILED_CLOSED.marker'}
9  IGNORE_EXTS = {'.pdf', '.tex', '.png', '.jpg', '.jpeg', '.zip', '.tar', '.gz', '.log', '.aux', '.fls', '.fdb_latexmk', '.out', '.toc', '.xdv', '.synctex(busy)'}
10
11 # Exclude from code block inclusion, but keep in tree
12 SKIP_CONTENT_FILES = {'LICENSE'}
13
14 PRIORITY_FILES = [
15     'README.md',
16     'agent.md',
17     'devref.md',
18     'diagrams.md',
19     'toggle.sh',
20     'recover.sh',
21     'setup.sh',
22     'scripts/toggle-linux.sh',
23     'scripts/recover-linux.sh',
24     'scripts/setup-linux.sh',
25     'docker-compose.yml',
26     'scripts/common.sh',
27     'scripts/watcher.sh',
28     'scripts/crypto.sh'
29 ]
30
31 def get_language(filename):
32     ext = os.path.splitext(filename)[1].lower()
33     if ext == '.sh' or ext == '.applescript':
34         return 'bash'
35     elif ext == '.py':
36         return 'Python'
37     elif ext == '.go':
38         return 'Go'
39     elif ext in {'.yaml', '.yml'}:
40         return 'bash'
41     elif ext == '.plist':
42         return 'XML'
43     elif ext == '.md':
44         return ''
45     return ''
46
47 def escape_tex(text):
48     return text.replace('_', '\\_')
49
50 def generate_tree(dir_path, prefix=""):
51     tree_str = ""
52     entries = sorted(os.listdir(dir_path))
53     entries = [e for e in entries if e not in IGNORE_DIRS and e not in IGNORE_FILES and os.path.splitext(e)[1].lower() not in IGNORE_EXTS]

```



```

132     filepath = os.path.join(root, file)
133     relpath = os.path.relpath(filepath, PROJECT_ROOT)
134     all_relpaths.append(relpath)
135
136
137     def sort_key(path):
138         if path in PRIORITY_FILES:
139             return (0, PRIORITY_FILES.index(path))
140         return (1, path)
141
142     all_relpaths.sort(key=sort_key)
143
144     with open(tex_path, 'w', encoding='utf-8') as f:
145         f.write(preamble)
146         f.write(tree)
147         f.write(midamble)
148
149     for relpath in all_relpaths:
150         lang = get_language(relpath)
151         lang_attr = f"[language={lang}]" if lang else ""
152
153         f.write(f"\\section{{{escape_tex(relpath)}}}\n")
154         f.write(f"\\begin{{{lstlisting}}}{lang_attr}\n")
155         try:
156             with open(os.path.join(PROJECT_ROOT, relpath), 'r', encoding='utf-8', errors='ignore') as src:
157                 content = src.read()
158                 import re
159                 # Strip all non-ASCII characters (emojis, box drawings, em dashes, etc)
160                 content = re.sub(r'[\x00-\x7F]+', '', content)
161                 f.write(content)
162                 if not content.endswith('\n'):
163                     f.write('\n')
164             except Exception as e:
165                 f.write(f"Error reading file: {e}\n")
166             f.write("\\end{lstlisting}\n\n")
167
168         f.write("\\end{document}\n")
169
170     print(f"Generated {tex_path}")
171
172 if __name__ == '__main__':
173     main()

```

37 scripts/logger/go.mod

```

1 module nullexit/logger
2
3 go 1.22

```

38 scripts/logger/main.go

```

1 package main
2
3 import (
4     "bufio"
5     "encoding/json"
6     "fmt"
7     "io"
8     "log"
9     "os"
10    "regexp"
11    "strings"
12    "time"
13 )
14
15 const (
16     queryLogPath = "/adguard_work/data/querylog.json"
17     procKmsgPath = "/proc/kmsg"
18     outputLogPath = "/app/blocked.log"
19 )
20
21 func logMessage(msg string) {

```

```

22 timestamp := time.Now().Format("2006-01-02 15:04:05")
23 formattedMsg := fmt.Sprintf("%s %s\n", timestamp, msg)
24 fmt.Print(formattedMsg)
25
26 f, err := os.OpenFile(outputLogPath, os.O_APPEND|os.O_CREATE|os.O_WRONLY, 0644)
27 if err != nil {
28     log.Printf("Error writing to log file: %v\n", err)
29     return
30 }
31 defer f.Close()
32 f.WriteString(formattedMsg)
33 }
34
35 func tailFile(filepath string, out chan<- string) {
36     for {
37         if _, err := os.Stat(filepath); os.IsNotExist(err) {
38             time.Sleep(1 * time.Second)
39             continue
40         }
41         break
42     }
43
44     f, err := os.Open(filepath)
45     if err != nil {
46         log.Printf("Error opening file %s: %v\n", filepath, err)
47         return
48     }
49     defer func() {
50         if f != nil {
51             f.Close()
52         }
53     }()
54
55     // Seek to end
56     f.Seek(0, io.SeekEnd)
57     reader := bufio.NewReader(f)
58
59     for {
60         line, err := reader.ReadString('\n')
61         if err != nil {
62             if err == io.EOF {
63                 stat, err := os.Stat(filepath)
64                 if err == nil {
65                     currentOffset, _ := f.Seek(0, io.SeekCurrent)
66                     if stat.Size() < currentOffset {
67                         // File truncated or rotated
68                         f.Close()
69                         for {
70                             if _, err := os.Stat(filepath); os.IsNotExist(err) {
71                                 time.Sleep(1 * time.Second)
72                                 continue
73                             }
74                             break
75                         }
76                         f, _ = os.Open(filepath)
77                         reader = bufio.NewReader(f)
78                         continue
79                     }
80                 }
81                 time.Sleep(500 * time.Millisecond)
82                 continue
83             }
84             log.Printf("Error reading file %s: %v\n", filepath, err)
85             time.Sleep(1 * time.Second)
86             continue
87         }
88         out <- line
89     }
90 }
91
92 type adguardLogEntry struct {
93     IP      string `json:"IP"`
94     QH      string `json:"QH"`
95     QT      string `json:"QT"`
96     Result struct {
97         IsFiltered bool `json:"IsFiltered"`
98         Reason      int  `json:"Reason"`
99         Rules       []struct {
100             Text string `json:"Text"`
101         } `json:"Rules"`
102     } `json:"Result"`

```

```

103 }
104
105 func monitorDNS() {
106     logMessage("[System] Starting DNS block logger...")
107     lines := make(chan string)
108     go tailFile(queryLogPath, lines)
109
110     reasonMap := map[int]string{
111         2: "CustomRule",
112         3: "BlockList",
113         4: "SafeBrowsing",
114         5: "ParentalControl",
115         6: "SafeSearch",
116         7: "BlockedService",
117     }
118
119     for line := range lines {
120         var data adguardLogEntry
121         if err := json.Unmarshal([]byte(line), &data); err != nil {
122             continue
123         }
124
125         res := data.Result
126         if res.IsFiltered || (res.Reason != 0 && res.Reason != 1) {
127             qh := data.QH
128             if qh == "" {
129                 qh = "unknown"
130             }
131             qt := data.QT
132             if qt == "" {
133                 qt = "unknown"
134             }
135             clientIP := data.IP
136             if clientIP == "" {
137                 clientIP = "unknown"
138             }
139
140             ruleText := "unknown rule"
141             if len(res.Rules) > 0 && res.Rules[0].Text != "" {
142                 ruleText = res.Rules[0].Text
143             }
144
145             reasonStr, ok := reasonMap[res.Reason]
146             if !ok {
147                 reasonStr = fmt.Sprintf("Reason-%d", res.Reason)
148             }
149
150             logMessage(fmt.Sprintf("[DNS] Blocked %s (Type: %s) for client %s | Reason: %s | Rule: %s", qh, qt,
151                 clientIP, reasonStr, ruleText))
152         }
153     }
154 }
155
156 func monitorIPs() {
157     logMessage("[System] Starting IP block logger...")
158
159     prefixRe := regexp.MustCompile(`^IP_BLOCK_[A-Z_]+\:`)
160     srcRe := regexp.MustCompile(`^SRC=([0-9a-fA-F\.:]+)\`)
161     dstRe := regexp.MustCompile(`^DST=([0-9a-fA-F\.:]+)\`)
162     protoRe := regexp.MustCompile(`^PROTO=([A-Z0-9]+\`)
163     sptRe := regexp.MustCompile(`^SPT=(\d+)\`)
164     dptRe := regexp.MustCompile(`^DPT=(\d+)\`)
165     inRe := regexp.MustCompile(`^IN=([a-zA-Z0-9\.\_]+\`)
166     outRe := regexp.MustCompile(`^OUT=([a-zA-Z0-9\.\_]+\`)
167
168     f, err := os.Open(procKmsgPath)
169     if err != nil {
170         if os.IsPermission(err) {
171             logMessage("[System] ERROR: Insufficient permissions to read /proc/kmsg. Enable CAP_SYSLOG.")
172         } else {
173             logMessage(fmt.Sprintf("[System] ERROR in IP logger: %v", err))
174         }
175         return
176     }
177     defer f.Close()
178
179     reader := bufio.NewReader(f)
180     for {
181         line, err := reader.ReadString('\n')
182         if err != nil {
183             if err == io.EOF {

```

```

183     time.Sleep(100 * time.Millisecond)
184     continue
185 }
186 logMessage(fmt.Sprintf("[System] ERROR reading kmsg: %v", err))
187 time.Sleep(1 * time.Second)
188 continue
189 }
190
191 if strings.Contains(line, "IP_BLOCK") {
192     match := prefixRe.FindStringSubmatch(line)
193     if len(match) > 1 {
194         prefix := match[1]
195
196         src := "unknown"
197         if m := srcRe.FindStringSubmatch(line); len(m) > 1 {
198             src = m[1]
199         }
200
201         dst := "unknown"
202         if m := dstRe.FindStringSubmatch(line); len(m) > 1 {
203             dst = m[1]
204         }
205
206         proto := "unknown"
207         if m := protoRe.FindStringSubmatch(line); len(m) > 1 {
208             proto = m[1]
209         }
210
211         spt := ""
212         if m := sptRe.FindStringSubmatch(line); len(m) > 1 {
213             spt = m[1]
214         }
215
216         dpt := ""
217         if m := dptRe.FindStringSubmatch(line); len(m) > 1 {
218             dpt = m[1]
219         }
220
221         inIf := ""
222         if m := inRe.FindStringSubmatch(line); len(m) > 1 {
223             inIf = m[1]
224         }
225
226         outIf := ""
227         if m := outRe.FindStringSubmatch(line); len(m) > 1 {
228             outIf = m[1]
229         }
230
231         direction := "inbound"
232         if strings.HasSuffix(prefix, "DST") {
233             direction = "outbound"
234         }
235
236         listType := "MALICIOUS"
237         if !strings.Contains(prefix, "MALICIOUS") {
238             parts := strings.Split(prefix, "_")
239             if len(parts) >= 3 {
240                 listType = "GEO_" + parts[2]
241             }
242         }
243
244         portStr := ""
245         if dpt != "" {
246             portStr = ":" + dpt
247         }
248
249         srcPortStr := ""
250         if spt != "" {
251             srcPortStr = ":" + spt
252         }
253
254         ifStr := ""
255         if inIf != "" && outIf != "" {
256             ifStr = fmt.Sprintf("IF: %s->%s", inIf, outIf)
257         } else if inIf != "" || outIf != "" {
258             ifStr = fmt.Sprintf("IF: %s%s", inIf, outIf)
259         }
260
261         logMessage(fmt.Sprintf("[IP] Blocked %s to %s%s (Proto: %s) from %s%s (%s) | List: %s", direction, dst,
262             portStr, proto, src, srcPortStr, ifStr, listType))
263     }

```

```

263     }
264 }
265 }
266
267 func main() {
268     logMessage("[System] Nullexit Block Logger starting...")
269
270     go monitorDNS()
271     monitorIPs()
272 }

```

39 scripts/pf.conf

```

1 # scripts/pf.conf - Packet Filter configuration for nullexit killswitch
2 # Dynamically parameterized by pfctl macro override
3
4 # ext_if must be passed via command line, e.g. -D ext_if=en0
5 # If not passed, default to en0
6 ext_if = "en0"
7 ts_if = "utun+"
8
9 # Tables populated dynamically via pfctl
10 table <vpn_endpoints> persist
11 table <derp_relays> persist
12
13 # Core Rules
14 set skip on lo0 # Ensure local loopback is never blocked
15 scrub out all max-mss 1160 # TCP MSS Clamping: Prevents fragmentation on host TCP flows (like Tailscale
    control plane)
16 pass quick on $ts_if all # Allow all Tailscale traffic (essential to prevent SSH lockout)
17
18 # Default deny outbound WAN traffic
19 block out on $ext_if all
20
21 # Block inbound Screen Sharing (VNC) and SSH from local Wi-Fi, forcing them to use Tailscale
22 block in on $ext_if proto tcp from any to any port { 22, 5900 }
23
24 # Exceptions for tunnel handshakes and coordination
25 pass out quick on $ext_if proto udp to <vpn_endpoints> port 2408
26 pass out quick on $ext_if proto udp to <derp_relays>
27 pass out quick on $ext_if proto tcp to 192.200.0.0/24 port 443
28
29 # Allow local LAN traffic (AirDrop, local Tailscale P2P, etc)
30 pass out quick on $ext_if proto tcp to { 192.168.0.0/16, 10.0.0.0/8, 172.16.0.0/12 }
31 pass out quick on $ext_if proto udp to { 192.168.0.0/16, 10.0.0.0/8, 172.16.0.0/12 }
32 pass out quick on $ext_if proto icmp to { 192.168.0.0/16, 10.0.0.0/8, 172.16.0.0/12 }
33
34 # Allow DHCP outbound (client port 68 to server port 67) so we can obtain/renew IP address
35 pass out quick on $ext_if proto udp from any port 68 to any port 67

```

40 scripts/reinstall-host-tailscale.sh

```

1 #!/usr/bin/env bash
2 #
3 # scripts/reinstall-host-tailscale.sh
4 #
5 # Complete uninstall + reinstall of the host-side Homebrew Tailscale.
6 #
7 # Use this when the brew LaunchAgent is wedged, Login Items entries are
8 # duplicated, "Tailscale is stopped" persists despite
9 # `brew services restart tailscale`, or you simply want a clean baseline
10 # before re-engaging the gateway as exit node.
11 #
12 # Idempotent. Safe to re-run. Use --dry to preview every step with zero
13 # changes. Use --yes to skip the per-step confirmation prompts.
14 #
15 # This script ONLY touches the HOST-side Tailscale (the one installed by
16 # Homebrew `brew install tailscale` and managed by launchd as
17 # homebrew.mxcl.tailscale). The nullexit gateway container's tailscaled
18 # (which lives inside the Colima VM at 100.100.21.8 and offers the exit
19 # node to this host) is unaffected.
20

```

```

21 set -u
22
23 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
24
25 DRY=0
26 YES=0
27
28 for arg in "$@"; do
29     case "$arg" in
30         --dry) DRY=1 ;;
31         --yes) YES=1 ;;
32         --help|-h) # NOTE: heredoc is UNQUOTED so $SCRIPT_DIR expands in the body. If you
33             # add new $-style placeholders here, they'll expand the same way.
34             cat <<USAGE
35 Usage: bash "$SCRIPT_DIR/reinstall-host-tailscale.sh" [--dry] [--yes]
36
37 --dry Print every step, make zero changes. Use to audit before
38 committing.
39 --yes Auto-confirm the prompts at every destructive step. Useful in
40 CI / fully-scripted recovery; do not use until you've read
41 the script.
42
43 After this script completes cleanly, you still must (MANUALLY):
44
45 1. Open System Settings Privacy & Security Local Network and
46 enable Tailscale. macOS will prompt automatically the first time
47 the fresh install of `tailscale` is invoked.
48
49 2. Re-engage the tailnet + exit node:
50     sudo tailscale up --ssh=true --accept-dns=false \
51     --exit-node="$(cat .gateway_ip | tr -d '\r' | awk 'NR==1{print $1; exit}')" \
52     --exit-node-allow-lan-access=true
53
54 3. Re-run ./toggle.sh from the project root.
55
56 4. Final verification:
57     curl -s https://www.cloudflare.com/cdn-cgi/trace | grep -E '^(warp|ip|colo)='
58     expected: warp=on, ip=<Cloudflare>, colo=YYZ/YUL
59 USAGE
60     exit 0
61     ;;
62     *)
63     echo "Unknown flag: $arg" >&2
64     exit 2
65     ;;
66     esac
67 done
68
69 # ----- helpers -----
70
71 confirm() {
72     if [ "$YES" = 1 ] || [ "$DRY" = 1 ]; then
73         return 0
74     fi
75     printf "      %s [y/N] " "$1"
76     read -r ans
77     case "$ans" in
78         y|Y|yes|YES) return 0 ;;
79         *)
80         echo "      aborted by user"
81         exit 1
82         ;;
83     esac
84 }
85
86 header() {
87     printf '\n %s\n' "$1"
88 }
89
90 # with_timeout <seconds> <cmd...>
91 # Run a command in the background and kill it if it exceeds <seconds>.
92 # Returns the command's exit code if it finished in time, 124 (matching
93 # GNU `timeout` convention) if it timed out. macOS doesn't ship GNU
94 # coreutils `timeout`, so this is the bash-3.2-friendly replacement.
95 # stdout + stderr are discarded (we only care about exit code); redirect
96 # or pipe externally if you need the output.
97 with_timeout() {
98     local timeout_s="${1:-30}"
99     shift
100     "$@" >/dev/null 2>&1 &
101     local cmdpid=$!

```

```

102 local elapsed=0
103 while kill -0 "$cmdpid" 2>/dev/null; do
104     if [ "$elapsed" -ge "$timeout_s" ]; then
105         kill -9 "$cmdpid" 2>/dev/null
106         wait "$cmdpid" 2>/dev/null
107         return 124
108     fi
109     sleep 1
110     elapsed=$((elapsed+1))
111 done
112 wait "$cmdpid" 2>/dev/null
113 return $?
114 }
115
116 # ----- 0. Pre-flight -----
117 header "Pre-flight current host tailscale state"
118 echo " brew: $(command -v brew >/dev/null 2>&1 && brew --version | head -1 || echo 'not on
PATH')"
119 echo " tailscaled running: $(pgrep -lf tailscaled 2>/dev/null | wc -l | tr -d ' ') process(es)"
120 echo " brew service tailscale: $(brew services list 2>/dev/null | awk '$1=="tailscale"{print $2, $3}' | tr
'\n' ' ' | sed 's/ $//')
121 echo " LaunchAgent plists (under ~/Library/LaunchAgents/):"
122 ls -l ~/Library/LaunchAgents/ 2>/dev/null | grep -i 'tail' | sed 's/~/ /' || echo " (none)"
123
124 # Detect Tailscale's macOS Network Extension (residual from prior App Store
125 # Tailscale.app installs). Lives under /Library/Apple/SystemExtensions/ and
126 # is managed by `systemextensionsctl` NOT launchd. If loaded, brew
127 # tailscaled will NOT engage the Network Extension framework slot because
128 # the SE still has it claimed. Symptom: 'sudo tailscale status' returns
129 # "Tailscale is stopped" even when the daemon is alive and listening.
130 se_found=0
131 if command -v systemextensionsctl >/dev/null 2>&1; then
132     # Grab any line matching tailscale and network-extension
133     se_line=$(systemextensionsctl list 2>/dev/null | grep -iE '(io|com)\.tailscale\..*network-extension' | head -
n 1)
134     if [ -n "$se_line" ]; then
135         se_found=1
136         # Extract the teamID (10-char uppercase/numeric word)
137         se_teamID=$(echo "$se_line" | tr -s ' \t' '\n' | grep -E '^[A-Z0-9]{10}$' | head -n 1)
138         # Extract the bundleID (starts with io.tailscale or com.tailscale)
139         se_bundleID=$(echo "$se_line" | tr -s ' \t' '\n' | grep -E '^(io|com)\.tailscale\.' | head -n 1 | sed 's
/(.*)/')
140
141         # If teamID is not found, fallback to '-'
142         if [ -z "$se_teamID" ]; then
143             se_teamID="-"
144         fi
145     fi
146 fi
147
148 if [ "$se_found" = 1 ] && [ -n "$se_bundleID" ]; then
149     echo " Tailscale Network Extension System Extension is loaded (residual from App Store Tailscale.app
install)"
150     echo " Team ID: $se_teamID, Bundle ID: $se_bundleID"
151     echo " brew tailscaled cannot engage the NE slot while this SE has it claimed."
152     if [ "$DRY" = 1 ]; then
153         echo " [dry] (would prompt) sudo systemextensionsctl uninstall $se_teamID $se_bundleID"
154     else
155         confirm "Run 'sudo systemextensionsctl uninstall $se_teamID $se_bundleID'? (frees the NE slot for brew
tailscale; non-fatal if it errors)"
156         sudo systemextensionsctl uninstall "$se_teamID" "$se_bundleID" 2>&1 \
157         || echo " ! uninstall returned non-zero (may require reboot, non-fatal)"
158     fi
159 else
160     echo " no Tailscale System Extension loaded"
161 fi
162
163 # Pattern used by the Step-1 bootout loops, in both dry + real branches.
164 # Widened from a bare /tailscale/ so the same loops also pick up any
165 # nullexit-labeled LaunchAgents (e.g., com.nullexit.wake-recovery, the
166 # caffeinate + post-wake recovery hook). The wake-recovery LaunchAgent
167 # is re-installed by ./toggle.sh / setup.sh on re-engage, so wiping it
168 # here is non-fatal and is part of restoring a clean baseline.
169 TAILSCALE_LABEL_RE='tailscale'
170 NULLEXIT_LABEL_RE='nullexit'
171
172 # ----- 1. brew services stop -----
173 header "Step 1 Stop the brew-managed service + unload LaunchAgents"
174 # Detect whether the brew-managed service is registered as $USER (gui/$UID
175 # domain) or as root (system domain). NOPASSWD sudoers can leave a stale
176 # root-owned registration after a prior 'sudo brew services ...' round;

```

```

177 # 'brew services stop' without sudo refuses to touch a root-owned plist
178 # (Error: Service 'tailscale' is started as `root`).
179 brew_status_line="$(brew services list 2>/dev/null | awk '1=="tailscale"')"
```

```

180 brew_status_user="$(echo "$brew_status_line" | awk '{print $3}')"
181 if [ -n "$brew_status_line" ]; then
182     if [ "$DRY" = 1 ]; then
183         echo "      [dry] brew services stop tailscale (user=$brew_status_user)"
184     else
185         if [ "$brew_status_user" = "root" ]; then
186             confirm "Run 'sudo brew services stop tailscale' (service is registered as root)?"
187             sudo brew services stop tailscale 2>&1 || echo "      (sudo brew services stop returned non-zero non-fatal)"
188         else
189             echo "      brew services stop tailscale (user=$brew_status_user, no sudo needed)"
190             brew services stop tailscale 2>&1 || echo "      (brew reported it wasn't actually running)"
191         fi
192     fi
193     echo "      unload any tailscale-related LaunchAgents (prevents launchd from respawning after kill)"
194     if [ "$DRY" = 1 ]; then
195         echo "      [dry] for each 'tailscale|nullexit'-matched label in gui/$UID domain:"
196         while IFS= read -r label; do
197             [ -z "$label" ] && continue
198             echo "      [dry] launchctl bootout gui/$UID/$label"
199             done < <(launchctl list 2>/dev/null | awk -v ts="$TAILSCALE_LABEL_RE" -v nx="$NULLEXIT_LABEL_RE" 'NR > 1 && ($3 ~ ts || $3 ~ nx) {print $3}')
200             echo "      [dry] for each 'tailscale|nullexit'-matched label in system domain (root-owned brew plist):"
201             while IFS= read -r label; do
202                 [ -z "$label" ] && continue
203                 echo "      [dry] (would prompt) sudo launchctl bootout system/$label"
204                 done < <(sudo -n launchctl list 2>/dev/null | awk -v ts="$TAILSCALE_LABEL_RE" -v nx="$NULLEXIT_LABEL_RE" 'NR > 1 && ($3 ~ ts || $3 ~ nx) {print $3}')
205                 if launchctl print system/com.tailscale.ipn.macos >/dev/null 2>&1; then
206                     echo "      [dry] (would prompt) sudo launchctl bootout system/com.tailscale.ipn.macos"
207                 fi
208                 if launchctl print system/com.tailscale.ipn.macsys >/dev/null 2>&1; then
209                     echo "      [dry] (would prompt) sudo launchctl bootout system/com.tailscale.ipn.macsys"
210                 fi
211             else
212                 # Bootout every gui-domain LaunchAgent whose label matches 'tailscale'
213                 # OR 'nullexit'. The nullexit arm catches com.nullexit.wake-recovery,
214                 # which holds caffeinate/watcher.sh alive across a daemon reset not
215                 # desired while the script is establishing a fresh baseline. Awake-
216                 # recovery is re-installed by ./toggle.sh / setup.sh on re-engage, so
217                 # wiping here is non-fatal. The pattern stays narrower than a bare
218                 # /tail/ or /null/ so unrelated Apple services are not unloaded.
219                 while IFS= read -r label; do
220                     [ -z "$label" ] && continue
221                     echo "      launchctl bootout gui/$UID/$label"
222                     launchctl bootout "gui/$UID/$label" 2>/dev/null || true
223                     done < <(launchctl list 2>/dev/null | awk -v ts="$TAILSCALE_LABEL_RE" -v nx="$NULLEXIT_LABEL_RE" 'NR > 1 && ($3 ~ ts || $3 ~ nx) {print $3}')
224                 # Bootout every system-domain LaunchDaemon/Agent whose label matches
225                 # 'tailscale' OR 'nullexit'. This catches the root-owned brew plist left
226                 # behind by prior 'sudo brew services ...' invocations that was the
227                 # case that left tailscaled PID 47447 57734 respawning between sudo
228                 # pkill runs. Defensive nullexit catch in case the user previously
229                 # `sudo ln -s`'d the wake-recovery plist into /Library/LaunchDaemons.
230                 while IFS= read -r label; do
231                     [ -z "$label" ] && continue
232                     confirm "Run 'sudo launchctl bootout system/$label'?"
233                     sudo launchctl bootout "system/$label" 2>/dev/null || true
234                     done < <(sudo -n launchctl list 2>/dev/null | awk -v ts="$TAILSCALE_LABEL_RE" -v nx="$NULLEXIT_LABEL_RE" 'NR > 1 && ($3 ~ ts || $3 ~ nx) {print $3}')
235                 # Legacy Tailscale.app system-domain entries. Each one is checked and
236                 # confirmed separately so we don't sudo-prompt or sudo-bootout for nothing.
237                 if launchctl print system/com.tailscale.ipn.macos >/dev/null 2>&1; then
238                     confirm "Run 'sudo launchctl bootout system/com.tailscale.ipn.macos'?"
239                     sudo launchctl bootout system/com.tailscale.ipn.macos 2>/dev/null || true
240                 fi
241                 if launchctl print system/com.tailscale.ipn.macsys >/dev/null 2>&1; then
242                     confirm "Run 'sudo launchctl bootout system/com.tailscale.ipn.macsys'?"
243                     sudo launchctl bootout system/com.tailscale.ipn.macsys 2>/dev/null || true
244                 fi
245             fi
246             echo "      waiting up to 10s for tailscaled to exit"
247             for i in 1 2 3 4 5 6 7 8 9 10; do
248                 if [ "$DRY" = 1 ]; then
249                     echo "      [dry] sleep 1; loop check would terminate here"
250                     break
251                 fi
252                 sleep 1

```

```

253     if ! pgrep -lf tailscaled >/dev/null 2>&1; then
254         echo "        tailscaled exited in ${i}s"
255         break
256     fi
257     [ "$i" = 10 ] && echo "        tailscaled still alive after 10s Step 2 will deal with it"
258 done
259 else
260     echo "        brew reports no tailscale service loaded"
261 fi
262
263 # ----- 2. Kill any orphan tailscaled -----
264 header "Step 2 Kill any orphan tailscaled process(es) (may escalate to sudo)"
265 if ! pgrep -lf tailscaled >/dev/null 2>&1; then
266     echo "        no orphans"
267 else
268     echo "        Stray process(es):"
269     pgrep -lf tailscaled | sed 's/^/ /'
270     echo "        no-sudo: pkill tailscaled"
271     if [ "$DRY" = 1 ]; then
272         echo "        [dry] pkill tailscaled on failure: sudo pkill -9 tailscaled"
273     else
274         if pkill tailscaled 2>/dev/null; then
275             echo "        no-sudo pkill succeeded"
276         else
277             echo "        ! no-sudo pkill insufficient; cannot escalate yet"
278         fi
279         sleep 1
280     fi
281
282 # Recheck; if still alive, offer sudo escalation
283 if { [ "$DRY" = 0 ] && pgrep -lf tailscaled >/dev/null 2>&1; } || [ "$DRY" = 1 ]; then
284     if [ "$DRY" = 1 ]; then
285         echo "        [dry] sudo pkill -9 tailscaled (escalation would fire here)"
286     else
287         confirm "Run 'sudo pkill -9 tailscaled' to escalate?"
288         sudo pkill -9 tailscaled 2>&1
289         sleep 1
290     fi
291 fi
292
293 if [ "$DRY" = 0 ]; then
294     if pgrep -lf tailscaled >/dev/null 2>&1; then
295         echo "        Failed to kill stragglers bailing out"
296         pgrep -lf tailscaled | sed 's/^/ /'
297         exit 3
298     fi
299     echo "        all tailscaled processes gone"
300 fi
301 fi
302
303 # ----- 3. brew uninstall -----
304 header "Step 3 brew uninstall tailscale"
305 if ! brew list tailscale >/dev/null 2>&1; then
306     echo "        not currently installed via brew skip"
307 else
308     # Detect whether the keg directory is owner=$USER or owner=root. If the
309     # keg is root-owned (typically a residual of some prior 'sudo brew ...'
310     # cycle that did perms fixups), 'brew uninstall' without sudo refuses
311     # to remove the files and prints: 'Error: Could not remove tailscale
312     # keg! Do so manually: sudo rm -rf /opt/homebrew/Cellar/tailscale/<v>'.
313     # Detect that and escalate to a sudoed uninstall, with a manual rm -rf
314     # fallback if `sudo brew uninstall` itself errors out (e.g., partial
315     # state where the keg dir is gone but brew still thinks it's installed).
316     # Detect keg state. Treat `(dir absent)` AND `(dir present + empty)` as the
317     # same `<unknown>` bucket so both escalate to sudo. The empty-parent case
318     # is what you get after a manual `sudo rm -rf /opt/homebrew/Cellar/tailscale/<v>`
319     # from inside: brew's parent dir is left present-but-empty and brew still
320     # thinks the package is installed.
321     keg_dir="/opt/homebrew/Cellar/tailscale"
322     if [ -d "$keg_dir" ] && [ -n "$(ls -A "$keg_dir" 2>/dev/null)" ]; then
323         keg_owner="$(stat -f '%Su' "$keg_dir" 2>/dev/null || echo '<unknown>')"
324     else
325         keg_owner="<unknown>"
326     fi
327     if [ "$DRY" = 1 ]; then
328         echo "        [dry] brew uninstall tailscale (keg_owner=$keg_owner)"
329     else
330         # 'root' AND '<unknown>' both require sudo escalation. The '<unknown>'
331         # case is the partial-state path: keg dir was already removed (manually
332         # or by a prior failed uninstall) but brew's internal state still says
333         # installed. Without this branch the script would fall through to a

```

```

334 # non-sudo brew uninstall that fails again with the same 'Could not
335 # remove tailscale keg!' error.
336 if [ "$keg_owner" = "root" ] || [ "$keg_owner" = "<unknown>" ]; then
337     confirm "Run 'sudo brew uninstall tailscale' (keg=$keg_owner)?"
338     sudo brew uninstall tailscale 2>&1 || {
339         echo " ! sudo brew uninstall failed; offering manual rm -rf fallback"
340         # Guard the glob so an empty/absent dir doesn't literal-expand the
341         # '*' and confuse the user with a sudo error about a non-existent path.
342         if [ -d /opt/homebrew/Cellar/tailscale ] && [ -n "$(ls -A /opt/homebrew/Cellar/tailscale 2>/dev/null)"
]; then
343             confirm "Run 'sudo rm -rf /opt/homebrew/Cellar/tailscale/*'?"
344             sudo rm -rf /opt/homebrew/Cellar/tailscale/* 2>&1
345         else
346             echo "      keg dir already empty or absent nothing more to rm"
347         fi
348     }
349 else
350     confirm "Run 'brew uninstall tailscale' (removes LaunchAgent plist + linked symlinks)?"
351     brew uninstall tailscale 2>&1
352 fi
353 fi
354 fi
355
356 # ----- 4. Wipe state directories -----
357 header "Step 4 Wipe stale state"
358 # User-owned
359 declare -a USER_PATHS
360 USER_PATHS=(
361     "$HOME/.local/share/tailscale"
362     "$HOME/.local/run/tailscaled.socket"
363     "$HOME/.local/run/tailscaled.state"
364     "$HOME/.local/state/tailscale"
365     "$HOME/Library/Application Support/Tailscale"
366     "$HOME/Library/Caches/com.tailscale.ipn.macos"
367     "$HOME/Library/Caches/Tailscale"
368     "$HOME/Library/Logs/Tailscale"
369 )
370 echo "      User-owned paths (no sudo):"
371 for p in "${USER_PATHS[@]}; do
372     if [ -e "$p" ]; then
373         if [ "$DRY" = 1 ]; then
374             echo "      [dry] rm -rf '$p'"
375         else
376             rm -rf "$p"
377             echo "      rm -rf '$p'"
378         fi
379     fi
380 done
381
382 # System
383 declare -a SUDO_PATHS
384 SUDO_PATHS=(
385     "/var/lib/tailscale"
386     "/var/cache/tailscale"
387 )
388 echo "      System paths under /var (will prompt for sudo if any are present):"
389 eligible_sudo=0
390 for p in "${SUDO_PATHS[@]}; do
391     if [ -e "$p" ]; then
392         eligible_sudo=1
393         if [ "$DRY" = 1 ]; then
394             echo "      [dry] sudo rm -rf '$p'"
395         else
396             if confirm "Run 'sudo rm -rf $p'?" ; then
397                 sudo rm -rf "$p" && echo "      rm -rf '$p'"
398             fi
399         fi
400     fi
401 done
402 [ "$eligible_sudo" = 0 ] && echo "      (none of /var/lib/tailscale, /var/cache/tailscale are present)"
403
404 # ----- 5. Wipe stale LaunchAgent plists -----
405 header "Step 5 Wipe stale LaunchAgent plist files"
406 declare -a PLIST_FILES
407 PLIST_FILES=(
408     "$HOME/Library/LaunchAgents/homebrew.mxcl.tailscale.plist"
409     "$HOME/Library/LaunchAgents/com.tailscale.ipn.macos.plist"
410     "$HOME/Library/LaunchAgents/homebrew.mxcl.tailscale.plist.bak"
411     "$HOME/Library/LaunchAgents/com.nullexit.wake-recovery.plist"
412 )
413 for f in "${PLIST_FILES[@]}; do

```

```

414 if [ -e "$f" ]; then
415     if [ "$DRY" = 1 ]; then
416         echo "    [dry] rm -f '$f'"
417     else
418         rm -f "$f"
419         echo "        rm -f '$f'"
420     fi
421 fi
422 done
423 echo "    Remaining tailscale|nullexit plists (should be empty):"
424 ls -l ~/Library/LaunchAgents/ 2>/dev/null | grep -i 'tail' | sed 's/~/' || echo "    (none)"
425
426 # ----- 5.5. Drain Background-Items (Login Items) -----
427 header "Step 5.5 Drain Background-Items (Login Items)"
428 # macOS 13+ keeps a parallel "Background Items" database alongside
429 # classic LaunchAgents. Even when we wipe ~/Library/LaunchAgents/homebrew.mxcl.tailscale.plist
430 # in Step 5, the .btm file at
431 # ~/Library/Application Support/com.apple.backgroundtaskmanagementagent.backgrounditems.btm
432 # can still hold a Background-Items entry pointing at the (now-dead) plist
433 # path. Next login: macOS sees the entry, tries to bootstrap, finds the
434 # plist missing, and either silently regenerates it via brew's post-install
435 # hook (causing tailscaled respawn at next login) or refuses outright.
436 # `sfltool reset-login-items` is Apple's canonical API for wiping the
437 # entire .btm file. It DOES wipe all Login Items the user re-adds what
438 # they want via System Settings General Login Items. Acceptable here
439 # because the goal of this script is "fresh baseline for Tailscale".
440 BTM="$HOME/Library/Application Support/com.apple.backgroundtaskmanagementagent/backgrounditems.btm"
441 if [ -f "$BTM" ]; then
442     if [ "$DRY" = 1 ]; then
443         echo "    [dry] found $BTM; would prompt sfltool reset-login-items"
444     else
445         confirm "Run 'sfltool reset-login-items' (wipes ALL Login Items; rebuild via System Settings General
446             Login Items)?"
447         if sfltool reset-login-items 2>&1; then
448             echo "        Login Items drained"
449         else
450             echo "        ! sfltool reset-login-items failed (non-fatal; do manually: System Settings General Login
451                 Items)"
452         fi
453     fi
454 else
455     echo "    no backgrounditems.btm present"
456 fi
457
458 # ----- 6. brew install -----
459 header "Step 6 brew install tailscale (fresh)"
460 if brew list tailscale >/dev/null 2>&1; then
461     echo "    already installed skip"
462 else
463     confirm "Run 'brew install tailscale'?"
464     if [ "$DRY" = 1 ]; then
465         echo "    [dry] brew install tailscale"
466     else
467         brew install tailscale 2>&1
468     fi
469 fi
470
471 # ----- 6.5. Strip quarantine xattr from keg -----
472 header "Step 6.5 Strip quarantine xattr from tailscale install"
473 # brew's freshly poured bottles don't carry com.apple.quarantine (the
474 # bottles are content-addressed and Homebrew-verified), but the upstream
475 # installer (Tailscale.app from App Store) and any manual download do.
476 # Strip the xattr from the entire keg as a defensive safety net so the
477 # freshly-installed tailscaled binary and any helper binaries are not
478 # Gatekeeper-blocked on first launch. Idempotent no-op if no quarantine
479 # xattr is present.
480 quarantine_count="$(xattr -lr /opt/homebrew/Cellar/tailscale 2>/dev/null | grep -c '^[^]* com.apple.quarantine
481     ' || echo 0)"
482 if [ "${quarantine_count:-0}" -gt 0 ]; then
483     if [ "$DRY" = 1 ]; then
484         echo "    [dry] found $quarantine_count file(s) with com.apple.quarantine; would prompt xattr -dr"
485     else
486         confirm "Run 'xattr -dr com.apple.quarantine /opt/homebrew/Cellar/tailscale'? (strips quarantine xattr from
487             $quarantine_count file(s); non-fatal if it errors)"
488         if xattr -dr com.apple.quarantine /opt/homebrew/Cellar/tailscale 2>&1; then
489             echo "        quarantine xattr stripped"
490         else
491             echo "        ! xattr -dr failed (non-fatal; open /opt/homebrew/Cellar/tailscale in Finder, right-click
492                 tailscaled Open Anyway)"
493         fi
494     fi
495 fi

```

```

490 else
491     echo "        no com.apple.quarantine xattr on tailscale install"
492 fi
493
494 # ----- 7. brew services start (NO sudo!) -----
495 header "Step 7 Start the service as $USER (NO sudo, with multi-tier self-healing)"
496 if ! command -v tailscale >/dev/null 2>&1; then
497     echo "        tailscale not on PATH after install  brew install errored; aborting"
498     exit 4
499 fi
500
501 # Aliveness fan-out: ANY one of these is taken as proof of life. None of
502 # them alone is reliable  pgrep misses tailscaled if it does setproctitle
503 # or fork-detach; launchctl's reported PID can be a shim that exited
504 # within 1s; tailscale-cli hangs while macOS Local-Network permission is
505 # unresolved. Combined, they cover each other's blind spots.
506
507 tailscaled_api_up() {
508     # Strategy A: tailscale CLI can talk to the local API. Canonical proof.
509     # Bound at 8s  tailscale status itself returns quickly on success OR
510     # failure, but if the post-install state is wedged it can hang.
511     with_timeout 8 /opt/homebrew/bin/tailscale status >/dev/null 2>&1
512 }
513
514 tailscaled_proc_up() {
515     # Strategy B: any process has 'tailscaled' substring in argv.
516     pgrep -lf tailscaled >/dev/null 2>&1
517 }
518
519 tailscaled_lc_up() {
520     # Strategy C: launchctl-managed service has a non-dash, non-zero PID.
521     local lc_pid
522     lc_pid="$(launchctl list 2>/dev/null | awk '$3 == "homebrew.mxcl.tailscale" {print $1; exit}')"
523     if [ -n "$lc_pid" ] && [ "$lc_pid" != "-" ] && [ "$lc_pid" -gt 0 ] 2>/dev/null; then
524         return 0
525     fi
526     return 1
527 }
528
529 tailscaled_is_alive() {
530     tailscaled_api_up && return 0
531     tailscaled_proc_up && return 0
532     tailscaled_lc_up && return 0
533     return 1
534 }
535
536 confirm "Run 'brew services start tailscale'?"
537 if [ "$DRY" = 1 ]; then
538     echo "        [dry] brew services start tailscale"
539     echo "        [dry] (then poll up to 30s for tailscaled_is_alive)"
540     echo "        [dry] (recovery tier 1: launchctl kickstart)"
541     echo "        [dry] (recovery tier 2: launchctl bootout + bootstrap of explicit plist)"
542     echo "        [dry] (recovery tier 3: brew services restart (stop+start cycle))"
543     echo "        [dry] (recovery tier 4: direct /opt/homebrew/bin/tailscaled spawn (bypasses launchd))"
544 else
545     brew services start tailscale 2>&1
546     # Primary poll  tailscaled_is_alive covers all 3 detection strategies.
547     echo "        polling for tailscaled to come up (up to ~30s)"
548     tailscaled_up=0
549     for i in 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15; do
550         if tailscaled_is_alive; then
551             tailscaled_up=1
552             echo "        tailscaled alive after ${i}*2s"
553             break
554         fi
555         sleep 2
556     done
557
558     # Recovery tier 1: kickstart the launchd-managed service. This is the
559     # cheapest and most-aligned-with-launchd fix when brew has loaded the
560     # LaunchAgent but tailscaled didn't fork. Kickstart tells launchd
561     # "re-spawn if not running" without forcing a teardown.
562     if [ "$tailscaled_up" = 0 ]; then
563         echo "        ! Recovery tier 1: launchctl kickstart"
564         if launchctl kickstart -kp "gui/$UID/homebrew.mxcl.tailscale" 2>&1; then
565             for i in 1 2 3 4 5 6 7 8 9 10; do
566                 if tailscaled_is_alive; then
567                     tailscaled_up=1
568                     echo "        tailscaled alive after kickstart (${i}*2s)"
569                     break
570                 fi

```

```

571     sleep 2
572     done
573     else
574         echo "          ! kickstart returned non-zero (LaunchAgent may not be loaded)"
575     fi
576 fi
577
578 # Recovery tier 2: explicit plist bootout + bootstrap. Forces a fresh
579 # launchd registration cycle for the brew-managed service, which fixes
580 # the wider class of "brew says started but daemon never spawned"
581 # wedge states that kickstart alone can't dig out of.
582 if [ "$tailscaled_up" = 0 ]; then
583     echo "          ! Recovery tier 2: launchctl bootout + bootstrap of explicit plist"
584     launchctl bootout "gui/$UID/homebrew.mxcl.tailscale" 2>/dev/null || true
585     sleep 1
586     # Brew installs the canonical plist template at $HOMEBREW_PREFIX/Library/.
587     # On Apple Silicon that's /opt/homebrew/Library/LaunchAgents/. On Intel
588     # it's /usr/local/Library/LaunchAgents/. Detect which and pick the
589     # right one so this script works across both architectures.
590     if [ -f /opt/homebrew/Library/LaunchAgents/homebrew.mxcl.tailscale.plist ]; then
591         plist_path="/opt/homebrew/Library/LaunchAgents/homebrew.mxcl.tailscale.plist"
592     elif [ -f /usr/local/Library/LaunchAgents/homebrew.mxcl.tailscale.plist ]; then
593         plist_path="/usr/local/Library/LaunchAgents/homebrew.mxcl.tailscale.plist"
594     else
595         plist_path=""
596     fi
597     if [ -n "$plist_path" ]; then
598         if launchctl bootstrap "gui/$UID" "$plist_path" 2>&1; then
599             for i in 1 2 3 4 5 6 7 8 9 10; do
600                 if tailscaled_is_alive; then
601                     tailscaled_up=1
602                     echo "          tailscaled alive after bootstrap (${i}*2s)"
603                     break
604                 fi
605                 sleep 2
606             done
607         else
608             echo "          ! explicit bootstrap returned non-zero"
609         fi
610     else
611         echo "          ! could not locate brew plist template; skipping bootstrap tier"
612     fi
613 fi
614
615 # Recovery tier 3: brew services restart (forces a stop + start cycle,
616 # which unsticks LaunchAgents with stale 'started' state from prior
617 # half-cycles). This is the canonical brew-doc fix for 'services don't
618 # actually start even though brew says started'.
619 if [ "$tailscaled_up" = 0 ]; then
620     echo "          ! Recovery tier 3: brew services restart (forces stop+start cycle)"
621     if brew services restart tailscale 2>&1; then
622         for i in 1 2 3 4 5 6 7 8 9 10; do
623             if tailscaled_is_alive; then
624                 tailscaled_up=1
625                 echo "          tailscaled alive after brew restart (${i}*2s)"
626                 break
627             fi
628             sleep 2
629         done
630     else
631         echo "          ! brew services restart returned non-zero"
632     fi
633 fi
634
635 # Recovery tier 4: last-resort direct spawn of the daemon binary,
636 # bypassing launchd entirely. If tailscaled crashes here too, we'll
637 # capture its actual stderr in /tmp for the user to read (NOPASSWD
638 # sudoers often have local-network permission denied the captured
639 # stderr makes it obvious what to fix in System Settings).
640 if [ "$tailscaled_up" = 0 ]; then
641     echo "          ! Recovery tier 4: direct spawn (bypasses launchd; tail stderr to /tmp)"
642     /opt/homebrew/bin/tailscaled >/tmp/nullexit-tailscaled-spawn.log 2>&1 &
643     spawned_pid=$!
644     disown "$spawned_pid" 2>/dev/null || true
645     sleep 3
646     if tailscaled_is_alive; then
647         tailscaled_up=1
648         echo "          tailscaled alive via direct spawn (pid=$spawned_pid)"
649         echo "          (NOTE: not under launchd supervision a reboot, re-run of"
650         echo "          this script, or ./toggle.sh is required to recreate."
651         echo "          Last 5 lines of stderr captured to /tmp/nullexit-tailscaled-spawn.log:)"

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```

652     tail -n 5 /tmp/nullexit-tailscaled-spawn.log 2>/dev/null | sed 's/~/ /' || true
653 fi
654 fi
655
656 if [ "$tailscaled_up" = 0 ]; then
657     echo "    HARD FAIL: tailscaled refuses to stay alive across every recovery tier"
658     echo "    launchctl-managed state (if available):"
659     launchctl print "gui/$UID/homebrew.mxcl.tailscale" 2>/dev/null \
660         | grep -E 'state =|last exit code =|runs =|path =' \
661         | sed 's/~/ /' \
662         || echo "    (no launchctl state available for homebrew.mxcl.tailscale)"
663     echo "    Last 10 lines of /tmp/nullexit-tailscaled-spawn.log (if any):"
664     tail -n 10 /tmp/nullexit-tailscaled-spawn.log 2>/dev/null | sed 's/~/ /' || echo "    (no log)"
665     echo "    Most likely remaining cause: macOS Local Network permission denied."
666     echo "    System Settings Privacy & Security Local Network toggle Tailscale ON."
667     exit 7
668 fi
669 fi
670 sleep 3
671
672 # ----- 8. Verify -----
673 header "Step 8 Verify fresh install"
674 echo "    brew service tailscale: $(brew services list 2>/dev/null | awk '$1=="tailscale"{{print $2, $3}}' | tr
675     '\n' ' ' | sed 's/ $//')
676 echo "    tailscaled process:      $(pgrep -lf tailscaled 2>/dev/null | head -1 || echo 'NONE (check brew logs)
677     ')"
678 ts_ver=""
679 if [ -x /opt/homebrew/bin/tailscale ]; then
680     ts_ver="$(/opt/homebrew/bin/tailscale version 2>/dev/null | head -1 | tr -d '\n')"
681 else
682     ts_ver="$(tailscale version 2>/dev/null | head -1 | tr -d '\n')"
683 fi
684 echo "    tailscale version:      ${ts_ver:-binary missing}"
685 ts_status=""
686 if [ -x /opt/homebrew/bin/tailscale ]; then
687     ts_status="$(/opt/homebrew/bin/tailscale status 2>/dev/null | head -3)"
688 fi
689 if [ -n "$ts_status" ]; then
690     echo "    tailscale status:"
691     printf '%s\n' "$ts_status" | sed 's/~/ /'
692 else
693     echo "    tailscale status:      (binary missing or errored)"
694 fi
695
696 if { [ "$DRY" = 0 ] && /opt/homebrew/bin/tailscale status 2>/dev/null | grep -q '^Tailscale is stopped'; };
697 then
698     echo
699     echo "    NOTICE: 'tailscale status' still reports stopped after a fresh install."
700     echo "    Two equally likely causes:"
701     echo "    (a) macOS hasn't yet shown the first-run 'Local Network' prompt for"
702     echo "        Tailscale open System Settings Privacy & Security Local"
703     echo "        Network. Toggle Tailscale ON."
704     echo "    (b) macOS denied Local Network permission at some prior point."
705     echo "    Either way: System Settings Privacy & Security Local Network."
706     echo "    If the entry isn't present, invoking any tailscale command will"
707     echo "    re-trigger the prompt automatically."
708 fi
709
710 # ----- 9. Done -----
711 header "Done"
712 if [ "$DRY" = 1 ]; then
713     echo "    (dry-mode finished; no follow-up commands were issued.)"
714     exit 0
715 fi
716
717 cat <<'DONE'
718
719 Manual follow-ups (all of these; the script stops short of running sudo
720 tailscale up so you can verify the brew install is actually live first):
721
722 1. System Settings Privacy & Security Local Network Tailscale ON
723    macOS will normally pop the prompt the first time you run `tailscale`.
724    If it didn't, trigger it via the menu bar check.
725
726 2. Re-engage the tailnet + exit node:
727     sudo tailscale up --ssh=true --accept-dns=false \
728         --exit-node="$(cat .gateway_ip | tr -d '\r' | awk 'NR==1{print $1; exit}')" \
729         --exit-node-allow-lan-access=true
730
731 3. Re-run: ./toggle.sh
732     (re-installs the com.nullexit.wake-recovery LaunchAgent wiped

```

```

730         in Step 5 to clear the slate and re-engages the gateway)
731
732     4. Final verification:
733         curl -s https://www.cloudflare.com/cdn-cgi/trace | grep -E '^(warp|ip|colo)=|
734         expect: warp=on, ip=<Cloudflare IP>, colo=YYZ/YUL
735
736     Container-side (gateway / 100.100.21.8) Tailscale is unaffected by this
737     script. Other tailnet devices (S24, iPhone, Windows) are unaffected.
738
739 DONE

```

41 scripts/routing-fix.sh

```

1  #!/bin/sh
2  # routing-fix: Maintains routing for SOCKS5 proxy traffic through WARP (tun0)
3  # and FORWARD rules for Tailscale exit-node return traffic.
4  #
5  # Runs inside the warp container's network namespace. This script ensures:
6  # 1. Table 200 has default via tun0 (for SOCKS5 proxy traffic from 172.18.0.2)
7  # with the WARP endpoint exception via eth0 (prevents tunnel loop)
8  # 2. The CGNAT ip rule (100.64.0.0/10 table 52 tailscale routes) is maintained for Tailscale
9  # 3. FORWARD RELATED,ESTABLISHED rule allows return traffic for exit-node
10 # forwarded connections (S24 tailscale0 eth0 host internet)
11 #
12 # CRITICAL: Does NOT touch the main table's default route. Gluetun handles that
13 # internally via its own routing rules and WireGuard fwmark (0xca6c table 51820).
14 # Overriding the main table default would create a loop: encrypted WireGuard
15 # packets exiting tun0 would match default dev tun0 and re-enter the tunnel.
16
17 set -e
18
19 trap 'echo "routing-fix: Caught signal, exiting..."; exit 0' TERM INT QUIT
20
21 sleep 15 &
22 wait $! || true
23
24 echo "routing-fix: Setting up routing for SOCKS5 proxy through WARP (tun0)..."
25
26 # Dynamically detect Docker subnet from eth0
27 DOCKER_NET=$(ip route show dev eth0 | grep -v default | head -1 | awk '{print $1}')
28 DOCKER_GW=$(ip route show default | head -1 | awk '{print $3}')
29 WARP_ENDPOINT="${WARP_ENDPOINT}"
30 TAILSCALE_ALLOW_LAN_P2P="${TAILSCALE_ALLOW_LAN_P2P:-false}"
31 IP_BLOCKLIST_FILE="/userfilters/ip_blocklist.ipset"
32 LAST_IP_MTIME=""
33 echo "routing-fix: Docker network: ${DOCKER_NET}, gateway: ${DOCKER_GW}"
34
35 # Table 200: Used by ip rule 100 (from 172.18.0.2) for SOCKS5 proxy traffic
36 # WARP endpoint goes via eth0 to avoid tunnel loop, everything else via tun0
37 ip route add "${WARP_ENDPOINT}" via "${DOCKER_GW}" dev eth0 table 200 2>/dev/null || \
38 ip route replace "${WARP_ENDPOINT}" via "${DOCKER_GW}" dev eth0 table 200 2>/dev/null || true
39 ip route replace default dev tun0 table 200 2>/dev/null || true
40
41 # Main table: Just ensure the Docker subnet route exists via eth0
42 # (gluetun owns the default route here do NOT touch it)
43 ip route replace "${DOCKER_NET}" dev eth0 2>/dev/null || true
44
45 # FORWARD RELATED,ESTABLISHED: Allow return traffic for exit-node connections.
46 # The container has BOTH iptables (nftables backend) and iptables-legacy
47 # loaded. Gluetun's post-rules.txt uses the nftables backend, while Tailscale
48 # uses the legacy backend. Packets go through BOTH stacks, so we add the rule
49 # to both to ensure it survives regardless of which backend has policy DROP.
50 #
51 # Packet flow: S24 outgoing (tailscale0->eth0) ACCEPTed by first rule.
52 # Return traffic (eth0->eth0 via table 52 to host) needs RELATED,ESTABLISHED
53 # to match the conntrack entry created by the outgoing flow.
54 add_fwd_related_established() {
55     # nftables backend (used by gluetun's post-rules.txt)
56     if ! iptables -C FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT 2>/dev/null; then
57         iptables -A FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT 2>/dev/null || true
58     fi
59     # legacy backend (used by Tailscale's ts-forward)
60     if command -v iptables-legacy >/dev/null 2>&1; then
61         if ! iptables-legacy -C FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT 2>/dev/null; then
62             iptables-legacy -A FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT 2>/dev/null || true
63         fi

```

```

64 fi
65 }
66
67 add_fwd_related_established
68
69 # Policy Routing for Tailscale P2P:
70 # Tailscale sends its encrypted mesh UDP packets from port 41642.
71 # If these go out tun0 (WARP), Cloudflare's strict NAT drops the returning
72 # hole-punch packets, causing Tailscale to fail P2P and fall back to DERP relays (high latency).
73 # We mark packets originating from port 41642 and force them out the main table (eth0).
74 #
75 # LAN P2P is controlled by two sources (in priority order):
76 # 1. /app/.lan_p2p_detected written by watcher.sh on macOS on every network change
77 #    using airport -I (802.1x detection) + AP isolation probe. Overrides .env.
78 # 2. TAILSCALE_ALLOW_LAN_P2P env var (from .env) static fallback.
79 # When disabled, we explicitly DROP local RFC1918-destined Tailscale UDP so the
80 # phone cleanly falls back to DERP rather than getting poisoned by macOS gvproxy SNAT.
81 add_tailscale_p2p_bypass() {
82     # Dynamic override from watcher.sh (network auto-detection) takes priority over .env
83     local allow_p2p="${TAILSCALE_ALLOW_LAN_P2P:-false}"
84     if [ -f "/app/.lan_p2p_detected" ]; then
85         allow_p2p=$(cat "/app/.lan_p2p_detected" 2>/dev/null | tr -d '[:space:]')
86     fi
87
88     # Always allow P2P directly to the Docker host gateway (the Mac running this container).
89     # The gateway container is always co-located with the host Mac on the same machine,
90     # so direct Tailscale P2P via the Docker bridge is always safe and avoids DERP overhead.
91     # This RETURN rule must be inserted BEFORE the general DROP rules so it takes priority.
92     local default_gw
93     default_gw=$(ip route show default | awk '{print $3}')
94     if [ -n "$default_gw" ]; then
95         if ! iptables -t mangle -C OUTPUT -p udp --sport 41642 -d "${default_gw}/32" -j RETURN 2>/dev/null; then
96             iptables -t mangle -I OUTPUT 1 -p udp --sport 41642 -d "${default_gw}/32" -j RETURN 2>/dev/null || true
97         fi
98     fi
99
100     # Also explicitly whitelist all physical IPs of the host Mac (e.g. 172.17.52.223).
101     # The Tailscale control plane registers the Mac using its physical IP, so the
102     # container will attempt P2P handshakes using that physical IP instead of the
103     # Docker bridge IP. If this physical IP falls within 172.16.0.0/12, it would be dropped.
104     if [ -f "/app/.host_ips" ]; then
105         while IFS= read -r host_ip; do
106             [ -z "$host_ip" ] && continue
107             if ! iptables -t mangle -C OUTPUT -p udp --sport 41642 -d "${host_ip}/32" -j RETURN 2>/dev/null; then
108                 iptables -t mangle -I OUTPUT 1 -p udp --sport 41642 -d "${host_ip}/32" -j RETURN 2>/dev/null || true
109             fi
110         done < "/app/.host_ips"
111     fi
112
113     if [ "${allow_p2p}" != "true" ]; then
114         # Drop Tailscale UDP packets destined for local subnets to prevent macOS gvproxy SNAT
115         # endpoint poisoning. When disabled (campus/enterprise WPA2-Enterprise with AP isolation),
116         # dropping these forces the S24 to use DERP relays reliably.
117         for subnet in "192.168.0.0/16" "10.0.0.0/8" "172.16.0.0/12"; do
118             if ! iptables -t mangle -C OUTPUT -p udp --sport 41642 -d "$subnet" -j DROP 2>/dev/null; then
119                 iptables -t mangle -A OUTPUT -p udp --sport 41642 -d "$subnet" -j DROP 2>/dev/null || true
120             fi
121         done
122     else
123         # Remove DROP rules if they exist (switching from restricted to allowed)
124         for subnet in "192.168.0.0/16" "10.0.0.0/8" "172.16.0.0/12"; do
125             while iptables -t mangle -D OUTPUT -p udp --sport 41642 -d "$subnet" -j DROP 2>/dev/null; do :; done
126         done
127     fi
128
129     # Bypass STUN and P2P packets to public IPs (DERP relays and remote peers) out eth0
130     if ! iptables -t mangle -C OUTPUT -p udp --sport 41642 -j MARK --set-mark 0x8888 2>/dev/null; then
131         iptables -t mangle -A OUTPUT -p udp --sport 41642 -j MARK --set-mark 0x8888 2>/dev/null || true
132     fi
133     if ! ip rule show | grep -q 'fwmark 0x8888'; then
134         ip rule add fwmark 0x8888 lookup 254 pref 98 2>/dev/null || true
135     fi
136 }
137
138 add_tailscale_p2p_bypass
139
140 create_log_drop_chain() {
141     local ipt="$1"
142     local chain="$2"
143     local prefix="$3"
144

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145 if ! $ipt -N "$chain" 2>/dev/null; then
146     $ipt -F "$chain" 2>/dev/null || true
147 fi
148 $ipt -A "$chain" -j LOG --log-prefix "$prefix"
149 $ipt -A "$chain" -j DROP 2>/dev/null || true
150 }
151
152 add_country_block() {
153     # To add a country to the blocklist, simply define the 'BLOCKED_COUNTRIES' variable in .env with 2-letter ISO
154     # country codes.
155     # You can find the country code by looking at the filename on http://www.ipdeny.com/ipblocks/data/countries/
156     # For example, to block China (cn) and Russia (ru), set: BLOCKED_COUNTRIES="cn ru" in your .env file
157     for zone in $BLOCKED_COUNTRIES; do
158         set_name="block_${zone}"
159         zone_upper=$(echo "$zone" | tr 'a-z' 'A-Z')
160         chain_name="LOG_DROP_${zone_upper}"
161         log_prefix="IP_BLOCK_${zone_upper}: "
162
163         # Create the ipset if it doesn't exist
164         if ! ipset list "$set_name" >/dev/null 2>&1; then
165             ipset create "$set_name" hash:net 2>/dev/null || true
166             echo "routing-fix: Downloading ${zone_upper} IPs for blocklist..."
167             curl -sS "http://www.ipdeny.com/ipblocks/data/countries/${zone}.zone" | grep -v '^#' | while read -r
168             subnet; do
169                 [ -n "$subnet" ] && ipset add "$set_name" "$subnet" 2>/dev/null || true
170             done
171         fi
172
173         # Apply LOG_DROP rule to both backends
174         for ipt in iptables iptables-legacy; do
175             command -v "$ipt" >/dev/null 2>&1 || continue
176
177             create_log_drop_chain "$ipt" "$chain_name" "$log_prefix"
178
179             # Clean up old plain DROP rules to ensure we hit the log-and-drop chains
180             while $ipt -D FORWARD -m set --match-set "$set_name" dst -j DROP 2>/dev/null; do ;; done
181
182             if ! $ipt -C FORWARD -m set --match-set "$set_name" dst -j "$chain_name" 2>/dev/null; then
183                 $ipt -I FORWARD -m set --match-set "$set_name" dst -j "$chain_name" 2>/dev/null || true
184             fi
185         done
186     done
187 }
188
189 add_ip_blocklist() {
190     # Only reload when the compiled file has actually changed (mtime check).
191     # This prevents a pointless ipset restore on every 30-second loop tick.
192     if [ ! -f "$IP_BLOCKLIST_FILE" ]; then
193         return 0
194     fi
195
196     CURRENT_MTIME=$(stat -c %Y "$IP_BLOCKLIST_FILE" 2>/dev/null || echo "0")
197     if [ "$CURRENT_MTIME" != "$LAST_IP_MTIME" ]; then
198         echo "routing-fix: IP blocklist changed, reloading..."
199         # ipset restore handles the atomic swap internally:
200         # create_new populate swap with live destroy_new
201         if ipset restore < "$IP_BLOCKLIST_FILE" 2>/dev/null; then
202             LAST_IP_MTIME="$CURRENT_MTIME"
203             echo "routing-fix: IP blocklist loaded ($(ipset list block_malicious | grep -c '[0-9]') entries)."
204         else
205             echo "routing-fix: Warning: ipset restore failed. Retrying next cycle."
206             return 1
207         fi
208     fi
209
210     # Apply FORWARD rules in BOTH iptables backends.
211     for ipt in iptables iptables-legacy; do
212         command -v "$ipt" >/dev/null 2>&1 || continue
213
214         create_log_drop_chain "$ipt" LOG_DROP_MALICIOUS_DST "IP_BLOCK_MALICIOUS_DST: "
215         create_log_drop_chain "$ipt" LOG_DROP_MALICIOUS_SRC "IP_BLOCK_MALICIOUS_SRC: "
216
217         # Clean up old plain DROP rules to ensure we hit the log-and-drop chains
218         while $ipt -D FORWARD -m set --match-set block_malicious dst -j DROP 2>/dev/null; do ;; done
219         while $ipt -D FORWARD -m set --match-set block_malicious src -j DROP 2>/dev/null; do ;; done
220
221         if ! $ipt -C FORWARD -m set --match-set block_malicious dst -j LOG_DROP_MALICIOUS_DST 2>/dev/null; then
222             $ipt -I FORWARD -m set --match-set block_malicious dst -j LOG_DROP_MALICIOUS_DST 2>/dev/null || true
223         fi
224
225         if ! $ipt -C FORWARD -m set --match-set block_malicious src -j LOG_DROP_MALICIOUS_SRC 2>/dev/null; then

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224     $ipt -I FORWARD -m set --match-set block_malicious src -j LOG_DROP_MALICIOUS_SRC 2>/dev/null || true
225 fi
226 done
227 }
228
229 # Start background block logger (DNS + IP)
230 if ! pgrep -f "nullexit-logger" >/dev/null; then
231     echo "routing-fix: Starting background block logger..."
232     nullexit-logger &
233 fi
234
235 add_country_block
236 add_ip_blocklist
237
238 echo 'routing-fix: Routes applied.'
239
240 UNHEALTHY_COUNT=0
241
242 # Re-assert loop (every 30 seconds)
243 while true; do
244     # Table 200 routes
245     ip route replace "${WARP_ENDPOINT}" via "${DOCKER_GW}" dev eth0 table 200 2>/dev/null || true
246
247     # Tunnel health check
248     if ! curl -m 3 -sS --interface tun0 https://cloudflare.com/cdn-cgi/trace >/dev/null 2>&1; then
249         UNHEALTHY_COUNT=$((UNHEALTHY_COUNT + 1))
250     else
251         UNHEALTHY_COUNT=0
252         if [ -f "/app/TUNNEL_FAILED_CLOSED.marker" ]; then
253             rm -f "/app/TUNNEL_FAILED_CLOSED.marker"
254         fi
255     fi
256
257     if [ "$UNHEALTHY_COUNT" -ge 3 ]; then
258         echo "routing-fix: Tunnel health check failed $UNHEALTHY_COUNT times. Failing closed."
259         ip route del default dev tun0 table 200 2>/dev/null || true
260         ip route replace blackhole default table 200 2>/dev/null || true
261         touch "/app/TUNNEL_FAILED_CLOSED.marker"
262     else
263         ip route replace default dev tun0 table 200 2>/dev/null || true
264     fi
265
266     # Main table: keep Docker subnet route
267     ip route replace "${DOCKER_NET}" dev eth0 2>/dev/null || true
268
269     # CGNAT ip rule for Tailscale
270     if ! ip rule show | grep -q '100.64.0.0/10'; then
271         ip rule add to 100.64.0.0/10 lookup 52 pref 99 2>/dev/null || true
272         echo 'routing-fix: CGNAT rule missing, re-injected'
273     fi
274
275     # FORWARD RELATED, ESTABLISHED: Allow return traffic in both iptables
276     # backends. Gluetun may reset the nftables ruleset on VPN reconnect,
277     # and Tailscale may reset the legacy ruleset on auth refresh.
278     add_fwd_related_established
279
280     # Ensure P2P packets bypass WARP to maintain low latency
281     add_tailscale_p2p_bypass
282
283     # Enforce country blocklist
284     add_country_block
285
286     # Enforce IP blocklist (reloads automatically when file changes)
287     add_ip_blocklist
288
289     sleep 30 &
290     wait $! || true
291 done

```

42 scripts/rule-compiler/go.mod

```

1 module nullexit/rule-compiler
2
3 go 1.22

```

43 scripts/rule-compiler/main.go

```
1 package main
2
3 import (
4     "bufio"
5     "bytes"
6     "crypto/md5"
7     "encoding/base64"
8     "encoding/hex"
9     "fmt"
10    "io"
11    "net/http"
12    "net/netip"
13    "os"
14    "os/exec"
15    "path/filepath"
16    "regexp"
17    "sort"
18    "strings"
19    "sync"
20    "time"
21 )
22
23 const (
24     BlackListPath = "black_list.txt"
25     WhiteListPath = "white_list.txt"
26     OutputPath     = "adguard/work/userfilters/compiled_rules.txt"
27     IpOutputPath   = "adguard/work/userfilters/ip_blocklist.ipset"
28     IpCacheDir     = "adguard/work/userfilters/cache/ip"
29     CacheDir       = "adguard/work/userfilters/cache"
30 )
31
32 var CoreLists = []string{
33     "https://raw.githubusercontent.com/anudeepND/blacklist/master/adservers.txt",
34     "https://raw.githubusercontent.com/anudeepND/blacklist/master/facebook.txt",
35     "https://raw.githubusercontent.com/crazy-max/WindowsSpyBlocker/master/data/hosts/spy.txt",
36     "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/adblock/native.samsung.txt",
37     "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/adblock/native.apple.txt",
38     "https://abp.oisd.nl/basic/",
39     "https://adaway.org/hosts.txt",
40     "https://pgl.yoyo.org/adservers/serverlist.php?hostformat=adblockplus&showintro=1&mimetype=plaintext",
41     "https://raw.githubusercontent.com/nextdns/cname-cloaking-blocklist/master/domains",
42 }
43
44 var MediumAdditions = []string{
45     "https://raw.githubusercontent.com/lightswitch05/hosts/master/docs/lists/facebook-extended.txt",
46     "https://someonewhocares.org/hosts/zero/hosts",
47     "https://urlhaus.abuse.ch/downloads/hostfile/",
48 }
49
50 var HeavyAdditions = []string{
51     "https://raw.githubusercontent.com/StevenBlack/hosts/master/hosts",
52     "https://raw.githubusercontent.com/Perflyst/PiHoleBlocklist/master/SmartTV-AGH.txt",
53     "https://raw.githubusercontent.com/DandelionSprout/adfilt/master/GameConsoleAdblockList.txt",
54 }
55
56 var IpBlockLists = []string{
57     "https://feodotracker.abuse.ch/downloads/ipblocklist.txt",
58     "https://www.spamhaus.org/drop/drop.txt",
59     "https://rules.emergingthreats.net/fwrules/emerging-Block-IPs.txt",
60     "https://cinsscore.com/list/ci-badguys.txt",
61 }
62
63 func getProfiles() map[string][]string {
64     profiles := make(map[string][]string)
65
66     light := append([]string(nil), CoreLists...)
67     light = append(light, "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/domains/light.txt")
68     profiles["light"] = light
69
70     medium := append([]string(nil), CoreLists...)
71     medium = append(medium, MediumAdditions...)
72     medium = append(medium, "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/domains/multi.txt")
73     profiles["medium"] = medium
74
75     heavy := append([]string(nil), CoreLists...)
76     heavy = append(heavy, MediumAdditions...)
77     heavy = append(heavy, HeavyAdditions...)
```

```

78 heavy = append(heavy, "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/domains/pro.txt")
79 profiles["heavy"] = heavy
80
81 return profiles
82 }
83
84 func loadEnvProfile() string {
85     profile := "medium"
86
87     if bytesData, err := os.ReadFile(".env"); err == nil {
88         scanner := bufio.NewScanner(strings.NewReader(string(bytesData)))
89         for scanner.Scan() {
90             line := strings.TrimSpace(scanner.Text())
91             if line != "" && !strings.HasPrefix(line, "#") {
92                 parts := strings.SplitN(line, "=", 2)
93                 if len(parts) == 2 {
94                     key := strings.TrimSpace(parts[0])
95                     val := strings.TrimSpace(parts[1])
96                     if key == "GATEWAY_RULE_PROFILE" {
97                         val = strings.Trim(val, "\"")
98                         profile = strings.ToLower(val)
99                     }
100                 }
101             }
102         }
103     } else if !os.IsNotExist(err) {
104         fmt.Printf("Warning: Failed to read .env file for profile configuration (%v). Using default.\n", err)
105     }
106
107     if envVal := os.Getenv("GATEWAY_RULE_PROFILE"); envVal != "" {
108         profile = strings.ToLower(envVal)
109     }
110
111     profiles := getProfiles()
112     if _, exists := profiles[profile]; !exists {
113         fmt.Printf("Warning: Profile '%s' is invalid. Falling back to 'medium'.\n", profile)
114         profile = "medium"
115     }
116     return profile
117 }
118
119 func loadDomains(filepath string) map[string]bool {
120     domains := make(map[string]bool)
121     if bytesData, err := os.ReadFile(filepath); err == nil {
122         scanner := bufio.NewScanner(strings.NewReader(string(bytesData)))
123         for scanner.Scan() {
124             line := strings.TrimSpace(scanner.Text())
125             if line != "" && !strings.HasPrefix(line, "#") {
126                 domains[strings.ToLower(line)] = true
127             }
128         }
129     }
130     return domains
131 }
132
133 func parseDomainsFromContent(content string) map[string]bool {
134     domains := make(map[string]bool)
135     scanner := bufio.NewScanner(strings.NewReader(content))
136     ipRegex := regexp.MustCompile(`^\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}$`)
137     agRegex := regexp.MustCompile(`^\|([a-z0-9.-]+)\|` )
138     domainRegex := regexp.MustCompile(`^[a-z0-9.-]+$`)
139
140     for scanner.Scan() {
141         line := strings.TrimSpace(scanner.Text())
142         if line == "" || strings.HasPrefix(line, "!") || strings.HasPrefix(line, "[") {
143             continue
144         }
145         parts := strings.SplitN(line, "#", 2)
146         line = strings.TrimSpace(parts[0])
147         if line == "" {
148             continue
149         }
150
151         fields := strings.Fields(line)
152         var rule string
153         if len(fields) >= 2 {
154             if ipRegex.MatchString(fields[0]) {
155                 rule = fields[1]
156             } else {
157                 rule = fields[0]
158             }
159         }
160     }

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```

159     } else {
160         rule = fields[0]
161     }
162
163     rule = strings.TrimSpace(strings.ToLower(rule))
164
165     var domain string
166     if m := agRegex.FindStringSubmatch(rule); m != nil {
167         domain = m[1]
168     } else if domainRegex.MatchString(rule) {
169         domain = rule
170     } else {
171         domain = rule
172     }
173
174     if domain != "" && domain != "localhost" && domain != "0.0.0.0" && domain != "127.0.0.1" && domain != "
175         broadcasthost" {
176         domains[domain] = true
177     }
178     return domains
179 }
180
181 // WARNING: Parsing AdGuardHome.yaml using regex instead of a proper YAML parser
182 // is brittle and assumes the exact format currently emitted by AdGuard.
183 // If an AdGuard version bump updates the configuration schema format,
184 // this parser will fail silently or loudly and should be refactored to use gopkg.in/yaml.v3.
185 func getAdguardNativeLists() []string {
186     yamlPath := "adguard/conf/AdGuardHome.yaml"
187     var enabledUrls []string
188     if _, err := os.Stat(yamlPath); os.IsNotExist(err) {
189         return enabledUrls
190     }
191
192     contentBytes, err := os.ReadFile(yamlPath)
193     if err != nil {
194         fmt.Printf("Warning: Failed to parse AdGuardHome.yaml for native lists (%v)\n", err)
195         return enabledUrls
196     }
197     content := string(contentBytes)
198
199     filtersRegex := regexp.MustCompile(`(?ms)^filters:(.*)"?(?:[a-zA-Z_]+|\z)`
200     match := filtersRegex.FindStringSubmatch(content)
201     if len(match) > 1 {
202         filtersText := match[1]
203         blocks := strings.Split(filtersText, "\n - ")
204         urlRegex := regexp.MustCompile(`url:\s*(\S+)`)
205
206         for _, block := range blocks {
207             if strings.TrimSpace(block) == "" {
208                 continue
209             }
210             if strings.Contains(block, "enabled: true") {
211                 urlMatch := urlRegex.FindStringSubmatch(block)
212                 if len(urlMatch) > 1 {
213                     url := urlMatch[1]
214                     if !strings.Contains(url, "compiled_rules.txt") {
215                         enabledUrls = append(enabledUrls, url)
216                     }
217                 }
218             }
219         }
220     }
221     return enabledUrls
222 }
223
224 func getAdguardCompiledRulesCachePath() string {
225     yamlPath := "adguard/conf/AdGuardHome.yaml"
226     if _, err := os.Stat(yamlPath); os.IsNotExist(err) {
227         return ""
228     }
229
230     contentBytes, err := os.ReadFile(yamlPath)
231     if err != nil {
232         fmt.Printf("Warning: Failed to resolve compiled_rules.txt cache path (%v)\n", err)
233         return ""
234     }
235     content := string(contentBytes)
236
237     filtersRegex := regexp.MustCompile(`(?ms)^filters:(.*)"?(?:[a-zA-Z_]+|\z)`
238     match := filtersRegex.FindStringSubmatch(content)

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239 if len(match) > 1 {
240     blocks := strings.Split(match[1], "\n - ")
241     idRegex := regexp.MustCompile(`id:\s*(\d+)`)
242     for _, block := range blocks {
243         if !strings.Contains(block, "compiled_rules.txt") {
244             continue
245         }
246         idMatch := idRegex.FindStringSubmatch(block)
247         if len(idMatch) > 1 {
248             return fmt.Sprintf("adguard/work/data/filters/%s.txt", idMatch[1])
249         }
250     }
251 }
252 return ""
253 }
254
255 func handleFetchError(urlStr string, cacheFile string, err error) map[string]bool {
256     if _, statErr := os.Stat(cacheFile); statErr == nil {
257         fmt.Printf(" -> Warning: Failed to fetch %s (%v). Falling back to expired local cache.\n", urlStr, err)
258         content, readErr := os.ReadFile(cacheFile)
259         if readErr == nil {
260             domains := parseDomainsFromContent(string(content))
261             fmt.Printf(" -> Loaded from expired cache (%d domains).\n", len(domains))
262             return domains
263         }
264         fmt.Printf(" -> Warning: Failed to read expired cache (%v). Using local lists only.\n", readErr)
265     } else {
266         fmt.Printf(" -> Warning: Failed to fetch %s (%v). Using local lists only for this source.\n", urlStr, err)
267     }
268     return make(map[string]bool)
269 }
270
271 func fetchRemoteDomains(urlStr string) map[string]bool {
272     os.MkdirAll(CacheDir, 0755)
273
274     hash := md5.Sum([]byte(urlStr))
275     urlHash := hex.EncodeToString(hash[:])
276     cacheFile := filepath.Join(CacheDir, fmt.Sprintf("%s.txt", urlHash))
277
278     if stat, err := os.Stat(cacheFile); err == nil {
279         fileAge := time.Since(stat.ModTime()).Seconds()
280         if fileAge < 86400 {
281             fmt.Printf("Loading remote blacklist from cache: %s\n", urlStr)
282             content, err := os.ReadFile(cacheFile)
283             if err == nil {
284                 domains := parseDomainsFromContent(string(content))
285                 fmt.Printf(" -> Loaded from cache (copy is %.1f hours old, %d domains).\n", fileAge/3600.0, len(domains))
286             }
287             return domains
288         }
289         fmt.Printf(" -> Warning: Failed to read cache file %s (%v). Re-fetching...\n", cacheFile, err)
290     }
291
292     fmt.Printf("Fetching remote blacklist from: %s ...\n", urlStr)
293
294     req, err := http.NewRequest("GET", urlStr, nil)
295     if err != nil {
296         return handleFetchError(urlStr, cacheFile, err)
297     }
298     req.Header.Set("User-Agent", "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)")
299
300     client := &http.Client{Timeout: 15 * time.Second}
301     resp, err := client.Do(req)
302     if err != nil {
303         return handleFetchError(urlStr, cacheFile, err)
304     }
305     defer resp.Body.Close()
306
307     if resp.StatusCode != 200 {
308         return handleFetchError(urlStr, cacheFile, fmt.Errorf("HTTP %d", resp.StatusCode))
309     }
310
311     bodyBytes, err := io.ReadAll(resp.Body)
312     if err != nil {
313         return handleFetchError(urlStr, cacheFile, err)
314     }
315
316     content := string(bodyBytes)
317     domains := parseDomainsFromContent(content)
318

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```

319     if len(domains) < 10 {
320         return handleFetchError(urlStr, cacheFile, fmt.Errorf("Sanity check failed: only found %d domains. Possible
321             404 or bad URL", len(domains)))
322     }
323     err = os.WriteFile(cacheFile, bodyBytes, 0644)
324     if err != nil {
325         fmt.Printf("-> Warning: Failed to save cache file (%v)\n", err)
326     }
327
328     fmt.Printf("-> Successfully fetched and cached %d domains.\n", len(domains))
329     return domains
330 }
331
332 func optimizeSubdomains(domains map[string]bool, listName string) map[string]bool {
333     fmt.Printf("Optimizing %s by removing redundant subdomains...\n", listName)
334     rawCount := len(domains)
335     if rawCount == 0 {
336         return make(map[string]bool)
337     }
338
339     optimized := make(map[string]bool)
340     for domain := range domains {
341         if strings.ContainsAny(domain, "|~@$/") {
342             optimized[domain] = true
343             continue
344         }
345
346         parts := strings.Split(domain, ".")
347         hasParent := false
348
349         for i := 1; i < len(parts)-1; i++ {
350             parent := strings.Join(parts[i:], ".")
351             if domains[parent] {
352                 hasParent = true
353                 break
354             }
355         }
356
357         if !hasParent {
358             optimized[domain] = true
359         }
360     }
361
362     saved := rawCount - len(optimized)
363     reduction := float64(0)
364     if rawCount > 0 {
365         reduction = (float64(saved) / float64(rawCount)) * 100
366     }
367     fmt.Printf("-> Reduced %s from %d to %d domains (-%d / %.1f%% reduction).\n", listName, rawCount, len(
368         optimized), saved, reduction)
369     return optimized
370 }
371
372 func parseIpsFromContent(content string) map[string]bool {
373     ips := make(map[string]bool)
374     scanner := bufio.NewScanner(strings.NewReader(content))
375
376     for scanner.Scan() {
377         line := strings.TrimSpace(scanner.Text())
378         if line == "" || strings.HasPrefix(line, "#") || strings.HasPrefix(line, ";") {
379             continue
380         }
381
382         fields := strings.Fields(line)
383         if len(fields) > 0 {
384             token := strings.TrimRight(fields[0], ";,")
385             if _, err := netip.ParsePrefix(token); err == nil {
386                 ips[token] = true
387             } else if _, err := netip.ParseAddr(token); err == nil {
388                 ips[token] = true
389             }
390         }
391     }
392     return ips
393 }
394
395 func handleIpFetchError(urlStr string, cacheFile string, err error) map[string]bool {
396     if _, statErr := os.Stat(cacheFile); statErr == nil {
397         fmt.Printf("-> Warning: fetch failed (%v). Falling back to stale cache.\n", err)
398         content, readErr := os.ReadFile(cacheFile)

```

```

398     if readErr == nil {
399         ips := parseIpsFromContent(string(content))
400         fmt.Printf(" -> %d IPs loaded from stale cache.\n", len(ips))
401         return ips
402     }
403     fmt.Printf(" -> Warning: stale cache also failed (%v).\n", readErr)
404 } else {
405     fmt.Printf(" -> Warning: %s failed (%v). No cache available. Skipping.\n", urlStr, err)
406 }
407 return make(map[string]bool)
408 }
409
410 func fetchRemoteIps(urlStr string) map[string]bool {
411     os.MkdirAll(IpCacheDir, 0755)
412
413     hash := md5.Sum([]byte(urlStr))
414     urlHash := hex.EncodeToString(hash[:])
415     cacheFile := filepath.Join(IpCacheDir, fmt.Sprintf("%s.txt", urlHash))
416
417     if stat, err := os.Stat(cacheFile); err == nil {
418         fileAge := time.Since(stat.ModTime()).Seconds()
419         if fileAge < 86400 {
420             fmt.Printf("Loading IP feed from cache: %s\n", urlStr)
421             content, err := os.ReadFile(cacheFile)
422             if err == nil {
423                 ips := parseIpsFromContent(string(content))
424                 fmt.Printf(" -> %d IPs (cache is %.1fh old).\n", len(ips), fileAge/3600.0)
425                 return ips
426             }
427             fmt.Printf(" -> Warning: cache read failed (%v). Re-fetching...\n", err)
428         }
429     }
430
431     fmt.Printf("Fetching IP feed: %s ...\n", urlStr)
432     req, err := http.NewRequest("GET", urlStr, nil)
433     if err != nil {
434         return handleIpFetchError(urlStr, cacheFile, err)
435     }
436     req.Header.Set("User-Agent", "Mozilla/5.0")
437
438     client := &http.Client{Timeout: 15 * time.Second}
439     resp, err := client.Do(req)
440     if err != nil {
441         return handleIpFetchError(urlStr, cacheFile, err)
442     }
443     defer resp.Body.Close()
444
445     if resp.StatusCode != 200 {
446         return handleIpFetchError(urlStr, cacheFile, fmt.Errorf("HTTP %d", resp.StatusCode))
447     }
448
449     bodyBytes, err := io.ReadAll(resp.Body)
450     if err != nil {
451         return handleIpFetchError(urlStr, cacheFile, err)
452     }
453
454     content := string(bodyBytes)
455     ips := parseIpsFromContent(content)
456     if len(ips) < 1 {
457         return handleIpFetchError(urlStr, cacheFile, fmt.Errorf("Sanity check failed: only %d IPs found. Possible 404", len(ips)))
458     }
459
460     err = os.WriteFile(cacheFile, bodyBytes, 0644)
461     if err != nil {
462         fmt.Printf(" -> Warning: cache write failed (%v)\n", err)
463     }
464
465     fmt.Printf(" -> %d IPs fetched and cached.\n", len(ips))
466     return ips
467 }
468
469 func compileIpBlocklist() int {
470     fmt.Println("\n IP Blocklist Compilation ")
471
472     allIps := make(map[string]bool)
473     var mu sync.Mutex
474     var wg sync.WaitGroup
475
476     maxThreads := 8
477     if len(IpBlockLists) < 8 {

```

```

478     maxThreads = len(IpBlockLists)
479 }
480 semaphore := make(chan struct{}, maxThreads)
481
482 for _, urlStr := range IpBlockLists {
483     wg.Add(1)
484     go func(u string) {
485         defer wg.Done()
486         semaphore <- struct{}{}
487         defer func() { <-semaphore }()
488
489         ips := fetchRemoteIps(u)
490         mu.Lock()
491         for ip := range ips {
492             allIps[ip] = true
493         }
494         mu.Unlock()
495     }(urlStr)
496 }
497 wg.Wait()
498
499 fmt.Printf("Total unique entries before filtering: %d\n", len(allIps))
500
501 reservedStr := []string{
502     "10.0.0.0/8",
503     "172.16.0.0/12",
504     "192.168.0.0/16",
505     "127.0.0.0/8",
506     "169.254.0.0/16",
507     "100.64.0.0/10",
508     "0.0.0.0/8",
509 }
510 var reserved []netip.Prefix
511 for _, s := range reservedStr {
512     p, _ := netip.ParsePrefix(s)
513     reserved = append(reserved, p)
514 }
515
516 cleanIps := make(map[string]bool)
517 for entry := range allIps {
518     var prefix netip.Prefix
519     p, err := netip.ParsePrefix(entry)
520     if err != nil {
521         addr, err2 := netip.ParseAddr(entry)
522         if err2 != nil {
523             continue
524         }
525         prefix = netip.PrefixFrom(addr, addr.BitLen())
526     } else {
527         prefix = p
528     }
529
530     overlaps := false
531     for _, r := range reserved {
532         if r.Overlaps(prefix) {
533             overlaps = true
534             break
535         }
536     }
537     if !overlaps {
538         cleanIps[entry] = true
539     }
540 }
541
542 removed := len(allIps) - len(cleanIps)
543 if removed > 0 {
544     fmt.Printf(" -> Removed %d private/reserved entries.\n", removed)
545 }
546 fmt.Printf(" -> Final IP blocklist: %d entries.\n", len(cleanIps))
547
548 os.MkdirAll(filepath.Dir(IpOutputPath), 0755)
549
550 f, err := os.Create(IpOutputPath)
551 if err != nil {
552     fmt.Printf("Error writing IP output file: %v\n", err)
553     return len(cleanIps)
554 }
555 defer f.Close()
556
557 f.WriteString("# nullexit Compiled IP Blocklist\n")
558 f.WriteString("# Sources: Feodo Tracker, Spamhaus DROP/EDROP, Emerging Threats, CINS\n")

```

```

559 f.WriteString(fmt.Sprintf("# Entries: %d\n\n", len(cleanIps)))
560
561 f.WriteString("create block_malicious_new hash:net maxelem 200000 -exist\n")
562
563 var sortedIps []string
564 for ip := range cleanIps {
565     sortedIps = append(sortedIps, ip)
566 }
567 sort.Strings(sortedIps)
568
569 for _, ip := range sortedIps {
570     f.WriteString(fmt.Sprintf("add block_malicious_new %s -exist\n", ip))
571 }
572
573 f.WriteString("create block_malicious hash:net maxelem 200000 -exist\n")
574 f.WriteString("swap block_malicious block_malicious_new\n")
575 f.WriteString("destroy block_malicious_new\n")
576
577 fmt.Printf("IP blocklist written to %s\n", IpOutputPath)
578 return len(cleanIps)
579 }
580
581 func main() {
582     profile := loadEnvProfile()
583     fmt.Printf("Active memory profile: '%s'\n", strings.ToUpper(profile))
584
585     localBlackList := loadDomains(BlackListPath)
586     whiteList := loadDomains(WhiteListPath)
587
588     blackList := make(map[string]bool)
589     for k := range localBlackList {
590         blackList[k] = true
591     }
592
593     profiles := getProfiles()
594     urlsToFetch := profiles[profile]
595
596     fmt.Printf("\nStarting concurrent downloads for %d lists...\n", len(urlsToFetch))
597     startTime := time.Now()
598
599     var mu sync.Mutex
600     var wg sync.WaitGroup
601
602     maxThreads := 16
603     if len(urlsToFetch) < 16 {
604         maxThreads = len(urlsToFetch)
605     }
606     semaphore := make(chan struct{}, maxThreads)
607
608     for _, urlStr := range urlsToFetch {
609         wg.Add(1)
610         go func(u string) {
611             defer wg.Done()
612             semaphore <- struct{}{}
613             defer func() { <-semaphore }()
614
615             domains := fetchRemoteDomains(u)
616             mu.Lock()
617             for d := range domains {
618                 blackList[d] = true
619             }
620             mu.Unlock()
621         }(urlStr)
622     }
623     wg.Wait()
624
625     fmt.Printf("Finished concurrent downloads in %.2f seconds using %d threads.\n", time.Since(startTime).Seconds(), maxThreads)
626
627     adguardNativeUrls := getAdguardNativeLists()
628     var adguardNativeDomains map[string]bool
629     if len(adguardNativeUrls) > 0 {
630         fmt.Printf("\nFetching %d AdGuard native list(s) for deduplication...\n", len(adguardNativeUrls))
631         adguardNativeDomains = make(map[string]bool)
632
633         maxNativeThreads := 4
634         if len(adguardNativeUrls) < 4 {
635             maxNativeThreads = len(adguardNativeUrls)
636         }
637         nativeSemaphore := make(chan struct{}, maxNativeThreads)
638

```

```

639     for _, urlStr := range adguardNativeUrls {
640         wg.Add(1)
641         go func(u string) {
642             defer wg.Done()
643             nativeSemaphore <- struct{}{}
644             defer func() { <-nativeSemaphore }()
645
646             domains := fetchRemoteDomains(u)
647             mu.Lock()
648             for d := range domains {
649                 adguardNativeDomains[d] = true
650             }
651             mu.Unlock()
652         }(urlStr)
653     }
654     wg.Wait()
655
656     if len(adguardNativeDomains) > 0 {
657         fmt.Println("Deduplicating compiled rules against AdGuard native lists...")
658         originalSize := len(blackList)
659         for d := range adguardNativeDomains {
660             delete(blackList, d)
661         }
662         fmt.Printf(" -> Removed %d redundant rules already covered by AdGuard.\n", originalSize-len(blackList))
663     }
664 }
665
666 var contradictions []string
667 for d := range whiteList {
668     if blackList[d] {
669         contradictions = append(contradictions, d)
670     }
671 }
672
673 if len(contradictions) > 0 {
674     fmt.Printf("Detected %d contradictions. Whitelist taking priority.\n", len(contradictions))
675     sort.Strings(contradictions)
676     for _, domain := range contradictions {
677         if localBlackList[domain] {
678             fmt.Printf(" -> Removed local blacklist domain: %s\n", domain)
679         }
680         delete(blackList, domain)
681     }
682 }
683
684 blackList = optimizeSubdomains(blackList, "blacklist")
685 whiteList = optimizeSubdomains(whiteList, "whitelist")
686
687 os.MkdirAll(filepath.Dir(OutputPath), 0755)
688 outFile, err := os.Create(OutputPath)
689 if err != nil {
690     fmt.Printf("Error creating output file: %v\n", err)
691     return
692 }
693 defer outFile.Close()
694
695 outFile.WriteString("! Custom Compiled Rules (Auto-Generated)\n")
696 outFile.WriteString(fmt.Sprintf("! Memory Profile: %s\n", strings.ToUpper(profile)))
697 outFile.WriteString(fmt.Sprintf("! Total Block Rules: %d\n", len(blackList)))
698 outFile.WriteString(fmt.Sprintf("! Native AdGuard Rules: %d\n", len(adguardNativeDomains)))
699 outFile.WriteString(fmt.Sprintf("! Total Whitelist Rules: %d\n", len(whiteList)))
700
701 outFile.WriteString("! --- Blacklist Rules ---\n")
702
703 var sortedBlacklist []string
704 for d := range blackList {
705     sortedBlacklist = append(sortedBlacklist, d)
706 }
707 sort.Strings(sortedBlacklist)
708
709 for _, domain := range sortedBlacklist {
710     if strings.Contains(domain, "$") {
711         parts := strings.SplitN(domain, "$", 2)
712         dom, mod := parts[0], parts[1]
713         if strings.HasPrefix(dom, "|") {
714             if strings.HasSuffix(dom, "~") {
715                 outFile.WriteString(fmt.Sprintf("%s%s\n", dom, mod))
716             } else {
717                 outFile.WriteString(fmt.Sprintf("%s~%s\n", dom, mod))
718             }
719         } else {

```

```

720     outFile.WriteString(fmt.Sprintf("||%s~%s\n", dom, mod))
721 }
722 } else if strings.HasPrefix(domain, "/") || strings.HasPrefix(domain, "|") || strings.HasSuffix(domain, "|")
723 ) || strings.HasSuffix(domain, "~") {
724     outFile.WriteString(fmt.Sprintf("%s\n", domain))
725 } else {
726     outFile.WriteString(fmt.Sprintf("||%s~\n", domain))
727 }
728 }
729 outFile.WriteString("\n! --- Whitelist Rules ---\n")
730 var sortedWhitelist []string
731 for d := range whiteList {
732     sortedWhitelist = append(sortedWhitelist, d)
733 }
734 sort.Strings(sortedWhitelist)
735
736 for _, domain := range sortedWhitelist {
737     if strings.HasPrefix(domain, "/") || strings.HasPrefix(domain, "|") || strings.HasSuffix(domain, "|") ||
738 strings.HasSuffix(domain, "~") || strings.HasPrefix(domain, "@@") {
739         rule := domain
740         if !strings.HasPrefix(domain, "@@") {
741             rule = "@@" + domain
742         }
743         outFile.WriteString(fmt.Sprintf("%s\n", rule))
744     } else {
745         outFile.WriteString(fmt.Sprintf("@@||%s~\n", domain))
746     }
747 }
748
749 fmt.Printf("\nSuccessfully compiled %d block rules and %d allow rules to %s\n", len(blackList), len(whiteList),
750 OutputPath)
751
752 cachedFilterPath := getAdguardCompiledRulesCachePath()
753 if cachedFilterPath != "" {
754     input, err := os.ReadFile(OutputPath)
755     if err == nil {
756         err = os.WriteFile(cachedFilterPath, input, 0644)
757         if err == nil {
758             fmt.Printf("Updated AdGuard filter cache: %s\n", cachedFilterPath)
759         } else {
760             fmt.Printf("Warning: Could not update AdGuard filter cache %s (%v)\n", cachedFilterPath, err)
761         }
762     } else {
763         fmt.Printf("Warning: Could not read OutputPath to copy to cache (%v)\n", err)
764     }
765 } else {
766     fmt.Println("Warning: Could not determine compiled_rules.txt cache path from AdGuardHome.yaml; skipping
767 cache update.")
768 }
769
770 adguardReachable := false
771
772 dockerComposeYamlExists := false
773 if _, err := os.Stat("docker-compose.yml"); err == nil {
774     dockerComposeYamlExists = true
775 }
776
777 dockerPath, err := exec.LookPath("docker")
778 if err == nil && dockerComposeYamlExists {
779     cmd := exec.Command(dockerPath, "compose", "ps", "--status", "running", "-q", "adguardhome")
780     out, err := cmd.Output()
781     if err == nil && strings.TrimSpace(string(out)) != "" {
782         fmt.Println("Force-recreating AdGuard Home container to drop persisted filter state...")
783         upCmd := exec.Command(dockerPath, "compose", "up", "-d", "--force-recreate", "--no-deps", "adguardhome")
784         err := upCmd.Run()
785         if err == nil {
786             fmt.Println("AdGuard Home force-recreated successfully.")
787             time.Sleep(8 * time.Second)
788             adguardReachable = true
789         } else {
790             fmt.Println("Failed to restart AdGuard Home. Is it running?")
791         }
792     }
793 }
794
795 if adguardReachable {
796     agPass := "nullexit"
797     if bytesData, err := os.ReadFile(".env"); err == nil {
798         scanner := bufio.NewScanner(strings.NewReader(string(bytesData)))
799         for scanner.Scan() {
800             line := scanner.Text()

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```

797     if strings.HasPrefix(line, "ADGUARD_PASSWORD=") {
798         parts := strings.SplitN(line, "=", 2)
799         if len(parts) == 2 {
800             agPass = strings.Trim(strings.TrimSpace(parts[1]), "\\\"'")
801         }
802         break
803     }
804 }
805 }
806
807 creds := base64.StdEncoding.EncodeToString([]byte(fmt.Sprintf("admin:%s", agPass)))
808 req, err := http.NewRequest("POST", "http://127.0.0.1:3000/control/filtering/refresh", bytes.NewBuffer([]
809     byte("{}")))
810 if err == nil {
811     req.Header.Set("Authorization", fmt.Sprintf("Basic %s", creds))
812     req.Header.Set("Content-Type", "application/json")
813     client := &http.Client{Timeout: 15 * time.Second}
814     resp, err := client.Do(req)
815     if err == nil {
816         fmt.Printf("Triggered AdGuard filter refresh via REST API (HTTP=%d).\n", resp.StatusCode)
817         resp.Body.Close()
818     } else {
819         fmt.Printf("Warning: AdGuard filter refresh API call failed (%v). New whitelist will not load until
820             next refresh tick (~24h) or manual refresh.\n", err)
821     }
822 }
823 compileIpBlocklist()
824 }

```

44 scripts/setup-common.sh

```

1  #!/bin/bash
2  # 1. Docker
3  step "Checking Docker"
4
5  if ! command -v docker >> output.log 2>&1; then
6      echo ""
7      if [[ "$OSTYPE" == "darwin"* ]]; then
8          echo " Docker is not installed."
9          # Note: Homebrew aggressively rolls forward and discourages version pinning.
10         # This setup is confirmed stable on Colima v0.10.3 and Docker v29.6.1.
11         read -rp " Would you like to automatically install Colima & Docker via Homebrew? [y/N]: "
12         INSTALL_DOCKER
13         if [[ "$INSTALL_DOCKER" == "y" || "$INSTALL_DOCKER" == "Y" ]]; then
14             step "Installing Colima and Docker..."
15             brew install colima docker docker-compose
16             step "Starting Colima VM..."
17             colima start --memory 0.6 --vm-type vz --network-address --network-mode bridged
18         else
19             echo ""
20             echo " Please install Docker manually. Two options:"
21             echo " A) Colima (Recommended): brew install colima docker docker-compose && colima start --memory
22             0.6 --vm-type vz --network-address --network-mode bridged"
23             echo " B) Docker Desktop: https://www.docker.com/products/docker-desktop/"
24             die "Install Docker, then re-run this script."
25         fi
26     else
27         echo " Docker is not installed."
28         # Note: The Docker convenience script always fetches the latest stable release.
29         # This setup is confirmed stable on Docker Engine v29.6.1.
30         read -rp " Would you like to automatically install Docker? (Requires sudo) [y/N]: " INSTALL_DOCKER
31         if [[ "$INSTALL_DOCKER" == "y" || "$INSTALL_DOCKER" == "Y" ]]; then
32             step "Installing Docker..."
33             curl -fsSL https://get.docker.com | sh
34             sudo usermod -aG docker "$USER"
35             echo -e "\n\033[0;32m[OK] Docker installed successfully!\033[0m"
36             echo -e "\033[0;33m[!] CRITICAL: You must log out of your SSH session and log back in for Docker
37             permissions to apply.\033[0m"
38             die "Please log out, log back in, and re-run setup.sh."
39         else
40             echo " Please install Docker manually:"
41             echo " curl -fsSL https://get.docker.com | sh"
42             echo " sudo usermod -aG docker $USER && newgrp docker"
43             die "Install Docker, then re-run this script."
44         fi
45     fi
46 }

```

```

41     fi
42 fi
43 fi
44
45 if ! docker info >> output.log 2>&1; then
46     echo ""
47     if [[ "$OSTYPE" == "darwin"* ]]; then
48         echo " Docker is installed but not running."
49         echo " If using Colima:          colima start --memory 0.6 --vm-type vz --network-address --network-mode
         bridged"
50         echo " If using Docker Desktop: open the app."
51     else
52         echo " Docker is installed but not running."
53         echo " Run: sudo systemctl start docker"
54     fi
55     die "Start Docker, then re-run this script."
56 fi
57
58 if ! docker compose version >> output.log 2>&1; then
59     die "Docker Compose v2 not found. Update Docker or install the compose plugin:\n https://docs.docker.com/
        compose/install/"
60 fi
61
62 ok "Docker $(docker --version | grep -oE '[0-9.]+' | head -1) is running."
63
64 # 1.5 Python 3
65 step "Checking Python 3"
66
67 if ! command -v python3 >> output.log 2>&1; then
68     echo ""
69     if [[ "$OSTYPE" == "darwin"* ]]; then
70         echo " Python 3 is not installed."
71         echo " macOS includes it via Xcode Command Line Tools, or you can install it via Homebrew:"
72         echo "          brew install python3"
73         die "Install Python 3, then re-run this script."
74     else
75         echo " Python 3 is not installed."
76         echo " Please install it using your system's package manager. For example:"
77         echo "   Debian/Ubuntu:  sudo apt install python3"
78         echo "   Fedora:         sudo dnf install python3"
79         echo "   Arch:           sudo pacman -S python"
80         die "Install Python 3, then re-run this script."
81     fi
82 fi
83
84 # Note: This setup is confirmed stable on Python 3.9.6 (macOS default).
85 PYTHON_VER=$(python3 --version 2>&1 | head -n 1)
86 ok "$PYTHON_VER is installed."
87
88 # 2. Tailscale (host)
89 step "Checking Tailscale (host)"
90 # The host needs its own Tailscale installation separate from the Docker
91 # container so Toggle-Gateway can run 'tailscale down' before stopping
92 # containers and 'tailscale up' after starting them. Without this, the
93 # exit-node deadlock that setup.sh is designed to prevent can still occur.
94
95 # Resolve the tailscale binary: CLI (brew/package) or bundled macOS app
96 TAILSCALE_BIN=""
97 if command -v tailscale >> output.log 2>&1; then
98     TAILSCALE_BIN="tailscale"
99 elif [[ -x "/Applications/Tailscale.app/Contents/MacOS/Tailscale" ]]; then
100     TAILSCALE_BIN="/Applications/Tailscale.app/Contents/MacOS/Tailscale"
101 fi
102
103 RECOMMENDED_TS_VERSION="1.98.5"
104
105 if [[ -z "$TAILSCALE_BIN" ]]; then
106     warn "Tailscale not found on host installing..."
107
108     if [[ "$OSTYPE" == "darwin"* ]]; then
109         if command -v brew >> output.log 2>&1; then
110             step "Installing Tailscale CLI formula via Homebrew (standalone, no .app GUI)..."
111             # Note: We install Tailscale via Homebrew, targeting the stable version (recommended: 1.98.5)
112             brew install tailscale -q
113             TAILSCALE_BIN="tailscale"
114
115             step "Starting tailscaled as a per-user LaunchAgent (auto-starts at login)..."
116             if brew services start tailscale 2>&1 | grep -qE 'Successfully|started'; then
117                 ok "tailscaled LaunchAgent registered."
118             else
119                 warn "Could not auto-start tailscaled. Run 'brew services start tailscale' manually."

```

```

120         warn "For a system-wide LaunchDaemon that runs before login, run instead:"
121         warn "      sudo brew services start tailscale"
122     fi
123     echo ""
124     else
125         die "Homebrew not found. Install Tailscale manually (recommended version: ${RECOMMENDED_TS_VERSION}):\n https://tailscale.com/download/mac\n Then re-run this script."
126     fi
127
128     elif [[ "$OSTYPE" == "linux-gnu"* ]]; then
129         curl -fsSL https://tailscale.com/install.sh | sh
130         sudo systemctl enable --now tailscaled 2>> output.log || true
131         TAILSCALE_BIN="tailscale"
132
133     else
134         die "Unsupported OS. Install Tailscale manually (recommended version: ${RECOMMENDED_TS_VERSION}):\n https://tailscale.com/download\n Then re-run this script."
135     fi
136
137     TAILSCALE_VER=$((${TAILSCALE_BIN} version 2>> output.log | head -n 1)
138     if [[ "$TAILSCALE_VER" != "$RECOMMENDED_TS_VERSION" ]]; then
139         warn "Tailscale installed, but version (${TAILSCALE_VER}) differs from recommended version ($RECOMMENDED_TS_VERSION)."
140     else
141         ok "Tailscale installed (version $TAILSCALE_VER)."
142     fi
143 else
144     TAILSCALE_VER=$((${TAILSCALE_BIN} version 2>> output.log | head -n 1)
145     if [[ "$TAILSCALE_VER" != "$RECOMMENDED_TS_VERSION" ]]; then
146         warn "Tailscale is already installed (${TAILSCALE_VER}), but recommended version is $RECOMMENDED_TS_VERSION for host/container consistency."
147     else
148         ok "Tailscale is already installed at recommended version (${TAILSCALE_VER})."
149     fi
150 fi
151
152 # Authenticate the host machine if it isn't already connected.
153 # This is an interactive step it opens a browser login URL.
154 # We do it now, before containers start, so Toggle-Gateway can
155 # safely run 'tailscale down/up' from day one.
156 if ! ${TAILSCALE_BIN} status >> output.log 2>&1; then
157     echo ""
158     echo " Your host machine needs to join your Tailscale network."
159     echo " A browser window or login URL will appear authenticate, then"
160     echo " return here. The script will continue automatically."
161     echo ""
162     if [[ "$OSTYPE" == "linux-gnu"* ]]; then
163         sudo ${TAILSCALE_BIN} up
164     else
165         ${TAILSCALE_BIN} up
166     fi
167     ok "Host machine connected to Tailscale."
168 else
169     ok "Host machine is already connected to Tailscale."
170 fi
171
172 # 3. wgcf
173 step "Checking wgcf (Cloudflare WARP key tool)"
174
175 if ! command -v wgcf >> output.log 2>&1; then
176     warn "wgcf not found installing..."
177
178     if [[ "$OSTYPE" == "darwin"* ]]; then
179         command -v brew >> output.log 2>&1 || die "Homebrew not found.\nInstall wgcf manually: https://github.com/ViRb3/wgcf/releases"
180         brew install wgcf -q
181
182     elif [[ "$OSTYPE" == "linux-gnu"* ]]; then
183         ARCH=$(uname -m)
184         case $ARCH in
185             x86_64) WGCF_ARCH="amd64" ;;
186             aarch64) WGCF_ARCH="arm64" ;;
187             armv7l) WGCF_ARCH="armv7" ;;
188             *) die "Unsupported architecture: $ARCH.\nInstall wgcf manually: https://github.com/ViRb3/wgcf/releases" ;;
189         esac
190
191         echo " Fetching latest release..."
192         WGCF_VERSION=$(curl -sf https://api.github.com/repos/ViRb3/wgcf/releases/latest \
193             | grep 'tag_name' | cut -d'"' -f4)
194         [[ -z "$WGCF_VERSION" ]] && die "Could not fetch wgcf version. Check your internet connection."

```

```

195     sudo curl -sfl \
196         "https://github.com/ViRb3/wgcf/releases/download/${WGCF_VERSION}/wgcf_${WGCF_VERSION#v}_linux_${
197         WGCF_ARCH}" \
198         -o /usr/local/bin/wgcf
199     sudo chmod +x /usr/local/bin/wgcf
200 else
201     die "Unsupported OS. Install wgcf manually: https://github.com/ViRb3/wgcf/releases"
202 fi
203 fi
204
205 ok "wgcf is ready."
206
207 # 4. WARP profile
208 step "Generating Cloudflare WARP profile"
209
210 if [[ -f "wgcf-profile.conf" ]]; then
211     warn "wgcf-profile.conf already exists skipping key generation."
212 else
213     wgcf register --accept-tos 2>> output.log || die "wgcf register failed. Check your internet connection."
214     wgcf generate 2>> output.log || die "wgcf generate failed."
215     ok "WARP profile generated."
216 fi
217
218 # Parse keys (IPv4 address only gluetun doesn't need the IPv6 one)
219 PRIVATE_KEY=$(awk '/PrivateKey/{print $3}' wgcf-profile.conf)
220 PUBLIC_KEY=$(awk '/PublicKey/{print $3}' wgcf-profile.conf)
221 ADDRESSES=$(awk '/Address/{print $3}' wgcf-profile.conf | grep -v ':' | head -1)
222
223 [[ -z "$PRIVATE_KEY" ]] && die "Could not parse PrivateKey from wgcf-profile.conf"
224 [[ -z "$PUBLIC_KEY" ]] && die "Could not parse PublicKey from wgcf-profile.conf"
225 [[ -z "$ADDRESSES" ]] && die "Could not parse IPv4 Address from wgcf-profile.conf"
226
227 ok "WARP keys extracted."
228
229 # 5. Tailscale auth key
230 step "Tailscale authentication"
231
232 TS_AUTHKEY=""
233 if [[ -f ".env" ]]; then
234     EXISTING_TS_KEY=$(read_env_var TS_AUTHKEY)
235     if [[ -n "$EXISTING_TS_KEY" ]]; then
236         TS_AUTHKEY="$EXISTING_TS_KEY"
237         warn "Found existing TS_AUTHKEY in .env. Skipping prompt."
238     fi
239 fi
240
241 if [[ -z "$TS_AUTHKEY" ]]; then
242     echo ""
243     echo "Generate a key at: https://login.tailscale.com/admin/settings/keys"
244     echo "'Generate auth key' enable Reusable if you plan to re-run setup"
245     echo ""
246     read -rp "Paste your Tailscale auth key: " TS_AUTHKEY
247     echo ""
248 fi
249
250 [[ -z "$TS_AUTHKEY" ]] && die "Tailscale auth key is required."
251 [[ "$TS_AUTHKEY" != tskey-auth-* ]] && warn "Key doesn't look like a Tailscale auth key proceeding anyway."
252 ok "Auth key accepted."
253
254 # 6. AdGuard Home admin password
255 step "AdGuard Home admin account"
256
257 # Hardcoded default credentials to prevent lockout
258 # Username: admin
259 # Password: nullexit
260 if [[ -f ".env" ]]; then
261     EXISTING_AG_PASS=$(read_env_var ADGUARD_PASSWORD)
262     if [[ -n "$EXISTING_AG_PASS" ]]; then
263         ADGUARD_PASSWORD="$EXISTING_AG_PASS"
264         ADGUARD_PWD_ESC="$EXISTING_AG_PASS"
265         ok "Found existing ADGUARD_PASSWORD in .env."
266     else
267         ADGUARD_PASSWORD="nullexit"
268         ADGUARD_PWD_ESC="nullexit"
269         ok "Default password set to 'nullexit'."
270     fi
271 else
272     ADGUARD_PASSWORD="nullexit"
273     ADGUARD_PWD_ESC="nullexit"
274     ok "Default password set to 'nullexit'."

```

```

275 fi
276
277
278 # 7. Write .env
279 step "Writing .env"
280
281 if [[ -f ".env" ]]; then
282     read -rp " .env already exists. Overwrite? [y/N]: " OVERWRITE
283     if [[ "$OVERWRITE" != "y" && "$OVERWRITE" != "Y" ]]; then
284         warn "Keeping existing .env."
285     else
286         WRITE_ENV=1
287     fi
288 else
289     WRITE_ENV=1
290 fi
291
292 if [[ "${WRITE_ENV:-0}" == "1" ]]; then
293     # Preserve existing configuration flags if .env already exists
294     EXISTING_PROFILE="medium"
295     EXISTING_MSS="1120"
296     EXISTING_HIJACK="true"
297     EXISTING_EXIT="true"
298     EXISTING_THRESH="6"
299     EXISTING_KILL="false"
300     EXISTING_BLOCKED="kp il"
301     EXISTING_HOST_MTU="1200"
302
303     if [[ -f ".env" ]]; then
304         [[ -n "$(read_env_var GATEWAY_RULE_PROFILE)" ]] && EXISTING_PROFILE=$(read_env_var GATEWAY_RULE_PROFILE)
305         [[ -n "$(read_env_var GATEWAY_MSS)" ]] && EXISTING_MSS=$(read_env_var GATEWAY_MSS)
306         [[ -n "$(read_env_var GATEWAY_HIJACK_HOST)" ]] && EXISTING_HIJACK=$(read_env_var GATEWAY_HIJACK_HOST)
307         [[ -n "$(read_env_var GATEWAY_USE_EXIT_NODE)" ]] && EXISTING_EXIT=$(read_env_var GATEWAY_USE_EXIT_NODE)
308         [[ -n "$(read_env_var WARP_FAIL_THRESHOLD)" ]] && EXISTING_THRESH=$(read_env_var WARP_FAIL_THRESHOLD)
309         [[ -n "$(read_env_var KILL_SWITCH)" ]] && EXISTING_KILL=$(read_env_var KILL_SWITCH)
310         [[ -n "$(read_env_var BLOCKED_COUNTRIES)" ]] && EXISTING_BLOCKED=$(read_env_var BLOCKED_COUNTRIES)
311         [[ -n "$(read_env_var HOST_MTU)" ]] && EXISTING_HOST_MTU=$(read_env_var HOST_MTU)
312     fi
313
314     cat > .env <<EOF
315 WIREGUARD_PRIVATE_KEY=${PRIVATE_KEY}
316 WIREGUARD_PUBLIC_KEY=${PUBLIC_KEY}
317 WIREGUARD_ADDRESSES=${ADDRESSES}
318 TS_AUTHKEY=${TS_AUTHKEY}
319 GATEWAY_RULE_PROFILE=${EXISTING_PROFILE}
320 # Safe MSS for double-tunneled traffic (Tailscale + WARP). Change to 1180 if you experience slow speeds and
321 # have a healthy path.
322 GATEWAY_MSS=${EXISTING_MSS}
323 # Lower the host Tailscale MTU to 1200 to prevent fragmentation when passing through the 1280-byte WARP tunnel.
324 HOST_MTU=${EXISTING_HOST_MTU}
325 # Set to false to run as a 'Headless Server' (Docker acts as an exit node, but the host Mac/Linux's own
326 # internet is NOT hijacked).
327 GATEWAY_HIJACK_HOST=${EXISTING_HIJACK}
328 GATEWAY_USE_EXIT_NODE=${EXISTING_EXIT}
329 WARP_FAIL_THRESHOLD=${EXISTING_THRESH}
330 KILL_SWITCH=${EXISTING_KILL}
331 ADGUARD_PASSWORD=${ADGUARD_PASSWORD}
332 BLOCKED_COUNTRIES="${EXISTING_BLOCKED}"
333 EOF
334     ok ".env written."
335 fi
336
337 # 8. Prepare directories
338 step "Preparing AdGuard directories"
339
340 mkdir -p adguard/conf adguard/work/userfilters
341
342 # Remove empty AdGuardHome.yaml that would cause a crash loop (same logic as Toggle-Gateway scripts)
343 if [[ -f "adguard/conf/AdGuardHome.yaml" && ! -s "adguard/conf/AdGuardHome.yaml" ]]; then
344     rm "adguard/conf/AdGuardHome.yaml"
345     warn "Removed empty AdGuardHome.yaml to prevent crash loop."
346 fi
347
348 ok "Directories ready."
349
350 # 9. Compile DNS filter rules
351 step "Compiling DNS filter rules (this may take a minute)"
352
353 # 10. Start containers

```

```

353 step "Starting containers"
354 # Note: tailscale depends on adguardhome being healthy (port 5335 up),
355 # so Docker Compose will hold it back automatically until the wizard is done.
356 docker compose up -d
357 ok "Containers starting."
358
359 # 11. Wait for AdGuard Home setup wizard
360 step "Waiting for AdGuard Home setup wizard"
361 echo " (up to 60 seconds...)"
362
363 MAX=60; ELAPSED=0
364 until docker exec routing-fix curl -sf http://127.0.0.1:3000 >> output.log 2>&1; do
365     sleep 2; ELAPSED=$((ELAPSED + 2))
366     [[ $ELAPSED -ge $MAX ]] && die "AdGuard Home didn't come up.\nDebug with: docker compose logs adguardhome"
367 done
368
369 ok "AdGuard Home is up."
370
371 # 12. Configure AdGuard Home via API
372 step "Configuring AdGuard Home"
373
374 # Run all API calls in a single docker exec sh -c so the session cookie file
375 # is shared across calls without leaving the container.
376 docker exec routing-fix sh -c "
377     set -e
378
379     # Step 1: Complete the setup wizard (sets admin creds + DNS port to 5335)
380     curl -sf -X POST http://127.0.0.1:3000/control/install/configure \
381         -H 'Content-Type: application/json' \
382         -d '{"web":{"ip":"0.0.0.0"},"port":3000},"dns":{"ip":"0.0.0.0"},"port":5335},"username":"admin","password":"'${ADGUARD_PWD_ESC}'" \
383         >/dev/null || true
384
385     # Step 2: Wait for AdGuard to restart after wizard
386     sleep 4
387     WAITED=0
388     until curl -sf http://127.0.0.1:3000 >/dev/null 2>&1; do
389         sleep 2; WAITED=$((WAITED + 2))
390         [ $WAITED -ge 30 ] && { echo 'AdGuard did not restart in time'; exit 1; }
391     done
392
393     # Step 3: Login to get session cookie
394     curl -sf -X POST http://127.0.0.1:3000/control/login \
395         -H 'Content-Type: application/json' \
396         -d '{"name":"admin","password":"'${ADGUARD_PWD_ESC}'" \
397         -c /tmp/agh.cookie >/dev/null
398
399     # Step 4: Point upstream DNS back through the WARP tunnel
400     curl -sf -X POST http://127.0.0.1:3000/control/dns_config \
401         -H 'Content-Type: application/json' \
402         -b /tmp/agh.cookie \
403         -d '{"upstream_dns":["127.0.0.1:53"],"bootstrap_dns":["1.1.1.1","8.8.8.8"],"upstream_mode":"load_balance"}' \
404         >/dev/null
405
406     # Step 5: Register the compiled rules file as a filter list
407     curl -sf -X POST http://127.0.0.1:3000/control/filtering/add_url \
408         -H 'Content-Type: application/json' \
409         -b /tmp/agh.cookie \
410         -d '{"name":"Custom Compiled Rules","url":"/opt/adguardhome/work/userfilters/compiled_rules.txt"}' \
411         >/dev/null || true
412
413     # Step 6: Force a filter refresh to load the rules immediately
414     curl -sf -X POST http://127.0.0.1:3000/control/filtering/refresh \
415         -H 'Content-Type: application/json' \
416         -b /tmp/agh.cookie \
417         -d '{"whitelist":false}' \
418         >/dev/null || true
419
420     rm -f /tmp/agh.cookie
421 "
422
423 ok "AdGuard Home configured."
424
425 # 13. Wait for Tailscale
426 step "Waiting for Tailscale to authenticate"
427 echo " (Tailscale only starts once AdGuard's healthcheck passes up to 90s)"
428
429 MAX=90; ELAPSED=0; TS_IP=""
430 until [[ -n "$TS_IP" ]]; do

```

```

431 TS_IP=$(docker exec tailscale tailscale ip -4 2>> output.log || true)
432 if [[ -z "$TS_IP" ]]; then
433     sleep 3; ELAPSED=$((ELAPSED + 3))
434     if [[ $ELAPSED -ge $MAX ]]; then
435         warn "Tailscale hasn't authenticated yet."
436         warn "Once it does, re-run the toggle script it will resolve the IP dynamically."
437         break
438     fi
439 fi
440 done
441
442 if [[ -n "$TS_IP" ]]; then
443     ok "Tailscale IP: ${TS_IP} (resolved dynamically by the toggle script on each startup)"
444 fi

```

45 scripts/unlock-files.sh

```

1 #!/bin/bash
2 # Resolve project root relative to this script's location, so the script
3 # works regardless of the caller's CWD.
4 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
5 PROJECT_ROOT="$(cd "$SCRIPT_DIR/.." && pwd)"
6
7 echo "Unlocking .env..."
8 if [ -f "$PROJECT_ROOT/.env" ]; then
9     sudo cat "$PROJECT_ROOT/.env" > "$PROJECT_ROOT/.env.tmp" && mv "$PROJECT_ROOT/.env.tmp" "$PROJECT_ROOT/.env"
10    echo ".env unlocked"
11 fi
12
13 echo "Unlocking AdGuard cache files..."
14 for f in "$PROJECT_ROOT"/adguard/work/userfilters/compiled_rules.txt \
15         "$PROJECT_ROOT"/adguard/work/userfilters/ip_blocklist.ipset \
16         "$PROJECT_ROOT"/adguard/work/userfilters/cache/*.txt \
17         "$PROJECT_ROOT"/adguard/work/userfilters/cache/ip/*.txt \
18         "$PROJECT_ROOT"/adguard/work/data/filters/*.txt; do
19     if [ -f "$f" ]; then
20         cat "$f" > "$f.tmp" && mv "$f.tmp" "$f"
21         echo "Unlocked $f"
22     fi
23 done
24
25 echo ""
26 echo "All locked files have been successfully bypassed via atomic rename!"

```

46 tor-bridges.txt

47 white_list.txt

```

1 0.client-channel.google.com
2 1drv.com
3 2.android.pool.ntp.org
4 akamaihd.net
5 akamaitechnologies.com
6 akamaized.net
7 amazonaws.com
8 android.clients.google.com
9 api-adservices.apple.com
10 api.ipify.org
11 api.rlje.net
12 app-api.ted.com
13 appleid.apple.com
14 apps.skype.com
15 appsbackup-pa.clients6.google.com
16 appsbackup-pa.googleapis.com
17 appspot-preview.l.google.com
18 apt.sonarr.tv
19 aspnetscdn.com

```

```

20 attestation.xboxlive.com
21 ax.phobos.apple.com.edgesuite.net
22 books-analytics-events.apple.com
23 brightcove.net
24 c.s-microsoft.com
25 cdn.cloudflare.net
26 cdn.embedly.com
27 cdn.optimizely.com
28 cdn.vidible.tv
29 cdn2.optimizely.com
30 cdn3.optimizely.com
31 cdnjs.cloudflare.com
32 cert.mgt.xboxlive.com
33 clientconfig.passport.net
34 clients1.google.com
35 clients2.google.com
36 clients3.google.com
37 clients4.google.com
38 clients5.google.com
39 clients6.google.com
40 connectivitycheck.android.com
41 connectivitycheck.gstatic.com
42 continuum.dds.microsoft.com
43 cpms.spop10.ams.plex.bz
44 cpms35.spop10.ams.plex.bz
45 cse.google.com
46 ctldl.windowsupdate.com
47 cws.conviva.com
48 d2c8v52ll5s99u.cloudfront.net
49 d2gatte9o95jao.cloudfront.net
50 dashboard.plex.tv
51 dataplicity.com
52 def-vef.xboxlive.com
53 delivery.vidible.tv
54 dev.virtualearth.net
55 device.auth.xboxlive.com
56 display.ugc.bazaarvoice.com
57 displaycatalog.mp.microsoft.com
58 dl.delivery.mp.microsoft.com
59 dl.dropbox.com
60 dl.dropboxusercontent.com
61 dns.msftncsi.com
62 download.sonarr.tv
63 drift.com
64 driftt.com
65 dynupdate.no-ip.com
66 ecn.dev.virtualearth.net
67 edge.api.brightcove.com
68 eds.xboxlive.com
69 fonts.gstatic.com
70 forums.sonarr.tv
71 g.live.com
72 geo-prod.do.dsp.mp.microsoft.com
73 geo3.ggpht.com
74 gfwsl.geforce.com
75 giphy.com
76 github.com
77 github.io
78 googleapis.com
79 gravatar.com
80 gstatic.com
81 help.ui.xboxlive.com
82 hls.ted.com
83 i.ytimg.com
84 il.ytimg.com
85 iadsdk.apple.com
86 imagesak.secureserver.net
87 img.vidible.tv
88 imgix.net
89 imgs.xkcd.com
90 instantmessaging-pa.googleapis.com
91 intercom.io
92 jquery.com
93 jsdelivr.net
94 keystone.mwbsys.com
95 lastfm-img2.akamaized.net
96 licensing.xboxlive.com
97 live.com
98 livepassdl.conviva.com
99 login.live.com
100 login.microsoftonline.com

```

101 manifest.googlevideo.com
102 meta-db-worker02.pop.ric.plex.bz
103 meta.plex.bz
104 meta.plex.tv
105 microsoftonline.com
106 msftncsi.com
107 my.plexapp.com
108 nexusrules.officeapps.live.com
109 nine.plugins.plexapp.com
110 no-ip.com
111 node.plexapp.com
112 notes-analytics-events.apple.com
113 notify.xboxlive.com
114 npr-news.streaming.adswizz.com
115 ns1.dropbox.com
116 ns2.dropbox.com
117 o1.email.plex.tv
118 o2.sg0.plex.tv
119 ocsp.apple.com
120 office.com
121 office.net
122 office365.com
123 officeclient.microsoft.com
124 om.cbsi.com
125 onedrive.live.com
126 outlook.live.com
127 outlook.office365.com
128 pings.conviva.com
129 placeholder.it
130 placeholderit.imgix.net
131 players.brightcove.net
132 pricelist.skype.com
133 products.office.com
134 proxy.plex.bz
135 proxy.plex.tv
136 proxy02.pop.ord.plex.bz
137 pubsub.plex.bz
138 pubsub.plex.tv
139 raw.githubusercontent.com
140 redirector.googlevideo.com
141 res.cloudinary.com
142 s.gateway.messenger.live.com
143 s.marketwatch.com
144 s.youtube.com
145 s.ytimg.com
146 s1.wp.com
147 s2.youtube.com
148 s3.amazonaws.com
149 sa.symcb.com
150 secure.avangate.com
151 secure.brightcove.com
152 secure.surveymonkey.com
153 services.sonarr.tv
154 skyhook.sonarr.tv
155 spclient.wg.spotify.com
156 ssl.p.jwpcdn.com
157 staging.plex.tv
158 status.plex.tv
159 t.co
160 t0.ssl.ak.dynamic.tiles.virtualearth.net
161 t0.ssl.ak.tiles.virtualearth.net
162 tawk.to
163 tedcdn.com
164 themoviedb.com
165 thetvdb.com
166 tinyurl.com
167 title.auth.xboxlive.com
168 title.mgt.xboxlive.com
169 traffic.libsyn.com
170 tvdb2.plex.tv
171 tvthemes.plexapp.com
172 twimg.com
173 ui.skype.com
174 video-stats.l.google.com
175 videos.vidible.tv
176 vidtech.cbsinteractive.com
177 weather-analytics-events.apple.com
178 widget-cdn.rpxnow.com
179 win10.ipv6.microsoft.com
180 wp.com
181 ws.audioscrobbler.com

```

182 www.dataplicity.com
183 www.googleapis.com
184 www.msftconnecttest.com
185 www.msftncsi.com
186 www.no-ip.com
187 www.youtube-nocookie.com
188 xbox.ipv6.microsoft.com
189 xboxexperiencesprod.experimentation.xboxlive.com
190 xflight.xboxlive.com
191 xkms.xboxlive.com
192 xsts.auth.xboxlive.com
193youtu.be
194 youtube-nocookie.com
195 yt3.ggpht.com
196 zee.cws.conviva.com
197
198 # YouTube family base domains so every subdomain (video stream servers on
199 # *.googlevideo.com, image hosts on *.yting.com, profile thumbnails on
200 # *.ggpht.com, etc.) is automatically allowlisted via the AdGuard '||domain^'
201 # wildcard. The specific s.youtube.com / manifest.googlevideo.com entries
202 # above are kept for clarity but will be optimized away by rule-compiler
203 # because their parents are now in this list.
204 youtube.com
205 googlevideo.com
206 yting.com
207 ggpht.com
208 gvt1.com
209 gvt2.com
210 gvt3.com
211 gvt1.net
212 gvt2.net
213 gvt3.net
214
215 edge-mqtt.facebook.com
216 gateway.instagram.com
217 i.instagram.com
218
219 facebook.com
220 fb.com
221 fbcdn.net
222 fbsbx.com
223 messenger.com
224 whatsapp.com
225 web.whatsapp.com
226 instagram.com
227 cdninstagram.com

```